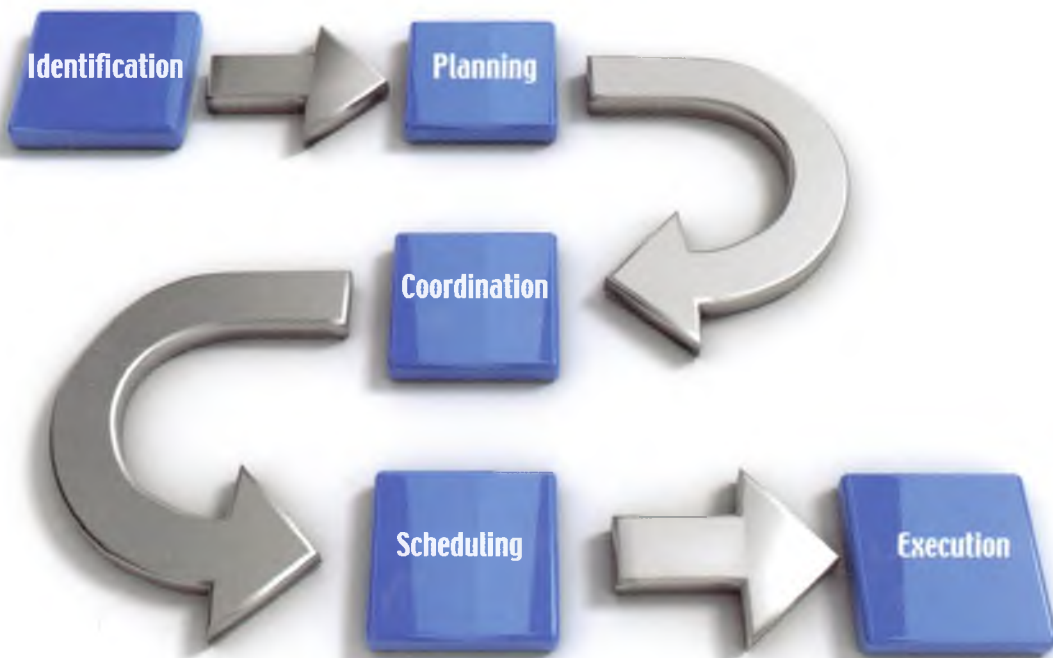


Maintenance Planning, Coordination, Scheduling

SECOND
EDITION



Don Nyman & Joel Levitt

Maintenance Planning, Coordination and Scheduling

Second Edition

Don Nyman

Joel Levitt

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DEDICATION

We each dedicate this book to our respective families for all their love, patience, understanding, support, and sacrifices during the several decades we have pursued our careers—often away from home a majority of the weeks in each calendar year.

Barbara Spender Nyman, Don's wife, has put up with 54 years of career and travel while raising their wonderful family of 4 children, 10 grandchildren, and 5 great-grand children. How's that for dedication?

Because this book chronicles a wide array of maintenance management basics, it builds on and elaborates on experiences developed not only by the authors, but also by others. We are therefore indebted to many associates, clients, and others who have worked in the field of maintenance through the decades.

Finally, this book is also dedicated to you maintenance professionals who labor to keep our factories running, our airports open, and our trucks on the road. You do your magic without visibility, often without a lot of status, and sometimes without even a thank you. We would like to acknowledge your dedication, your long hours, and your sense of duty.

We are grateful for the support of our friends at Industrial Press: John Carleo, Robert Weinstein, and Janet Romano.

Don Nyman
Joel Levitt
June 2010

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TABLE OF CONTENTS

<i>About the Authors</i>	v
<i>Preface</i>	xi
<i>The Purpose of this Book</i>	xiii
<i>Introduction</i>	xv
1. Selling Planning, Coordination, and Scheduling to Management and Operations	1
Reactive Maintenance	2
Bottom Line Impact of Maintenance/Reliability Excellence	7
Contribution of Planning Coordination, and Scheduling	10
Symptoms of Ineffective Job Planning	18
Convey the Many Benefits that Accrue to Each Stakeholder	19
Emphasize the Maintenance Deliverables	23
2. Prerequisites to Success Preparation	25
Environment	25
Organization	26
Informational Database	27
Materials Support	28
Reliability	29
Planning, Coordination, and Scheduling	30
Job Execution	31
3. The Nature of Maintenance Activities and Related Organizational Structure	35
Organization by Work Type	36
4. Where Planning Fits into Good Maintenance Practices	41
Should Work Preparation Be a Separate and Distinct Function?	43
The Proven Answer	45
Channels of Coordination and Communication	47
Working Liaisons	49

Should Planning Be Separated from Scheduling?	49
Relationship with Other Functions	52
5. Roles and Responsibilities	55
Factors Influencing Planner/Scheduler Control Span	59
6. Managing the Planning and Scheduling Function	65
Management of Planners	65
Managing Planning	66
Position Benefits—Manager of Maintenance/Reliability Support Services	67
7. Backlog Management	69
Backlog Management	70
Job Status	73
8. Development of Work Programs and the Maintenance Calendar	77
Annual Master Schedules	81
9. Sizing the Maintenance Staff	83
Staffing Processes	83
Temporary Staffing Demands	92
10. The Proactive Process of Work Preparation	95
Steps of the Work Preparation Process	97
11. Detailed Planning of Individual Jobs	101
Screening Work Requests	101
Assessing and Scoping the Job	104
Dealing with Job “Creep”	106
Job Research	107
Detailed Build-Up of Job Steps and Requirements	109
Assembly of Job Package	111
Equipment Access, Safety and Statutory Permits	113

12. Materials Support	119
Necessity	119
Materials Management	121
Tools of Materials Management	122
Roles and Responsibilities	135
 13. Work Measurement	 137
History	137
Applications of Maintenance Work Measurement	139
Levels of Maintenance Work Methodology	141
Best Methodology for Each Application	147
 14. Analytical Estimating	 151
A Recommended Approach to Maintenance Work Measurement	151
Comparative Time Estimating	160
Building an Estimate	167
 15. Coordination with Operations	 171
Set Up of the Coordination Process	173
Preparation for the Weekly Coordination Meeting	173
 16. Scheduling Maintenance Work	 181
The Weekly Expectation	181
Instructions for Preparing Schedules	189
Other Approaches to Scheduling	199
Additional Offerings	199
 17. Job Execution and Supervision	 203
Linking the Schedule to Supervisor Responsibilities	203
The Morning Meeting	208
 18. Job Feedback, Close Out, Analysis, and Schedule Compliance	 211
Job Feedback	211
Job Closeout	212
Analysis	213

Schedule Compliance	214
Reasons for Non-Compliance	214
Calculation of Schedule Compliance	215
19. Computerized Maintenance Management Information System (CMMIS) in Support of Planning, Coordination, and Scheduling	219
Where Does Your Work Come From?	220
20. Planning Major Maintenance Projects	227
Planning Process for Major Maintenance Effects	228
Who Should apply Project Management Techniques?	241
21. Network Analysis-Development and Depiction	243
General Discussion of the Critical Path	244
Appendices	251
A. Position Description – Manager of Maintenance Support	253
B. Recognizing the Pitfalls in Planning for Others	255
C. Job Description – Maintenance Planner/Scheduler	259
D. Labor Library – Example	265
E. Alternate Approach to Scheduling	267
F. Typical Sources of Planned Maintenance Work	275
G. Planner Activity Log Sheet	277
H. Determination Worksheet Ratio of Technicians to Planner/Schedulers	279
I. Assessment – Maintenance Organization	281
J. Assessment – Maintenance Computer Information Support	287
K. Assessment – Maintenance Work/Job Planning	289
L. Assessment – Maintenance Work Measurement	295
M. Assessment – Maintenance Material Support & Control	297
N. Assessment – Maintenance Coordination & Scheduling	303
O. Assessment Summary & Comparison to Benchmarks	309
Glossary	311
Index	314

PREFACE

The overriding objective of business entities worldwide is to achieve world-class stature—being competitive with the best in the world in every aspect of the business. In this state no other organization has a competitive advantage. (This assumes no governmental support in violation of international trade agreements.) To become world-class is dependent upon asset reliability and maintenance excellence, neither of which is attainable without comprehensively preparing for the effective execution of maintenance work. Such preparation encompasses *planning, coordination, and scheduling* which are the focus of this book.

Regardless of the industry in which they are engaged, organizations are responsible for optimized utilization of their installed asset capacity. The goal is for capital assets to yield the capacity for which they were designed, and on which they were economically justified, thus, allowing the business entity to thrive. This quest is crucial to Return on Invested Capital. So, one intention of this book is to help organizations pursue Maintenance/Reliability Excellence, which is that state of maintenance management and performance that effectively applies leading edge policies, procedures, systems, structures, methods, and technologies to the realization of optimal reliability.

The key to achieving maintenance/reliability excellence is nothing new. It has always been and still remains: get the basics right and make reliability a goal of the entire organization. Foremost among the basics commonly contained within world-class programs are planning, parts acquisition, work measurement, coordination, and scheduling. Together these basics constitute the preparation required for effective execution of maintenance work. Throughout this book we will use the term job preparation when speaking of these distinct tasks in their totality.

Well-planned, effectively communicated, and properly scheduled jobs accomplish more work, more efficiently, and at lower cost. Work properly prepared in this fashion disturbs operations less frequently, requires less equipment downtime, and is accomplished with higher quality—which in combination equal reliability.

Proper preparation also yields greater job satisfaction for maintenance technicians, and higher overall organizational morale. In fact, W.E. Deming said that pride in a job well done is the greatest motivator of workers. Planning, coordination, and then scheduling remove the barriers

to workers performing their jobs with the right tools and materials and with strong motivation.

Without proper coordination and scheduling, the crucial proactive routines optimized through other vital techniques (RCM, Predictive Maintenance, and Condition Based Maintenance) most likely will not be performed when due. Therefore, regardless of size, every organization must prepare for effective execution of its maintenance/reliability workload.

THE PURPOSE OF THIS BOOK

Since writing the first edition of *Maintenance Planning, Coordination & Scheduling* in 2001, the authors have individually written additional maintenance/reliability associated books. For this second edition, we have drawn from those books to help clarify the posture of planning, coordination, and scheduling within the broader quest for asset reliability supportive of operational and enterprise excellence. Still, we retain our focus on the preparatory tasks that lead to effective utilization and application of maintenance resources. As necessary, other aspects of Maintenance/Reliability Excellence are discussed to clarify the scope, responsibilities, and contributions of the Maintenance Planner/Scheduler as well as other functions impacted by or supportive of job preparation, execution, and completion.

This book tells the whole story of maintenance planning, coordination, and scheduling from beginning to end. It is written for four groups:

- It is a vital training document for planners as well as an educational document for those to whom planners are responsible. They should read it from beginning to end.
- For experienced maintenance professionals, this book can be used as a reference for clarifying steps of the preparation process or for gaining insight into procedures, forms, and control structures.
- It is also important for those who interface with the planning and scheduling function and are dependent upon the many contributions of planning and scheduling for operational excellence.
- Finally, this book contains many important insights for upper management, which in turn must create the proactive culture and environment that are essential to realizing and preserving World-Class Reliability.

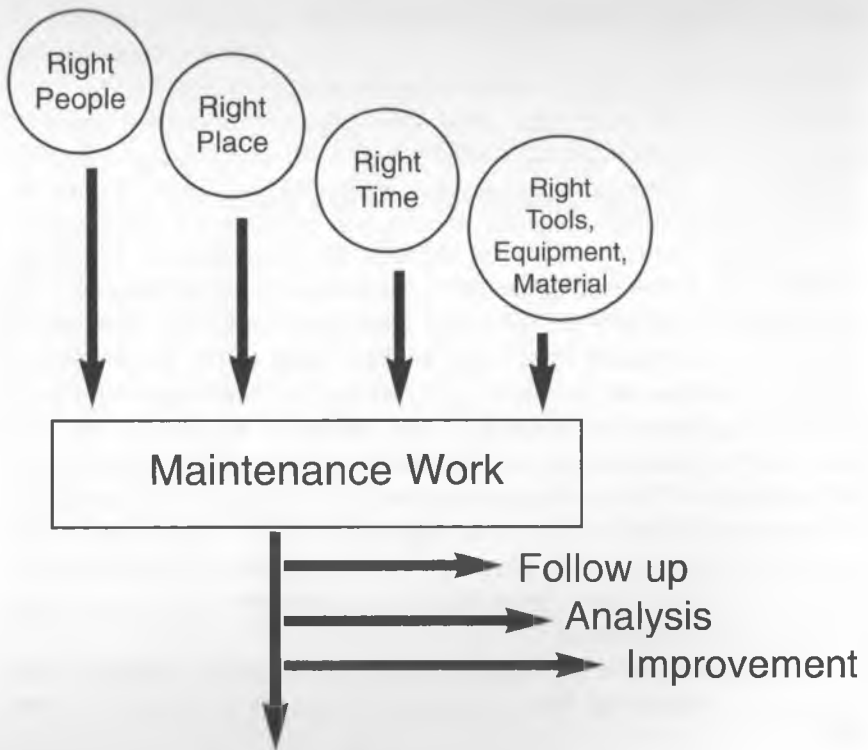


Figure 1. The Planner's Role

INTRODUCTION

Organizations the world over are responsible for the optimal utilization of the output capacity of their installed capital assets. Only in this manner can they achieve world-class business objectives. Product quality, on time delivery, regulatory compliance, profitability, and return-on-invested capital are all dependent upon reliable capacity to the limits of design capacity. Realization of these objectives requires a proactive culture and environment; in contrast to the reactive culture and environment in which a significant majority of organizations now operate (focused on short-term survival and quarterly financial results). Organizations must be successful in this quest, if they are to survive and thrive.

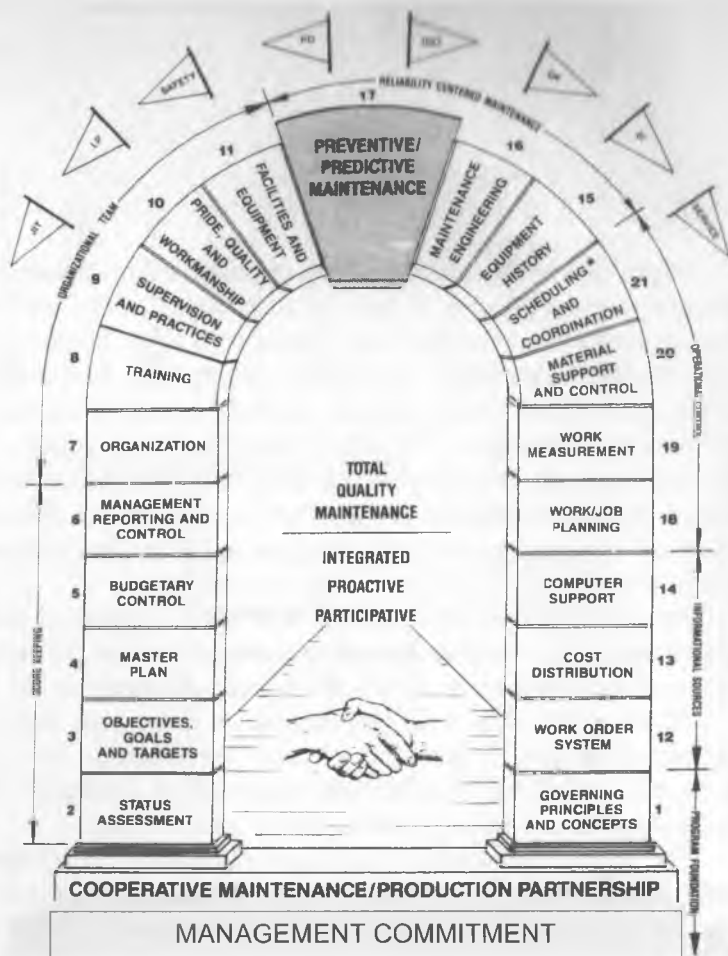
Therefore, before addressing the workings of planning, coordination, and scheduling, we must first address the culture and environment that senior management must nurture throughout the organization.

World-class operational performance is dependent upon optimized asset reliability. To deliver such asset performance, the maintenance function must spend each maintenance dollar optimally. This is where planning, coordination, and scheduling come to play.

The holistic approach to Maintenance/Reliability Excellence, the essential cultural change, and "integrated maintenance management partnership" are illustrated as the Maintenance Arch in Figure 1. Of the many activities and functions depicted, planning, coordination, and scheduling have the greatest impact on timely and effective accomplishment of maintenance work and thereby make the most profound contribution toward world-class operational excellence.

The Maintenance Arch

The components that make up the maintenance arch represent the integration of twenty-one building blocks that are essential to the achievement of maintenance/reliability excellence in support of world-class operations through asset/capacity reliability. The concept is built upon a bedrock of sustained management commitment, support, and involvement. It is built upon a footer of a cooperative operations/maintenance



Maintenance Arch (Gateway to Maintenance/Reliability Excellence)

partnership. Maintenance cannot achieve the objective alone. Realization requires commitment by the entire organization.

The building blocks of the arch are interdependently linked. It is impossible to reap full benefit of any single block without harmoniously addressing each of the other blocks. That said, the focus of this book will be on the four blocks (18, 19, 20, 21) that constitute essential preparation for effective execution of the maintenance workload. They are Planning, Materials Support, Work Measurement, and Coordination and

scheduling. Essential relationships with other building blocks are clarified, but they are not covered comprehensively. A quick synopsis is presented below. The twenty-one building blocks are presented in the six groupings shown around the periphery of the arch:

1. FIRM FOUNDATION

Successful processes begin with shared beliefs (Governing Principles and Concepts).

2. BOOKKEEPING

Status Assessment measures current operational state relative to shared beliefs.

The Master Plan sets forth actions, responsibilities, time lines, and essential resources required to close the gap between the current state and established Objectives, Goals, and Targets.

Diligent refinement is required to support the Master Plan. Management Reporting is the feedback vehicle by which sustained management commitment is earned.

3. INFORMATIONAL SOURCES

Twelve building blocks are dependent upon complete, reliable, and timely information. The source of such information is a well-specified, designed, and installed Computerized Work Order System. The information within that system is worthless without complete, reliable, and timely Cost Distribution.

4. ORGANIZATIONAL TEAM

The Organization must be structured for pro-action rather than for reactive response. Skills Training and Facilities, Tools, and Equipment are essential to organizational Pride and Effectiveness if Supervision is to assure Job Quality, including adherence to Planned Job Packages.

5. RELIABILITY CENTERED MAINTENANCE

Preventive/Predictive Maintenance is the keystone of the Arch and the vehicle by which reliability is assured. It requires strong Maintenance Engineering utilizing Equipment History and

deploying Root Cause Failure Analysis to optimize the PM/PdM program.

6. OPERATIONAL CONTROL

The preparatory trio, Planning, Coordination, and Scheduling, assure: that the right work, is accomplished at the right time, in a safe and effective manner that contributes substantially to optimized utilization of asset capacity. Work Measurement and Materials Support are integral to Preparation.

The Planning and Scheduling functions are the hub from which all proactive maintenance activity is prepared and coordinated.

Planning must be at the core of the maintenance effort because it provides for reliable delivery of all the other proactive programs. Combined with PM/PdM engineered through RCM, these programs produce quantum benefits that accrue to the enterprise's bottom line.

The several banners that fly from the arch convey the significant contribution that the maintenance function makes to reliability objectives of the overall business enterprise. This view is directly opposite of the common misconception of maintenance as a "necessary evil and a drain on profitability." The banners (acronyms and words) represent several management initiatives that are maintenance dependent—Just In Time, Loss Prevention, Employee Safety, Avoidance of Property Damage, Certification, Quality Assurance, Avoidance of Business Interruption, and Customer Service among others.

Clarification of Preparatory Activities

There is often confusion among the terms planning, coordination, and scheduling. These three distinct activities are closely related. Confusion stems from the fact that they are often compressed together and performed by the same individual, particularly in smaller organizations or on smaller jobs. However, they require different skill sets, and are performed at different points of the preparatory cycle. Accordingly, they are often performed by different individuals if the maintenance organization is large enough to justify several staff support positions.

Planning (how to do the job — Chapters 7 through 11): Planning is the development of a detailed process to achieve an end, e.g., a mainte-

nance repair or rebuild. It is the advanced preparation for a specific job so it can be performed in an efficient, reliable, and safe manner (Figure 2).

Work Measurement is a critical element of Job Planning. It is the development of a standard or estimate for a specific job; including crew size, labor-hours, and duration time.

Material Requirements is another critical element of Job Planning. It involves determining the parts, materials, tools, and equipment necessary to complete a specific job. It also includes initiation of the associated reservation and/or procurement process.

Planning is a process of detailed analysis that determines and describes the work to be performed, the sequence of associated tasks, methods to be used for their performance, and the required resources—including skills, crew size, labor-hours, parts, materials, special tools, and equipment, plus an estimate of total cost. It also includes identification of safety precautions, required permits, communication requirements, and reference documents such as drawings and wiring diagrams. It addresses essential pre-shutdown preparation, shutdown, execution and post shutdown start-up efforts.

Coordination (Chapter 15): Coordination ensures that all necessary logistics have been coordinated for job execution at a future date. There are two phases. Both occur after Planning and before Scheduling. The first phase commences as Planning is completed. It encompasses logistical efforts to assemble all required, non-labor resources so the job can be coded "Ready to be Scheduled" (as a status code within the computerized work order system). It is accomplished in harmony with Purchasing, Receiving, and Stores.

The second phase of Coordination occurs within a weekly coordination meeting that immediately precedes scheduling. operations, engineering, and maintenance review all "ready to be scheduled" jobs and reach agreement as to which are the most important to be performed during the coming schedule week. This phase fully considers: the limit of maintenance resources, feasibility of releasing the involved asset to Maintenance at a specific point of the schedule week, and feasibility of committing the specifically required maintenance labor-resources at the same point of the schedule week.

The agreement becomes a contract between Operations and Maintenance. The resulting Weekly Schedule has holding power because both parties participated in the development of it; therefore, both have ownership of it. It can no longer be referred to as: "that worthless maintenance schedule."

Scheduling (when to do the job—Chapter 16): Because it yields the earliest visible benefits, Scheduling is the “marketing arm” by which the essential, cultural conversion from a reactive to a pro-active environment is sold to Operations and Management. It is the documented allocation of labor resources and support equipment to specific jobs at times when Operations can make the associated assets available to Maintenance.

As part of the scheduling process, the Planner/Scheduler optimally allocates resources to specific jobs. However, as part of job execution (Chapter 17), the responsible maintenance supervisor makes the actual assignments to allocated individuals as documented on the schedule.

The resulting schedule reflects the anticipated work to be completed during the schedule week—given the limitation of maintenance resources expected to be available and paid for during the coming week. “We all accomplish more when working to achieve a published expectation.”

Considered together, Planning, Coordination, and Scheduling constitute Job Preparation. They are supportive functions distinct from line supervision (which is responsible for oversight of job execution) and are best performed by para-professional personnel. They are predicated on the principle that “the maintenance function achieves best results when each mechanic is given specific tasks to be completed in a definite time period (scheduling) in a specific manner (planning).” Mechanics do not plan for their own efficiency! The supervisor, the worker, and their colleagues should each know what is expected—including the goal and target for completion of each job.

“The productivity of work is not the responsibility of the worker but of the manager.” (Peter Drucker)

Reactive response puts speed ahead of reliability. Disregard the organizational vision, and the job will likely need to be done over. Doing work properly the first time requires effective Preparation. If there is insufficient time to do a job properly, when will there be time to do it over?



Selling Planning, Coordination, and Scheduling to Management and Operations

How can the crucial maintenance functions of planning, coordination, and scheduling be sold to management? Furthermore, how can all organizational units (Operations, Purchasing, Storeroom, Engineering, etc.) be truly convinced and committed to procedures necessary to capture the full benefit of planning and scheduling? Maintenance begins with two strikes because the function is poorly understood and underappreciated. Why invest in a function that is viewed only as a “necessary evil?” In the eyes of many, maintenance is a necessary evil that does not add value or contribute to the “bottom line.”

Let’s deconstruct this necessary evil conversation because it is a major barrier to our contribution to the company’s success. What impact does such a conversation have? How do you act if you are a necessary evil? Is this the basis for a healthy relationship? How do you contribute as a necessary evil; indeed, why would you even want to? If you want to be all you can be, how far can you go when everyone says that you are a necessary evil?

The misconception is that maintenance doesn’t contribute directly to the manufacture or delivery of anything. In modern parlance, the

common belief is that maintenance does not add value to the product. What we offer in this book is a new way of looking at maintenance. One such new viewpoint is to call maintenance "Capacity Assurance." We can prove that good maintenance practices actually sustain manufacturing capacity. The value of this preserved capacity dwarfs the cost of delivering reliable maintenance services.

So in fact, investments in maintenance yield significant returns and do add value. We must sell our contributions—this idea of capacity assurance with high quality, reliability, and customer satisfaction at lower unit cost.

Before the preparatory trio can be sold, management and the operating organization must be convinced as to the essentialness of a proactive culture and environment. Many organizations believe they are pursuing World-Class Reliability. Yet they continue to operate in a reactive mode, perpetually responding to never-ending equipment failures and doing little to eliminate root causes. Understanding precedes appreciation. Therefore, the first challenge is educational. We must educate before we can sell. Many senior and operational managers have little-to-no experience, interest, understanding, or appreciation of maintenance and its profound contributions to bottom line results. So, let us begin by clarifying the ills of reactive maintenance in contrast to the benefits of proactive maintenance.

REACTIVE MAINTENANCE

Today's quest for near-term profits often leads to downsized head count. As a result, thinly-staffed maintenance departments are all too common today. The maintenance labor resources that remain are consumed largely by asset failures that demand their urgent response. Although these urgent tasks require instant reaction, they are seldom the most important work regarding long-term objectives of the business entity. All they accomplish is perpetuation of the *reactive status quo*! On the other hand, important tasks (e.g., Preventive/Predictive Maintenance and Planned Backlog Relief) seldom need to be performed today and possibly not even this week.

To achieve reliability goals and, thereby, business objectives, there must be adequate maintenance resources to execute all three types of work performed by the maintenance function:

- Prompt response to true emergencies and urgencies
- Timely relief of all other maintenance workload (backlog)

that can be planned, coordinated, and scheduled prior to execution

- Reliable routines services such as preventive/predictive maintenance

Important work must be distinguished from simply urgent work. Beyond those resources required to respond to critical failures, sufficient resources must remain for performance of essential proactive work designed to reduce or eliminate the failures, yielding **Reliability**. The required resources must be appropriately distributed; we cannot allow the bulk of resources to be consumed by reactive fires and organizational indifference to the false inevitability of reactive maintenance. Chapter 7 speaks to the process by which backlog should be managed. Figure 1.1 compares common consumption of maintenance resources in a reactive operational environment to their distribution in a proactive environment/culture.

Nature of Work	Reactive Environment	Proactive Environment
Planned Backlog Relief	20 %	75 %
Preventive/Predictive Maintenance	25 %	15 %
Urgent Response	55 %	10 %

Figure 1.1 Comparative Distributions of Maintenance Resources

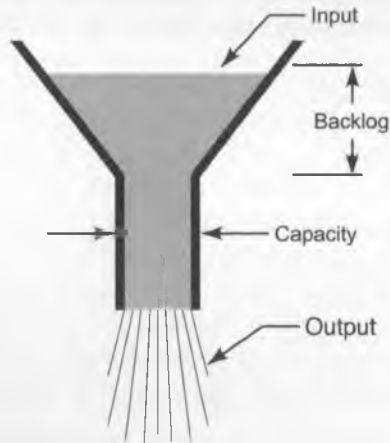
Failure to preserve the required balance between workload and properly distributed resources will mitigate all efforts to make the transition from reactive repair to proactive maintenance. Asset reliability becomes infeasible. This phenomenon is illustrated as the maintenance workload funnel in Figure 1.2.

Adage: *Output must be equal to Input. Otherwise, backlog will progressively increase to the point where assets fail with resultant inability to provide timely customer service.*

Insufficient proactive resources contribute to the next chronic problem associated with reactive operational environments: **Deferred Maintenance**. Assets have a limited life span. That life can be prolonged

Law:

Output must be equal to or greater than Input
or Backlog increases



Therefore, resources must be balanced with workload or transition from reactive to proactive maintenance is not feasible.

Figure 1.2 The Workload Funnel

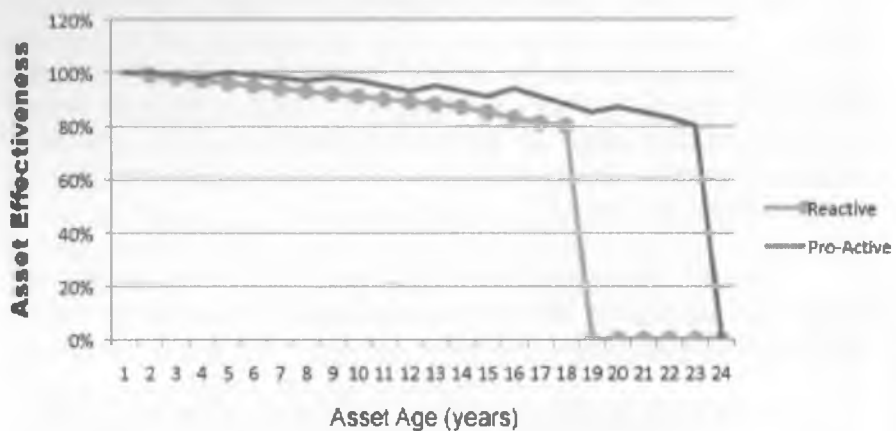


Figure 1.3 Proactive vs. Reactive Maintenance

by periodic restorative maintenance. Likewise, the production processes comprised of these assets have a theoretical capacity. This capacity diminishes in relation to the duration between proactive restorations that are designed to preserve actual capacity as near as possible to theoretical capacity. Because of budgets and operating schedules, Maintenance is often denied access to assets when periodic maintenance is due. The result is an increasing backlog of Deferred Maintenance that results in diminished condition, capacity, and life of assets. This common phenomenon is depicted in Figure 1.3.

When deferred maintenance is excessive, Maintenance and Operations fall into a deep hole of repetitive failures (Figure 1.4). Deferred Maintenance makes associated assets progressively less reliable. Resultant failures consume a progressively greater portion of maintenance resources. Loss of operating capacity has an adverse impact on bottom-line results. When profits go down, Management commonly responds with a campaign for headcount reduction. If maintenance is already in the reactive deferred maintenance pit, and maintenance resources are reduced, what is happening?

Consider this:

**FIRST RULE
OF HOLES:**

**When in one,
stop digging!**

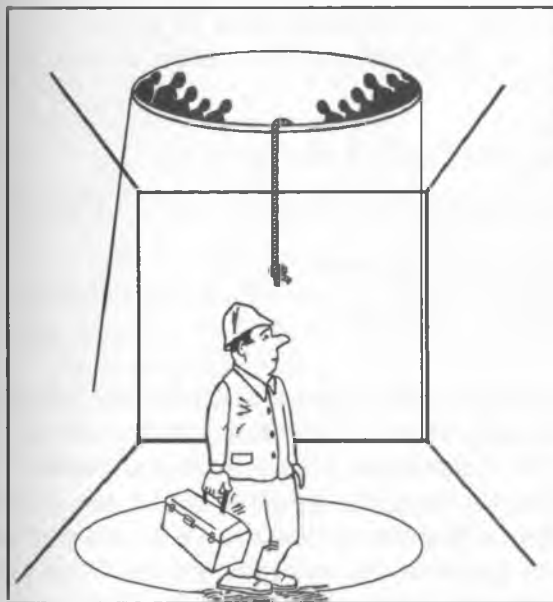


Figure 1.4 Deferred Maintenance Hole

Before progress toward reliability can occur, Maintenance must first climb vertically out of the "pit of repetitive failures" stemming from deferred maintenance. Equipment must be bootstrapped back to a state of maintainable reliability. Ultimately, Maintenance excellence can deliver both operational reliability and maintenance cost reduction—provided reliability is pursued first. Then, as reliability is achieved, cost reduction will follow.

In summary, to accept the Reactive/Deferred Maintenance status quo is to blatantly discard all the broadly sought management initiatives, several of which are depicted as banners flying from the Maintenance/Reliability Arch (Figure 1 in the Introduction):

- Reliable Capacity
- Quality Assurance
- Safety
- Regulatory Compliance
- Just In Time
- Customer Service
- ISO Certification
- Property Damage
- Loss Prevention
- Business Interruption
- Asset Preservation
- Return on Capital Investments
- Support of Short Range Business Plan
- Support of Long Range Business Plan

How can anyone think of Maintenance as a necessary evil or a non-contributor to profitability? Those who do think so are associating Maintenance only with the direct costs generated by the function as reflected in the Maintenance budget; depicted as the tip of the Maintenance Iceberg (Figure 1.5) above the waterline. How about all the contributions Maintenance makes to the reduction and elimination of costs and losses reflected elsewhere in the Operating and Capital budgets and depicted below the waterline of the iceberg?



Figure 1.5 The Maintenance/Reliability Iceberg

Reactive Maintenance and the associated breakdown costs significantly more than Proactive Maintenance does (initially below the waterline, ultimately throughout the entire iceberg).

Costs stemming from breakdowns (beneath the waterline) can be significant; typically 10-to-20 times more than the direct cost to repair (above the waterline). Unfortunately for the respect in which maintenance is held, many of the above indirect costs (below the waterline) are shrouded or invisible within most accounting systems.

BOTTOM-LINE IMPACT OF MAINTENANCE/RELIABILITY EXCELLENCE

Up to this point, all discussion has related to a broad organizational understanding that fosters an appreciation for the contributions of Maintenance/Reliability Excellence. But, within most organizations, the battle for funding is extremely competitive. It is necessary to communicate with and sell management in terms they relate to.

So now, the challenge is twofold. First, we always want to speak about the benefits of proactive maintenance in the terms and interests of management. Second, we want to quantify potential economic gains in order to obtain approval for essential funding (Planner/Scheduler positions, staff training, informational databases, etc.). Facts and quantification are necessary.

In Figure 1.6 we can see the potential benefits of proactive main-

Reduced downtime / increased uptime	T, O, M
Reduced number, size, and scale of repairs	O, M
Increased equipment availability	O, S
Improved identification of problem areas	M
Increased asset life	M, A
Improved information available for equipment specification	M
Increased safety (operators, maintenance technicians, and public)	All
Increased quality of output	O, S
Decreased liability exposure	L, T
Reduced maintenance and operating overtime (if not reassigned)	A, M
Lower overall maintenance costs—better use of labor and materials	M, A
Increased lead time for parts requirements	P, A
Increased control over parts and reduced inventory level	A, P
Reduce the need for expensive insurance policy spares	T, A
Reduce investment by not needing spare or stand-by units	A
Insurance that all parts are used for authorized purposes	A
Increased accountability for all cash spent	A
Lower cost/unit (cost per ton, cost per widget, cost per student)	T, O, A, S

T — Top management

M — Maintenance and Plant engineering

O — Operations

A — Accounting

S — Sales and marketing

L — Legal

Figure 1.6 Benefits of Proactive Maintenance

tenance. In the right column we can see who in the company would be most interested in a discussion of proactive maintenance translated into these terms.

The process for quantifying the potential stems from identification of each organization's iceberg. Most of the cost elements will be identical to those shown in the generic iceberg (Figure 1.5). However, many organizations will identify elements unique to their own operation. Identification of these elements is relatively simple. Yet, determining the historical cost incurred for each is often more challenging, especially for elements below the waterline. Gather traceable values where possible.

Sources of the values will include the accounting system, payroll records, downtime reports, purchasing records, and the computerized work order system. If the work order system is new or weak, the informational database may not be sufficiently established to yield some of the essential values. This would be reason to refine the CMMIS (Chapter 17). In the meantime, use agreed-upon estimates where necessary.

Figure 1.7 offers a generic rendering of the process described above. The magnitude of the potential is usually a managerial eye opener. Management often thinks of the potential only in terms of head count reduction of 5–20% of the Maintenance Budget itself (above the waterline). The full potential within the full iceberg is several multiples of their current thinking. Return on the proactive investment is well above normal thresholds.

The potential shown does not stem from any single initiative such as Planning, Coordination, or Scheduling. Nor is it to the credit of any sin-

Potential Contribution				
		Annual Potential		
Budget Item		Total (\$MM)	%	Profit Improvement (\$MM)
Maintenance Budget (Above Waterline)				
Labor		\$4.0	20%	\$0.80
Materials & Parts		\$4.0	15%	\$0.60
Overtime		\$1.0	40%	\$0.40
Overhead & Benefits		\$1.5	15%	\$0.23
Contract Support		\$2.0	30%	\$0.60
	Subtotal:	\$12.5		\$2.63
Operating Budget (Below Waterline)				
Equipment Failures		\$6.0	40%	\$2.40
Idling & Minor Stoppages		\$0.8	40%	\$0.32
Off Standard Speed		\$0.5	40%	\$0.20
Process Defects		\$2.0	20%	\$0.40
Yield Losses		\$3.0	20%	\$0.60
Recovery Costs (Remelt etc.)		\$0.5	20%	\$0.10
Set Ups, Change Overs & Adjustments		\$20.0	20%	\$4.00
Start Up Costs & Losses		\$9.0	20%	\$1.80
Inventory Carrying Cost		\$0.7	20%	\$0.14
Express Delivery Costs		\$0.3	40%	\$0.12
Demurrage		\$0.2	40%	\$0.08
Waste Disposal		\$0.1	40%	\$0.04
Energy Losses		\$0.5	40%	\$0.20
Impact of Late Deliveries & Other Customer Compliants		\$0.6	40%	\$0.24
Lost Business (Profit) Opportunity		\$3.0	40%	\$1.20
Product Liability Costs		\$0.7	60%	\$0.42
Cost of Environmental Upsets		\$0.3	60%	\$0.18
Clean Up of Spills		\$0.1	40%	\$0.04
Costs Associated with Accidents		\$0.2	80%	\$0.16
	Subtotal:	\$48.5		\$15.00
Capital Avoidance		\$20.0	40%	\$8.00

Figure 1.7 Build-up of Bottom Line Potential

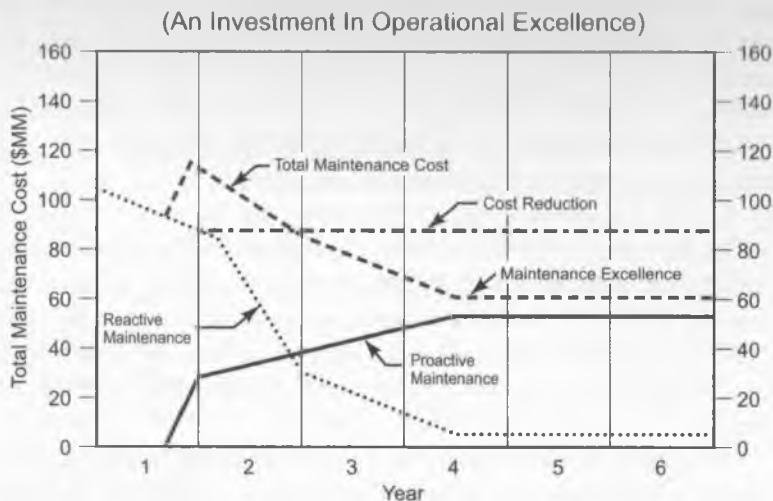


Figure 1.8 Proactive Maintenance Timeline

gle organization or function such as Maintenance. Any attempt to assign credit is inconsistent with the entire premise of the *Integrated Maintenance Management Partnership*, as introduced in the Preface of this book. It is impossible to individually justify each building block of the maintenance improvement process (see Figure 1, the Maintenance/Reliability Arch, from the Introduction). Indeed, the justification for any one of the twenty-one building blocks individually is in fact ... *Zero*. Individually, they are just tools. Integrated together, they provide a process for asset reliability and continuous improvement. Jointly, their justification is ample to meet typical ROI (return on investment) hurdles. Assigning credit to individual organizational units also undermines the Partnership that is critical to establishing and sustaining the proactive culture.

It is important that the investment as well as the potential return be quantified. Like all investments, the outgo occurs before the return is realized. But initial returns (partial) begin in short order. Figure 1.8 depicts the initial investment, the relative improvement, and the associated timeline.

CONTRIBUTION OF PLANNING, COORDINATION, AND SCHEDULING

The job of Planners/Schedulers entails clear description and scope of all proactive maintenance jobs; determination of required

resources; initiation of necessary procurements; development of planned job packages to document how jobs are to be performed in the most effective, safe, and environmentally-conscious manner; coordination of all resources to ensure availability when job execution is scheduled; coordination with Operations or requestor as to when the job can be scheduled; and follow up to capture lessons learned in order to improve future performance. These responsibilities are crucial because supervisors are consumed by execution of active jobs and technicians do not plan for their own efficiency. This last statement has been proven many times using the study technique of Activity Sampling.

Sampling is frequently used to determine the amount of technician capacity lost for lack of proper preparation of maintenance jobs. If maintenance is still in the reactive mode, results will approximate those shown in Figure 1.9, which compares typical sampling results for reactive maintenance with results for pro-active environments. The percentages presented reflect a number of studies in a variety of industries.

ACTIVITY	Reactive without Planning and Scheduling	Proactive with Planning and Scheduling
Receiving Instructions	5%	3%
Obtaining Tools and Materials	12%	5%
Travel to and from Job	15%	10%
Coordination Delays	8%	3%
Idle at Job Site	5%	2%
Late Starts and Early Quits	5%	1%
Authorized Breaks and Relief	10%	10%
Excessive Personal Time (extra breaks, phone calls, smoke breaks, slow return from lunch and breaks, etc.)	5%	1%
Subtotal	65%	35%
Net Accomplishment of Direct Work (wrench time at the job site)	35%	65%

Figure 1.9 Typical Maintenance Worker's Day—Reactive versus Proactive

Should any readers or organizations doubt that presented percentages reflect their own situation, a site-specific sampling should be conducted.

The 35% direct work in reactive mode versus the 65% direct work in proactive mode provides a simple but clear justification (Figure 1.10) for establishing a Planning, Coordination, and Scheduling function.

Two supervisors, each with a ten-person crew but without planner support, will have only seven fulltime equivalents "pulling the wrenches" to complete work.

The same two supervisors with planner support (even if it was established without a headcount increase) will have 12.4 fulltime equivalents pulling wrenches. This is a net productivity improvement of 77%, which is a fine return from any investment.

Two Crews Without Planning		Two Crews With Planning	
2	Supervisors	2	Supervisors
0	Planners	1	Planner
20	Total Craftspeople both crews	19	Total Craftspeople both crews
35%	Direct work percentage	65%	Direct work percentage
7.0	Equivalent fulltime workers	12.4	Equivalent fulltime workers
Improved Output from Planning and Scheduling: 77%			

Figure 1.10 Relative Improvement and Associated Timeline

Another way to appreciate the advantage of job planning is to depict what happens within an individual job without planning. Technicians jump into the work without forethought. Shortly they encounter a delay for lack of a spare part, tool, or authorization. This sequence may be repeated several times before the job is completed. In the planned mode, the needs are anticipated and provided for before a technician is assigned. This comparison is graphically presented in Figure 1.11.

Each dollar invested in planning typically saves three-to-five dollars during work execution. This improvement should be reinvested into Maintenance Excellence and Asset Reliability instead of taking it out in the form of maintenance cost or head count reduction. As established above (Figures 1.5 and 1.7), the greatest benefits of planning, coordination, and scheduling do not show up in the maintenance budget. There are far greater bottom-line contributions to be gained from asset reliability than from mere maintenance cost reduction. This approach is also more saleable to the maintenance work force. By committing themselves to asset reliability, they protect the future of their jobs. If the managerial focus is on cost reduction, the maintenance crew will likely lose overtime, if not their job. This is labor's common fear.

The duration of a planned, coordinated, and scheduled job is commonly half as long as when the same situation is reactively responded to

Under all circumstances, whenever maintenance is performed, it is planned. It is simply a question of who is doing the planning, when they are doing it, to what degree, and how well. Frequently, the wrong person does what little planning is done, at the wrong time, and usually under the pressure of a reactive maintenance environment.

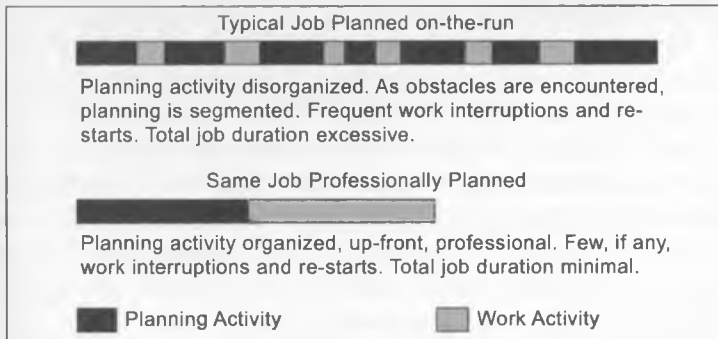


Figure 1.11 Professional Planning Versus Planning on the Run

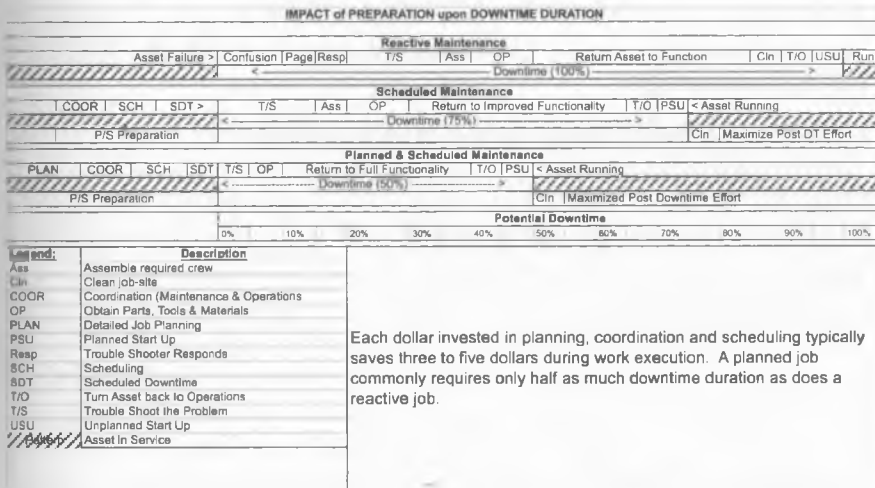


Figure 1.12: Impact of Preparation Upon Downtime Duration

without prior preparation. This reality is depicted in Figure 1.12.

The comparison is presented in three tiers: Reactive Response; Coordinated and Scheduled Maintenance without Job Planning; and Planned, Coordinated and Scheduled Maintenance (the full Preparatory Trio). Downtime Duration is displayed across the horizontal axis. To the

left of the downtime scale, any pre-shutdown preparatory effort is shown. To the right of the scale, any post-shutdown effort is shown. A legend of abbreviations used appears in the lower left. Let us walk through each of the three approaches to the same job.

Reactive Response

The process begins upon asset failure. Confusion reigns as Operations realizes what has happened and considers what must be done. Once Operations have recovered their senses, Maintenance is paged. It takes time to respond because all technicians are effectively engaged. Upon arrival, a technician troubleshoots the situation. It is determined that the assistance of a larger crew or another skill is required. They too need some time to respond. Upon some degree of disassembly, the need for replacement parts and/or special tools becomes apparent and triggers travel to either the shop or the storeroom to obtain them. Upon return, the crew then restores the asset to some state of operation. Usually this state is the same state that existed prior to failure because Operations is under pressure for daily output. Maintenance cleans up the job site, and turns the asset back to Operational control. Operation then goes through an unplanned start up process. The maximum amount of potential downtime (100%) is incurred.

Coordinated and Scheduled Maintenance

Taking advantage of Preventive/Predictive Maintenance, this process is proactive rather than reactive. Alerted by the results of a PM/PdM inspection, the Planner/Scheduler coordinates with Operations and schedules necessary shutdown and maintenance in time to avoid failure. Two of the three preparatory efforts occur before the asset is taken down. The first activity after the equipment is taken down is troubleshooting. However, because planning for the job has not been performed proactively (prior to the equipment being taken down), the crew is not adequate and part/tool requirements must be determined and obtained internal to the period of downtime. The asset is returned to a somewhat improved state of operability because additional state of operability because additional work to be performed while the asset is down was identified within the coordination phase. The asset is then turned back to Operations, which performs a planned start up. Post start up, Maintenance performs the necessary cleanup work. Downtime is reduced by approximately 25%.

Planned, Coordinated, and Scheduled Maintenance

The complete preparatory trio is accomplished pre-shutdown.

Troubleshooting is significantly reduced; the crew is of the proper size and composition; parts and tools have been kitted and delivered to the job site. Still, there may be some shortage of required items. The asset is returned to full functionality. Turnover and start-up are performed in a planned manner. Job-site cleaning is performed post-startup. Downtime is reduced an additional 25%; for a total of 50% reduction. The overall improvement stems from three changes:

- ☐ Better utilization of maintenance resources
- ☐ Joint Operation/Maintenance ownership of the maintenance schedule—leading to high schedule compliance
- ☐ Properly performing the most important maintenance work—the first time

Preparation must be at the core of any proactive maintenance effort. It provides for reliable delivery of all proactive efforts and, combined with PM/PdM engineered through RCM and RCA, it eliminates repetitive failure, and produces quantum benefits that accrue to the enterprise's bottom line.

Why Plan, Coordinate, and Schedule Maintenance Jobs?

Most maintenance departments do not plan to fail; they simply fail to plan and therefore do indeed fail. The major reason behind failure to plan is that putting out today's fires is given priority over planning for tomorrow. This approach all but ensures that future equipment failures will require reactive responses (ample supply of kindling to be consumed in future reactive fires). Secondary jobs are constantly put first by default. Doing secondary jobs may not be bad—but when we reach the point of putting the twelfth job first, we are in trouble. Reactive maintenance is simply a vicious circle, a continuous downward spiral.

We plan because planned maintenance reduces waiting and delay times that mechanics inevitably encounter when performing work that has not been properly prepared for. Poor utilization of a mechanic's time is usually not his or her fault. Nothing is more detrimental to maintenance performance and morale than unplanned or poorly planned jobs. All enthusiasm is lost when a crew arrives at a job site to find that the work can neither be started nor completed due to lack of proper preparation. Mismanagement is highly visible to the workers on the front lines. Conversely, with effective preparation, maintenance crews can take pride

in their work, and organizational performance is improved.

Most of us think we work best under pressure. Unconsciously we procrastinate until pressure forces us into reaction. But, do we really work best under pressure, or do we simply work faster and less effectively due to lack of preparation and hasty judgments?

Although urgency can engulf managers, urgent tasks are not always the most important tasks to be performed. The tyranny of urgency lies in distortion of priorities (subtle cloaking of minor projects with major status under the guise of crisis). But, the measure of true management is the ability to distinguish important from urgent and, in turn, refusal to be tyrannized by the urgent and refusal to manage by crisis. Time must not be simply consumed; it must be effectively utilized.

Granted, urgent tasks (repair of breakdowns) call for instant action, whereas important tasks (PPMs and planned backlog relief) rarely must be done today, or even this week. The momentary appeal of urgent tasks and instant action is irresistible and so they devour our resources and energy. But, in time, their deceptive prominence fades. With a sense of loss, we recall the important tasks that were pushed aside and realize that we are slaves to urgency.

A major obstacle to organizational success is allowing urgency alone to determine how resources will be consumed without considering the importance of the activity relative to the broad objectives of the enterprise. To climb out of this reactive pit, resources must remain to perform the important work of reliable routine services (PM/PdM) and timely backlog relief—they make us better in the future, thereby reducing future failures that require emergency response and consumption of critical maintenance resources that are commonly in short supply.

Advanced planning, coordination, scheduling, and the pursuit thereof are proactive skills. Planners must be allowed to focus on tomorrow rather than being caught up in today. Well-planned, properly scheduled, and effectively coordinated jobs can be accomplished more efficiently, at lower cost, with fewer disturbances of operations, higher quality (reduced variability of processes), improved morale with greater job satisfaction, increased longevity of equipment, and reduced parts usage. More work is completed more promptly with fewer resources, thereby increasing customer service. Studies show that \$3-to-\$5 can be saved during job execution for every \$1 invested in job preparation. Properly prepared jobs require only half as much downtime to execute as that required for unplanned response to equipment failures (as depicted in Figure 1.12).

When long-range objectives become obscured, it is easy to replace

them with much shorter-range and even hopelessly misplaced goals, such as speed of repair. This argument does not deny being efficient in the right things at the right time, but speed, as an end in itself, is futile.

Therefore, avoid overstressing efficiency while neglecting the careful selection of areas in which efficiency is to be sought.

Efficiency's myth lies in the assumptions that "fast" is, ipso facto, effective; and the harder one works, the more one accomplishes. The allure of speed seems irresistible—who could be against it? Rapid performance of the wrong task is not effective at all. Therefore, the hazard of emphasizing speed without regard to effectiveness must be avoided at all costs.

Action that puts speed ahead of results but disregards organizational objectives, in the long run, will be totally ineffective and likely will need to be done over. Doing the job right the first time requires proper preparation. If you don't have time to do it right, when will you have time to do it over?

Learn to Say "No"

The Rush Job

I am a rush job.

I belong to no age, for man has always hurried.

I prod all human endeavors. Men believe me necessary—but falsely.

I rush today because I was not planned yesterday.

I demand excessive energy and concentration.

I override obstacles, but at great expense.

I illustrate the old saying "Haste makes waste."

My path is strewn with the evils of overtime, mistakes, and disappointment.

Accuracy and quality give way to speed.

Ruthlessly I rush on ...I am a rush job.

Anonymous

Every job or project has an advocate of priority somewhere. Because people do not like to say no, everybody's urgency takes priority and nothing of importance is accomplished. The problem is not usually

with priorities one through three, but with "posteriorities" four through infinity. One simply cannot achieve performance excellence without concentrating effort in the critical areas.

One has to say about a suggested, but secondary project, "this should be done, but it's not our first priority. If it must be done, we'll have to let somebody else do it." Do not be timid about using the greatest time-saving word in the English language, that little two-letter word "No." Decide what not to do. Bill Cosby has been heard to say "I don't know the key to success, but the key to failure is trying to please everybody."

Keeping overall objectives in the forefront allows us to schedule time based on long-term objectives rather than on immediate crises.

In contrast to the benefits of proactive maintenance, the following sonnet effectively conveys the curse associated with reactive maintenance.

SYMPTOMS OF INEFFECTIVE JOB PLANNING

The following symptoms show clearly the ineffectiveness of maintenance resources when an organization is operating in a reactive rather than proactive mode. Delays encountered by mechanics are the norm while:

- Gaining detailed knowledge of the required work
- Obtaining permits
- Identifying and obtaining material, blueprints, tools, and skills required for completion of assigned work
- Delivering the above inputs to the job site
- Waiting for required spare parts that are not in stock

These delays occur frequently. Mismanagement is highly visible on a daily basis. This reality should be a wake-up call for management. The following scenarios are common:

Due to insufficient thought when requests are written, operating personnel must often alter jobs in progress, causing delays and productiv-

ity losses. Technicians may arrive at a job site on time, only to wait for their supervisor or someone else to contact the requestor or other operating person for clarification of the requested work. Often the operations department does not have the equipment ready or refuses to release it, despite having previously agreed to the schedule.

The number of craftsmen does not match the magnitude of the job (possibly too many and possibly too few). Participation of support craft is not anticipated, or workmen are of the wrong craft skill, or have arrived in the wrong sequence (e.g., millwrights arrive before electricians have disconnected equipment).

Craftsmen have no prior knowledge of the detailed job task breakdown. Instead, they are left to decide how to do the job and what materials are required. They then either leave the job site to source materials from the storeroom or stand by and wait for them to be delivered. Even more disruptive is when after a job is started, it is discovered that some required parts or materials are not in stock and must be purchased outside the plant with extra cost for express delivery. The job is halted in a state of disassembly waiting for the items to arrive.

Craftsmen cannot develop work rhythms because of interruptions and delays encountered, causing them to hop from crisis to crisis without completing jobs already in progress.

Maintenance works in standby mode waiting for the next emergency while supervisors become dispatchers of response to breakdowns. Communication is faulty and incomplete throughout the process; it takes many guises including failure to satisfy the originator (what was wanted was not what was done), frayed tempers, frustration over wasted efforts, downtime delays, makeshift repairs, and prolonged quality problems. Each of these can result in higher overall costs. Maintenance personnel in the trenches know the frequency of these occurrences, but managers are often unaware.

CONVEY THE MANY BENEFITS THAT ACCRUE TO EACH STAKEHOLDER

Stakeholders are people in positions impacted by maintenance effectiveness. To sell them on Planning, Coordination, and Scheduling, convey the many benefits that accrue to each.

Planning, Coordination, and Scheduling Provide Significant Benefits to Management

They provide these benefits by:

- Working to achieve the optimal economic level of maintenance in support of both short and long range operational needs
- Improving regulatory compliance
- Improving employee safety
- Reducing total unit cost of maintenance while improving customer service, condition of equipment, and facilities through better use of labor and materials
- Providing a central source of information concerning the condition of equipment, the associated maintenance workload, and available resources to perform it
- Increasing the useful life of fixed assets
- Accurately forecasting labor and material needs, thus permitting immediate recognition of labor shortages and excesses with steps taken to level peak workloads
- Establishing expectations for what is to be accomplished each week by the maintenance payroll investment (forty hours of productive work from each maintenance worker) and analysis of variations from expectations
- Improving preparation, management, and control of major shutdowns, outages, and turnarounds
- Improving efficiency through anticipation of needs and avoidance of potential delays
- Providing factual data, measurement, analysis of efficiency, and identification of variations from expected performance
- Challenging the need for work requests of questionable value and justification
- Providing information identifying problem areas that require focused attention

To capture these benefits, management must lead the cultural transition from a reactive operating environment to a proactive operating environment.

Planning, Coordination, and Scheduling Confer Significant Benefits on Operations and Production

These efforts confer these benefits when they:

- Increase equipment availability
- Minimize downtime and interruptions to operations
- Provide more accurate estimates of when an asset will be returned to service
- Apply technical knowledge and experience to the analysis of each planned job
- Maintain accurate backlog status
- Facilitate anticipation of required repairs before they become emergencies
- Provide close and continual coordination between operations and maintenance with a single point of contact for all emergency and scheduled work—whether pending, in-process, or completed
- Provide orderly procedures for requesting, preparing, executing, and closing out maintenance support
- Coordinate the cooperation necessary to provide essential maintenance resources at times when operations can best relinquish the associated equipment capacity

To gain these benefits, Operations must commit to the Operating/Maintenance Partnership and Covenant.

Planning, Coordination, and Scheduling Are a Boon to Maintenance Supervisors

They are a boon when they:

- ☐ Provide more time for direct job site leadership
- ☐ Identify the best methods and procedures
- ☐ Increase quality of output
- ☐ Define and measure workload, permitting advanced determination of staffing required in a given area or location
- ☐ Preclude delays that would otherwise occur after work has begun due to waiting for information, materials, equipment, other skills, tools, etc.
- ☐ Monitor job status
- ☐ Establish expectations for the maintenance work force through provision of a detailed job schedule for the entire week, with individual timelines as appropriate, thereby furnishing a control vehicle by which maintenance supervisors can monitor progress throughout the work day and work week
- ☐ Coordinate manpower, materials, and equipment including:
 - Coordinated crafts
 - Parts and materials
 - Special tools and equipment
 - Shop and other support
 - Equipment access: permits, lock outs, preparation of asset for maintenance work
 - Off-site job preparation to minimize duration of downtime

Integrate preventive/predictive maintenance into the overall schedule

- Establish realistic priorities
- Anticipate and preclude bottlenecks and interruptions
- Provide accurate promises that can be fulfilled
- Control overtime

To gain these benefits, maintenance supervisors and their team members must also accept the proactive transition and commit to full usage of the proactive preparation with constructive feedback supportive of continuous improvement.

Both Purchasing and Stores Win

Purchasing and Stores both win when Planning, Coordination, and Scheduling:

- Optimize maintenance inventory
- Insure that parts are ordered with adequate lead time allowing Purchasing to follow proper procedures to obtain best prices
- Reduce the number of emergency purchases and cost of express freight
- Improve accountability for all parts and material
- Improve information available for equipment specification

To gain these benefits, Materials Management must endorse procurement based on life-cycle cost versus minimal cost.

Credit for the many benefits documented above is not exclusive to planning, coordination, and scheduling. Maintenance/Reliability Engineering, Maintenance Supervision and Workforce, and others also make major contributions.

EMPHASIZE THE MAINTENANCE DELIVERABLES

Hopefully by now any thought of Maintenance as a “necessary evil” has been dispelled. Indeed, maintenance contributes in many ways that are crucial to success of the business entity. Few drivers of maintenance workload are discretionary. If maintenance is not funded and sized, employees are maimed or killed, the environment is destroyed, jobs are lost, management is replaced, and the entity fails. Figure 1.13 illustrates a proper build-up of the Maintenance Budget.

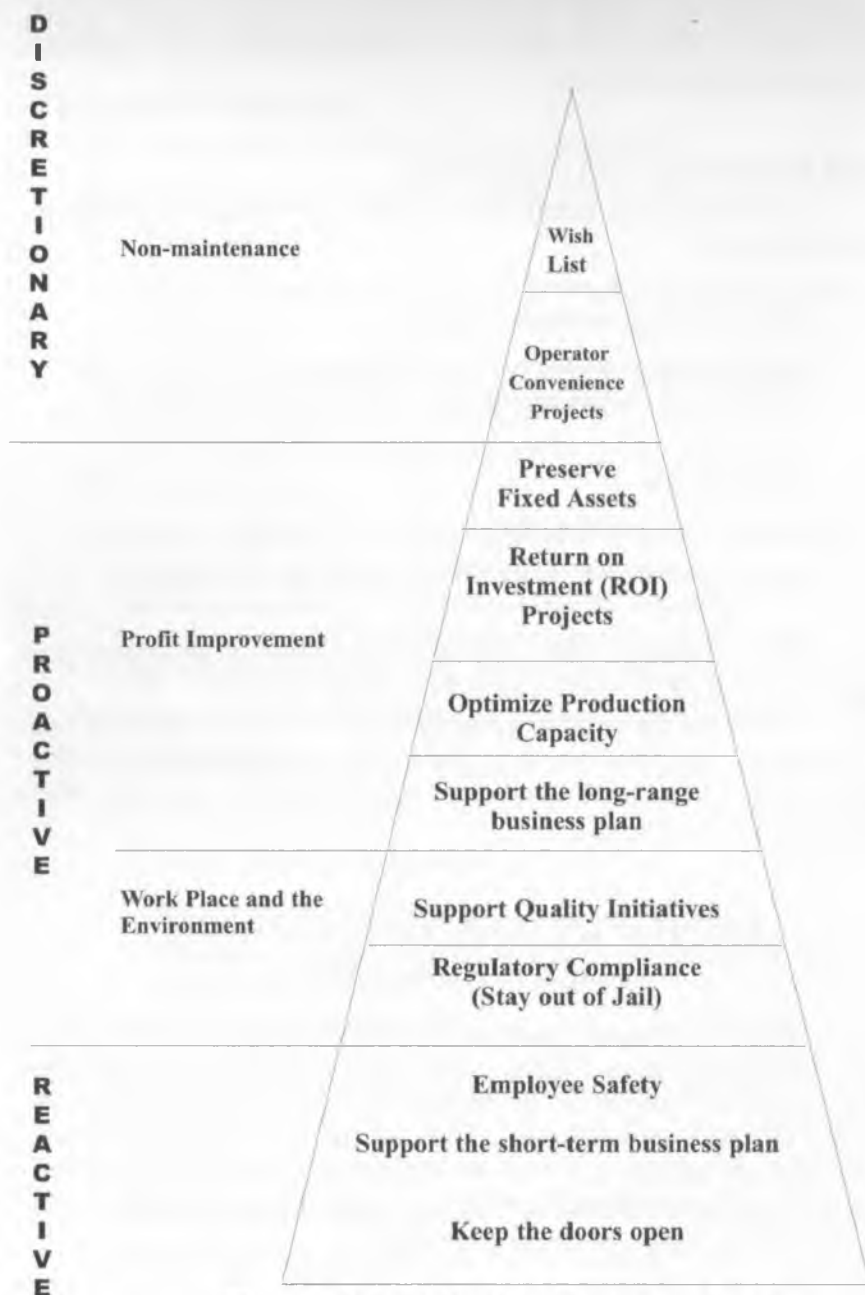


Figure 1.13 Build-Up of the Maintenance Budget

Prerequisites to Success Preparation

Now that the return on the Planning, Coordination, and Scheduling investment has been clearly established, we need to address the associated prerequisites. Investment usually must be made before preparation can be effective. Maintenance needs upgrades to its command and control structures. Like everything else, investment precedes return (ROI).

A number of proven prerequisites must be in place before work preparation can yield optimal results. They are grouped and discussed below by nature.

ENVIRONMENT

A proactive environment is essential to Reliability. When the local environment is reactive in nature, cultural change is essential. Achieving such change falls largely on senior management and is one of the most difficult prerequisites.

- The entire organization must honor an established set of shared, proactive beliefs.
- A covenantal partnership must be nurtured among functional organizational units.
- Organizational expectations must be aligned to proactive reliability rather than to reactive response.

- The mission of Maintenance must be supportive of ultimate objectives of the enterprise. The importance of each demand upon maintenance resources must be evaluated from that perspective, allowing the scheduling of such resources to be based on long-term objectives rather than immediate crisis. When long-range objectives become obscured, short-range and even hopelessly misplaced goals easily replace them.
- Operations must be supportive in establishing a practical level of maintenance whereby both long-term and short-term operating plans are met while maintenance needs are accomplished through anticipation, planning, and scheduling. The facilities must then be maintained at specified levels of operating efficiency, at the lowest possible cost consistent with the goals of producing quality product as economically as possible.
- Maintenance must regard Production as an internal customer and provide them the customer service they deserve, within the framework of a Production/Maintenance partnership that replaces excessive use of "the customer is always right" syndrome.
- Production deserves to have its work requests performed in a timely manner.

Therefore, maintenance resources must be balanced with maintenance workload on a quarterly basis. Backlogs must be kept within reasonable limits. If they are below minimum and remain there for very long, there is an insufficient volume of work to accommodate the smooth scheduling of resources. If they are above maximum limits, work turns so slowly that response and customer service is unacceptable. Urgent work is performed at the expense of scheduled jobs. In turn, the scheduled jobs which are displaced result in work being carried over to the next schedule period unless addressed by a temporary increase in resources, such as overtime.

ORGANIZATION

The maintenance organizational structure must foster pro-action. Many maintenance departments are organized primarily for immediate response to daily failures. How do they expect to ever get out of the reactive abyss?

- ☐ The organizational structure must provide for performance of all three maintenance work types (prompt response to failures, reliable PM/PdM routines, and timely backlog relief).
- ☐ Resources should be distributed in the following proportions (10% response, 30% Programmed Maintenance including inspections that require shutdown and 60% backlog relief including Corrective Maintenance).
- ☐ Resources must be adequate to meet demands of optimized maintenance plans for all assets. This demand encompasses essential skills dictated by the complexity of the operational process.
- ☐ The four maintenance support functions should be established with realistic control spans (Supervisors, Planner/Schedulers, Reliability Engineers, and Administrative Support Personnel).
- ☐ Planner/Schedulers must devote 100% of their time to planning, coordination, and scheduling. Their capacity should not be diluted by
 - Administrative routines
 - Supervisory Relief
 - Reliability Engineering
- ☐ Strong functions for maintenance buying, storeroom management, and receiving should be established.
 - All functional units must effectively meet their responsibilities.

INFORMATIONAL DATABASE

The informational database (Computerized Maintenance Management System) must be fully loaded, complete, and reliable. This includes:

- ☐ A strong computerize work order system must be instituted. All control of maintenance work originates from work orders.
- ☐ Work Requests must be written by the originators with proper descriptive information and sufficient lead time to enable

effective preparation. Needed work must be identified as far in advance as possible so that backlog is known and jobs can be effectively planned prior to being scheduled. Of course, urgent response is the exception in regard to lead time.

- ☐ The maintenance information database must be complete and reliable, including:
 - An up-to-date Maintenance Technical Library
 - A comprehensive Planning Library including Planned Job Packages and a Catalog of Job Estimates
 - A complete History of Equipment Repairs to enhance pursuit of reliability from date-in-service if possible
- ☐ Upon job completion, meaningful feedback from supervisors and technicians is essential to enhancement and refinement of Planned Job Packages and Equipment History.

MATERIALS SUPPORT

To sustain a cost-effective, responsive, and reliable operation, maintenance departments are dependent upon reliable and timely logistical support from Purchasing, Receiving, and Stores. Associated prerequisites are:

- ☐ Timely procurement, expediting, assembly, delivery, and availability of required materials and spare parts
- ☐ Well-designed and centrally located Storerooms with a reliable inventory (on-premise or readily available locally) to:
 - Meet reactive demands when equipment fails.
 - Support routine maintenance efforts designed to avoid future failures.
 - Assure availability of infrequently used parts that are either critical to operation of plant and equipment or subject to long delivery times (Insurance Spares).
 - Provide frequently-used parts and materials that can be purchased less expensively in bulk than in minor quantities.

- ☐ An effective stock location system facilitating management and prompt location of all inventory items essential to efficient and effective job execution
- ☐ Effective storeroom support so that planners need only identify required withdrawals by preparing necessary stores requisitions, and need not do stock picking, job kitting, order staging, order security, or order delivery to the job site
- ☐ Purchasing must provide a responsive process for timely procurement of all parts and materials directly purchased or authorized to be held in inventory
- ☐ Planners should only need to prepare Purchase Order Requests; Purchasing should source, purchase, prepare Purchase Orders, track, and expedite delivery
- ☐ Receiving must reliably alert planners when purchased items are received
- ☐ Overhaul and Rebuild capabilities should be established
- ☐ Vendor relationships must be effective, reliable, and timely partnerships
- ☐ Maintenance must provide the Purchasing department sufficient lead time on all proactive work to enable JIT procurement of required parts and materials
- ☐ Plant Engineering and Project Engineers must use established direct purchase channels rather than reliance upon inventory held for reactive maintenance needs
- ☐ Material and parts cost must be captured by work order to which applied; otherwise, Equipment History is not a reliable management/engineering tool

RELIABILITY

The function of Maintenance/Reliability Engineering is distinct from the Planning, Coordination, and Scheduling function. If the engineering function is not effectively established, efforts of the planner/scheduler function will be diluted. Therefore, maintenance/reliability engineering is vital:

- Through the Reliability Centered Maintenance (RCM) process, the function must strive toward an Optimal Maintenance Plan for each asset.
- Each large or repetitive equipment failure should be analyzed using Root Cause Failure analysis.
- PM/PdM routines must be comprehensive and thorough. They must be conducted as controlled experiments and continuously refined until optimized.
- Corrective Maintenance must be performed promptly to gain the return on the PM/PdM investment.

PLANNING, COORDINATION, AND SCHEDULING

Required maintenance work must be originated, prepared, and executed on time and with quality workmanship. It is essential to know what is to be done, when it is to be done, how best to do it, and then do it right the first time. Related prerequisites are:

- Every effort must be made to identify all needs for performance of maintenance jobs as well as all potential delays that may be encountered; and then perform the coordination necessary to meet the needs and avoid the delays. This is the Planner role.
- All jobs requiring shutdown must be reviewed with key operating personnel to gain their input.
- Averages and estimates must be continuously refined until they are worthy of being termed true standards.
- Maintenance and operating management must commit to holding structured weekly coordination and scheduling sessions to reach agreement about the work that is most important to being performed during the approaching schedule week, given the limits of available maintenance resources.
- The entire organization must understand the importance of sticking to the resultant weekly schedule.
- Schedule Compliance must be aggressively pursued.

- Effective planning, coordination, and scheduling of maintenance work must be completed far enough in advance to permit Operations to plan for equipment to be out -of-service, and to minimize nonproductive time and output shortfalls.
- Periodic Activity Sampling is necessary to identify ongoing avoidable delays that necessitate refinement of Planned Job Packages.

JOB EXECUTION

Prior to, during, and upon completion of job execution, supervisors and technicians are responsible for review, reference, and use, as well as enhancement of preparatory efforts and documents.

- Supervisors must review all preparatory efforts with planners to approve, assure nothing is lost in transition, and add any further insights and safety precautions.
- Supervisors must periodically be in attendance at job sites to provide effective leadership during execution, follow-up, and feedback; assuring that crews make full use of preparatory efforts, perform work correctly, and provide proper and documented feedback.
- Constant effort is needed to improve work methods, completeness, neatness, quality, and efficiency.
- Regular feedback regarding status of work requests and completion promises must be provided to planners/schedulers.
- Supervisors must be relieved of non-supervisory responsibilities so they can focus on their job-site duties.

One other item that is essential to the success of the effort is for the company to have clear objectives for planning and scheduling—a mission, so to speak. Figure 2.1 is a working document from a major manufacturer. You can see the level of detail and the results that they expected.

Figure 2.1

**Sample Top Ten List of MWO Planning Objectives for Company
Implementing Planning and Scheduling (SAP shop)**

A. Daily Review of Notifications Created in the Previous Twenty Four Hours:

1. Validation
2. Prioritization
3. Start Dates
4. Functional Location and Equipment Number
5. Creation of Work Orders

B. Define MWO Job Scopes.

1. Visit the Job Site.
2. Identify the Requirements for Planning the Work Order.
 - a. Identify Crafts Skills Required to Complete the Task.
 - b. Identify Crew Size and Duration Requirements.
 - c. Identify Safety and Environmental Requirements.
 - d. Identify Special Equipment and Tools Required to Complete the Task.

C. Revise and update MWO's with Details.

1. Labor, Skills and Duration
2. Start Date
3. Detailed Location
4. Detailed Task List
5. Detailed Safety and Environmental Requirements
6. Detailed Material List
7. Detailed Special Equipment and Tools
8. Detailed Plan for Testing when the Task is Complete

D. Acquire Materials.

1. Create Reservations.
2. Create Purchase Requisitions.
3. Order with Purchasing Card.
4. Order from Supply Chain Vendor.

E. Backlog Management

1. Review the Backlog for Open Work Order Status.
2. Review Start Dates.
3. Review Priorities.
4. Review Revision Codes if any.
5. Move Work Orders to "Ready" Status when the Planning is Complete and Materials are Accounted for.

F. Review the Ready Backlog with Process Owner.

1. Review the Ready Backlog with the Process Owner and the Maintenance Work Center Supervisor.
2. Identify Tasks that may Require External Services.
3. Review Priorities.
4. Review Start Dates.
5. Review Details.

G. Schedule Tasks.

1. Schedule 100% of Available Internal Maintenance Manpower.
2. Schedule MWOs:
 - a. Schedule approximately 20% to 30% PMRs.
 - b. Schedule approximately 30% High Priority Corrective Work Orders from the Ready Backlog.
 - c. Schedule approximately 30% Low Priority Corrective Work Orders from the Ready Backlog.
 - d. Schedule approximately 10% Maintenance and CAPX work.
 - e. Work Center Maintenance Supervisor to select the Correct Maintenance Craftsmen for the Required Task, and Schedule Work Orders to Craftsmen.
 - f. Plan all MWO's Prior to Scheduling (if not already complete).

H. Lead the Daily Planning Meeting.

1. Play a Lead Roll in Communication and Coordination.
2. Discuss and Promote the MAC/SAP and RCM Process Standards. Standardize MAC/SAP.
3. Gain Commitment for the Weekly Maintenance Schedule.
4. Discuss and Record Reactive Work and its Affect on the Schedule.
5. Create and Discuss Action Logs for Improvement Initiatives.

I. Completion of Work Orders

1. Audit Notification As Found As Left Information
2. Audit Parts Usage and Materials Returned to Warehouse
3. Audit Planned verses Actual Costs
4. TECO Work Order

J. Manage BOMs, PMRs, and SAP Master Data.

The Nature of Maintenance Activities and Related Organizational Structure

Most maintenance departments are organized only for reaction to urgent demands, yet they wonder why they live in a reactive environment with a reactive culture. Organizationally, there must be recognition of and provision for the three broad types of work performed by the maintenance department: prompt emergency response, reliable routine service, and timely backlog relief. To be organized primarily for the first type (urgent response) is obviously detrimental to the latter two types.

To provide reliable asset capacity, maintenance resources must be balanced with essential workload and organized in a manner that assures effective performance of all three work types. Deferring critical maintenance inevitably leads to further breakdowns and a perpetuation of reactive maintenance that ultimately leads to failure of the business entity.

Managers must not allow urgency alone to consume all available resources. Resources must be provided for the important work that improves future reliability, thereby reducing future urgencies. Although breakdowns and other suddenly developing conditions require prompt response, they represent only the urgent portion of the maintenance workload. Preoccupation with urgent response perpetuates the reactive status quo and causes the future of an organization to look like the past—failure to meet and sustain organizational objectives.

ORGANIZATION BY WORK TYPE

Effective control of the maintenance function depends upon clear accountability for each type of demand placed upon the organization. One form of maintenance organization is structured to facilitate control of each work type. Although there are other structures that can be successfully deployed, this structure (Figure 3.1) best facilitates the understanding sought within this chapter.

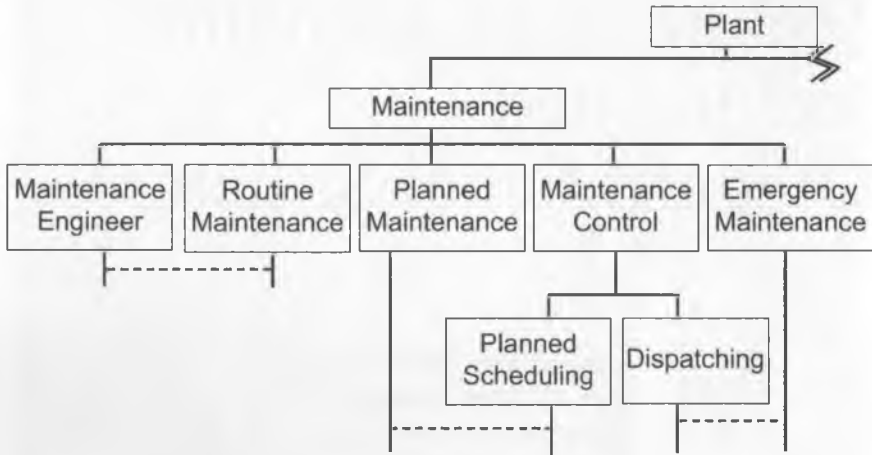


Figure 3.1 Organization by Work Type

Such a structure is composed of three major maintenance groups covering the three principal types of demand. The concept establishes two minimally sized crews to meet routine and emergency demands with a third large group devoted to planned relief of the maintenance backlog. Let us reach an understanding of the three groups and the work types they perform. They require different forms of control and personal assignments.

The Routine Group Provides Reliable Routine Service

This group has responsibility for performing all management-approved routine tasks in accordance with detailed schedules and established quality levels. Most prevalent within this workload is the preventive/predictive maintenance process (PM/PdM). An optimized PM/PdM process, exclusive of major shutdown inspections, commonly requires 15–20% of maintenance labor resources. Because major inspections

require a crew of several people to disassemble, change-out, reassemble, etc., they are scheduled for performance by the large backlog relief group. Including these major inspections, the PM/PdM program normally consumes 30–35% of maintenance resources. Neither emergency nor scheduled backlog work should be allowed to interrupt the Routine Group, thereby protecting integrity of the PM/PdM schedule.

Routine work is specifically defined, of known and consistent content, and requires a predictable amount of time. It is performed in a defined manner, according to a known schedule. Routine work is proactive and, therefore, should be planned, coordinated, and scheduled. The planning should be thorough when each routine is established; any further planning should not be required until the routine is re-engineered (refined). Through the CMMIS, scheduling should be automatic when each routine falls due.

Routines include all PM/PdM and other inspections, as well as lubrication, calibration, tests, cleaning, adjustment, tightening, etc.

Routine crews carry tools and materials for small deficiencies commonly found. Time is provided for minor short repairs or called corrective actions, of less than 30 minutes. Corrective actions should be noted as part of PM/PdM feedback. It is important to code the time spent on short repairs as Corrective Maintenance instead of part of the Preventive Maintenance routine itself. A well-designed CMMIS should accommodate this important differentiation.

Findings that require more than 30 minutes trigger *Corrective Work Orders* that are entered into the CMMIS and fed into the backlog. If inspection frequency is established properly, there should be sufficient lead time to plan and schedule the work by the backlog relief crew at a future time following proper preparation.

A healthy ratio is one correction per six inspections; whether correction is performed as an *action* at time of inspection or performed as a planned and scheduled *work order*. If the ratio is greater than 1:6, a reduction of inspection frequency may be in order. If the ratio is less than 1:6, an increase to frequency of inspection may be in order because the risk of failure between inspections is too great.

The PM/PdM Program must be conducted as a controlled experiment. This means that PM/PdM Schedule Compliance must exceed 95%. The most logical technicians to assign this work to are very methodical. They like to work quietly on their own. Others can be driven crazy by their pace. They are a perfect fit for routine work.

The Emergency Group Provides Prompt Response to True Urgent Needs

The Emergency Group has responsibility for handling essentially all urgent demands such as breakdowns, requesting assistance only when necessary. In other words, this group protects the other two groups from interruption. Unless staffed for peak demand, it cannot fulfill this objective 100% of the time, which is not the optimization sought. If properly sized, they should only need to request help about 10% of the time. Because the Routine Crew should not be interrupted, assistance is provided by the Planned Backlog Relief Crew. In a proactive maintenance environment, prompt response to urgent work requires approximately 10% of maintenance labor resources—assuming the Emergency Group is staffed by well-trained, multi-skilled troubleshooters.

Sources of Planned Work	Percent of Total Planned Work
Results of PM/PdM inspections	30%
Programmed component replacements	20%
Overhauls/rebuilds	15%
Internal customer requests	10%
Engineering project support	8%
Safety and environmental work	5%
Root Cause Failure analysis	5%
Management directed work	4%
Service requests	2%
Accidental damage	1%
Total Planned Work	100%

Figure 3.2 Sources of Planned Work

While the organization is still transitioning from a reactive to proactive culture, urgent work offers little opportunity for planning or scheduling, except in a most rudimentary way. It should be radio-dispatched, mobilized, and well equipped, making it a lean and highly-skilled troubleshooting team. Over time, as planners build libraries of planned job packages, they can begin to develop job pre-plans. After all, how equipment fails is known. We just don't know when it will fail. Job pre-plans can be developed and loaded into the CMMIS. When failure occurs, the responding troubleshooter can access the pre-plan and gain many benefits of proper planning.

The Planned Backlog Relief Group Provides Timely Relief of those Work Requests with Adequate Lead Time to be Planned

Backlog consists of all planned-work (non-emergency or urgent) that are still open (fully or partially) (Figure 3.2). As used in this book, it does not connote delinquency. In a proactive environment, the bulk of maintenance workload should be planned. Sixty-five to seventy-five percent (65–75 %) of maintenance resources should work in this mode, as part of the Planned Backlog Relief Group. Schedule compliance for this group should exceed 90%. As stated previously, the Planned Group is called upon to support the Response Group whenever it encounters peak demands (10% of the time by design). This support still allows the Planned Group (constituting 75% of the maintenance workforce) to be scheduled and assigned to well-prepared jobs and not be interrupted 90% of the time. This approach is far more proactive than most maintenance departments anywhere.

Backlog work originates from PM/PdM/TPM inspections—which can be performed by maintenance, operations, or other suitably trained personnel—in the form of Corrective Work Orders. The work also comes from projects (both capital and expense) as well as from non-urgent requests originated by sources throughout the organization. Once backlog work orders have been properly prepared, they form (together with PM/PdMs that are due) the basis for coordinated Weekly Schedules (see Chapter 16).

The remaining technicians comprising the maintenance team are best applied to planned backlog relief. They are neither the best troubleshooters nor the methodical inspectors. They might be classified as our

typical, average technicians. But we want to put our average people (still learning their trade) on jobs with the fewest surprises and interruptions. That describes planned backlog relief.

The important portions of maintenance workload are reliable routine service and timely backlog relief. Proper attention here provides asset reliability, minimizes future emergencies, leads to a proactive environment, and enables sustained viability of the organizational entity.

Whatever organizational structure you decide to deploy, it must be equally responsive to the three work types. Otherwise, a large portion of the work force will be found in standby mode at fixed or semi-fixed areas of assignment, waiting for the next emergency. Workforce utilization and asset reliability will be extremely poor. Planning and scheduling will be ineffective because few resources will be available to work on the jobs diligently prepared (planned, coordinated, and scheduled). At best, Schedule Compliance will be only 50–60%.

**If the Organization Strives for Proactive Reliability,
It Must Be Organizationally Structured
for Proactive Reliability.**

Smaller maintenance departments often visualize a problem trying to deploy the Work Type organizational structure presented above, especially if they have a small crew distributed between two or three shifts. First, do not get hung up on separate supervisors for each work type. A single supervisor could well distribute several technicians between the three work types. With only three technicians on a given shift, you could consider a point person for each work type. Point persons for emergency response could be the first responders to failures. They would be assigned to interruptible jobs (such as bench work) between emergencies. Point persons for routine work would start every day doing scheduled PM/PdMs. When all scheduled routine work is complete, routine technicians would join the backlog relief technicians.

Where Planning Fits Into Good Maintenance Practices

When a maintenance planning and scheduling function is being established, the first questions that usually arise are where and how it fits into the organization. The first answer is that it is structured within the maintenance organization, not outside it. Second, it should be organizationally independent of the specific maintenance supervisors it is both tasking and supporting. This arrangement is based on the principle of checks and balances. If planning reports to first-line maintenance supervisors, there is a tendency to establish lenient expectancies in terms of estimates and schedules. There is also a tendency to use the planner for daily expediting, clerical work, and other reactive and inappropriate duties.

Planners, therefore, should be on the same organizational level as the supervisors they support on a week-to-week basis. Neither should be superior or subordinate to the other. Unless the maintenance department is large, both supervisors and planners should report to a common maintenance manager. If the department is large and has several support positions including planners, maintenance engineers, and administrators (clerks), then a Manager of Maintenance Support Services position is recommended to lead all support functions. In this scenario, this manager and the Maintenance Manager (to whom line supervisors are responsible) should both report to a common Director. See Figure 4.1 for organizational positioning of the Manager of Maintenance Support Services and Appendix A for an associated position description.

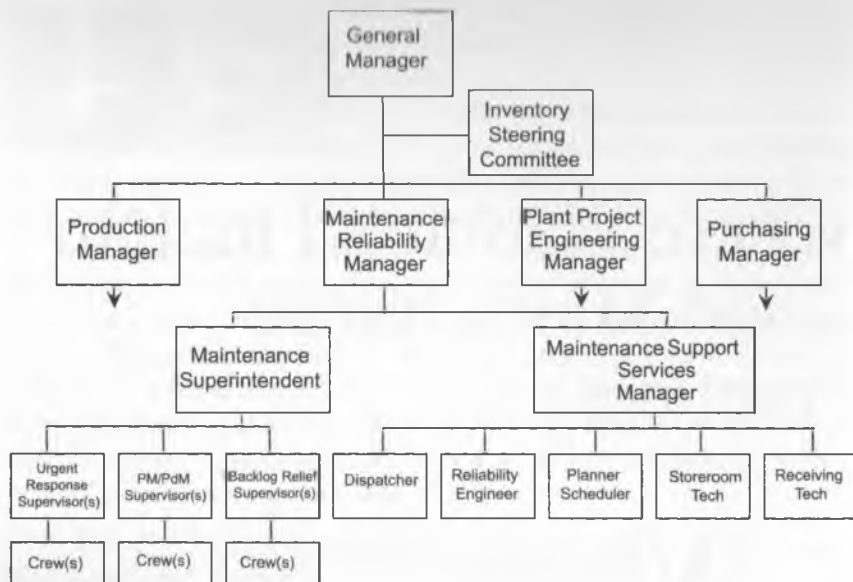


Figure 4.1 Organizational Positioning of the Manager of Maintenance Support Services

The maintenance supervisor and maintenance planner form the most important partnership within the maintenance department. This relationship is sensitive, at least at the beginning. It is beholden upon planners to remember that they are staff in support of line supervision. Staff supports, counsels, recommends, and convinces line but does not issue orders to line. This rule must not be broken if the partnership is to prosper.

Appendix B contains an article entitled *Recognizing the Pitfalls of Planning for Others*. All planners should read and digest this article.

In terms of job grade, the planner position should be one grade higher than a junior first-line supervisor and one grade lower than a senior first-line supervisor. There is a need to attract experienced, although junior, supervisors into planning and subsequently to attract experienced planners to senior supervisory positions as they pursue the career path to Manager of Maintenance Support Services or Maintenance Manager.

The Planner job is preferably salaried rather than hourly. Planning, coordination, and scheduling are managerial responsibilities. The first allegiance of planner/schedulers must be to the goals of the organization, not to the labor movement. Yet, the best maintenance plan-

ners come from the craft ranks. They have usually performed similar jobs themselves and can visualize the job being planned. Determining and estimating job steps and material requirements is thus easier.

Sooner or later, the question arises as to whether planners should be decentralized or centralized. Should their workstation be decentralized in the operating departments they support or centralized with the maintenance shops? The best answer is they need a desk in both places. Planning begins in the field at future job sites. By contrast, coordination and scheduling are best accomplished centrally where planners and internal customers gather to resolve distribution of available resources in the best interest of the overall enterprise. Regardless of decentralization or centralization, a quiet location conducive to research and thought is critical. This location should be adequately separated from supervisor offices with their heavy traffic and noise.

SHOULD WORK PREPARATION BE A SEPARATE AND DISTINCT FUNCTION?

The functions of planning, coordination, and scheduling provide the preparation essential to effective execution of maintenance work and thereby to achievement of maintenance excellence, asset reliability, and world-class operations. As Peter Drucker writes, "The productivity of work is not the responsibility of the worker but of the manager. A worker will not plan for his own efficiency." Craftsmen and supervisors are primarily concerned with job performance, not with preparation for optimal performance. The three preparatory functions are best performed by dedicated, para-professional, management personnel.

Unfortunately, under modern principles of human behavior and the arrival of lean, participative, organizational philosophies, staff support is being drastically reduced in favor of reliance upon high involvement work-teams. These teams are expected to prepare for as well as execute their own work—the common belief being that separation of preparation from execution is unnecessary. Consistent with concepts of worker involvement and self-direction, job preparation is being forced to the lowest organizational level possible.

In today's high technology world, good maintenance mechanics are a precious resource. They are the primary resource aimed at preserving asset reliability and production capacity. In most unionized facilities, only mechanics are contractually authorized to use tools when performing

maintenance work. When they are engaged in planning, they are not using their tools. Numerous studies (see Figure 1-9) show that maintenance mechanics spend, on average, only two or three hours per day applying tools of their trades. It is imperative that activities which consume mechanic time and are not related to tool usage be held to a minimum.

Furthermore, people preparing for their own work, one job at a time, are less effective than staff professionals who are trained, specialized, and focused on coordination of the resources (labor, materials, parts, tools, equipment, technical documentation, support personnel, and transportation) required for performing all jobs in the maintenance backlog. Participative worker involvement is a powerful vehicle for continuous improvement; however, dedicated, structured, and focused preparation by well chosen and trained planners remains an essential requirement. Without such preparation, maintenance resources are applied ineffectively. Still, many general managers are reluctant to acknowledge the requirement for dedicated planners. The options and related fallacies are discussed next.

Assigned Craftsman

If planning is left to technicians, it is rarely performed well. It results in delay, wasted effort, and inefficiency. The worker engages in activities that reduce time available for direct work. Indirect activity and travel tend to be very high in proportion to direct work time.

Because of their position in the organizational hierarchy, craftsmen are not well-postured for many of the liaison and relationships associated with the planning coordination and scheduling roles.

Responsible Supervisor or Team Leader

First-line supervisors should be focused on leadership, instruction, oversight, and control, as well as training associated with methods, workmanship, quality, feedback, and housekeeping. Given the demands of daily maintenance execution, supervisors are forced to concentrate on the immediacy of today's problems. They have little time left to focus on effective preparation for future activities. If they are tasked to address both preparation and execution, planning for future jobs is almost always neglected due to the pressure of daily demands.

Due to the demands of execution, supervisors typically plan just prior to the start of each job. Thus, they have little time to consider methods and to identify and acquire required materials, tools, support crafts, etc. The result is improper crew size, missing items, ineffective methods, delays, unfinished jobs, and dissatisfied customers. Any time that super-

vision does spend on planning reduces the amount of time that would otherwise be devoted to field supervision with associated benefits, such as:

- Supervisory follow-through promotes workmanship and productivity.
- Instruction and training develops apprentices into craftsmen of the highest caliber—capable of maintaining reliability of high technology equipment.
- Through prompt handling of employee issues, grievances are avoided, good employee relations are fostered, and damage to organizational morale is minimized. The resultant high morale and team spirit contribute to increased output, efficiency, and reliability.

THE PROVEN ANSWER

Experience shows that the functions of preparation, supervision, and maintenance engineering are best separated into three individual positions rather than being combined into the responsibilities of one or more individuals. These three functions require different skills, which are rarely possessed by a single person. Therefore, Planning, Coordination, and Scheduling should be a para-professional, staff function, separate from Supervision and Maintenance/Reliability Engineering.

Preparation by a focused staff group having broad prospective of all maintenance needs and commitments is the proven answer. Focused planners can prepare several jobs more effectively and efficiently than craftsmen can plan for themselves, one job at a time, in the midst of their primary responsibility of job execution.

Separation of preparation from execution does not exclude maintenance technicians from involvement in the planning portion of preparation. Mechanics, supervisors, and planners all contribute to the planning process. Craftsmen can contribute in several important ways:

- First, some jobs either by design (minor jobs) or by default (urgent jobs) are not covered by planned job packages. For these jobs, it is significantly preferable for any assigned mechanics to give the job some forethought of their own rather than charging into it without benefit of any preparation at all. Accordingly, it is beneficial that mechanics be trained in the basics of job planning. With the basics ingrained, even urgent

work orders gain some benefit of being planned.

- Second, there are situations where the backlog waiting to be planned becomes out of control. This situation is inevitable unless the planning function is staffed for peaks, which is not optimal. When the situation does occur, the use of selected mechanics to help the planner catch up is recommended. The most ludicrous place to allow the backlog to bottleneck is the "Waiting to be Planned" status. Better to use a small portion of the mechanic staff as temporary planners than to lose control of the backlog, forcing the entire resource of mechanics to work without benefit of planning.
- Third, there are situations where a given mechanic's knowledge simply exceeds that of incumbent planners or supervisors. Such knowledge should be collected and committed to planned job packages for use by all mechanics. It is foolish not to draw upon the best knowledge available.
- Fourth, mechanics can be utilized as planning assistants on a temporary, rotating basis. This approach is particularly appropriate when the planner's span of control (ratio of technicians supported by each planner) is excessive due to budgetary limitations. A number of organizations use this approach with outstanding results. Normally, the rotation period is three-to-six months. Four benefits are derived:
 - Additional planning capacity increases planned job coverage and reduces the backlog waiting to be planned. Planners are infused into the organization to expedite the effective completion of requested work. Backlog should not be held pending planning. The longest period that jobs should sit in the planner queue is three days. There are many delays within the maintenance arena. Do not allow them to be compounded by tardy preparation.
 - Sharing the mechanics' current knowledge with the planners helps to keep the planners' knowledge abreast of field conditions.
 - Mechanics gain perspective regarding the planning process and benefits thereof. This insight improves their use of planned job packages and promotes improved feedback, leading to improvement of the packages for future usage.

- Mechanics return to their crew or team as disciples of the planning, coordination, and scheduling process. This experience enhances training and development of the entire crew.
- A fifth possible involvement relates to participative team concepts, wherein one member of the team is designated as planner for a period of time. The planner assignment is rotated through the crew. This concept replaces a fixed staff position and complies with the idea of self-directed, participative team.
- The irrefutable involvement is that maintenance technicians are always responsible for continuous and constructive feedback to improve future planning.
- Some maintenance organizations have chosen to divide planning responsibility into two positions: field planner and office planner. The field planner, usually a senior mechanic, is responsible to visit the job site to establish scope of the work, determine the best way to perform the job, and identify safety and environmental concerns. The field planner then feeds information to the office planner who interfaces with the CMMIS. The authors do not endorse this approach as a long-term solution. Anybody selected as a planner/scheduler should have the aptitude and desire to learn computerized information systems.

CHANNELS OF COORDINATION AND COMMUNICATION

The Planning and Scheduling group is the hub of intra- and inter-functional coordination and communication. Planners are the principle point of contact and liaison among maintenance, operations, and other supported departments. If they do their job well and are unbiased moderators among functions, supported departments will feel that the maintenance planners are as much a part of each supported department as they are a part of the maintenance department. This direct relationship format is depicted in Figure 4.2.

Direct Liaison provides two-way, closed loop communication within the maintenance department as well as between maintenance and internal customers. The primary linkage with customers is through plan-

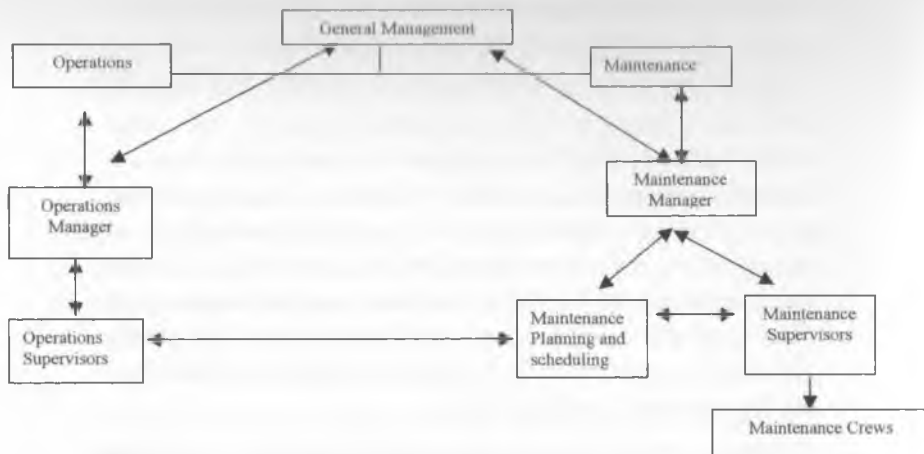


Figure 4.2 Direct Liaison

ners, but this does not restrict direct communication between managers and supervisors and their operating counterparts.

The antithesis is presented in Figure 4.3. In this relationship, planners support the maintenance manager as a staff assistant. All other intra- and inter-department liaison is accomplished by line personnel (managers

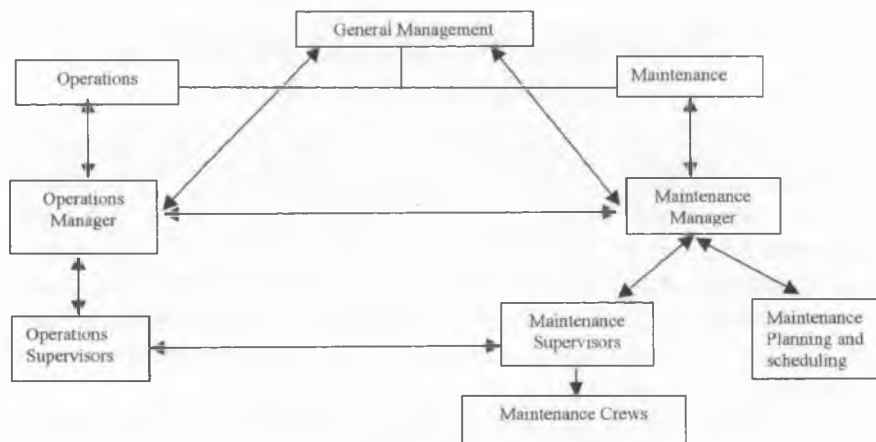


Figure 4.3 Indirect Liaison

and supervisors), as time allows, which, as we have already discussed, is insufficient. Do not allow your installation to pursue this course.

WORKING LIAISONS

In large organizations—in order to improve coordination among maintenance, operations, and other internal customers—consideration should be given to the identification of primary points of contacts (Maintenance Liaisons). Within each major operational unit, such working liaisons are essential throughout the workweek, especially during the weekly coordination and scheduling process. Let's assume there are six operating departments in a continuous operation requiring five shifts to cover 168 operating hours per week. Such a requirement implies 30 operating supervisors who place demands upon maintenance.

Planners cannot coordinate effectively with this many individuals. Therefore, it is recommended that large organizations establish a specific person in each operating department as a focal point for communication and liaison with maintenance. These responsibilities for liaison with maintenance are frequently assigned to a process engineer within each operating department, or to an assistant superintendent. Prior to each weekly coordination meeting, these individuals bring themselves up-to-date in regard to all open work orders originated by personnel within their given department. In this manner, they are prepared to effectively represent the interest of their department at the coordination meeting.

SHOULD PLANNING BE SEPARATED FROM SCHEDULING?

Planning and scheduling can be combined responsibilities of one planner/scheduler preparing all planned work within an organizational unit (operating area, maintenance skill, or maintenance team). They can also be separated responsibilities, with given individuals responsible for specific activities (planning, material coordination, or scheduling).

Planner deployment by operating area is preferable in order to establish strong operating/maintenance partnerships. Figure 4.4 illustrates the combined planner/scheduler option by operating area.

Many maintenance organizations are too small to consider anything other than the combined planner/scheduler option—they have no

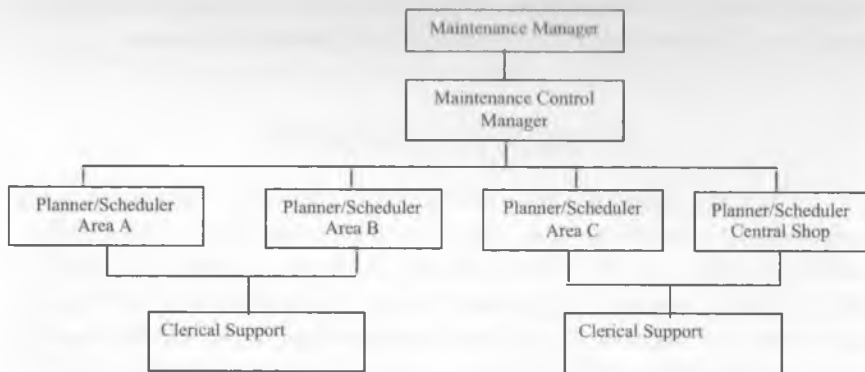


Figure 4.4 Combined Responsibility Option (Planner/Scheduler)

more than a few authorized planner/scheduler positions. It is also difficult to identify planner candidates with sufficient multi-skill knowledge to plan electrical as well as mechanical work. The shortage of positions dictates that the few positions that are authorized be structured as combined planner/schedulers. The next question becomes how should they be deployed: by operating area or by craft skill?

Sometimes, a functional structure is the answer. Planning requires more craft knowledge than scheduling does (the best planners come off the tools). Why not deploy the best planners to that portion of the job that

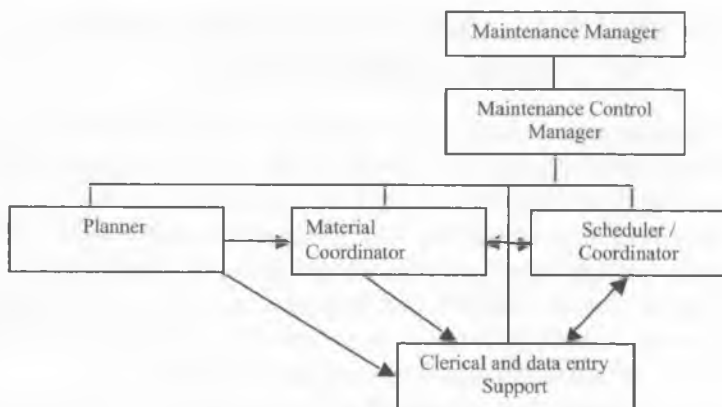


Figure 4.5 Functional Responsibility Option

requires their knowledge? Estimating and scheduling require time management skills. Materials coordination can be performed by persons with good stores and procurement experience. Acceptance of this logic leads to the Functional Responsibility Option (Figure 4.5).

Organizations with 60 or more maintenance technicians warrant at least three planner/scheduler positions and can consider this option. If incumbents have focused, functional strengths rather than general craft knowledge, then the option of horizontally separating planners from schedulers may be essential. When the material management system is complex and planners have limited knowledge of that system, the material coordinator position should be considered. Do not confuse the Materials Coordinator with the coordination aspect of Job Preparation (planning, **coordination**, and scheduling).

As stated above, the best planners come off the shop floor. However, those people may not be good time managers (scheduling). The best schedulers might come from Industrial Engineering or Production Scheduling. If some people have been inherited from the storeroom or purchasing, they may not be candidates for either planning or scheduling, but may be very strong contributors in regard to sourcing, procuring, and expediting (Materials Coordinator).

Other common reasons to structure in this manner are size of the facility and remoteness of the work centers at which maintenance takes place. If the work centers are widespread, this structure should be given strong consideration. Locate planners in the field where work occurs. Locate schedulers centrally with responsibility to distribute resources based upon location of the workload. Also locate material coordinators centrally.

Planners detail the work and identify material needs, and then hand the job off to Material Coordinators who source, procure, and expedite the needed materials. When all materials are on hand, jobs are passed to Schedulers who coordinate with operations and schedule maintenance resources to perform the job at a time when operations can best release the associated asset.

In summary, selection of the most appropriate structure is a local determination based on size of the organization as well as size of the facility and degree of craft knowledge possessed by the incumbent planners and schedulers. Regardless of the option selected, the control span should be held consistent to one planner/scheduler per twenty technicians (1:20). This span is most difficult to achieve within the functional option. Assume five authorized positions in a maintenance organization of 100 techni-

cians. The recommended split would be one material coordinator, one scheduler, and three planners ($5/100 = 1/20$).

RELATIONSHIP WITH OTHER FUNCTIONS

To be effective, maintenance planners/schedulers require:

- ☐ A proper work station conducive to forethought
- ☐ Recognition as important contributors to the maintenance/reliability mission
- ☐ Recognition that they are planner/schedulers, not supervisors, relief supervisors, or "go-fers"
- ☐ Clear definition of their relationships with:
 - Maintenance managers, supervisors, and technicians
 - Operating managers, supervisors, and operators
 - Buyers, so that planners need only prepare purchase order requests; buyers source, prepare purchase order, track, and expedite delivery
 - Storeroom attendants, so that planners need only to identify and reserve required withdrawals; Stores receives, stocks, picks, kits, stages, secures, and delivers to the job site
 - Maintenance/Reliability engineers, so that planners do not have to develop standard operating and safety procedures, and do not have to devote attention to recurring reliability problems
- ☐ Participation of specified operating representatives at weekly coordination meetings to agree upon maintenance jobs to be scheduled during the upcoming week
- ☐ Cooperation from maintenance supervisors and technicians in the effective use and application of planned job packages
 - Feedback from technicians and supervisors regarding specific shortfalls in planned job packages so that improvement of future packages is facilitated

- Feedback from maintenance supervisors regarding compliance with and exceptions to weekly schedules

As previously mentioned, when there are more than five maintenance staff support positions (including planners, schedulers, material coordinators, dispatchers, administrators, and maintenance/reliability engineers), they should report to a position such as Manager of Maintenance Support Services. This position provides the group: coordination, functional discipline, and integrity as well as managerial acumen and clout. These benefits are not always within the capacity of maintenance managers consumed by demands of leadership, supervision, budgetary control, and response to upper management. When responsibility for formal planning and scheduling are added, the burden often becomes too much for the maintenance manager to cover effectively.

The Manager of Maintenance Support Services position is discussed in further detail in Chapter 6 and a position description is provided in Appendix A. When this position is established, it is a broader and more senior replacement for the Maintenance Control Manager position shown in Figures 4.4 and 4.5.



Roles and Responsibilities

Duties that are not the responsibilities of planner/schedulers require as much clarification as those that are. The responsibilities of the four distinct functions that must be provided for in the management of maintenance are:

1. Supervision of work execution (supervisors or team leaders)
2. Engineering dedication to elimination of repetitive failures (maintenance/reliability engineers).
3. Preparation of planned work (planners/schedulers)
4. Administrative support (administrators or clerks)

Control spans for these functions must be in keeping with Good Maintenance Practices, as conveyed in Figure 5.1:

Function	Ratio to Technicians	
	Most Common	Range Based on Conditions
Supervisors	1:10	8 to 15
Planner/Schedulers	1:20	12 to 30
Maintenance/Reliability Engineers	1:40	30 to 60
Administrators	1:40	30 to 60
Composite	1:5	

Figure 5.1 Functional Control Spans (GMP)

In very small departments (under eight maintenance technicians), the maintenance supervisor might perform all four functions, although probably not very well. As the department gets larger, each of these functions needs to be covered by an individual and, eventually, by several individuals.

Each ratio assumes that the other ratios also comply with GMP. If any of the control spans are significantly greater than those listed in Figure 5.1, one or more of the other functions is diluted and functional performance suffers. Maintenance effectiveness rapidly suffers the consequence and the cultural transition from reactive to proactive maintenance with resultant asset reliability is not achieved or, if achieved, is not preserved.

Supervisor Responsibilities

Maintenance Supervisors are responsible for the well being, training, and leadership of team members. They also have responsibility for control and follow up regarding:

- Review, refinement, and finalization of labor, parts, priorities, and methods identified by the planner; the best of plans can be improved
- Having the next job (at least) always ready and for assignment to each mechanic
- Proper, safe, and efficient execution of maintenance work
- Quality, duration, cost, and thoroughness of the completed job
- Time lost between jobs, at breaks, and during shift changes
- Tactical reaction to overruns and interruptions
- Tactical decisions necessary to remain as close to schedule as possible; tactics often must be established after job start (with approvals required before adding resources, contractors, overtime)
- Communication with planners/schedulers, operations, and maintenance/reliability engineering as frequently as possible
- Assurance of reasonably accurate distribution of time and materials to specific jobs

In exercise of these responsibilities, Supervisors also must:

- Balance motivation and discipline by interfacing with each mechanic at least once daily and visiting significant jobs two or three times daily
- Give adequate time and attention to formal and on-the-job training, never neglecting development of each team member; by this means, Supervisors lead the training process identifying, providing, or obtaining the skills training needed by each team member
- Act upon requests for support of team members and provide prompt and fair handling of their issues—effective listening reduces grievances!
- Control of tardiness, absenteeism, and vacations

Maintenance/Reliability Engineer Responsibilities

Maintenance/Reliability Engineers are responsible for applying technical skills and engineering ingenuity to avoid and correct equipment problems that cause repetitive downtime and quality variations. Their responsibility includes:

- ☐ Maintainability of new installations
- ☐ Identification and correction of chronic, costly, and dangerous equipment problems
- ☐ Technical advice to maintenance and operations
- ☐ Designing, monitoring, and continually refining effective and economically justified preventive/predictive maintenance programs, including:
 - Proper operation and care of equipment
 - Comprehensive lubrication program
 - Inspections and overhauls of equipment
 - Vibration and other predictive analyses
 - Protection from the environment
- ☐ Maintaining and analyzing equipment data and historical records to reduce or eliminate repetitive failures

- ☐ Ultimate establishment of an optimized plan for the maintenance of each asset

Planner/Scheduler Responsibilities

Planners/Schedulers are responsible for logistical support to remove all avoidable barriers standing in the way of effective completion of maintenance work. Planners perform the bulk of job preparation prior to coordination, scheduling, and execution of each job. This function encompasses:

- Job plans and estimates
- Assurance that all necessary logistics are identified and provided for
- Customer liaison for all planned maintenance work
- Continuous communication with all involved and impacted parties
- Coordination of manpower, parts and materials, equipment, and access to the asset in preparation for effective work execution
- Cooperative and coordinated scheduling of jobs in the order of agreed-upon importance
- Full day of work, each day of the schedule week, for each maintenance technician
- Arrangement for delivery of materials to the job sites
- Assurance that even low-priority jobs are accomplished in a reason period of time
- Maintenance of records, metrics, and control charts
- Management reports regarding actual performance versus established goals and targets

Maintenance Administrator Responsibilities

Administrators are responsible for all routine data entry and withdrawal associated with maintenance activity, most of which is associated with the CMMIS. The purpose is to conserve the capacity within the other three functions.

Factors impacting control spans for the preparatory trio are discussed below.

FACTORS INFLUENCING PLANNER/SCHEDULER CONTROL SPAN

Control span ratios should be used only as guidelines. Planner staffing depends upon a number of factors and should be tailored to fit the specific local situation, considering:

- Current state of the CMMIS installation has major impact on planner/scheduler efficiency and, thereby, the realistic control span. Is the software fully loaded? This includes: record of fixed information for each asset, bill of material for each asset, perpetual inventory for stock items with associated provision for reservation, equipment history, library of planned job packages, library of labor estimates, on-line library of prints, etc.
- Complexity of the facility to be maintained.
- Complexity of the maintenance organization itself (centralized, decentralized, combined, work type, process teams, etc.)
- Complexity of craft structure. For instance, some unionized organizations still have several distinct crafts, as opposed to multi-skill concepts. The more that crews are specialized, the more likely additional planners will be needed (see previous chapter).
- Structure of the preparatory staff is also discussed within the previous chapter. Also, other maintenance support in place as discussed above (e.g., maintenance buyers, storeroom attendants, CMMIS administrators). What degree of support is being provided by traditional Purchasing and Storeroom functions? If these functions are neglecting their traditional responsibilities supportive of Maintenance, inappropriate duties fall to planners/schedulers including one or more of: sourcing, procurement, expediting, receiving, stocking, picking, kitting, staging, securing, and delivery to the job site.
- Where the lines of responsibility are drawn between planners/schedulers and: other staff-support functions, super-

visors, requestors of maintenance support, and maintenance/reliability engineers.

- Number of craft personnel performing planned and scheduled work as opposed to urgent response and routine service, which have minor impact on workload of planner/schedulers.
- Current state of the maintenance/reliability culture and environment. If still in reactive mode rather than proactive mode, it is extremely difficult to keep planner/scheduler resources focused on future work (next week) as they should be. They are constantly pulled into daily urgencies.
- Level of planning and scheduling needed. There are maintenance environments that dictate a high proportion of urgent response versus planned maintenance and routine PM/PdM (hospitals, schools, offices, small/simple plants, etc.). Think about it this way—the planner/scheduler ratio (1:20) need be applied only to that portion of the total maintenance workforce that engages in planned and scheduled work. In a proactive environment this is typically 75%, but in some of the industries mentioned it may be less than 50%.
- Method of estimating to be used. There is significant difference between the time to make a gross estimate of labor-hours required to perform a job versus the time to engineer a standard by applying standard data (building blocks of individual steps). Sophistication of the measurement technique applied increases as the installation matures. Because creation of planner reference libraries also increases, the time required to develop estimates actually diminishes.
- Other peripheral responsibilities placed upon the planners (timekeeping, vacation control, absenteeism, management reporting, etc.). These time-consuming routines should be performed by administrative support personnel.
- A Worksheet is appended (H) to help determine a given organization's appropriate ratio for planner/scheduler to technicians.

Planner/Scheduler Responsibilities to the Material Management Process

Here is a good example regarding essentialness of “partnership. Both parties are dependent upon support of the other (a classic two-way street). Purchasing and Stores cannot perform their job without the support of planners/schedulers. One Purchasing executive put a printed banner over the entrance to her area. It read, “Your lack of planning does not necessitate an emergency for me.”

Reactive Maintenance drains Purchasing capacity. One reactive environment the authors encountered ran a \$330,000 annual airfreight bill due to the lack of proactive preparation.

Purchasing, Stores, Maintenance, Engineering, and Operations all have responsibilities regarding material support and control. Responsibilities specific to planner/schedulers include:

- ☐ Identify Material Requirements of each Planned Job.
 - Prepare Bill of Materials.
 - Establish Acquisition Plan.
 - Determine what items are in stock, reserve them, and initiate Stock Requisitions.
 - Identify items that must be directly ordered and prepare associated Purchase Order Requests, sourcing, procuring, and expediting these items are the responsibility of Purchasing.
 - Prepare Work Orders for in-house fabrications.
 - Prepare Purchase Order Requests, with reference to associated work orders, for contractor support and equipment rentals.
- ☐ Request direct purchase of items with adequate lead-time so that purchasing can be performed effectively and economically.
- ☐ Suggest inclusion of items to be added to authorized stock, with recommended minimum and maximum quantities.
- ☐ Review and recommend adjustments to inventory parameters (Min, Max, reorder point).
- ☐ Notify Purchasing when unusual demand may necessitate adjustment of restocking parameters.

- ☐ Notify the Storeroom of schedule delays that might influence care of staged materials.
- ☐ Provide Schedule information with sufficient lead-time, allowing the Storeroom to pick parts, and stage them for pickup or delivery to drop points and job sites.
- ☐ Assure that unused materials are returned to the Storeroom in good and clean condition for restocking (supervisor).
- ☐ Correct work order information and planned job packages when unneeded material is returned.
- ☐ Notify the Storeroom of stock items that will become obsolete or in excess (due to reduced operations or equipment replacement). Earliest possible notification is needed in order to prevent buildup of surplus inventory. The process of identifying parts that will become obsolete should be an ongoing routine, (Maintenance/reliability engineer).

Material Management Requirements in Support of Reliability

As major internal customers of Purchasing and Stores, maintenance craftsmen are the primary users of supply materials and spare parts. Performance standards must define the levels of materials management support required if the reliability mission of Maintenance is to be fulfilled. These standards should be consistent with the following characteristics. Purchasing must:

- Process purchase order requests promptly with minimum in-house lead-time.
- Obtain and communicate lead-time commitments from vendors and evaluate their performance in meeting these commitments.
- Track materials from placement of the Purchase Order Request to delivery to the originator.
- Follow-up availability and delivery of parts required to perform planned work orders.
- Notify Maintenance of delivery delays as soon as they are known.
- Expedite as necessary.

Storeroom Requirements in Support of Reliability

- Keep inventory orderly, with easily identified location.
- Keep adequate quantities of each stock item on hand to meet the day-to-day needs of Maintenance.
- Promptly reorder materials that are at their reorder point so stocks can be replenished before they run out.
- Maintain unique item identification of spare parts, materials, supplies, and tools.
- Provide an up-to-date catalog listing parts and supplies in stock by location.
- Cross-reference the catalog by Stores Reference Number, Vendor Serial Number, Key Word Description, and Where Used.
- Apply inventory control in relation to item value. High-value items should be controlled most closely. Medium-value items should be moderately controlled, and low-value items (such as hardware) warrant the least control, such as free-bin stock.
- Provide a quick issue service at the storeroom window.
- Process non-stock receipts and back order receipts as quickly as possible and notify users of arrival.
- Provide timely delivery of materials to secured locations, either at the job site or at specified drop points at maintenance shops and other areas of the facility. Delivery should be synchronized with scheduled start of planned jobs and expedited to urgent jobs.



Managing the Planning and Scheduling Function

Two issues are associated with management of the planning function. First is management of planning as a function; second is management of the incumbent planners as people. Management of planning involves measuring and plotting trends, benchmarking relative to prior periods and Good Maintenance/Reliability Practices (GMRPs). Details about measuring functional effectiveness are provided in Chapter 18, *Job Feedback, Closeout, Analysis, and Schedule Compliance*. The people side of the equation is discussed below.

MANAGEMENT OF PLANNERS

Maintenance managers normally lead planning efforts as well as the technicians when the maintenance department is small, with less than twenty technicians and only one planner. Maintenance departments that are large enough to warrant several support positions (including planners, maintenance/reliability engineers, system administrators, etc.) should have a Manager of Maintenance/Reliability Support Services. These managers are responsible for supporting maintenance with a variety of services as listed above, plus the CMMIS and, in some cases, maintenance procurement and inventory management.

The Manager of Maintenance/Reliability Support Services is:

- ☐ Champion and chief advocate of the Maintenance Excellence process
- ☐ Marketer and seller of maintenance control throughout the organization

- ☐ As manager of all maintenance support services (staff, not line), the chief planner/scheduler, chief of maintenance engineering, head librarian for the maintenance and planning technical library, and CMMIS administrator
- ☐ Owner and controller of the CMMIS who, as such,:
 - Controls system security
 - Steers and continuously seeks system improvements (coding, user friendliness, application, and reporting)
 - Trains all parties regarding their responsibilities to the system and their usage thereof
 - Serves as compliance officer to see that data input is accurate, complete, and sound
- ☐ Leader of database development and refinement (such as improving the equipment records, job plans, and job estimates)
- ☐ Quality assurance and control leader of CMMIS, Planning, Scheduling, and RCM (PM/PdM and root cause analysis)
- ☐ Chief analyzer of system information and trends of improvement and maintenance metrics, including balancing of resources with workload
- ☐ Teacher of analytical skills such as analysis of database information, root cause failure analysis, and other modalities
- ☐ Developer and definer of the issues and recommendations stemming from the Maintenance/Reliability Support Services Team
- ☐ Manager who may also be responsible for the maintenance storeroom and for procurements
- ☐ Keeper of the annual schedule of major events
- ☐ Analyst of special studies requested by management (or self-generated)

MANAGING PLANNING

Managing the planning function usually boils down to regular assessment of continuous improvement. Are planning, coordination, and scheduling paying off? Is product produced per maintenance dollar con-

tinually increasing? The answers will be "Yes" if all the prerequisites are in place and the function is well managed.

POSITION BENEFITS—MANAGER OF MAINTENANCE/RELIABILITY SUPPORT SERVICES

Support services in a large maintenance organization become a very important and diverse group. Gaining their full potential requires full time direction by a leader with:

- Analytical, presentation, and report writing skills
- Managerial acumen
- High regard, respect, and clout throughout the organization

Resistance to the position commonly raises the question: Can't the Plant Engineer or Maintenance Manager fulfill these roles? Yes, provided they have the aptitude and time commitment. Most often, they do not have the time. Their primary responsibilities are managerial, line leadership of supervisors and their crews, and capital projects. When they come from a mechanical or engineering background, they often lack experience, aptitude, and inclination toward information analysis.



Backlog Management

Control of backlog is fundamental to successful management of the maintenance function and thereby to realization of reliability. It must be held within manageable control limits by preserving a balance between resources and workload. Backlog is defined as the net workload, measured in labor-hours, requested but not yet completed. Another definition is that the backlog is all identified work remaining to be performed. If a job has been started, only the portion of labor-hours still to be completed remains in quantified backlog. Some of the jobs in backlog are delinquent in terms of promise date, but most are not delinquent. Being part of backlog does not connote delinquency.

Maintenance work is continuously created over time as:

- Equipment is utilized and reliability diminishes
- Facilities are exposed to weather and usage
- PM/PdM inspections and other routines come due
- Operational processes are altered or replaced

Planning for Maintenance/Reliability Excellence begins with Macro-Planning, which is perpetual balancing of maintenance resources with maintenance workload. If balance is not preserved between these two variables, the operation cannot achieve reliability.

Misuse of assets and deferral of essential maintenance work accelerates deterioration, further increasing the requirement for maintenance resources. When workload exceeds resources on an ongoing basis, the maintenance organization falls behind the rate of deterioration and the

amount of reactive maintenance grows. You can walk through a plant in this state and see the deferred jobs, broken and rusted items, and general disarray that result when deterioration gets the upper hand. This situation is common and deleterious to morale, product quality, safety, and job fulfillment for maintenance workers.

When an organization fails to stay ahead of deterioration, it cannot maintain reliability and certainly cannot climb out of an existing reactive condition without influx of adequate resources. Without such influx, it is impossible for the facility to be proactive because response to failures detracts from resources available for backlog relief, PM/PdM inspections, and newly-identified corrective work continually adds to backlog. The resource-demand imbalance continually worsens! When an organization is in this state, any deficiencies found by proactive inspections are not likely to be corrected before failure. This condition renders proactive crews (PM/PdM and Backlog Relief) impotent, destroying morale and effectiveness of the entire maintenance organization. Think about that! Inspections are a proactive investment. Prompt correction of identified deficiencies provide the return (ROI). Failure to capture the return on investment is asinine.

The backlog assigned to each crew or trade (measured in weeks) must be calculated and displayed on a monthly basis. The calculation vehicle is the "Maintenance Work Program" discussed later in this chapter and the next. Smart managers use this vehicle to base their decisions upon facts. Decision areas include:

- Staffing by type of work and skill required.
- Maintaining required resources by adjusting overtime, contractor support, or staffing; in that order of flexibility.
- Evaluating capacity available to handle project work and/or major shutdowns.
- Deciding if and when to utilize contract support to meet abnormally high demands.

BACKLOG MANAGEMENT

Well-managed maintenance departments hold backlog within established upper and lower control limits. Backlog Ready to be Scheduled is part of but isolated within Total Backlog. Two-to-four weeks

of Ready Backlog and four-to-eight weeks of Total Backlog are commonly accepted control limits.

Ready Backlog is just that—jobs that are ready to be scheduled. All tools, parts, materials, drawings, and authorizations are in hand. The job can be scheduled any time Operations can release the asset to Maintenance and required maintenance resources can be made available. During the Weekly Coordination Meeting, the parties agree upon a specific day and time to schedule each job. Total Backlog includes Ready Backlog plus all the other open work orders still in one or more Status Holds. These latter jobs could be missing parts, authorization, budget, or some other element (see Figure 7.1).

Too much backlog means excessive delay in response to customer needs and ultimately leads to Deferred Maintenance, the dreaded D word. Deterioration gains the upper hand. Capacity, reliability, safety, and customer service suffer. People requesting lower priority jobs notice that their jobs never get scheduled. The resultant lack of customer satisfaction contributes to the low esteem in which the maintenance function is held in many organizations. A well-designed and administered Priority coding structure with an aging feature can prevent such jobs from becoming indefinitely delayed or lost in the backlog. Even medium- and low-priority jobs must be completed within a reasonable time frame.

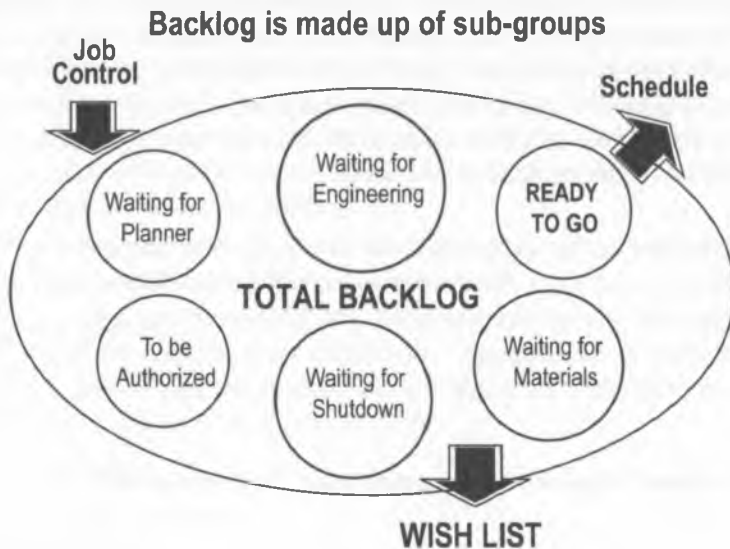


Figure 7.1

Backlog is the holding tank that helps regulate the flow of work into the maintenance shop (Figure 1.2 and 7.2). Jobs are scheduled in a smooth flow limited by available labor-hours. When Ready backlog increases beyond four weeks or Total backlog beyond eight weeks, the consequences can be severe (the holding tank overflows).

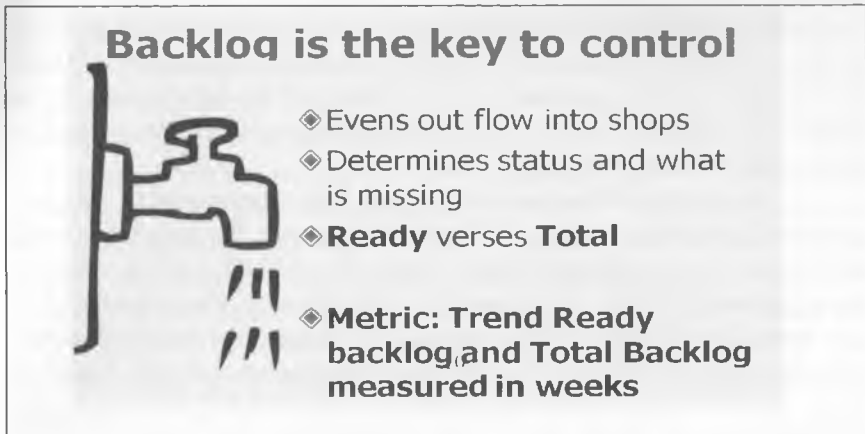


Figure 7.2

Too little backlog indicates insufficient workload to keep the existing maintenance staff effectively deployed and utilized. Work tends to be stretched to fill out the day and crucial maintenance resources are wasted. Backlog quantification (derived from Work Programs) should be regularly compared to available labor resources. Resources should be flexed (adjusted) up or down as analysis indicates necessity. There are three flexible resources for consideration.

1. **Overtime** is the easiest resource to adjust, assuming that it is not already excessive. An overtime range between 7% and 15% of paid straight time is recommended. Too little overtime is not sound maintenance strategy. Too much overtime is not healthy, safe, or productive for either the company or the employees.
2. **Contract Support** is the second most flexible resource.
3. **Permanent Staff** is the least flexible alternative. This form of adjustment should be deployed only when resources and workload

are outside the target range for backlog weeks, no relief is in sight, and the condition appears to be permanent (or at least semi-permanent).

EXAMPLE OF BACKLOG OUT OF CONTROL

At a steel mill in Western Pennsylvania, the total backlog was 96 weeks per tradesperson! That meant that, even if the plant were shut, the current resources would need almost two years to clean up the backlog. The consequence was a complete lack of management control. Additionally, there was a nasty work environment with low morale and self-esteem, a poor safety record, poor customer service, low profits, and a constant state of fatigue.

JOB STATUS

It is essential that planner/schedulers know the current status of each job in the backlog. Such status is documented via Status codes within the Computerized Maintenance Management Information System (CMMIS). The Status Code feature brings order out of chaos. It is the most important code in the entire Work Management System (WMS/CMMIS), especially for Planners/Schedulers. An example of status code structure is presented in Figure 7.3 on the following page.

The existence of a coherent backlog list by status answers questions generated from all levels of the organization. The CMMIS must enable planners to access backlog by:

- ☐ Planner supporting the area, trade, or job
- ☐ Job status
- ☐ Crew/supervisor
- ☐ Asset, machine, or production line
- ☐ Originator
- ☐ Age of work order
- ☐ Due date

Code	Description	Form of backlog
P	Waiting to be planned	Total
E	With Engineering	Total
A	Awaiting approval	Total
PF	Pending Funding	Total
DF	Deferred	Total
PO	Waiting for PO to be issued	Total
M	Awaiting receipt of materials	Total
FP	Further planning required	Total
DP	Requires downtime — programmed shutdown	Independent *
PW	Requires downtime — weekend	Ready
DS	Requires downtime — asset not scheduled	Ready
DO	Requires downtime — window of opportunity	Ready
R	Ready to be scheduled	Ready
FI	Ready for fill-in assignment	Ready
S	On the current schedule	Not in backlog
CO	All Maintenance work completed	Not in backlog
CM	WO pending material closeout	Not in backlog
C	Closed to Equipment History	Not in backlog
CP	Print Revisions not yet received	Not in backlog

*Major shutdowns several months off are held in an independant backlog

Figure 7.3 Work Order Status Codes

Each of the access paths to backlog answers different questions from different questioners. For example, sorting the backlog by machine identifies other jobs that might be performed during the access window to the given asset.

Backlog must be complete, current, pure, and reliable. Trends of Backlog Weeks are going to be presented to senior managers with a recommendation for force adjustment of one form or another. The basis of recommendations “better hold water.” How would the integrity of your current backlog stand up to the test shown in Figure 7.4?

Does your backlog include . . . ?

• Work Orders that are completed, but nobody has bothered to close them out?
• Duplicate jobs under different names or different work orders
• Jobs greater than a year or two old — excluding those awaiting a known, future shutdown?
• Jobs for which no one recognizes the originator or why the job was needed in the first place
• Small jobs supportive of larger jobs, but not recognized in backlog
• Poorly described jobs (no one knows what to do, or what to do it to)
• Job status not recorded or not updated and nobody recalls current status. Were parts required? Were they ordered? Were they delivered? If so, where are they now?
• Jobs needing to be done that are not recorded in backlog

Figure 7.4 Checklist for Validation of Backlog Integrity

A few suggestions regarding the backlog validation process are offered:

- Sort the backlog by asset or work center. Involve the asset custodian to participate in the validation process.
- When removing duplicate jobs, remove the lower priority or the most recently requested. If the estimated labor-hours differ, re-estimate them.
- Transfer all appropriate detail to the work order remaining in backlog.
- Do not eliminate aged jobs without determining if they are still required.
- Needed jobs beyond current budget must be brought to management attention with a case for approval.
- If there are two or more work orders for the same PM/PdM routine on the same asset, the oldest one should be closed with notation that it was missed (non-compliant). This effort is not necessary in a well-designed CMMIS because a shadowing feature catches the duality and automatically closes the oldest. It is important that the non-compliance be captured in

Equipment History so that PM/PdM Effectiveness in relation to Reliability can be accurately evaluated."

Jobs not captured in the backlog constitute a special problem. Internal customers may be conscious that Maintenance is already overloaded and see no point of compounding the situation. As a result, they simply stop inputting to the Work Order System, except for failures (urgent work). The reality is if the work needs to be performed, Maintenance must be made aware of it. Such work must be evaluated relative to other demands and assigned a priority/criticality ranking. An ignored maintenance problem might catapult to the forefront due to an accident, a production increase, a quality initiative, or ergonomics. If the job is not added into a controlled backlog, Maintenance does not know of it—at least officially. If a need exists, it should be identified and quantified as part of the backlog. Otherwise, it simply swirls around in the background as part of the amorphous mass of undefined deferred maintenance.

Development of Work Programs and the Maintenance Calendar

Work Programs, as previously stated, are the vehicle whereby maintenance resources are perpetually balanced with maintenance workload. Without this balance, maintenance/reliability excellence cannot be achieved. Work Programs consist of four sections (see Figure 8.1).

- ❑ The top section quantifies **Total Labor-Hours Authorized Per-Week** including, straight-time hours, overtime hours, and contractor support hours, minus any hours to be loaned to other departments. In the example, available hours total 896. These numbers should be readily available from the Maintenance Department Budget.
- ❑ The second section quantifies labor-hours associated with various **Indirect Commitments** such as vacations, absenteeism, training, and meetings. This subtotal (330 labor-hours) is subtracted from Total Labor-Hours (896) to establish how many labor-hours (566) are projected to be available for performance of **Direct Work**. These numbers should be obtainable from Payroll and Accounting Systems plus the Vacation Schedule.

Available Resources				
Crew size 20	Straight time labor hours per week		800	
	Planned overtime per week		96	
	Labor hours contracted or borrowed per week		0	
	Total Labor Hours Available per Week		896	
	Less indirect commitments (weekly average)			
	Lunch (if paid)		0	
	Vacation		120	
	Absence		24	
	Training		56	
	Meetings		40	
	Special Assignments		40	
	Average labor hours loaned to other areas		40	
	Other indirect		10	
	Total Indirect Labor Hours Projected per Week		330	
	Total Labor Hours per Week Available for Direct Work		566	
	Commitments Other Than Backlog Relief (Weekly Average)			
	Emergency/Urgent (unscheduled)		100	
	Routine PM		120	
	Other fixed routine assignments		0	
	Sub Total		220	
	Net Resources Available for Backlog Relief		346	
Backlog Data				
	Current	Backlog Weeks	Current	Target
Backlog Labor Hours Ready Status	3200	Ready	9.2	2 to 4
Total Labor Hours of Backlog	4800	Total	13.9	4 to 6

Figure 8.1 Work Program

- The third section quantifies typical weekly consumption of resources for Direct Work other than Backlog Relief — namely Urgent Response (100) and PM/PdM Routines (120). This subtotal (220) subtracted from 566 labor-hours leaves only 346 labor-hours available for **Planned Backlog Relief**. Urgent hours should be obtainable from the CMMIS via a sort of Weekly Activity by Priority Code. A four-week running average is rec-

ommended. PM/PdM hours can be obtained from the PM/PdM database (see section below regarding Annual Schedules). Determine the labor-hours required to fulfill the PM/PdM process as currently engineered. These hours should be averaged and balanced across 52 weeks per year. The PM/PdM program should place a level demand on resources week-to-week. If resources are not currently balanced with workload, required hours may be significantly greater than current average PM/PdM hours. Such insight is representative of the reality that Work Programs bring forth.

- ❑ In the fourth section, resources available for backlog relief (346) are divided into current levels of **Ready Backlog** (3200 labor-hours) and **Total Backlog** (4800 labor-hours) to establish Backlog Weeks (9.2 and 13.9 respectively) compared to **Established Control Limits** (2-to-4 and 4-to-8 weeks). Current backlog by status should be readily extractable from the CMMIS.

Few senior managers have career experience with maintenance/reliability. Therefore, they are dependent upon insights available from the CMMIS and Work Programs. Using data from the above example and assuming such insight is not regularly made available,

It is likely that only the top level information is available (4800 labor-hours of backlog and 896 authorized labor-hours per week). Management's perception would be that Total Backlog equals 5.4 Weeks (4800/896) — well within control limits of 4-to-8 weeks. All is apparently well! Management probably thinking resource reduction. But, as we have seen, reality is quite different (13.9 Weeks of Total Backlog). The real need is for resource increase.

Do not develop Work Programs exclusively for the total maintenance work force. The **average** condition can be very misleading. Balance must be preserved at the crew level, not simply for the entire maintenance organization. The average condition for the entire organization might appear to be under control only because the backlogs of specific teams, areas, skills, or shifts are out of control (one or more to the high side, with others to the low side). Such imbalance is common. There may be too many

mechanics and too few electricians. Perhaps too many technicians are assigned to the Primary Department and too few to the Secondary Department. Or, too many are working the day shift where there is little access to equipment while too few work the night shift when more access is feasible. Another common distribution problem is everybody working the normal weekday schedule with only overtime resources left to work weekends when production is not scheduled and equipment access is plentiful. In all these examples, workload is not under control, despite overall adequacy of resources. The problem is poor distribution of authorized resources.

Work Programs should be developed for the whole maintenance group on a monthly basis, and for each maintenance team at least quarterly. The resultant data should then be trend charted to trigger appropriate managerial action (resource increase, resource decrease, no adjustment necessary)—see Figure 8.2.

Situations as the above should trigger consideration of Multiple-Skills Training. Cross-trained technicians greatly enhance the ability to distribute resources according to what form of workload exists where.

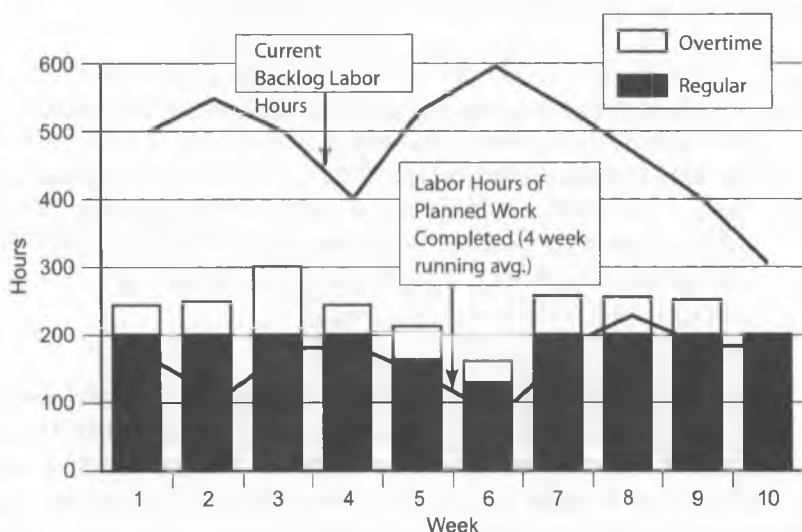


Figure 8.2 Trend Chart of Backlog Weeks

Steps of Work Program Development

- ☐ Make allowances for all commitments to indirect activity.
- ☐ Clarify the labor hours required for response to breakdowns and urgent jobs.
- ☐ Clarify the labor hours required to meet PM/PdM requirements.
- ☐ Quantify labor hours available for relief of backlog.
- ☐ Quantify backlog weeks (Total and Ready).

Analyses Stemming from Work Programs

- ☐ Ensure that expectations for backlog relief are realistic.
- ☐ Define the labor hours of work to be loaded to each crew's weekly schedule.
- ☐ Clarify resources available to absorb project work versus the need to utilize contract support.
- ☐ Ensure that resources are sufficient to meet requested completion dates (real or implied by assigned priority) and that even low priority jobs can be completed in a reasonable period of time.

ANNUAL MASTER SCHEDULES

Two annual schedules are essential; they significantly facilitate preparation of week-to-week schedules throughout the year.

The Annual PM/PdM and Programmed Maintenance

Master Schedule ¹

This schedule is synchronized with the Annual Operating Schedule in order for Maintenance to take advantage of windows of opportunity within the Operating Schedule. There may be periods of low demand for production output. There may be less heavily scheduled:

¹*Courtesy of Mark Goldstein*

- ☐ Shifts of the day
- ☐ Days of the week
- ☐ Weeks of the month
- ☐ Months or Quarters of the Year

Fit feasible maintenance requirements into such windows.

Balance daily and weekly maintenance requirements evenly across the 52 weeks of the year. There should be minimal PM/PdM peaks and valleys throughout the year.

The Annual PM/PdM Schedule feeds into the Annual Maintenance Calendar.

The Annual Maintenance Calendar²

This calendar documents major company, statutory, production, and maintenance events for the year. Adjustments and updates are made throughout the year with as much lead-time as possible. As stated in the previous chapter, workload associated with each major maintenance event is held in independent backlog buckets. See Chapter 20 for Planning and Management of Major Maintenance Shutdowns and Projects.

Preparation Steps

- ☐ Identify all the major maintenance events for the next year including: annual and quarterly PM/PdMs (see 52-week PM/PdM schedule above), scheduled shutdowns, and known large projects.
- ☐ Identify major production events for the coming period such as product year change over, large projects, expansions, product line changes and reductions.
- ☐ Identify events for individual maintenance technicians (vacations, training, temporary assignment, etc.).
- ☐ Identify any ongoing sickness or accident recovery in the maintenance work force.
- ☐ Identify events for the plant (bigwig tours, etc.).
- ☐ Identify holidays and community events.

² Courtesy of Les Kellogg

Sizing the Maintenance Staff

Frequently in a variety of situations, the size of maintenance staffing is questioned, “How many maintenance positions are necessary to adequately maintain this facility?” Everybody seems to have an opinion. The question may be raised during the annual budgeting process or at times when there is pressure to reduce cost or cut headcount. The reality is that nobody can possibly know the necessary staffing without “definition and measurement of required workload.”

To begin with, the appropriate question is, “How many maintenance technicians with necessary skills are required to effectively maintain this facility at a specified level of reliability?” The point of difference is that “adequately” has no place in World-Class Reliability. Shortage relative to “required” staffing inevitably results in deterioration of process reliability and facility condition.

The staff required to preserve an asset relates to size, usage, and condition. Surveys and benchmarking against other sites may provide some insight, but there are simply too many variables and unknowns when trying to compare one operation to others. All the variables identified in Figure 9.1 influence Workload (labor-hours—the common unit of measurement that dictates required staffing)!

STAFFING PROCESSES

Simply put; there are no commonly accepted processes by which to determine required maintenance staffing. Attempts to do so are usually

Influences
Facility Related
Geographic Location, Size and Complexity
Type and Quality of Construction
Age of Buildings, Equipment, and Vehicles
Condition and Magnitude of Deferred Maintenance
Laws, Building Codes, and Statutes
Industry Related
Industry Segment
Process Abrasiveness
Operating Hours Per Week (continuous or batch)
Demands Upon Installed Capacity (%)
Frequency of Product Changeovers
Local Practices
Reliable-Life of Procured Equipment, Parts and Materials
Culture and Environment (Proactive vs. Reactive)
Provisions to Facilitate Performance of Maintenance

Figure 9.1 Influences Upon Required Maintenance Staffing

inadequate and indefensible. Maintenance does little to justify current staffing, much less build a case for required staffing. Finance and Accounting have only actuarial data or they back into expense budgets from projected revenues. Resultant decisions by Management are very arbitrary. Even when there has been Sales Growth or Expansion of Capacity, it remains very difficult to gain approval for additional maintenance resources.

The best approach to take depends upon whether the facility is new or existing, and where existing facilities stand within the transition from a reactive to a proactive culture and environment. There are essentially four stages of transition, each with a best available approach offered.

1. Initial Approach

This approach is applicable to a new facility or to any facility that has yet to develop a Maintenance informational database. It is based on an academic formula that provides a quick and approximate method of estimating maintenance staffing (+/- 30%). It makes no provision for custodial services. However, every situation demands a starting point.

$$\text{MBS} = \frac{[(\text{RCB} \times \text{BMR}) + (\text{RCSE} \times \text{EMR})] \times \text{NMF} \times \text{LP}}{\text{AASTLR}}$$

Where:

☐ MBS = Maintenance Budgeted Staffing

☐ RCB = Replacement Cost of Buildings

Stationary Equipment such as building utilities are included. R.S. Means Costing Information may be helpful if this value is not readily available.

☐ BMR = Building Maintenance Ratio

BMR is the percentage of asset value that must be reinvested to ward off deterioration. For standard building types, the common range is 0.5%–2% per year. A 2% MR implies that the entire building would be replaced on a 50-year cycle. The exact amount depends on the issues brought up at the beginning of the chapter (including building use, climate, and type of construction.)

☐ RCE = Replacement Cost of Equipment

This factor includes process, production, packaging, warehouse, and mobile equipment. In capital-intensive industries, the RCE is of significantly greater magnitude than the RCB.

☐ EMR = Equipment Maintenance Ratio

For much equipment this ratio is 3%–15% per year of RCE. For example, service contracts for electronic equipment, including full labor and parts, are often priced at 12% of replacement cost (which includes both profit and travel).

☐ NMF = Non-Maintenance Factor

The relationship between total time and material spent on all activity performed by the maintenance function and the time and materials spent on true maintenance, as opposed to non-mainte-

nance work which the maintenance department performs (renovations, new construction, operating support such as set ups, change-over, and replacement of production expendables). This factor must be included in the equation. Although these efforts are commonly among the responsibilities of the maintenance department, they are not required to preserve the assets and, therefore, are not by definition true "maintenance."

When there are several categories of assets, consider applying separate factors for each (e.g., production equipment, mobile fleet equipment, buildings).

❑ LP = Labor Percentage

This value is the percentage of the total maintenance budget that is Direct Labor without fringes or overtime premiums. If these two latter items are included in LP, they must also be reflected in the denominator of the equation (AASCLR).

❑ AASCLR = Average Annual Straight Time Labor Rate

This factor is the average hourly labor rate of all maintenance employees annualized.

Example of the Calculation (Given the following assumptions)

RCB	=	\$50,000,000
RCE	=	\$160,000,000
BMR	=	2%
EMR	=	7%
NMF	=	1.25
LP	=	50%
AASCLR	=	\$41,600

Applying the formula, we find:

$$MBS = \frac{(RCB \times BMR) + (RCE \times EMR) \times NMF \times LP}{AASCLR}$$

$$\begin{aligned}
 MBS &= \frac{(\$50MM \times .02) + (\$160MM \times .07) \times 1.25 \times .50}{\$41,600} \\
 &= \frac{(\$1MM + \$11.2MM) \times 0.625}{\$41,600} = \frac{\$7.63MM}{\$41,600} \\
 &= 183 \text{ Positions}
 \end{aligned}$$

The RCB and RCE (Replacement Cost for Buildings and Equipment) should be readily available for new installations. For older facilities, they should be available from accounting records in the form of Gross Depreciable Value and Net Depreciable Value. Gross Value and Replacement Value should be comparable for new installations.

Net Depreciated Value is Gross Depreciated Value minus Depreciation Written Off (life to date). Application of Gross Value to the formula results in greater staffing than application of Net Value does. The problem with Gross value is that it lacks any inflation value for the many years that older facilities have been in service.

The problem with Net Value is that it implies that maintenance workload decreases with age. The reality is that maintenance workload increases with age.

Some financial systems periodically establish Current Replacement Value. Replacement Value is calculated using today's construction and procurement costs. People who concern themselves with it are involved with Risk Management. If available, this is the value to be applied.

2. Established Approach

This approach is applicable to maintenance organizations established in terms of time (several years) and that have a reasonably reliable CMMIS providing information and controls by which to manage (including sizing) the maintenance function. Figure 9.2 illustrates buildup of a business argument presentable to management in regard to maintenance staffing.

Let's review the buildup left to right and bottom to top. Initial staffing stems from quantification of workload experienced during the previous year. The quantified positions do not necessarily coincide with actual staffing during the previous year. Staffing progressively diminishes (left to right) as the culture/environment transitions from Reactive to Proactive.

Preventive/Predictive Maintenance Inspections

Begin with preventive/predictive inspections. Presumably, the PM/PdM process has been engineered (not yet optimally in this approach). The program has been presented and sold to management and the organization. The staffing required to perform the process as designed and approved is 15 positions.

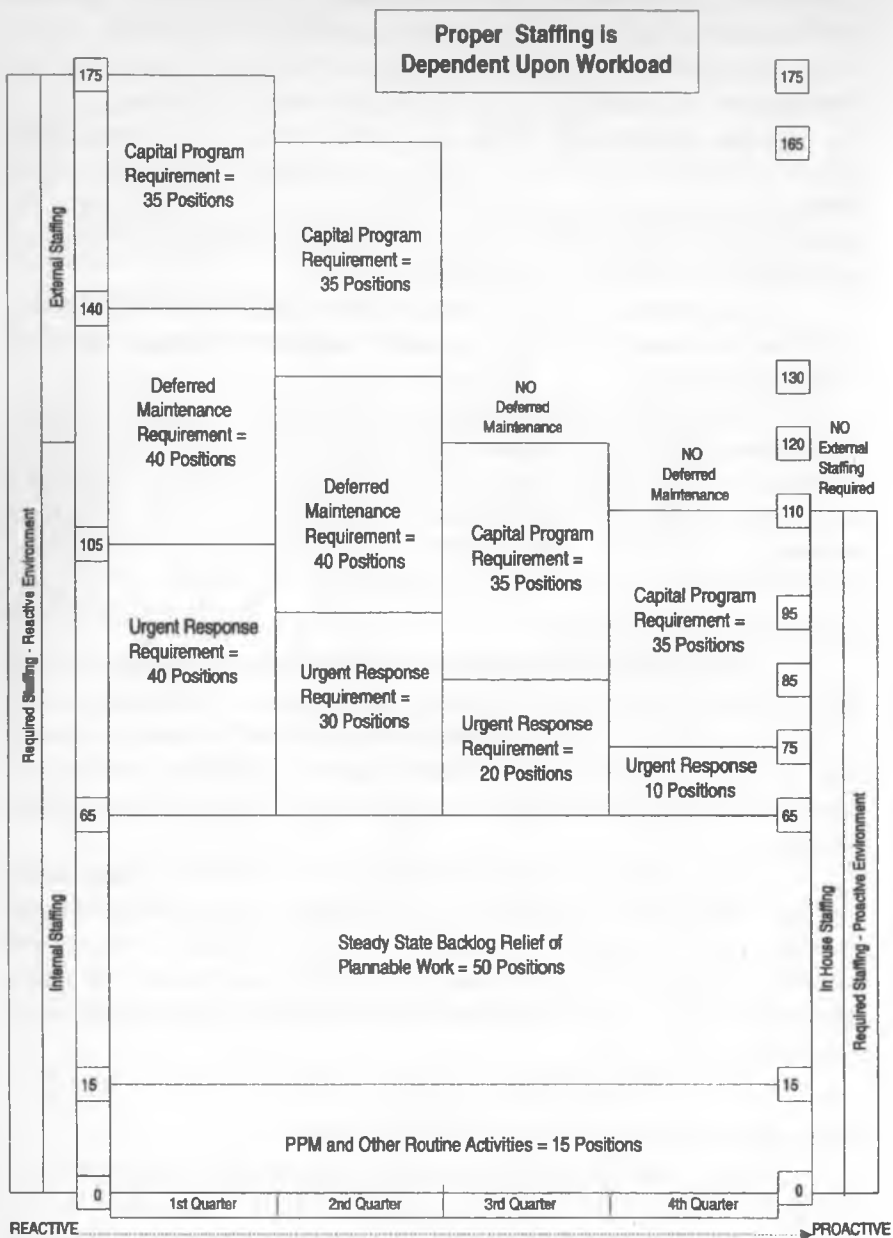


Figure 9.2

Once management accepts a recommended PM/PdM Process, these fifteen positions should be sacrosanct. The number should not be questioned without engineering reasons to modify the process. These resources will be applied to the detection of impending failures so that they can be corrected in a planned manner rather than under emergency conditions.

Do not develop this staffing number from history. History provides the proper staffing only if PM/PdM Compliance is 100%. The need is "required" resources, not historically committed resources, given reactive demands upon limited resources.

Frequency of Routines	Labor Hours / Occurrence	Repetitions per Year	Labor Hours per Year
Daily	?	365	?
Weekly	?	52	?
Monthly	?	12	?
Quarterly	?	4	?
Semi-Annual	?	2	?
Annual	?	1	?
Less than Annual	?	0.5 to 0.1	?
		Total Per Year:	27,000 (assumed)
		Required Staff @ 1800 hours per year	15

Figure 9.3 Buildup of Required Staffing for PM/PdM

Steady State Backlog Relief

This is the rate at which new requests for planned work keeps coming in. Access CMMIS information by priority code (possibly Priorities 3 and 4) for the previous twelve months to establish a supportable rate of input into backlog. Include all approved workload rather than completed work because work may have been deferred for lack of resources. Realize that the 15 PM/PdM positions identified in Figure 9.2 will generate additional backlog of Corrective Maintenance beyond the amount previously experienced. An adjustment will be appropriate.

Divide annual labor-hours of approved work by average hours each maintenance person works per year. This calculation will yield required staffing for the execution of Planned Work. For Figure 9.2, the calculation was:

$$90,000 \text{ labor-hours} / 1,800 \text{ hours per position-year} = 50 \text{ positions}$$

Steady state does not include deferred maintenance, which is addressed further up the illustration.

Urgent Response

Urgent response is the reactive workload forced upon maintenance when asset reliability is not established and sustained. Again, using CMMIS information covering the previous twelve months, sort by emergency and urgent priority codes (probably priorities 1 and 2). Divide annual labor-hours of reactive work by annual labor-hours per position.

In Figure 9.2, the assumption is 40 positions. Assume PM/PdM Schedule Compliance was poor last year due to insufficient commitment of required resources. However, in the future, the required resources (15 positions) will be committed. Therefore, the need for reactive response should diminish significantly. This diminishment is illustrated as three decreases of 10 positions with an ultimate need of only 10 positions by year-end. That is the relative level of return that can be expected from an effectively administered PM/PdM process (elimination of 30 reactive positions for an increase of 15 proactive positions).

Deferred Maintenance

The term deferred maintenance encompasses all work requested that remains in backlog, neither completed nor disapproved after a reasonable period of time (e.g., 3 months). It might be held in one of several status codes:

- ☐ Planning
- ☐ Engineering
- ☐ Approval
- ☐ Funding
- ☐ Materials
- ☐ Delivery
- ☐ Major Shutdown Access

Accessing CMMIS; sort backlog for all jobs over three months old. Divide the resultant labor-hour total by the hours worked per person in a six-month period (six months being selected as a reasonable time frame to overcome the deferred state). The assumption in Figure 9.2 is 40 required positions to achieve this objective. However, this provision is no longer needed after mid-year. Once an organization achieves this worthy objective and experiences the benefits of proactive maintenance, it should never be allowed to slide back into the deferred state. That would be false economy!

Capital Program Requirements

Translate the capital program for the coming year into terms of maintenance workers required. Our illustration (9.2) assumes a result of 35 positions.

Summary of Requirements

The five analyses discussed above quantify total maintenance staffing required for the coming year. The requirement is shown to be 175 at the beginning of the year with reduction to 110 by the end of the year.

The narrow second column from the left in Figure 9.2 suggests a proper internal/external scenario. Total staffing requirement calls for 175 positions throughout the first quarter, but only 110 positions throughout the fourth quarter and thereafter. Organizations should be staffed for normal workloads, not for temporary peak-loads. Current internal staffing of 125 positions is sufficient to handle PM/PdM, Backlog Relief, Urgent Response, and half (50%) of Deferred Maintenance. The balance of Deferred Maintenance and the Capital Program should be externally contracted, since they are not ongoing steady state requirements. The example assumes a natural attrition of 15 positions by year end (not depicted), which should bring internal staffing down to 110 positions. Therefore, the anticipated staffing decrease can be assimilated without forced layoffs.

A staffing buildup, logic, and managerial presentation such as this is the best approach when a maintenance organization has been established for several years and the required information can be extracted from the CMMIS.

3. Ultimate Approach

This approach becomes feasible once a Maintenance/Reliability Process is fully operational. The ultimate objective of Maintenance/Reliability Engineering is development of an optimal Maintenance/

Reliability Plan for each asset. The total Reliability Plan is the compilation of the asset plans.

Each asset plan represents the optimal combination of maintenance approaches:

- ☐ Preventive Maintenance Inspections
- ☐ Predictive Maintenance Inspections
- ☐ Programmed Maintenance Replacements
- ☐ Periodic Turnarounds
- ☐ Run Until Failure (in some cases)

This is the state of World-Class Maintenance/Reliability Excellence. Not until this condition is achieved is **Required Maintenance Staffing** truly determinable.

4. On-Going Balancing

Work Programs (introduced in Chapter 8) is the vehicle used dynamically to perpetually balance staffing with workload. The tool is applicable throughout the timeline associated with Approaches 2 and 3 above. Periodic adjustments are still required in Approach 2, far less frequently once Approach 3 is achieved.

Without records of backlog trends by craft there is no hard justification for resource adjustments. The maintenance leadership team is dependent upon anecdotal evidence and indirect measures of workload (rising customer complaints, increased downtime, growing use of contractors, etc.). Unfortunately, such indirect measures at best are lagging indicators—they inform that problems exist well after much of the damage has been done. Sometimes a catastrophic failure is the main indicator that staffing is inadequate.

In all cases, jobs must be properly planned and measured before labor-hour requirements and, therefore, staffing can be quantified and justified with a business-based argument.

TEMPORARY STAFFING DEMANDS

Unpredictable demands create temporary need for additional resources. They should not influence permanent staffing. That said, each

maintenance organization must identify reliable external sources of supplemental support (reasonably local). There are four categories of unpredictable demand:

1. Catastrophe Demand

If a plant is located in a river valley, it may infrequently be exposed to flooding. In a snow belt, an occasional blizzard will occur. Fire is a catastrophe that any facility might experience. These random occurrences should not be factored into permanent maintenance staffing. Make contingency plans with reliable external contractors.

2. Construction-Related Demand

Major construction on site can cause disruption to maintenance operations. Maintenance Techs are often detached from normal duties to escort or support contractors. Areas require extra cleaning, utilities are disrupted, and other problems occur. For project duration, these abnormal demands should be reflected in Work Programs. If resultant Backlog Weeks will go out of Control Limits, appropriate action should be taken. The likelihood is that overtime or contracting will be increased. Addition of staff is unlikely.

3. Social Demand

The maintenance demand created by visitors, such as tours and visits from company brass and outside VIPs, fit this category. This form of demand will diminish backlog relief for a week or two. It is possible that it will force backlog weeks out of control limits. Temporary action may be necessary, but change of permanent staff will not be appropriate.

4. Personal Service Demand

Maintenance personnel sometimes perform services such as package delivery, airport pick-ups, home remodeling, or working on company-sponsored community projects (such as building a playground). The community project should be handled as a planned and scheduled work order. Delivery and pick-up should be handled as Standing Work Orders and reflected as Indirect Commitments. Home remodeling should be disapproved unless the home is company owned, in which case it would be a planned and scheduled work order.

Most demands of this type are best handled through the CMMIS so their demand upon available resources is accounted for, their appropriateness questioned, their impact upon ongoing operations evaluated, and their demand if appropriate provided for.

The Proactive Process of Work Preparation

Work Preparation is the inclusive term, previously introduced, that encompasses:

- ☐ Detailed Planning of Individual Jobs (Micro-Planning)
- ☐ Identification, Procurement, and Reservation of Parts, Materials, Tools, and Equipment
- ☐ Work Measurement of Required Crew Size, Labor-Hours, and Job Duration
- ☐ Coordination between Maintenance and Operations as to Jobs, Dates, and Times for Next Week's Schedule
- ☐ Schedule Documentation by Job, by Day, and by Allocated Technician to Establish Expectation for Weekly Job Completions
- ☐ Transition of All Preparatory Efforts from Planner/Scheduler to Supervisors for Execution of Scheduled Jobs

In harmony, these distinct endeavors comprise the Work Preparation required for effective execution of maintenance work. They are basic elements contained within World-Class Maintenance/Reliability because they enable realization of asset reliability supportive of stable long- and short-term operational output. They are essential to safe, environmentally vigilant, effective, and efficient execution of maintenance work. They result in more work being accomplished more efficiently, at

lower cost, with fewer operating disruptions, reduced variability of processes, more consistency of high quality product, increased equipment-life, reduced usage of parts, and improved morale with greater job satisfaction.

When Preparation is effectively performed by a professional staff organization, maintenance supervisors are relieved of many commonly-misplaced responsibilities. They are thereby able to devote their full attention to effective leadership of job execution and on-going development of the maintenance team. Technicians become far more productive because delays, interruptions, and conflicts are proactively precluded (resolved on paper rather than on the shop floor) by professional planner/schedulers before the job is scheduled.

Although many maintenance organizations assign the six preparatory phases to a single function (Planner/Scheduler), herein we address each in an independent chapter (11–16).

Objectives of Work Preparation

Work Preparation has several primary objectives. It:

- ☐ Forms the communication center from which all maintenance activity is communicated and coordinated.
- ☐ Optimally supports operations by improving maintenance in the broadest sense, considering both the technical aspects and the service provided to the internal customer. Do not neglect customer relations. Loss of production capacity is minimized, and maintenance work is completed, when needed, in a safe, environmentally conscious environment, at optimal cost.
- ☐ Is requisite to effective performance of the maintenance function.
- ☐ Establishes equitable allocation of resources based on varying business needs, demands of customer service (internal as well as external), and other criteria.
- ☐ Coordinates use of the various input resources (material, man power, and equipment), as required, to complete each job in an orderly manner and at least overall cost.
- ☐ Achieves optimal, effective, utilization and application of all maintenance resources with minimal manpower delay and idleness.

- ☐ Reflects the foresight to anticipate and thereby forestall delays that commonly plague maintenance work performed reactively. Labor, materials, and equipment needs are identified on paper and provided for before resources are committed to the field.
- ☐ Allows a maintenance craftsman to prepare for, safely perform, and economically complete, each job to the satisfaction of the requestor with minimal interruption of operations.
- ☐ Establishes expectancy against which utilization of maintenance resources can be evaluated. Individuals know what is expected of them and both the output and productivity of maintenance workers are favorably impacted.
- ☐ Relieves supervisors of much indirect activity, enabling them to spend more time at the job site leading job execution efforts (a proven way to improve productivity by itself).

Without proper preparation, maintenance is haphazard, costly, and ineffective; it will consistently fail to meet promised completion dates. These failures will cause constant problems for Operations. When Maintenance consistently fails to put equipment back into service as promised, Operations will become increasingly reluctant to proactively release equipment for planned maintenance.

STEPS OF THE WORK PREPERATION PROCESS

Effective preparation for a planned and scheduled job involves a logical, step-by-step process with involvement of several organizational functions. To aid clarity, we will separate Administrator, from Planner, from Coordinator, from Scheduler. However, all phases may be performed by one Planner/Scheduler.

The process begins when a member of the local organization originates a Work Order Request for maintenance support. Each request is screened by an Administrator supporting a Planner. The first screening separates urgent from non-urgent. Urgent jobs go directly to a Maintenance Supervisor for immediate response. Planning is by-passed; without lead time there is no opportunity for preparation.

Work Orders to be planned are further screened: for complete and accurate entry of all data for which requestors are responsible; to assure that the request and requested completion date are reasonable; and that it

is not a duplicate of a WO already in the system.

If Planners are not familiar with the job, they discuss the job with the Requestor at the potential job site. Pinning down exactly what the requestor has in mind can be half the battle. The Planners refine job description, objectives, and scope, possibly with sketches or photos. Within this step of the process, the planners define "*what*" is to be done. This should be accomplished within a reasonable period of time from date of request (3 days max). One-third of the Planner Day should be spent in the field at future job sites.

Upon return to their work station (two-thirds of the day), Planners research reference files to determine if the job or a substantially similar job has been previously planned with a Planned Job Package already on the shelf. If so, the following steps are greatly simplified.

Taking into account physical location, movement of materials into and out of the area, and the working space, how many technicians can work efficiently? The job is broken into component tasks and steps, thus illustrating the manner in which the job is to be performed and defining "*how*".

Referencing the Material Library of the given asset, Planners identify required replacement parts and whether they are Authorized Inventory Items or Direct Purchase Items. If an inventory item, Planners determine if the required quantity is available in stock. If so, they reserve them to assure availability whenever the job is actually scheduled.

If a Direct Purchase Item, Planners initiate a Purchase Order Requests. In addition to parts, Planners document materials, hardware, special tools, protective equipment, and field equipment necessary to complete the job.

At this point, responsibility for further Preparation shifts to the Storeroom for inventory items and to Purchasing for direct purchase items. The Storeroom secures reserved items, either electronically in CMMIS (without physical removal from its storeroom location) or otherwise—physically picked, kitted by WO, and secured in the cage within the storeroom designated for accumulating and holding kits of items reserved for planned jobs until they are scheduled.

Purchasing sources Direct Purchase Items, orders them, tracks their delivery, and expedites as necessary. Receiving receives and inspects direct purchase items and delivers them to the Storeroom where, together with items from stock, they are accumulated in the secured cage for the appropriate planned job until that job is scheduled.

After initiating the Procurement Phase (above), the Planners transition into the work Measurement Phase—establishing labor needs and

job duration required to "perform" the job. This step includes determining the sequence of required skill sets, special licensing requirements (e.g., certified welder), and possibly the need for contractor support. These steps define "who" is to do the "what." Not until this point can "how much" be determined.

Preliminary approval is sought as early in the process as possible so that planner capacity is not consumed on jobs that will not be approved. If the job is routine, approval might be casual, without the need of prior planning. However, if approval requires an estimate of cost before management approval can be obtained, the estimate cannot be provided until this point of the Preparatory Process. Should approval be denied at any point of the preparatory process, the Storeroom must be informed; the reserved items can then be returned to stock or returned to the supplier for credit.

At this point, planner determine essential reference materials and assemble them with all job information into Planned Job Packages. The assembly should be electronic, but can still be in hard copy if CMMIS capability is limited.

When all items required for a given planned job have been accumulated, the Storeroom informs the responsible Planner. Job Status is changed to Ready for Scheduling.

The next phase of the process is Coordination. Maintenance, Operations, Engineering, and possibly other internal customers meet in a regular Weekly Coordination Meeting to reach accord regarding the most important jobs for deployment of available maintenance resources during the coming schedule week. This phase is crucial to the cultural transition from reactive to proactive maintenance, and to identify a mutually workable day and time when operations can release the asset and maintenance can make the necessary resources available. Without this coordination and participation, the resultant schedule remains "That Expletive Maintenance Schedule," It is easy for Operations to ignore it because it is not theirs. The objective of the meeting is to establish Joint Ownership with common reason to protect Schedule Compliance.

Immediately following the meeting, the Weekly Schedule is assembled and posted. Thus "when" each job will be performed is defined. Expectation for Maintenance achievement during the coming week is established—detailed by job, by day, and by allocated technician. As such, expectation is also established for each crew, supervisor, and technician level. We all work more effectively when challenged by realistic and fair expectations. Consistent with the Schedule, the Storeroom

delivers kitted items to the job site.

Finally, Preparation is not complete until everyone knows what is going to happen. The Planners communicate all preparatory information to the appropriate Supervisors, who communicate it to the assigned technicians. Nothing is lost in transition from preparation to execution. WHO is to do WHAT, HOW, and WHEN is communicated to all parties.

When all these steps have been accomplished, a job plan has been prepared with a schedule for execution according to priorities and equipment availability established with Operations. Supervisors are relieved of preparatory activity, so that they can devote effort to leading job execution. Technicians will be far more productive. Upon completion of job execution, Supervisors and Technicians are responsible for continuous improvement of the Preparatory Process in the form of constructive feedback, thus closing the proactive loop.

***Quality is a byproduct of successful planning
and timely availability of job requirements.***

What are the barriers to quality? The barriers on a routine job are listed during the planning process. We want to have the resources available on the job so that the workers can work efficiently, safely, and with the highest quality.

What happens when something is missing? Maintenance trades people will improvise, make do, work around any barriers. Improvisation is great when absolutely necessary; but is dangerous, costly, and potentially unsafe, and quality compromising as a steady diet.

Another Issue of Successful Planning

W.E. Deming (1900-1993, the foremost quality expert in the 20th century) said that the greatest motivator for workers was the pride in a job well done. The key to quality in maintenance work is to plan and manage work so that the maintenance workers have the opportunity to feel that pride. The job plan must insure that they have the right tools, the right parts, etc., and are not pulled off jobs so that they get that feeling of satisfaction from having done a job well. Quality flows from having removed the barriers to quality!

Detailed Planning of Individual Jobs

Job Planning is a seven-step process that includes:

- ☐ Screening Work Requests
- ☐ Assessing and Scoping the job to be performed
- ☐ Job Research to avoid redundant planning effort
- ☐ Detailed Build Up of Job Steps and Requirements
- ☐ Material Take-offs and Initiation of associated procurements (Chapter 9)
- ☐ Assembly of Planned Job Package
- ☐ Receipt of Feedback and Associated Refinement of planned job package (Chapter 18)

Material Take-offs and Feedback are addressed in subsequent chapters. The other five steps are addressed within this chapter.

SCREENING WORK REQUESTS

Work streams into the Planner's in-box from many sources including PM/PdM inspections and requests from various members of the organization. With support of Maintenance Administrators, Planners review all work requests (WOR) except those that must be performed on

the same day as requested. Because requests calling for immediate action contain no lead time, they cannot be planned. There is no reason to involve Planners. Instead, they are directed to and handled by Maintenance Supervisors and their crew.

All other Work Order Requests are reviewed and screened by Planners for redundancy, necessity, completeness, and accuracy. Within this review, the Planners confirm that:

- ☐ The request is not a Duplicate. Sometimes this task is particularly difficult when the backlog is large. A major reason for this difficulty is that the same job may be requested using very different descriptions. For example, "Remove and Replace a particular pipe" might be the same job as "Correct seepage into sump." It requires a sharp eye and good knowledge of the plant to spot likely duplicates. Effective CMMISs contain helpful features in this regard. Sorting all open work orders for the given asset facilitates the check for duplication.
- ☐ The Description is clear and complete, and the appropriate Equipment Code has been assigned. However, WOR descriptions are often found to be inadequate and equipment codes have not been entered or, if entered, are wrong for the job requested. Explanations go something like: "Everyone knows what we mean by the hissing spider" — the shop floor name for a case sealer that had a bunch of hoses that hissed when it moved. Accurate descriptions and specific identification of the involved asset make CMMIS output information reliable and useable. Everybody's job becomes easier. Of course, it begins with a well-designed equipment numbering system, readily cataloged, and available to all users. To facilitate initiation of WORs from the operating floor, equipment numbers should be stenciled on the equipment itself. Ideally, asset identification can be scanned. To facilitate initiation from offices or other off site locations, appropriate individuals should carry a pocket catalog of equipment numbers for their areas of responsibility. This situation might be facilitated with drop down menus.
- ☐ All fields the Requestor is responsible for have been entered and contain valid codes. Effective analysis of Equipment History is dependent upon good descriptions and proper cod-

ing. Otherwise completeness, reliability, and, therefore, usefulness of information are corrupted.

- ☐ Priority and requested completion dates are realistic and provide practical lead-time, so the job can be properly prepared for effective and efficient execution. Priority codes should be consistent throughout the facility. Their application should be based on equipment critically and true urgency of the work requested, not on the perceived suffering of the requestor.
- ☐ The requested work is needed. If so, does it need to be accomplished at this time? In this sense, the planner serves as a gate keeper of maintenance resources. If questioned about the need or priority, the issue is resolved with the requesting department or referred to the Maintenance Manager for resolution.
- ☐ Authorization has been obtained. Different job types and magnitude require different approval levels. Higher levels of approval often require a preliminary estimate that Planners must prepare. Engineering approval should be required for all alterations and modifications (before Maintenance processes the request any further). Some of the worst industrial accidents in the world have resulted from field modifications that were undertaken without benefit of formal engineering review.

Prioritization

Prioritization creates a dilemma because it is often used to express the customer's feeling of urgency when it really should be used to express urgency of the Work to the operation. Priority Code serves several purposes:

- ☐ Its first purpose is to identify "true" urgency of response.
- ☐ The second is to differentiate those jobs that will be urgently handled by the appropriate maintenance supervisor from those that will be handled by a planner/scheduler.
- ☐ The third is to indicate relative importance within the planning queue. Each priority code carries an implied "due date."
- ☐ The fourth pertains to the weekly coordination meeting. Going into the meeting, perceived importance is based on "aged-

priority.” Coming out of the meeting, “order of agreed upon importance” becomes the controlling influence (see Chapter 15, Coordination).

- ☐ A fifth usage may be to measure how well priority implied due dates are met.

Figure 11.1 summarizes a simple approach to prioritizing tasks.

Priority code	Explanation	Timing	Special
1	Emergency (Including breakdown, quality issue, and safety, environmental, and statutory issues that are truly urgent)	Dispatch Response Team immediately	If essential: Interrupt schedule Use overtime
2	Critical but less urgent work orders	Must be performed within current week	Interrupt schedule Sacrifice less important jobs at bottom of schedule
3	Planned Job – Rush	Force onto next week’s schedule	Sacrifice older requests as necessary
4	Planned Job – In order of Age	Schedule within four weeks	Priority goes up as it gets older
Note: Safety and environmental issued can be prioritized. Not all dictate Priority 1.			

Figure 11.1 Simple Prioritization Scheme

ASSESSING AND SCOPING THE JOB

Before progressing further, the required level of planning must be determined. Does the job warrant detailed planning or only cursory planning? Is the effort and cost of detailed planning worth the value to be gained? This question is appropriate during early installations of Maintenance Planning and Scheduling, when there is seldom sufficient capacity to plan all jobs. However, the planning workload becomes manageable as the library of previously planned job packages expands. Eventually, all jobs should be planned, even failures. Think about it. It is known that each unit of equipment will fail, how it will fail, and what work will be required. The unknown is when it will fail in each manner of failure. So, the associated jobs can be planned. However, they cannot be scheduled. Therefore, develop Planned Job Pre-Plams and add them to the online database. Then, when the asset fails at 3AM, the Coverage

Mechanic on duty will have access to well-prepared Job Plans.

One-third of a Planner's day should be spent visiting future job sites to analyze requested jobs. Planners want to maximize known facts and minimize use of assumptions, which often prove to be wrong, with serious adverse impact on Schedule Compliance. The best way to reduce assumptions is to get out of the office and visit future job sites to establish the facts and thereby improve planned job packages. Planners strive to identify all needs and to foresee potential delays or interruptions, which dramatically increase time required to complete jobs. The objective is to forestall those that are avoidable. Some may be unavoidable.

Job Assessment and Scoping should roughly follow the flow outlined in Figure 11.2.

• Confer with the requestor to clarify the desired result. By clarifying the end objective, the means to get there can often be simplified or the job can be expanded to solve several problems at once.
• Refine the description accordingly.
• Clarify the specific job location (building, floor, bay, etc.).
• List what needs to be done (job content).
• Define start and finish points (job scope).
• Finalize priority.
• Visualize job execution and outline the requirements.
• Record the steps necessary to execute the job.
• Prepare sketches or take photos to clarify the intent of the work order for assigned mechanics or simply as a reference during detailed planning.
• Take minor measurements (exactly). Complex measurements (e.g., for fabrication) should be left to the assigned technicians.
• Determine required conditions. Must this job be coordinated with Production?
• Is it necessary for equipment to be down (major or minor)?
• Define involved control loops.
• Determine if other equipment or adjacent areas will be impacted during performance of this job?
• Check for safety hazards.

Figure 11.2 Job Assessment and Scoping Checklist

DEALING WITH JOB “CREEP”

Job “creep” beyond the original scope or intention for the job is a common and serious occurrence in the maintenance arena. Technicians often find one or more additional things to be done while at the job site. In many cases, they receive accolades for their diligence. Although we do not want to discourage diligence, we must keep in mind the scope on which the “return to operation promise” was based. Suppose Maintenance told Operations that only four hours of downtime were required for performance of the job. Production Output and Customer Delivery is dependent upon the promise. Individual technicians cannot be allowed to make independent decisions regarding expansion of scope. Scope changes during job conduct lead to many conflicts and distrust between Operations and Maintenance.

The decision to expand scope must be cooperative. Sometimes, Operations can live with an extension of downtime in the interest of a more comprehensive and reliable restoration. Other times, they would rather not extend the downtime. In the latter situation, final resolution will depend upon how long the expansion of scope will extend downtime and how much longer will it keep the asset in working order. Even when the answer makes good sense, will Operations agree with extension of downtime?

If original scope will keep the asset operational for a reasonable period of time, Operations may deny scope extension. The job will be closed up at that point, and a Corrective Work Order created to perform the additional identified work at an agreed upon time. In either case, the planned job package should be reviewed to determine how the scope conflict might be avoided in the future. Learn from the errors and omissions.

Performing the job right, the first time, would appear to be the least costly and most reliable approach. Keep in mind though: the parts to complete the expanded job may not be in inventory. We don’t know because they were not part of the preparatory process.

Most important, Operations’ promised obligations to External Customers must be considered. Ultimately, success of the overall business is dependent upon sales. Therefore, decisions regarding deviations from job scope, and from Maintenance Schedule Compliance as well, must be cooperative within the Maintenance/Operations Partnership (see the footer of the Maintenance/Reliability Arch, Figure 1 in the Introduction). A familiar domestic example of Job Creep is offered below:

A young couple decided to replace a sink base cabinet in a mother-in-law suite soon to be occupied by a live-in babysitter. They assumed the job could be done over a weekend for about \$350 worth of materials.

They pulled out the cabinet and noticed the wall behind had never been installed correctly, so they ripped that out. With the wall removed they had the opportunity to pull wires for new electrical circuits and to add a GFI. At this point they noticed that a small change would make the kitchen more efficient. So the gas pipe to the range was moved to accommodate the new set up.

The wall was reconstructed and received a nice paint job. After the cabinet, range, and refrigerator were reinstalled the rest of the room looked shabby. So the remaining walls needed a fresh paint job. Now the range looked old and needed to be replaced. Presumably, the job was finished, except that the floor covering looked out-of-date and worn.

Eight weekends and \$1,750 later, the new babysitter was able to move in. This story illustrates how job scopes can creep without proper planning and definition, pricing, authorization, and budgetary control.

JOB RESEARCH

The remaining two-thirds of the planner day as well as the planning cycle are completed at the planner workstation. The third step of detailed planning is Job Research conducted to make full use of Planner Reference Libraries, which are assemblages of prior planning efforts. Libraries of Planned Job Packages, Job Estimates, Bills of Material for each Asset, Procedures, Equipment Manuals, etc., are referenced to determine if the job or portions thereof have been previously planned. Avoid redundant effort. It is seldom necessary to plan a job from scratch, especially as the maintenance planning installation matures. Within the Research step, Planners look at several things:

- ☐ Reference Equipment History to determine if the job has been previously performed. When was it previously done? Is this frequency excessive? If so, consider if anything can be done to avoid recurrence or such frequency.
- ☐ Consider alternative approaches. Is this the best solution to the problem? Should additional work be performed this time to ensure a more permanent solution? Should the item be replaced rather than repaired (the Repair/Replace decision)? Should the item be purchased rather than made (the Make/Buy decision)?
- ☐ Remain conscious of alternate plans for the involved equipment. If there are plans in next year's capital budget to replace this equipment, why perform a major overhaul at this time?
- ☐ Consult the procedures file to identify necessary tag outs, safety inspections, fire watches, and standby positions associated with open flames, fall protection for ladder and scaffold use, vessel entry, etc. Safety must **always** be a primary concern within the planning and execution processes. Planning is critical to safety and loss prevention because the planning phase is the logical time to think about and incorporate safety matters into Planned Job Packages, before pressure to finish jobs and return the asset to operation come into play.
- ☐ Contact other functions with involvement or potential input in the job. Planners use judgment regarding when to call upon the skills and knowledge of other parties?supervisors (maintenance and operation), technicians, operators, engineers, and other specialists (operations, safety, process, environment, quality, and contractors).

Use of these resources conserves planner capacity, prevents accidents, and avoids environmental degradation. Planners may only need to enter data specific to the given work order, fill in some gaps, and add current knowledge.

DETAILED BUILD UP OF JOB STEPS AND REQUIREMENTS

For now, assume that we are dealing with a job not previously planned. The Planner's task within this step of the preparatory process is to describe the job in terms of how it is to be performed and what resources will be required. This includes finalization of job description; documentation of job scope; detailing sequential steps of job performance; identifying and initiating procurement of all required parts, materials, tools and equipment; establishing required staffing and job duration; initiating required permitting; and notating all safety and environmental cautions. If the Planner does not fulfill these core responsibilities, the assigned crew will not know the expectation for a properly completed job. The job might then take a number of different shapes dependent upon the technician to whom it is assigned.

Within this process step, the Planner:

- ☐ Selects and describes the best method for job performance.
- ☐ Determines and sequences the job by specific, safe, and logical steps.
- ☐ Identifies and highlights safety and environmental-related steps (lockout/tag out, vessel entry, burning, fire watch, etc.).
- ☐ Identifies hazard and mitigation issues including personal protective equipment (PPE).
- ☐ Determines labor resource requirements, including crew size, duration, labor-hours, and skill-sets (craft and skill level) for each sequential step of the job. Technicians eventually assigned to the job must possess the physical and mental aptitude required to perform the job. These requirements must be made clear. See Chapter 13, Work Measurement, for methodology of labor-hour determination.
- ☐ Determines any requirements for fabrication (in-house or external contractor support). Creates associated Work Orders and Purchase Order Requests with reference to the associated WO in order to preserve completeness of Equipment History.
- ☐ Identifies step dependencies (prerequisites).

- ☐ Conveys linkages between multiple work groups mutually needed to perform a given job. Initiates appropriate Cross Work Orders if involvement of a support group is significant. If support is minor, distinguishes each group with individual steps (line items) on a common work order.
- ☐ Lists determinable parts, materials, supplies, and consumables. Prepares associated Bill of Materials for the job; establishes an acquisition plan for each item on the BOM (Storeroom Requisition with Reservation for items in Inventory or a Purchase Order Request for Direct Purchase Items). See Chapter 12 for Storeroom and Purchasing action steps.
- ☐ Identifies special tools that are not part of personal tools required for each trade.
- ☐ Identifies equipment requirements, including cranes, welders, compressors, and field lighting.
- ☐ Considers transportation and issues associated with access to the equipment.
- ☐ Considers disposal issues (asbestos, oils, other contaminants).
- ☐ Identifies and initiates all permit requirements.
- ☐ Defines necessary custody and control of involved assets.
- ☐ Defines safe access to assets (decontaminated, cooled down etc., safe work platforms, and humane working condition).
- ☐ Estimates total cost in terms of labor, material, and external charges.
- ☐ Obtains up-to-date drawings, wiring diagrams, and other information.
- ☐ Coordinates and expedites necessary authorizations based on final cost estimates including operational, financial, and engineering approval.
- ☐ Finally defines any testing or base lining (such as initial vibration signature) to be done at the conclusion of the job.
- ☐ Any additional factors that may delay or hinder safe, environmentally-conscious, and effective job completion should be anticipated with steps taken for their avoidance.

ASSEMBLY OF JOB PACKAGE

Next, the Planner assembles all the above planning efforts into a Planned Job Package. Given the data contained within the package, coupled with a thorough verbal exchange between Planner and Supervisor and a similar exchange between Supervisor and Technician, nothing should be lost between strategic preparation and tactical execution. Toward this end, a comprehensive Planned Job Package should include:

- ☐ Detailed Work Order with all required fields completed with valid codes, etc.
- ☐ Job Planning Sheet with Sequenced Steps detailed by craft and skill level. Contractors as well as in-house resources must be included. Plan to perform as much pre-shutdown fabrication and other preparation as possible. In like manner, plan to perform as much post-shutdown wrap-up as possible. Both of these actions are intended to optimally minimize asset out-of-service time.
- ☐ Attached safety and environmental procedures with **Cautions** highlighted.
- ☐ Duration and labor-hour Estimates for each Step. When beneficial, include a time-line by step to convey sequencing and simultaneous tasks to the assigned crew(s).
- ☐ Bill of Materials distinguishing among authorized stock items, direct purchases, and items to be fabricated.
- ☐ Clearances and all required permits (internal and governmental) processed to the point of safety. Final lock outs must be jointly performed by responsible mechanic and equipment operator.
- ☐ Other reference documents the assigned crew is likely to need (e.g., prints, sketches, photos, specifications, sizes, tolerances).
- ☐ Site Set-Down Plan if required disassembly is significant and space is limited.

Transition from Preparation to Execution

Planners hold the assembled Planned Job Package until all necessary materials are procured, and the job enters the coordination and scheduling phase. At that point, they review the package with the responsible Maintenance Supervisors, who in turn review it with the assigned technicians. None of the planning effort with associated insight and knowledge is lost in the transition between staff preparation and line execution.

Admonition is appropriate here. The Planned Job Package is a working document that is to be at the job site, open and referenced throughout execution. It is not to remain in the supervisor's office, or on the technician's work bench in the shop. There have been numerous accounts where accident investigators have found that a proper procedure existed that, if followed, would have prevented the accident. Unfortunately, it was not in possession of the assigned crew at the work site or, if was there, it was inadequately referenced and applied.

Feedback from Execution to Improve Future Preparation

Once the job is completed, constructive feedback to Planners from technicians, supervisors, and internal customers is vital to continuous improvement of the Preparatory Process. Such feedback must become second nature to all parties. Until that nature becomes entrenched, consider periodically inserting Job Planning Surveys into planned job packages and/or sending them to Work Order Originators. Survey questions might include (allowing space for explanations):

- ☐ Were job instructions clear?
- ☐ Was the work performed as specified?
- ☐ Was the estimate within +/- 15% (total hours as well as within individual crafts)?
- ☐ Did unusual problems occur?
 - Delays due to permit or permission problems?
 - Delays related to equipment access?
 - Delays caused by lack of craft coordination?
 - Trips for tools, parts, or supplies needed after the job was started?

EQUIPMENT ACCESS, SAFETY AND STATUTORY PERMITS

The permitting aspects of Preparation are worthy of further exploration. Safe access to equipment is addressed within the maintenance planning process. Legal and statutory permitting is mostly associated with major projects and therefore handled by Engineering and Legal, not by Maintenance Planners. Let's consider them separately:

Safety and Process Driven Permitting

During normal operation, assets are in custody of Operations. Maintenance must obtain clearance to interrupt production and take control of the asset (frequently by lockout) either to rehabilitate it or to replace it. Access can be simple or complex, but in all cases no work should begin without safe access to the asset. Clearance often requires some form of permit. As a bare minimum, there is a **lockout / tag out practice** to be observed. More complex jobs and environments may require temporary process changes involving rerouting and extensive coordination with several departments. Making such provisions involve thinking through access requirements: shutdown and lockout steps, individuals to be contacted, valves to be isolated, and other issues to be addressed before custody of the asset can be gained.

The following discussion reviews some of the formal and informal processes by which Maintenance gains permission to work on an asset and gains control of that asset. They are driven by safety, process, and regulatory issues.

Safety Driven Issues

Most large facilities have processes for obtaining permits for the performance of dangerous work. Safety permissions are closely related to legal permissions as many flow directly from laws. For example, proper lockout / tag out procedures are prescribed by law and reinforced by safety permits. Most Safety organizations have established procedures beyond legal requirements.

There are various forms of dangerous work permits. They include Open Flame / Hot Work Permits specifying a Safety Watch person with fire extinguisher; Confined Space Entry Permits (a legal requirement in the United States and Canada; Line-Open Permit where poison gases might be present; Entry Permit for clean rooms or other secured areas; and

other forms of permit specific to given industries.

Process Driven Issues

In complex process industries, substantial procedures are required to take an asset out of service with assurance that the area will be safe to work in, chemicals will not spill from an upstream process that wasn't shut down, or a downstream process doesn't run dry and crack a vessel that wasn't shut down. It is logical that operational and/or engineering coordination and sign off must be required. The procedure must be thoroughly documented and copiously followed throughout planning and execution. These procedures should be added to reference files because they will surely be applicable to future job situations. Maintenance has always been repetitive. Although Reliability Centered Maintenance strives to eliminate or at least reduce the frequency of given failures, build up of reference files will remain beneficial.

Three examples are offered below to illustrate the magnitude of problems that can occur when coordination and permitting procedures are by-passed:

Example 1: A large production facility had three generator sets for critical loads. The controls and settings on the ASCO transfer switch were fed power from separate 24-volt batteries charged by the engines of the three generator sets. The generator repair contractor did not understand the process and disconnected all batteries at once. The entire back-up system (including the UPS bank) was taken off-line and had to be reprogrammed. Fortunately, the power did not go off during that period. There is now a practice in place that only one generator can be service at a time.

Example 2: Another instance of poor notification, this one with tragic consequences, was the disaster in Bhopal, India. The reaction to produce methyl isocyanate was exothermic (it produced heat). If too much heat were generated, the process would get out of control.

continued

Water got into the reactor, causing the process to overheat. The plant was designed with four safety systems to deal with a potential leak if one started. Over a period of years and through a series of unrelated events, three of the four safety systems were taken out of service. This accident demonstrated significant deficiencies in maintenance, operations, design, and safety procedures. Reviews of the accident identified significant design defects, operational mistakes, and maintenance problems that all contributed to the tragic outcome.

With comprehensive safety permit systems in place, incidents like the above are much less likely. No more than one of the four safety systems should have been permitted to be down at any given time. Nothing short of certainty is acceptable when it comes to safety and the environment.

Regulatory Driven Issues (Statutory or Legal)

Statutory permission is gained from or in compliance with laws of a legal authority. Legal issues can flow from Federal law (confined space entry rules) to State laws down to the local zoning board. Usually jobs that require legal permits have engineers or architects (internal or external) that carry responsibility for obtaining such permits. Regardless of where responsibility lies, planning checklists should include permitting requirements to further assure compliance. Permits must be in hand prior to associated jobs being scheduled. See Example 3.

Example 3: Bulldozers and graders were delivered to a site to improve the flow of storm water. During heavy rain, the southwest corner of the plant would flood. A hydrologist working for a landscape architect's office had reworked the water flow into a series of attractive vegetated swales. Everyone was ready. The architect asked the contractor for a copy of the Permit from the Township. The contractor, with a sick look said, "I thought you pulled the Permit." How many times has this story been repeated? In this instance, it took three months and most of the construction season to obtain the permit.

Generally, statutory permits come in three levels.

- ☐ **Local:** zoning variances, building permits, electrical permits, plumbing permits, noise permits, etc.
- ☐ **State:** Department of environmental resources permits, and other agencies regulating individual industries such as mining.
- ☐ **Federal:** EPA permits, Coast Guard (dealing with waterways), Army Corps of Engineers, etc.

Someday there may be a fourth level for global permits designed for activities that would impact other countries. Changes in the law at any level might precipitate required changes in maintenance access procedures. Simple things such as lockouts tag-outs, and confined space entry, for instance, require steps to comply with law.

This book is a work about planned maintenance, but there is potential application of planning to breakdown maintenance. This potential is a side benefit of complete and effective planner libraries:

- ☐ Quickly scope the job, review the files and assign as many of the elements as possible in the limited time.
- ☐ This gives the crafts a head start without delaying the job
- ☐ After the emergency is over, develop a standard job package and ?put it in the can? for next time (most maintenance work repeats).

Hints for Planners

- ☐ Smart planners maintain their files.
- ☐ Close relationship with craft people will insure specific feed back for improvement.
- ☐ Digital cameras are great.
- ☐ Assembly and disassembly are frequently symmetric. Think through disassembly and just reverse the steps for reassembly to complete the job.
- ☐ Consider developing generic estimates for Motors, Fans, Pumps, Welding, Fabrication, etc. The steps to remove and replace a motor are similar irrespective of the size of the

motor. All that changes are the time, crew size, and tools/equipment.

Think of Planning as a Formal Thought Process

Visualize the first step and consult Figure 11.3 about what is needed for that step.

Then complete the second and subsequent steps until the job is done, turned over, and cleaned up.

Figure 11.3 Visualizing the First Step¹

Job Step Description	Labor Skills, Crew Size, Hours	Parts, Materials, Consumables	Tools, PPE, Scaffolding, Equipment	Permits, JSA, Other Resources or Information
Such that an understandable step is described and is actionable. We usually recommend steps between 15 minutes and 4 hours for ease of time estimating.	Labor is one of the critical elements of any job. Questions include skill set, crew size, and duration.	All items that are used in the job not limited to parts but including rags, oil dry compound, etc. Also includes getting materials into the job location and disposal of waste and used spares.	This category looks at specialized tools, heavy equipment, lifting gear, work platforms, and safety gear.	Safety steps, drawings and manuals, wiring diagrams, permits, other permissions, and communications needs.

¹ Courtesy of Mike Brown of New Standard Institute

Materials Support

This chapter discusses all elements of Materials Support in terms of necessity, responsibilities, variables, procedures, and approvals. Before addressing the functional steps associated with Preparation, a broader review of Materials Management related to maintenance is necessary.

NECESSITY

To fulfill its mission effectively, maintenance is dependent upon reliable and prompt logistical support. Lack of support creates reliability problems with serious economic consequences. To sustain a cost-effective, responsive, and reliable operation, the maintenance department needs dependable and timely availability of required parts and materials. In reactive operations, twenty-five percent of equipment downtime duration is due to lack of spare parts necessary to perform repairs correctly. Nothing is more frustrating to maintenance technicians than to start a job only to find that it cannot be completed due to lack of necessary spare parts. Downtime is exponentially extended when the job has to wait for parts that must be sourced, procured, and received.

The vast majority of work performed by the maintenance force requires specific spare parts and general maintenance materials. The required item may be an expensive part or component unique to a specific equipment unit, or it may be a common hardware item costing less than a few dollars. However, the magnitude of disruption is not always proportional to the cost of the missing item:

- ❑ On the low end of the range, for lack of a required item, there may be loss of maintenance productivity because resources must be shifted from an unfinished job to another job. Resultant loss will probably be measureable in hundreds of dollars.

- ☐ On the high end, consequences might measure in hundreds of thousands of dollars due to major disruption of quality output with delinquent customer deliveries.
- ☐ The potential for such losses are clear justification for maintaining a reliable inventory of parts and materials (either on-premise or readily available from a reliable, local source) supportive of the maintenance mission.

Regardless, it is critical that the item be readily available or be quickly obtained so technicians can complete the repair promptly and properly to bring the asset back to operability with minimal loss of output.

The following complaints are common when material support is ineffective:

- ☐ “We are tired of Band-Aid repairs. Reliability requires jobs to be done properly on the first attempt.”
- ☐ “Stores regularly runs out of spare parts and other materials urgently needed to keep critical equipment on-line.”
- ☐ “Purchasing buys on cost alone.” Reliability is denigrated when procurement is based on the lowest bid alone. The consequence of less reliable parts on up-time often costs Production multiples of what Purchasing saves. When goals of one functional unit of any organization are maximized, goals of the business entity become suboptimized. Jon Ruskin, an early economist, characterized this dichotomy in the following manner:

The Lowest Bid

“It is unwise to pay too much, but it is worse to pay too little. When you pay too much, you lose a little money—that is all. When you pay too little, you sometimes lose everything, because the item you bought was incapable of doing what it was bought to do (Reliably). Common law of business balance prohibits paying a little and getting a lot—it cannot be done. If you deal with the lowest bidder, it is well to add something for the risk that you run. If you do that, you have enough to pay for something better.”

- ☐ "When Direct Purchases for specific jobs are received, the originators of the request are not informed by Receiving; nor are they alerted when delivery will be delayed."
- ☐ Maintenance Technicians must go to the Storeroom in order to identify required parts. If a stores catalog exists, it is incomplete and inaccurate.
- ☐ With no awareness of criticality and without consulting Maintenance or Engineering, Stores will unilaterally discontinue stocking seldom-used spares despite their criticality when they are needed.

MATERIALS MANAGEMENT

There are two conflicting objectives regarding materials support of maintenance and reliability excellence:

- ☐ Get the right materials to the right place, at the right time, at optimal cost. Therefore, adequate inventories must be maintained to minimize delays associated with stock outages and, in turn, to protect the plant against related downtime elongation.
- ☐ Avoid excessive inventories. Therefore, inventories must be designed and controlled so that carrying costs do not exceed costs associated with downtime plus cost savings associated with loss of volume discounts.

Optimization of the two objectives lies in engineering economics (see Figure 12.1). Each stocking decision must reflect the lowest overall cost to the operation. Total unit cost varies greatly between stock keeping units (SKUs—single part numbers). The total unit cost is the summation of: delivered cost, inventory carrying cost, and the consequence of not having the item on hand when needed. Because maintenance-related inventory is a portion of working capital, the management of inventory must be optimized based on financial impact on the overall business entity.

Optimization also involves a proper balance between Just-In-Time (JIT) acquisition and Statistical Inventory Control (SIC). Both have application to the inventory optimization process leading to sustained reli-

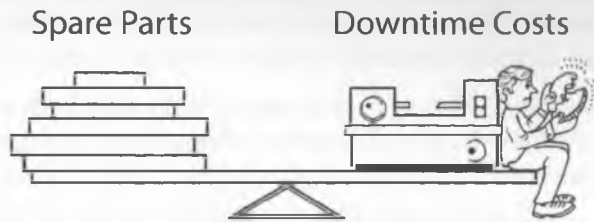


Figure 12.1 Inventory Balancing Act

ability. A basic prerequisite of JIT acquisition is a proactive culture and environment. Project work and planned maintenance jobs can rely on JIT because lead-time is available; such jobs should not be scheduled until all required parts and material are at hand. In contrast, emergency failures do not provide lead-time. Therefore, an organization in reactive mode must have adequate inventory. Otherwise, downtime increases, output and quality decrease, and customer service and profitability suffer.

A combination of JIT for proactive work and SIC for reactive demands is the answer. Use JIT, provided the need is not immediate, the job can be scheduled, and delivery lead-time is not excessive. For urgent needs inventory is necessary (use SIC) unless a dependable local supplier relationship can be established. Such a relationship might encompass a consignment inventory agreement.

TOOLS OF MATERIALS MANAGEMENT

Several tools are commonly deployed in well-managed Materials Management organizations (Purchasing, Stores, and Receiving); they effectively support the Maintenance/Reliability Mission. They include:

- ☐ Statistical Inventory Control
- ☐ Just-In-Time Delivery
- ☐ Inventory Classification
- ☐ ABC Analysis
- ☐ Inventory Cycle Counting

- ☐ Vendor Partnerships
- ☐ Consignment Stock
- ☐ Storeroom Delivery to Job Site

Each has been previously referenced and is further discussed below.

Control the Maintenance Storeroom with *Statistical Inventory Control (SIC)*

SIC is the historical management technique by which inventories of specific Stock Keeping Units (SKU) are optimized. It is illustrated below in terms of two primary variables (Rate of Usage and Lead-Time), plus five control parameters (Maximum Inventory, Reorder Point, Minimum Inventory, Economic Order Quantity, and Safety Stock):

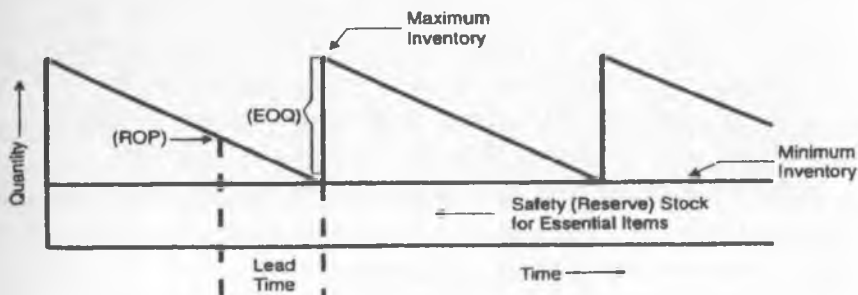


Figure 12.2 SIC Order Cycle

1. Rate of Use (ROU) — the average number of units of a given SKU issued in a period of time (usually days; however, ROU and Lead-Time are always stated in the same unit of time).

2. Lead Time (LT) — the elapsed time from identifying need to order (reorder trigger) until the ordered part or material is received into stock. There is internal as well as lead-time. Internal includes time in purchasing, transit to the supplier, and receiving (arrival at dock until placed into inventory). External is from time of order receipt by vendor until receipt at dock.

- A. Paperwork lead-time—from reorder trigger through issue and delivery of the purchase order to the vendor (internal).
- B. Vendor lead-time—from receipt of purchase order through arrival at receiving dock (external).
- C. Stocking time—from arrival at receiving dock through loading onto storeroom shelves and documenting the receipt into CMMIS so all parties know the item is available (internal).

3. Minimum Inventory (MIN)—quantity to which inventory is expected to fall immediately prior to receipt of the replenishment order (ideally zero unless lead-time is unreliable; see Safety Stock).

4. Reorder Point (ROP)—stock level at which a replenishment order should be generated, allowing necessary lead-time for replenishment to be received into inventory at the time stock drops to minimum (MIN). Think about this: if the replacement arrives as stock hits MIN, isn't this comparable to JIT? SIC has been in use for over 50 years, long before JIT came into popularity. Regardless, the parameter is calculated by multiplying the Rate of Use x Lead Time. ROP relates to Inventory Quantity Available for Issue, not to Inventory Quantity On-Hand, the difference being the quantity reserved for planned jobs. The calculation is:

$$\text{ROP} = \text{MIN} + (\text{ROU} \times \text{LT})$$

5. Economic Order Quantity (EOQ)—the order size at which the annual sum cost of purchase order processing, plus carrying cost of average inventory, plus the annual sum of lost volume discounts are minimized.

6. Maximum Inventory (MAX)—stock level that should not be exceeded, assuming predictable lead-time and rate of consumption.

7. Safety or Reserve Stock (SS)—quantity established as protection against stock-outs caused by delivery delays and/or abnormal variations in usage. This provision is normally deployed only on

critical items. Another means of protection on non-critical items is simply to increase MIN by a few units (1, 2, or 3 rather than 0).

If the maximum inventory level is exceeded or a stock out occurs, a system alert should trigger reexamination of established control parameters (ROP, MIN, and SS). ROP and EOQ should be dynamically adjusted by application of an algorithm programmed into the CMMIS. However, for the algorithm to remain effective, lead time, cost of carrying inventory, and cost of purchase order processing must be periodically updated.

Support Proactive Planned Jobs with *Just-In-Time* Concepts (JIT)

Debates often occur between the advocates of JIT and SIC. Optimization of the two is essential to realization of reliability. Apply JIT on proactive work and SIC on reactive work. Accordingly, Stores Inventory is exclusively, or at least primarily, for emergency needs. Project managers/engineers and maintenance planners should not rely on inventory to fulfill proactive needs. Exceptions to this general rule must be taken into account when authorizing, sizing, and budgeting inventory.

Once maintenance excellence and a proactive environment are established, no more than twenty-five percent (25%) of maintenance material needs should ultimately be dependent upon storeroom inventory. **Seventy-five percent (75%) should be procured through JIT.** The normal material/labor distribution range is 60/40% to 40/60% with a common average of 50/50%. Because emergency jobs should consume no more than 15% of total maintenance labor, they should consume less than 25% of maintenance materials [$15\% \times 1.5 (60/40) = 22.5\%$]. This calculation is based on the upper limit (60/40) of the labor/material range where material is fifty percent more costly than labor.

There is a common misapplication of JIT, when applied to procurement of maintenance material needs. Purchasing thinks that maintenance planners should convey when the job will be scheduled so that the required materials can be ordered JIT (Schedule Date – Lead-Time = Purchase Date). This concept is Reverse Scheduling. It may be reasonable for projects or annual maintenance turnarounds that are not scheduled to begin for several months. But, the goal for planned maintenance jobs is typically to perform them ASAP (at least before failure). If procurements are necessary, the planner activates a Purchase Order Request with the

expectation that Purchasing will immediately source and cut a Purchase Order to procure; then provide feedback to the planner as to promised delivery date.

Inventory Classification

Classification of items on the Bill of Materials (BOM) for each asset is another control technique deployed in well-managed organizations. Classification segregates items (stock and non-stock) by criticality. All items are not of the same level of criticality. The risk of not having a critical item readily available when needed is not acceptable. So, when the BOM for each asset is loaded to the CMMIS materials database, a degree of criticality (see below) should be assigned to each item. Therefore, the first application of classification is separation of items to be authorized as Stock Keeping Units from those that are to be Directly Purchased when needed.

Optimization of inventory requires variable service levels by SKU, depending on informed judgment regarding the acceptable degree of risk. Service level is measured at the storeroom. For example, 9 requisitions filled per 10 requisitions requested equals 90% service level. Therefore, the second application of classification is to establish an Essential Service Level for each item of authorized stock. Risk of stock out (non-availability of a required part) should vary based on consequence. Lead times for many critical spare parts are becoming increasingly longer, particularly when selected equipment is manufactured in other countries. On the other hand, standard replacement parts can be readily obtained from local supply houses. The following classifications and service levels are common:

Insurance spares—spares and components used on critical equipment. Not having insurance spares in stock can result in extended downtime of critical equipment and major loss of output critical to profitability and customer service. Although a number of these items can cost thousands of dollars, little risk can be taken, especially when long lead times are involved. The essential service level is 99.9%.

Insurance parts—parts used on critical equipment and components thereof. Such parts are purchased from the OEM and are generally used on only one type of asset. Their use is unpredictable because mean time between failures varies significantly.

A minimum amount of risk (98%) is acceptable because these items are available in distribution centers positioned around the country and can be obtained within a few days.

It is not uncommon for at least 80% of maintenance inventory value to be tied up in these top two classifications. They should not be included in the calculation of inventory turns per year because they are carried for insurance, whether they turn or not.

Standard replacement parts—parts that can be used on more than one component or piece of equipment. These parts are stocked by local supply houses for a number of users, or can be obtained with overnight delivery. Essential service level is 95%.

Hardware items—such as fasteners, hose and fittings, packing and gaskets, and other low unit cost items that are used throughout the facility and are readily available locally. Such items should be amply stocked, under loose control outside the secured portion of the storeroom in the form of free bin stock. Bins should be serviced by a local supplier, many of which will provide the bins in order to gain the contract. Withdrawals need not be charged to the specific WO on which applied. Essential service level is 90%.

Small Expendable Tools—all company-owned and furnished tools (and parts stocked for their repair) such as drills, blades, and batteries that do not require a withdrawal requisition but do require return of the expended item in order to obtain a replacement. Essential service level is 90%.

General supplies—all stocked items not fitting one of the other classifications—lubricants, cleaners, chemicals, filters, welding supplies, compressed gases, paints and supplies, operating supplies, janitorial supplies, safety apparel, gloves, charts, sanitary supplies, and office supplies. Essential service level is 85%.

A single expectation regarding service level (e.g., 90%) or regarding inventory turns (e.g., 4 turns per year) for all SKUs results in serious deprivation of operational reliability with debilitating business consequences.

ABC Analysis (Pareto's Law)

Vilafred Pareto was an 18th-century Italian economist who

proved that the majority of the wealth of Italy was controlled by a few families. We have taken that to mean:

Pareto's Law

A few things that we do are the most important.

Most things we do are not critical.

It can also be expressed as: "The significant few versus the insignificant many." Applicability of the law is nearly universal.

ABC Analysis is an analytical tool based upon Pareto's Law. It facilitates informed management decisions regarding a variety of issues. Related to current discussions regarding Materials Management, it is often used:

- ☐ In conjunction with stock classification
- ☐ To determine which items to be authorized as inventory (SKUs)
- ☐ To establish the degree of security to exercise over given SKUs
- ☐ To assign storeroom locations to given SKUs in order to locate the fastest movers closest to the point of issue or loading

As previously established, materials typically account for 40% to 60% of total maintenance cost. So, control and care of inventory is certainly warranted. However, control is costly and must be exercised wisely. Application of ABC analysis to establish a rationale for degree of control is effective. Force Rank any population, of any nature, by significance and it will be found that the top (most significant) 20% of the population (SKUs) represent 80% percent of the value (inventory value). These are the "A Items". The next 30% of the population represents the next 15% of the value (B items). The remaining 50% of the population represents the remaining 5% of the value (C items). Figure 12.3 below conveys common results of ABC Analysis:

Management and Control effort should focus on the significant A and B items. The insignificant C items (such as items previously classified as Hardware Items) should be staged as free bin stock outside the secured area of the storeroom with minimal control.

- ☐ **A Items**—Highest value, tightest control, close follow-up, accurate records.
- ☐ **B Items**—Normal control, good records.
- ☐ **C Items**—Limited control, free stock.

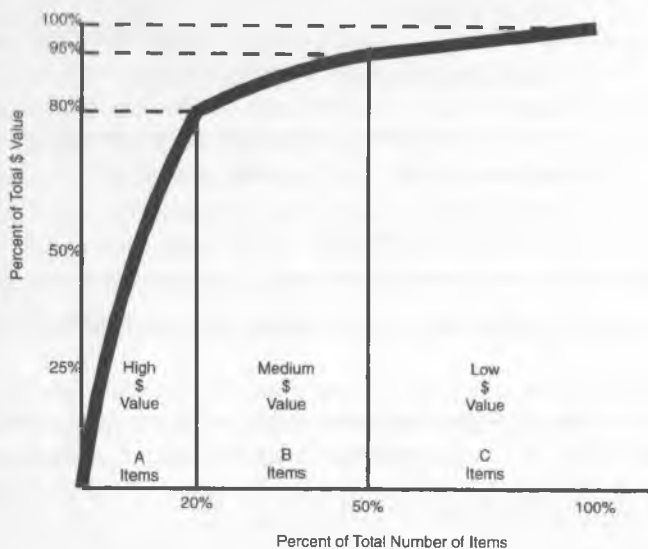


Figure 12.3 ABC Analysis Curve—Applied to Storeroom Inventory

While Classification of Stock Items and ABC Analysis are distinct techniques, they overlap, as shown in Figure 12. 4

Before rejecting the lofty Service Levels listed above, consider the number of items required to complete a repair right, the first time. If the answer should be three spare parts for which the specified service level is 90%; what is the likelihood of having at hand all three required parts? The answer lies in compounded probability ($90\% \times 90\% \times 90\% = 73\%$). Seventy-three percent likelihood of having the parts required to bring a critical production process back on-line is totally unacceptable in a world-class operation. A business entity that would accept such a standard will not survive in worldwide competition.

When considering reductions in inventory, bear in mind that insurance spares and parts represent 80 percent of inventory value. Appropriate authority have made an informed decision that adverse

Criticality	Stock Classification	Percent of Items	Value	Essential Service Level
A	Insurance Spares	20%	80%	100%
	Insurance Parts & Other			
	Critical Spares			98%
B	Standard Replacement Parts	30%	15%	95%
C	Hardware Items	50%	5%	90%
	Small Tools			90%
	General M&R Supplies			85%

Figure 12.4 Stock Classification, ABC Analysis, and Essential Service Levels

impact of not having these insurance items on hand when needed is an unacceptable risk. If considering a 10% reduction of overall inventory value, remember that the critical 20% of the items (A) is too risky to be included; the remaining 80% of the items (B&C) would have to be reduced by 50% to achieve the objective of a 10% reduction in overall inventory value. Do you believe that to be sustainable? Reliability and effective capacity would be seriously compromised.

Inventory Cycle Counting

Cycle Counting is a technique deployed by storeroom attendants to dynamically verify the accuracy of perpetual inventory counts against actual counts. Originally supplemental to fiscal inventories, now they are often accepted as suitably accurate replacements.

On a scheduled basis, the actual count of a specific SKU is compared to the perpetual count carried in CMMIS or some other computerized inventory management system preferred by Accounting. Frequency of the count for specific SKUs varies on rate of usage, value, or both. All SKUs must be counted at least once per year. Some may be counted as frequently as four times per year. Appropriate count and financial adjustments are made dynamically. As a result, inventory financial write-downs at year-end are minimal.

System-designed item counts per year are divided by the total shifts that the storeroom is staffed per year to determine how many counts

are to be made each shift.

This technique is also an effective management measure as to how well storeroom stocking, requisition, and return procedures are adhered to. Appropriate action is essential if Inventory Accuracy falls below 97%.

Vendor Partnerships

The common corporate climate these days pressures each business function to capture all possible cost reductions. In response, each function applies its own expertise to its own sphere of influence as encouraged by functional incentives. "Whenever individual functions maximize their own interests, the entity is sub-optimized" writes Peter Drucker. The search must be for "optimum" in all situations. There are five common material management initiatives that originate from these corporate pressures:

- ☐ National Contracts to capitalize on the purchasing power of the corporation. The corporation is able to negotiate better pricing than individual sites can negotiate.
- ☐ Single Source Supply to minimize administrative burden upon Purchasing and Accounts Payable.
- ☐ Consignment Stock where the vendor owns the stock but stores it in your stockroom. You pay for it when it is used. In very large operations, vendors may even erect their own store room on site.
- ☐ Partnership with Local Sources that provide exceptional service and technical support on engineered products (distinct from commodity products).
- ☐ Contract Storeroom Management to relieve internal resources from non-core duties.

Each of the five initiatives offers benefit. But an optimized hybrid of the five maximizes the positive and minimizes the negative of each when they are applied individually across the full spectrum of materials management. Pitfalls of singular application are explored below:

National Contracts with Single Source Suppliers

Often the top two initiatives are merged and serve the singular

interests of Procurement quite well, but often to the detriment of the overall enterprise. National contracts are frequently horizontal, single-source agreements, whereby a national supplier expands its product line to include the broad range of parts, components, assemblies, and materials required to support facility construction and plant maintenance; thereby becoming a single-source supplier. Such agreements often lack the critical service, response, and support necessary to sustain reliability and to support a just-in-time culture.

- ☐ National contracts are most appropriate for commodity-type hardware items such as standard fasteners, belts, and hoses. Unlike engineered parts, commodity items are less critical to reliable operation, safety of personnel, and protection of the surrounding environment. They are readily available locally should the national supplier be insufficiently responsive.
- ☐ They are often awarded on comparison of List Price from the national supplier versus Delivered Price from a local supplier. The unasked question is "Where will the National Supplier deliver from?" If the national supplier has a nationwide distribution network, its delivered price may still be lower than that of the local supplier. Otherwise, shipping and handling charges on frequent, small, and rush orders will exceed list price saving.
- ☐ How can a national supplier, without a broad network of distribution centers, respond to:
 - Midnight calls for 7:00AM delivery?
 - Prompt on-site technical support? It is unlikely that any single-source suppliers will have technical product knowledge across their (expanded) array of products. Such knowledge and support are critical in the case of engineered parts and materials, as opposed to commodity items.

Partnership with Local Sources

To the chagrin of local managers responsible for reliable output, their partnerships with local sources are often decimated when single-source national contracts become corporate edict. When striving to optimize procurement; the value of specialty distributors must not be undervalued in comparing cost with national vendors. Horizontal, national, single-source agreements weaken the position of local, specialized suppliers and

jeopardize the many value-added services they offer and that local sites depend upon in emergency situations.

Elimination of some local vendor relationships may be appropriate. But preservation of others can be critical. In many cases, the nature of required support is most effectively nurtured with high-quality local distributors postured to:

- ☐ Respond to emergency needs (24/7, even at 3:00AM if needed)
- ☐ Help evaluate failures through specialized knowledge and bring factory representatives to bear as needed.
- ☐ Keep local buyers, managers, engineers, project managers, and maintenance technicians abreast of the constant stream of technical advancements, new products, and other enhancements.
- ☐ Locally train the same parties in product application.
- ☐ Help source hard to find items.

It is unrealistic to expect these valuable local sources to continue responding to emergency situations without the opportunity to supply a portion of normal daily requirements. If they are called upon only for emergency support (24 hour, seven-day-a-week service, including holidays), their prices will drastically diminish list price savings associated with standard delivery terms of most national agreements.

Contract Storeroom Management

The fifth pitfall concerns Contract Storeroom Management. In addition to managing the storeroom, the Contract Manager may also be responsible for procurement of stock replenishments and possibly for procurement of direct purchase items (not authorized to be carried in inventory). This concept is viable provided the contractor is carefully selected and a clear, detailed contract is established. The contractor should:

- ☐ Own the stock until requisitioned.
- ☐ Manage all storeroom activity:
 - Process all transactions.
 - Release replenishment needs against system and blanket purchase orders, previously established by an independent, internal procurement body.

- Due to potential conflicts of interest, it is inadvisable for the contractor to independently procure from its parent company.
- ☐ The contractor has access to all current suppliers and their prices. With this knowledge they have an unfair bidding advantage.
- ☐ Likewise, the contractor should not establish specifications. Their specifications may be advantageous to product lines of their parent company or affiliates.

At least part of the answer to vendor partnerships lies in *vertical* (rather than horizontal) *national agreements*. Such agreements should be made directly with manufacturers within specific commodity groups, (e.g., hoses) that are highly regarded for time-tested quality. Local buyers, engineers, and maintenance managers should participate in the selection process. They know criticality (safety, environmental, quality, and reliability) of applications and past performance of parts produced by various manufacturers. Pricing should be established nationally by corporate purchasing directly with selected manufacturers.

Local buyers with input from maintenance and engineering should select distributors because they know which local distributors are most reliable and have demonstrated a spirit of partnership. If chosen manufacturers want the corporate-wide business, they will license and set up the locally-specified distributors to carry their product lines.

More than one manufacturer might be chosen within a core product group (those products with natural relationships), but the selected local distributor becomes the single-source for the entire product group. Within this structure, the purchasing power of the corporation is captured while unparalleled support of local distributors is preserved. The resultant economies derived from reliability, service, and technical support significantly exceed any cost increase that might be incurred in the processing of Payables.

Storeroom Delivery to Job Sites

The Storeroom is responsible for timely delivery of repair parts and project materials for both planned and emergency jobs to strategically located and secure drop points throughout the facility. The network of locations is based upon proximity to maintenance shops and normal job sites throughout the facility. Consideration is given to accessibility for

delivery personnel, safety of personnel in the area, and care of the delivered materials. The drop points have separate areas for deliveries and for unused items staged for return to Stores. Suitable delivery equipment is essential. Deliveries are synchronized with start of scheduled jobs.

ROLES AND RESPONSIBILITIES

Now, let's return to the preparatory process. When we left Job Planning (Chapter 11), the Planners had initiated and forwarded necessary Purchase Order Requests to Purchasing. They also placed appropriate reservations upon SKUs available in stock. In this age of technology, both actions should be transmitted electronically.

Assuming electronic reservation is reliable (the items will still be in stock when the job is scheduled). Stores need not take any further action until the associated job is scheduled. If electronic reservations are not faithfully protected, Stores must immediately pick, kit, secure, and hold the reserved items in a secured cage established for the holding of items held for planned jobs until they are scheduled.

As to Purchase Order Requests, acquisition of the associated items is on the critical path for completion of the specific work order. Until all required items are on-hand, the job will not be scheduled. It is therefore imperative that Purchasing sources and procures the items ASAP. The administrative procedure of Purchasing includes electronic feedback into CMMIS regarding promised delivery dates. Purchasing is then responsible for monitoring delivery status, expediting as necessary, and keeping Planning appraised of any change. As much of this as possible should be accomplished electronically. Well-designed CMMISs will accommodate all of it.

As Direct Purchase items are received, Stores receives and holds them in the secured cage referenced above. When all stock items and directly procured items are accumulated, Stores notifies Scheduling to that effect. At this point, the job is most likely Ready to Schedule, although it is possible that some status other than Awaiting Material has the job still on hold. This is the point at which the process transitions to Scheduling.

As soon as the schedule for next week is established, it is transmitted to the Storeroom. Stores picks and kits the reserved items, merges them with directly purchased items secured in the cage, and delivers them to the job site prior to job start or need.

The Planner's Role in Rebuildable Parts

Many heavy parts can be rebuilt. Rebuildables are a complex area for both the storeroom and for the planner. Your facility might have anything from a small bench for rebuilding hand tools, valves, and other small items to a completely equipped provision for rebuild of heavy-duty parts. Most maintenance departments use the small rebuilds as fill in work for those technicians who otherwise respond to breakdowns.

The larger rebuild shop is usually "owned" by the maintenance department, but the storeroom owns the core (item to be rebuilt, such as an engine for a generator). Complicating matters, large core units are sometimes removed and placed in the rebuild queue immediately (not even passing through stores). Complicating the picture even further, a supervisor might send a core unit to a vendor without consulting anyone from stores or planning. To increase control over these assets, some companies assign all rebuilding activity management to stores or purchasing, particularly if outside vendors are involved.

These core units floating around are one of the biggest areas of confusion when an organization is trying to maintain accurate accounts for a maintenance inventory. The issue is: how much value do we place on these core units before rebuilding and after rebuilding? Strategies for accounting for rebuildables are beyond the scope of this work.

What does concern us is that the planner might be responsible for managing some aspect of the rebuilding activity. At a minimum, the planner lists the part in the plan on the bill of material and includes paperwork that will deliver the part removed to the rebuilder—whether to an inside or outside shop. The planner might also note whether or not the replacement part can be a rebuilt.

Storeroom Benchmarks

- ☐ Service levels should be 85-to-100 percent, depending on classification of the item requisitioned (Figure 12.4).
- ☐ Inventory accuracy, as procedurally determined by cycle counting, should exceed 97 percent.

Work Measurement

If you can't measure it, you can't improve it!

Chapters 10 and 11 address how Work Measurement fits into the overall work preparation process. This chapter provides a broad education regarding the general subject of Work Measurement. Chapter 14 then provides training regarding the application of associated tools.

HISTORY

Work measurement can be traced as far back as the 18th century and the work of Coulomb. Modern work measurement (time and motion studies) goes back to the late 19th and early 20th centuries with pioneering work done by Harrington Emerson, Frederick Taylor, and Frank and Lillian Gilbreth. Much of the early efforts were applied to the work of construction tradesmen, such as bricklayers.

Measurement of maintenance work began in earnest shortly after WWII when the U.S. Navy developed and subsequently published *Engineered Performance Standards* for the Navy Public Works Department, which was responsible for all U.S. Navy bases and installations. That effort began with painting and carpentry and then expanded into other skills. Many of these standards relate to construction and maintenance of buildings. They have limited application to plant maintenance.

Development of the effort into a comprehensive "system" was a massive undertaking. At the peak of the effort there were more than 200 industrial engineers assigned to the project. Completion and initial release

of data took over 10 years. In the early 1970s, NAVFAC was assigned the job of keeping the standards up to date for use by all three branches of the military (Navy, Army, and Air Force). The last major enhancement was computerization of the entire effort in the early 1980s. The resulting library of maintenance standards is available for purchase from NAVFAC.

A related offshoot was applied to commercial industry during the 1950s, 60s, and 70s under the common name of *Universal Maintenance Standards* (UMS). Due to the cost and labor intensity of developing these standards, new UMS installations fell from favor by the 1980s. Due to the time and effort of maintaining established installations plus the cost of training a steady stream of new planners/schedulers, continued application also fell from common usage. New planners tended to revert back to gross estimates without reference and comparison to a systematic library of previously developed standards. Abandonment of established UMS systems, with the investment already made, was a most unfortunate decision (conscious or unconscious from a management standpoint). UMS installations did generate some of the best standards ever developed for maintenance work. Subsequent methodologies are not worthy of the term standards. They yield various levels of accuracy, consistency, and expectation. Management is left with the unknown: "Maintenance metrics look good on the surface, but how challenging are the expectations?"

The reason that Engineered Performance Standards and Universal Maintenance Standards are so labor intensive, time consuming and costly is due to their foundation methodology. They are based upon time and motion analysis with application of Predetermined Motion Times. Times for finite motions such as a "twelve-inch reach" are built up into "building blocks of universally applicable efforts" which can then be assembled into labor-hour estimates for specific "benchmark" jobs. Well-trained Planners/Applicators are essential if resultant standards are to be reliable. Once a sufficient library of benchmark jobs has been assembled into slotting tables, future job estimates can be developed by comparison to the benchmarks.

The basic motion standards and building blocks are universal. However, benchmarks and slotting tables must be developed locally due to facility peculiarities and distances, locally unique jobs, etc. The basic standards and essential training were usually obtained from an external consulting firm.

Another downside of UMS is the associated increase on the installation timeline for a comprehensive maintenance management program. Even with the commitment of several fulltime participants, UMS

normally consumed a full year of work measurement focus prior to start up of any actual planning, coordination, or scheduling. Management no longer has that much patience. They expect that the support tools will be developed as the installation progresses. UMS is not conducive to an on-the-run approach.

It should be apparent why
UMS has faded from the scene.

From here on, we will refer to “estimates” rather than “standards.”

APPLICATIONS OF MAINTENANCE WORK MEASUREMENT

Realistic job estimates are essential to the planning, coordination, scheduling, control, and motivation process. Before exploring other work measurement methodologies, we should clarify the several applications of maintenance estimates. They are used to:

- ☐ Balance manpower resources to workload (Macro-Planning and Work Programs).
- ☐ Establish meaningful maintenance schedules conveying the managerial expectation regarding what is to be accomplished with the resources expected to be available and paid during the approaching schedule week. Job Schedules are expressed in Duration Hours. **Duration hours** times crew size equals **Labor-Hours**. These two variables are both part of the scheduling process but should not be confused.
- ☐ Forward load preventive, predictive, and other programmed maintenance routines in a balanced manner to avoid peaks and valleys in the week-to-week PM/PdM demands upon labor resources. A common mistake is to have all quarterly, semi-annual, and annual PM/PdM tasks come due in clusters, rather than spreading them equally across each week of the year.
- ☐ Make realistic promises to Operations, particularly when jobs require that equipment be taken out of service. Such promises should include when the work will be started, how long the

work will take, and when the unit will be returned to service.

- ☐ Convey realistic expectancies by which employees can be motivated and both supervisory and crew performance can be evaluated. Employees perform best working in environments where they know what is expected of them and whether they are doing a good job when evaluated against fair, yet challenging, expectations.

The question becomes, "What level of expectation best serves each of the several applications?" True standards indicate how long jobs should take. Estimates frequently indicate how long jobs will most likely take. The difference is very significant when maintenance is still reactive with efficiency seldom higher than 50%. If a standard is used in such a situation, backlog weeks are understated, schedule compliance is extremely low, and completion promises are seriously missed. The challenge is to develop job estimates that establish a challenging expectancy on the one hand, but produce an achievable schedule on the other.

Maintenance personnel are often heard to say, "Maintenance jobs never go the same way twice." Although the statement is basically true, they add, "Therefore, maintenance cannot be measured." The latter portion of the statement is false. Maintenance work is repetitive. Many jobs occur weekly, some monthly or quarterly, and even annual jobs are repetitive to a degree. Therefore, maintenance jobs can be measured—with what degree of precision is the appropriate question.

Although issues encountered and resultant job content are not exactly the same each time the job is performed, the variability simply influences:

- ☐ Precision with which estimates can be set
- ☐ Methods of applying estimates to each application
- ☐ The number of jobs to be included in efficiency reports in order to level out fluctuations in individual jobs. Consequently, Maintenance Efficiency Reports normally cover a complete crew for a full week. Thus, the focus is not on job-by-job efficiency of individuals.

When setting production incentive standards, industrial engineers strive to achieve accuracy within five percent ($\pm 5\%$). In maintenance, the

accuracy goal is within fifteen percent ($\pm 15\%$), which is adequate for the applications (identified above) to which estimates are deployed within the maintenance arena. Although accuracy for individual jobs may not be precise, average accuracy over the dozens of jobs most maintenance crews complete within a week is more than adequate for the several maintenance applications—especially since improvement focus is on trends over time, rather than excessive emphasis on results within a specific time period.

Maintenance work often contains unpredictable elements, but those elements seldom influence the entire job. Consider the following. Even when disassembly time doubles due to working conditions; the associated troubleshooting, re-assembly, clean up, and feedback are not impacted proportionately. Most likely, the total job will not increase beyond the fifteen-percent range ($\pm$). The objective is to consistently achieve reasonable accuracy, not pinpoint accuracy.

When actual time differs from estimated time, a frequent reason is that the scope of the job encountered or performed differs from the scope on which the estimate was based. If the initial job description indicated a bearing change, but the job performed ends up being replacement of all bearings plus the shaft, impeller, and housing, it is not the same job. It was not the estimate that was wrong. It was the scope of that job that was wrong. It may be that the job scope was changed in midstream, or planners may not have scoped the job properly. They may have tried to plan the job without visiting the job site. Before closing the job out to Equipment History and Efficiency Reporting, properly describe the work actually performed and adjust the estimate accordingly.

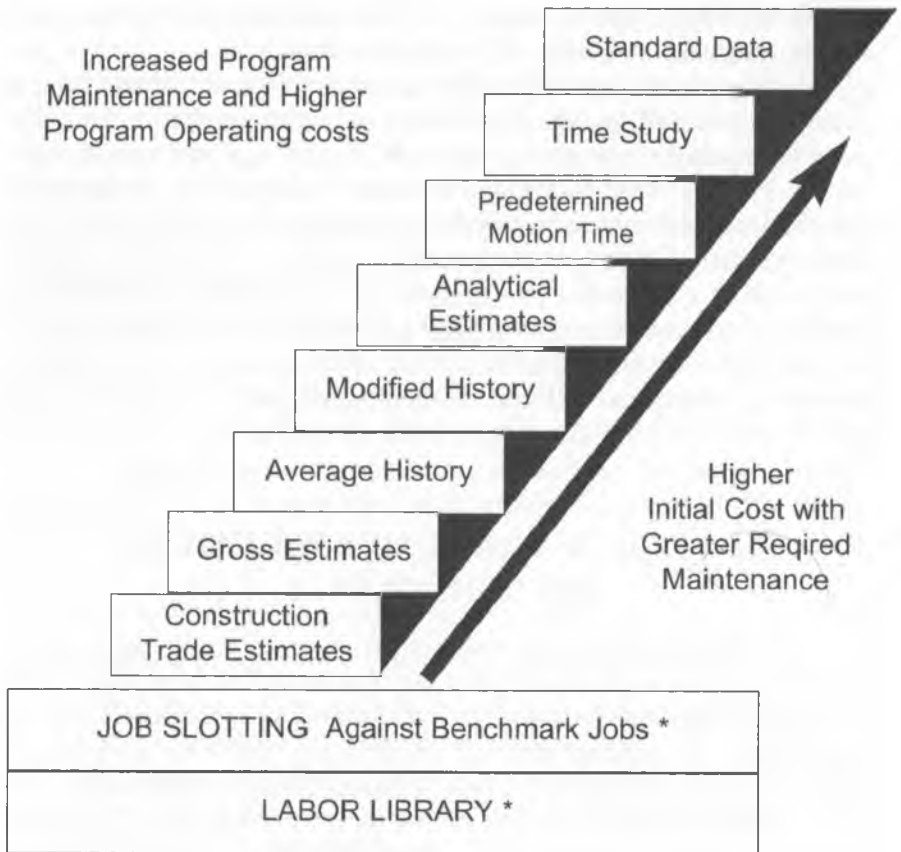
LEVELS OF WORK MEASUREMENT METHODOLOGY

Several forms of work measurement exist for use in maintenance applications. Having discussed why and how to use labor standards, the next phase of discussion focuses on what methods of estimate or standard development are most suitable and economically viable for each of the maintenance applications identified. A reasonable time frame and return on investment for the development effort must be considered. Also, there must be a means by which level of expectation can be maintained consistently between estimators (planners).

The level of work measurement sophistication is characterized as progress up the staircase in Figure 13.1. The level of sophistication to be

142 Chapter 13

deployed commonly increases with the maturity of the maintenance management installation, the level of precision required to raise maintenance performance to the next plateau, and the focus of the application (measurement of Efficiency or measurement of Schedule Compliance). An Efficiency focus ($\text{Standard Labor-hours} \div \text{Actual Labor-hours} = \% \text{ Efficiency}$) requires a more accurate form of measurement. A Planning and Scheduling focus requires a less precise form, but with greater certainty that actual will approximate estimate. Maintenance must not renege on a completion promise made to Operations.



* Recommended

Figure 13.1 Levels of Maintenance Work Measurement

Advantages and Disadvantages of Each Methodology

We will begin at the bottom of the staircase in Figure 13.1 and work up:

☐ Construction Trade Estimates (Published)

Construction Trade Estimates are shown as the lowest step because they are developed for contractors to use when bidding on construction jobs. They are not recommended for use regarding in-house maintenance. Construction is not among the most efficient industries and the estimates reflect that level of efficiency. They also include engineering safety factors in the interest of the bidder. Most of the estimates provided relate to construction jobs. Few of the estimates provided relate to plant maintenance. For these, you might consult manufacturers of equipment or components (seals, bearings, hoses, belts, etc.).

☐ Gross Estimates

Although they may be only educated guesses, most installations begin with Gross Estimates the second step on the desirability staircase. These estimates are the least costly and least time consuming to develop; they can be developed as the installation goes forth.

The disadvantage of gross estimates is they usually reflect the personal judgment of a particular supervisor, planner, or technician. Consistency seldom exists between estimators. Their estimates typically reflect how much time jobs will take rather than how much time they should take when properly planned and coordinated. Some estimators base their judgment on how long they themselves took to perform the job when they were craftsmen. Others base their estimates on how long a specific technician usually takes to perform the job. These estimates are especially poor when the estimator has not personally experienced the job.

However, if backlog is currently quantified only in terms of jobs or work orders, get it quantified in labor-hours now, even if use of gross estimates is temporarily necessary. "Jobs" is not a meaningful unit of measure because some require less than an hour, whereas others may require a hundred hours or more. The only common denominator is "labor-hours."

□ Historical Averages

Records of past actual labor-hours charged (Historical Times) are the third step in sophistication and the second most-common methodology. The labor-hours charged to previous work or individual jobs are recorded and accumulated, either in Labor Libraries or in Equipment History. Occurrences of a given job are averaged after elimination of skewed highs and lows. Some computer systems encompass this functionality. Resultant averages reflect the size and condition of the facility, condition of equipment, skill level of the maintenance work force, and the current state of job preparation including materials support, all of which are likely poor if the local culture is reactive.

On the other hand, if the average length of time that a job has been taking is known, what could possibly provide better estimates than Historical Averages? The answer is nothing, provided maintenance crews have been working at 100 percent efficiency and the state of maintenance excellence has already been achieved. Unfortunately, the efficiency of reactive maintenance tends to be in the 50% range. If world-class is to be achieved, such mediocrity cannot be the expectancy. Like gross estimates, use of historical averages should therefore be temporary. As the installation matures, scale up the sophistication level of estimating.

Because the work order system is the source of historical averages, it is often difficult to obtain reliable data on which to calculate the averages. If descriptions are not correct and time distribution is not complete and accurate, the averages will not be reliable. The averages will reflect the current environment, methods, and tooling rather than the standard methods and procedures with proper preparation (planning, coordination, and scheduling) by which effective and efficient execution is enabled.

Another problem when using historical averages is calculating efficiency. Efficiency equals standard labor-hours divided by actual labor-hours. If a historical average is used as the numerator, in essence the formula becomes average labor-hours divided by actual labor-hours. Efficiency always approximates 100%, but underservedly. Efficiency and schedule compliance may look favorable

using historical averages, but management will soon and rightly ask: "How demanding are the expectancies?"

❑ Modified Historical Averages

Modified averages are the first step of the staircase reflective of a "standard" as opposed to an estimate. There are two alternative approaches by which appropriate modification can be determined. Here we will address the most common. The alternative will be addressed in the next bulleted approach (Dynamic Adjustment of Current Averages).

If an effective CMMIS exists, historical averages can be extracted from Equipment History. If a CMMIS does not exist (a new installation) or an existing CMMIS is not effective, a base period of perhaps six months is necessary to accumulate sufficient data to strike averages for repetitive jobs. In either case, activity sampling is performed (throughout the base period in the latter case) to determine the efficiency of the various crews during the base period. If a given job averaged ten labor-hours and the crew averaged 70% efficiency during the base period, 7.0 labor-hours becomes the modified historical average.

This methodology provides quasi-standards. However, it is fairly expensive due to the cost of making sufficient observations to gain statistical reliability. It is several months before labor-hour measures are available for deployment. Human Resources (Personnel) may hesitate to endorse on-the-floor measurement of the workforce. In this regard, recognize that activity (or work) sampling is not time-study.

❑ Dynamic Adjusted Averages

One of the authors participated in this approach in the mid-1950s. After working with all the other approaches discussed here, he has come full circle and believes this to be the best approach. Reasons for this conclusion are clarified under the next major heading (Best Methodology for each Application).

This approach also begins with a base period for establishing initial times for each known maintenance job. Each of these averages is assigned an index of 1.00. During future measurement periods, the period average for each job is compared to the corre-

sponding base average to establish the current index for each job (e.g., 8.0 labor-hours base period / 7.0 hours current period = 1.14 current index). The current indexes are averaged (weighted) to establish overall improvement over time compared to the original base period. A common problem is that the workforce views it as 114% efficiency, which it is not. Assume that efficiency during the base period was only 50%. Fifty percent times an index of 1.14 implies that current efficiency is 57%, not 114%. The proper understanding must be conveyed repeatedly.

Each job is assigned a code to facilitate accumulation and analysis of data. Current averages for individual jobs are used for day-to-day planning and scheduling. Overall indexes flow into managerial metrics and progress trends.

☐ Analytical Estimates

Analytical estimates are fully discussed in the next chapter.

☐ Universal Maintenance Standards (UMS)

UMS is an industrial engineering approach to maintenance work measurement. Micro Motion Studies, Time Study, and Standard Data have been integrated into predetermined times for maintenance tasks—Universal Maintenance Standards. UMS takes work measurement to a level of precision unnecessary for the management of maintenance. They are too time consuming and expensive to set-up and to maintain. They have faded from common usage, despite being the most accurate method by which to develop maintenance standards.

Unless the associated investment has already been made and the standards are already available UMS is not recommended (for new installations). However, the UMS approach did spawn the two supplemental techniques introduced next.

☐ Job Slotting and Labor Libraries

Job slotting and labor libraries are effective techniques that can supplement any of the work measurement methodologies discussed. They too are fully addressed in the next chapter.

BEST METHODOLOGY FOR EACH APPLICATION

The work measurement challenge is to develop measures that establish challenging, yet fair expectations of performance (standards) while also facilitating achievable promises and schedules (estimates). Standards quantify how long jobs should take a properly trained technician working at a normal pace to perform—assuming effective job preparation. Estimates usually reflect how long jobs probably will take given current levels of performance. The difference is most significant when maintenance is still reactive, in which case, current efficiency is seldom higher than 50%. Figure 13. 2 facilitates further discussion of the challenge:

Form of Measurement	Application		
	Backlog	Scheduling	Efficiency
Current Average / 1.1*	X	X*	
Standard			X
*The 1.1 divisor puts 10% “pull” into schedules.			

Table 13.2 Form of Measurement Varies with Application

Chronologically within the Preparatory Process, Backlog Management is the first application to which job measures are applied. If standards that reflect 100% performance are used, Backlog might calculate as 4.0 weeks. However, if current efficiency is only 50%, it would take current maintenance resources 8.0 weeks to complete all the associated jobs. Therefore estimates reflective of current performance better serve the backlog application.

Scheduling is the second application of job measurements. Again, if a standard (100%) is used to make promises to Operations and to set the expectation for weekly job completions (Backlog Relief) when the maintenance crews is performing at only 50%, Schedule Compliance is going to be very low (50%). Even worse, half the promises made to Operations will be fumbled. Also, downtime durations will be twice as long as promised. Next time, Operations will be reluctant to release their assets to Maintenance. Because all management tools (budgets, goals, schedules,

etc.) are conceived to drive improvement, schedules often include pull, as denoted by the asterisk in Figure 13.2. "Pull" is the expectation of improvement above current performance levels. However, it is not realistic to expect efficiency to improve from 50% to 100% (or even 85%) in the near-term. In fact, 10% improvement within a quarter is a lofty goal (also reflected in the figure).

Worthy of consideration is a scenario where pull is not included on jobs involving "return to service" promises, but included on all other scheduled jobs.

Efficiency Reporting is the third application. As previously explained, if estimates reflective of average history are used as the numerator of the efficiency formula, the quotient always approximates 100% efficiency, even when true efficiency relative to a proper standard is only at 50%. So, Efficiency is the only one of the three applications that necessitates use of Standards.

Efficiency Reports should be developed weekly for the total crew or team responsible to a given supervisor. Using a one-week time frame means that the average efficiency for each crew is based on several dozen jobs, thereby balancing abnormally difficult jobs with less troublesome and trouble free jobs.

Individual (as opposed to crew) efficiency should be the exception rather than the rule. It should only be used constructively to guide development and training. Such usage can uncover specific training needs of individuals.

Efficiency of the individual should never be used to trigger or justify discipline, or as a **crutch** to cover-up for a lack of proper supervision and leadership. Understanding, acceptance, and respect for measured expectancies and the need for them must be preserved for the important applications of backlog control, work scheduling, and efficiency trends. These measurements convey management expectations. When technicians meet expectations, they know that their performance is up to par. Communication and feedback always improves morale.

Properly utilized, the measurement of individual and team efficiency is an important motivator. Without it, workers do not get sufficient and accurate feedback about their job performance. The feedback they do receive too often relates to supervisor or customer moods rather than quality or duration of the job performed.

Methodology Conclusion

With two metrics (true engineered standards and current efficiency), expectancy can be adjusted to serve each of the three applications. But the only methodology that yields an engineered standard is UMS; and UMS is no longer recommended as previously explained. Since Efficiency is the only application of the three that cannot be supported by the remaining methodologies (excluding UMS), the question becomes, "What do we do about Efficiency?"

Because Efficiency is an actuarial measure (after the job has been completed), it is the least important of the three applications. The other two applications are preparatory. They enable jobs to be performed more effectively, efficiently, safely, and regulatory compliant.

Furthermore, standards for emergency jobs should be post-applied (after the job is completed) based upon technician feedback regarding the nature of the problem and actions taken. During early phases of installation, planner capacity is better devoted to preparatory efforts than to post-completion reporting. Efficiency can therefore be sacrificed until the installation and reliability of estimates have matured. In the meantime, rely upon Schedule Compliance and Operational Uptime as the key metrics by which to monitor Maintenance/Reliability progress.

Until Engineered Standards are developed using Analytical Estimating (next chapter), Dynamically Adjusted Average is the most suitable of the Maintenance Work Measurement Methodologies that are based upon historical averages. Regardless of the methodology used to develop them, estimates should be continuously refined, based upon feedback, until they are worthy of being called Standards. As World-Class Maintenance/Reliability installations mature, the gap between Current Average Performance and Standard Performance should continually narrow until comparability is achieved.

Remember: In estimation, you are shooting for the most probable time for the job.



Analytical Estimating

A RECOMMENDED APPROACH TO MAINTENANCE WORK MEASUREMENT

The challenge within this chapter is to define an estimating process that yields reasonably accurate work measurement worthy of the term “standard” without being excessively costly and time consuming to establish and sustain. Such measurements must consistently reflect labor-hours within which a properly-trained technician working at a normal pace is able to perform specific maintenance jobs in a quality manner, provided jobs have been effectively prepared.

Analyzing and estimating maintenance work is indeed a challenge because unpredictable conditions are frequently encountered. Using *Analytical Estimating* as conveyed herein, persons with practical experience in performance of maintenance jobs are able to visualize steps and labor-hours required to perform jobs of familiar nature. Due to their background, people with experience as maintenance technicians are usually selected as planners. Of course, they must also possess other traits (see Appendix C for position description, including required characteristics).

Due to the following characteristics of maintenance jobs, Analytical Estimating meets the work measurement challenge:

- ❑ Pinpoint accuracy is not feasible because all the variables within maintenance jobs are not known at the time of estimation. In maintenance, therefore, we look for accuracy within $\pm 15\%$.
- ❑ The unknowns impact only a portion of the total job; they can be addressed by application of **Allowances** (discussed as we get further into this chapter).

Figure 11.1 The Common Job Sequence

Source	Task
P	Get ready and receive instructions for doing the job. Preparation includes receiving job instructions from the supervisor, collecting personal tools, obtaining parts from storeroom, and gathering required special tools and equipment
T	Travel to job site (outbound)
P	Listen to production input regarding symptoms
P	Make preliminary diagnosis and troubleshoot prior to shutdown
P	Shutdown and Lockout. This procedure is done jointly with the line supervisor, control room technician, and/or line operator. Equipment must be stopped in the proper sequence with proper lockouts
L	Partial or total disassembly to reach the problem area
P	Determination of full extent of problem
P	Identification of necessary replacement parts, obtaining them from the storeroom, or initiating a direct purchase for parts not held in inventory
L	Re-assembly of equipment using the replacement parts as needed
P	Check proper job completion, test operability of equipment, clean up the job site and put away tools
T	Travel back to shop (inbound)
P	Report on job and return unused parts, special tools and equipment
A	Allowances

Legend - Source of Estimate

P	Fixed Provision Table
T	Travel Time Table
L	Labor Library
A	Allowance Table

- ☐ It is relatively easy for trained planners to visualize and establish duration and labor-hour requirements for simple, short-duration jobs.
- ☐ Large complex jobs are most effectively estimated when divided into small segments, estimated at that level, and then summarized into an estimate for the total job.
- ☐ All maintenance jobs are performed in the general sequence shown in Figure 14.1.

Most maintenance departments perform thousands of jobs within each quarter, if not within each month. How many times (from scratch) should planners estimate job preparation, travel time, feedback, etc? Would it not be a wise investment to develop a series of reference tables to facilitate estimating for repetitive elements of job execution? Such tables are presented and discussed below.

P—Fixed Provision Table

All steps coded “P” are assembled within the Fixed Provision Table. The time provisions shown in Figure 14.2 are illustrative.

	Simple	Average	Complex
Receive Instructions	Minimal	Normal	Thorough
Gather Tools	Tool Belt	Personal Box	Special Shop Tools
Follow Safety Procedures	Normal	Lockout	+ Barriers and Protection
Preparation Sub Total:	0.1	0.2	0.3
Obtain Parts & Materials	None	Bin Stock	Stores Requisition
Reference Job Plan, & Drawings	None	Job Plan	Multiple References
Mid-Job Sub Total:	0.0	0.1	0.2
Feedback	WO and TDC	+ Findings	Detailed Feedback
Wrap Up Sub Total:	0.1	0.2	0.3
Total Hours Allowed / Person*	0.2	0.5	0.8
*All crafts except Painters. Add time for mixing, taping, drop cloths, etc.			

Figure 14.2 Fixed Provision Table

Appropriate estimates must be developed for each facility.

T—Travel Time Table

All steps coded “T” are assembled within the Travel Time Table (not unlike a point-to-point distance chart of a road map). Using plant drawings, measure distances from the maintenance shop and storeroom to each Operating Department or to each Work Center should a department area be very large. For each destination, determine round-trip travel times. Lacking another unit of measure, use a normal walking rate of 264 feet per minute. Then, develop the table further into simple, average, and complex jobs. Simple jobs can be done with the tools technicians carry in their tool pouch; a single round-trip is adequate. Average jobs often require an extra trip back to the shop or perhaps to the storeroom; allow two round-trips. Complex jobs may require a third round-trip. No job properly planned should need more than three round-trips, unless it requires more than one shift to perform. A simple travel-time table is shown in Figure 14.3.

		Allowed Hours Per Person		
From Shop To	Round Trip Hours	Simple	Average	Complex
Area A	0.5	0.5	1.0	1.5
Area B	0.4	0.4	0.8	1.2
Area C	0.3	0.3	0.6	0.9
Area D	0.2	0.2	0.4	0.6
All Other Areas	0.1	0.1	0.2	0.3
	Round Trips Provided	1	2	3

Figure 14.3 Travel-Time Table

L—Labor Library

All steps coded “L” are assembled within the Labor Library. A table similar to Figure 14.4 is developed for each asset, piece of equipment, or other grouping of work by nature. Labor Libraries address Direct Work (wrench time) occurring at the job site, detailed by job step or task. In many cases, the steps can be extracted from technical manuals for each equipment unit. Most manuals contain schematics detailing disassembly sequence (what comes off first, through what comes off last). Re-assem-

bly occurs in reverse order of disassembly. The next steps involve documentation of required skills, crew size, duration, and labor-hours required to perform each step.

Once a Labor Library has been established for each unit of equipment, planners need only to determine the depth of disassembly required to gain access to the failed component or assembly. An example of a Labor Library is presented in Figure 14.4 (see Appendix D for a filled in example).

LABOR LIBRARY										
Equipment Name:							Equipment No:			
Sequential Tasks		Craft:	Mechanic		E & I		Construction		Labor	
No.	Description		Crew	MH	Crew	MH	Crew	MH	Crew	MH
0.10										
0.20										
0.30										
0.40										
0.50										
0.60										
0.70										
0.80										
0.90										
1.00										

Figure 14.4 Labor Library

Library development in such detail for each equipment unit in a facility may seem to be an overwhelming task. However, if all the work is documented and cataloged, complete libraries will soon be built. To plan the first job that arises on a given unit, deep disassembly may not be needed. Capture the steps that are needed. The next time a failure occurs on the unit, the planner may need to go deeper. Add the additional depth to the library. It won't be long before the entire disassemble/re-assemble sequence has been captured and measured.

Perhaps only on occasion, but all maintenance departments define job steps, perform material take-offs, locate and reference prints and drawings, prepare clarifying sketches, and make labor-hour and job duration estimates. The problem is that they treat them as one time occasions and therefore make little effort to retain and catalog their preparatory efforts for future use or reference. Valuable information is simply discarded after jobs are completed. Lessons learned are not documented. As a result, errors tend to be repeated.

Take advantage of the repetitive nature of maintenance. Estimates show that in reactive environments, on average 80% or more of jobs performed are repeated each year. Catalog your planning efforts in a retrievable format. Planned Job Packages will be applicable time after time. As reference files are built over time, planning efforts become faster, easier, and better; and planners are able to stay abreast of their workload.

A—Allowance Tables

The above tables (Fixed Provision, Travel, and Labor Library) do not provide for activities such as authorized breaks and wash up, fatigue, unavoidable delays, or balancing delays common to multi-person and multi-crew jobs. These realities are provided for through application of allowance factors. Because it is unknown which jobs will encounter these situations, appropriate percentages are added to all jobs. Some jobs receive too much allowance; other jobs receive too little allowance; but they balance out across the day.

The most common allowances are referred to as Personal, Fatigue, and Delay (PF&D) allowances. Each is commonly quantified as five percent ($5\% + 5\% + 5\% = 15\%$). Labor Agreements or Employee Practices normally provide breaks and wash ups that amount to approximately five percent of the work day (Personal Allowance). However, fatigue occurs between break (another 5%—Fatigue Allowance). Although planners strive to identify and preclude potential delays, frequently there are delays that cannot be avoided even if anticipated. These are Unavoidable Delays and are also quantified at five percent (Delay Allowance).

Because many maintenance jobs require more than one technician or possibly more than one crew (different craft or skill) to be performed, there are two additional allowances that may be required:

- ☐ When more than one person is required to perform a job, it is unlikely that all are required throughout the course of the job. There are times that all can work together simultaneously. There are times that all can work independently. But there are also times that only some of the crew can be engaged; the others simply must stay out of the way. This latter condition requires a Crew Balancing Allowance (1 to 3%).
- ☐ When more than one crew is required (e.g., Mechanics and Electricians), planners do their best to predict when the first crew will be ready to be joined by the second crew, but frequently

one crew is forced to wait for the other. Perhaps the previous elec-planned. Jobs like this require a Multi-Crew Allowance (1 to 2%).

When jobs are well planned, these instances are minimized. But they do occur. A typical table of allowances is shown in Figure 14.5.

		Level of Inherent Fatigue		
Nature of Allowance	%	Light	Average	Heavy
Personal Time (Breaks and Clean Up)	5	5	5	5
Fatigue	5-10	5	7	10
Unavoidable Delay	5	5	5	5
Sub Total	15-20	15	17	20
Crew Balance—Multi Man	0-3	0-3	0-3	0-3
Crew Balance—Multi Craft	0-2	0-2	0-2	0-2
Total	15-25	15-20	17-22	20-25

Figure 14.5 Table of Allowances (Illustrative)

All the above conditions seldom occur in a single job or throughout an entire job. The **vast majority** of maintenance jobs need only 15% allowance. A great number of jobs require 17 or 18% because one or two abnormal conditions can be expected to occur. Some jobs require 20%. Very rarely, but occasionally complex jobs require as much as 25% allowance.

Figure 14.6 summarizes other specific conditions that enter decisions regarding percent allowance to be applied.

Fixed Provisions and Travel never need more than 15%. Additional allowances are warranted only on job steps that are Direct Work (at the job site using tools of the trade—i.e., wrench time).

Do not apply allowances to Gross Estimates or to Historical Averages. Those forms of work measurement already reflect conditions addressed by allowances. Use allowances only when engineering a standard using a process such as Analytical Estimating.

Allowances are typically added to the total estimated labor-hours, including direct work, fixed provisions, and travel time. However, there are times when it is imperative to add them at the level of individual tasks. First, it is unlikely that heavy fatigue or crew balancing is necessary throughout an entire job. Second, breaks, fatigue, and delays occur throughout jobs, not at the end of jobs. Therefore, when providing a job

time-line to facilitate job control or to predict when an asset will be brought back on-line, the need for allowances must be shown throughout the job, if the time-line is to be realistic. In such cases, add allowance factors to duration hours as well as to labor-hours.

Also be careful when defining labor requirements for a multi-person job. In quantifying the labor requirements for a job as requiring a crew of four with duration of four elapsed hours; does that necessarily equal sixteen labor-hours? Or, are the first two members of the crew required for the full four hours, whereas the other two are required for only one hour at the beginning of the job plus another one hour at the end? The latter scenario would equal twelve labor-hours rather than sixteen. This is a common miscommunication that inflates backlog and loads schedules with poorly utilized resources.

Figure 14.7 provides a job planning format that is consistent with Analytical Estimating.

COMPARATIVE TIME ESTIMATING

Absolute accuracy is neither possible nor necessary when establishing expectancies for the control of maintenance efforts. What is essential for control is consistency of estimates and ease of making them. Without consistency, technicians will feel that estimates are unfair and will fight against their usage.

It is in the nature of human mental processes to estimate by comparison. We compare the unknown with the known and then estimate the degree of similarity (identical, larger, smaller, and by how much?). We tend to do these comparisons automatically and very subjectively. Some people have an almost infallible sense of comparative magnitude whereas others find it hard to come anywhere near accuracy.

Consistent estimating requires that all planners refer to and apply a common library or catalog of job estimates. There are basically four structures that enable comparative job estimating:

- ☐ Systematic card files, spreadsheets or database software that can be sorted by: Skill, Nature of Work, Crew Size, Duration, Labor-Hours.
- ☐ Catalogs of standard data for the building of job estimates
- ☐ Labor Libraries
- ☐ Slotting Tables of Benchmark Jobs

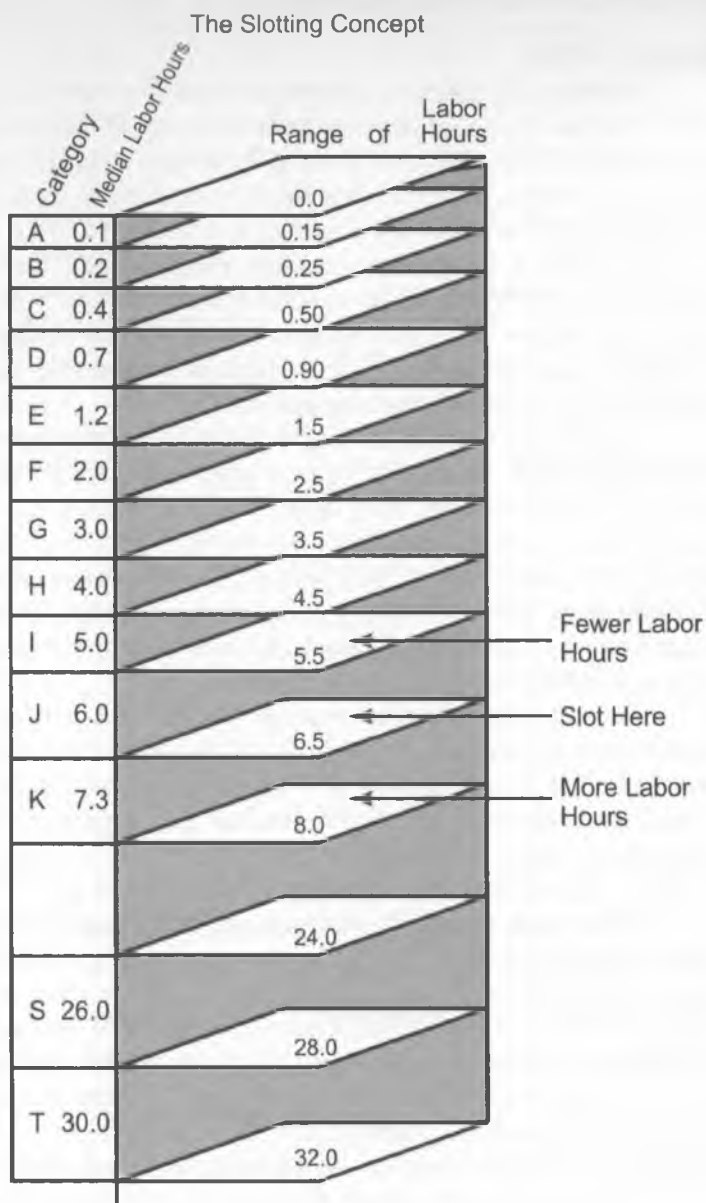


Figure 14.8 The Slotting Concept

Best practice rests between the latter two structures. Slotting tables are the current subject, discussed below.

Slotting Tables

Usually it is easier to determine whether a new job is larger or smaller than another job as opposed to determining, in isolation, how long the new job will take. The Job Slotting Technique compares new jobs to be evaluated with a group of benchmark jobs that are well known and have previously been carefully described and estimated.

To make a comparative estimate, planners must first define the scope of the job to be performed and prescribe the method to be used. This planner task requires good knowledge of processes and equipment, visits to job sites, and discussion with requestors, responsible maintenance supervisors, and knowledgeable technicians.

Back at their workstation, planners next consult appropriate sections of benchmark libraries to locate a benchmark job of similar nature and level of labor intensity. They use a process of bracketing. A judgment is made based on mental comparison of what is involved in doing a benchmark job (for which required labor-hours have been established) to what will likely be required to perform the job being estimated. Using the slotting process as illustrated in Figures 14.8 and 14.9, planners make judgments as to which slot new jobs best fit into.

Planner judgment may indicate that the new job is less labor intensive than the benchmark jobs in slot K, but more labor intensive than those in slot I. Although the new job may not be an exact match with any of the J benchmarks, it is nevertheless the best available comparison. Close enough! Make it another J job, which are known to range between 5.5 and 6.5 labor-hours with an average of 6.0 labor-hours.

Reasonable accuracy and consistency has been achieved expeditiously. Remember that the technique of comparative estimating involves comparison of jobs with those in the library, not the matching of jobs. This distinction is important because a library consisting of a few hundred carefully-selected benchmark jobs enables experienced Planners to produce consistent estimates for most maintenance work. Otherwise, thousands of jobs have to be described, planned, and estimated to cover the variety and multitude of work performed by most maintenance organizations.

Comparative estimating requires subjective thought by the planners, but the only decision they must make is a choice between one time interval and the next. Comparison with library jobs of known duration greatly reduces guesswork. Notice that the bracket increments in Figure 14.8 are consistent with a desired accuracy of +15 %.

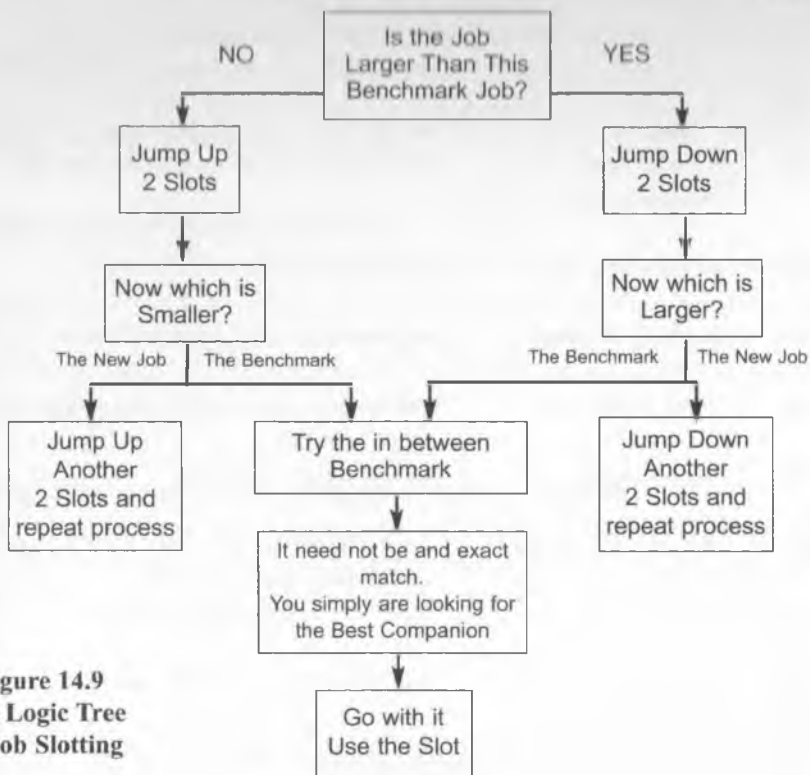


Figure 14.9
The Logic Tree
of Job Slotting

GROUP _____		GROUP CODE 01-04 _____	
E. Jobs Code <u>984</u>	F. Jobs Code <u>985</u>	G. Jobs Code <u>986</u>	H. Jobs Code <u>987</u>
2.5 HRS.	3.5 HRS.	4.5 HRS.	5.0 HRS.
2.00 HRS.	3.00 HRS.	4.00 HRS.	6.00 HRS.
7.00 HRS.			
Repair Door	Repair Calendar Drive	Repair Top Press Roll	Replace Wire on P.M.
Repair Valve - Up to 4"	Misc. Minor Repairs - Pulper	Replace Top Wei Press Roll	Replace Pump
Repair Hoist - 1 to 5 Ton	Replace PIV Valve	Replace Doctor on P.M.	Replace Top Size Press Roll
Repair Service Elevator	Replace Doctor Blade Holder	Repair Fan	
Repair Pump	Install Dandy Roll		
Misc. Minor Repairs - Langston Winder	Minor Repairs to Cylinders		
Repair Print-Weigh Scales	Repair Saveall		

Figure 14.10 Slotting Table — Common Format

Code 0290XX		
Code	Task Area Title	Coverage (includes replace/repair/clean the following:)
0290.XX		
0.01	Air Compressors, Vacuum Pumps	Reciprocating and rotary type vacuum pumps
0.02	Air Cylinders	Single action, double action
0.03	Bearings, Bushings	Anti-friction bearings, plain bushings
0.04	Chain, Belts, Cables	Roller, silent, and conveyor chain; V-type, fiat-type, Veelos belts
0.05	Bakes, Clutches	Manual electric, pneumatic
0.06	hives, Gear Reducers	Variable speed drives (Reeves & PIV); fixed ratio speed; shaft mounted reducers
0.07	Fans	Materials handling and air handling
0.08	Fasteners	Keys bolts, screws, taper & steer pins
0.09	Filters	Air, oil, and water
0.10	Fire Protection	Inspect, test, and recharge
0.11	Gears, Pulleys, Sprockets	Gears: spur, bevel, Helical, worm, Pulleys: V-type, fiat conveyor type
0.12	Motors	AC motors: single- and 3-phase; DC motors; hydraulic motors; gear-head motors
0.13	humps	Centrifugal, positive displacement, gear, reciprocating
0.14	Rolls, Rushes, Shafts	Rolls - Covers all types: s.s., rubber, felt, cork; shafts used as rolls; elliptical, etc.
0.19	PM Inspections	Covers PM inspections on all equipment ad machines

Figure 14.11 Family of Slotting Tables For Millwright Work

Figure 14.8 presents the slotting concept visually, but actual Slotting Tables commonly take the form illustrated in Figure 14.10, which contains a sample spreadsheet of Benchmarks cataloged by skills required, nature of work, and the task.

Cataloging of Slotting Tables

A series of slotting tables such as shown above are developed and cataloged for each skill. As the list of Benchmark Jobs expands, slotting tables become more specific and progressively improve. Ultimately they result in families of Slotting Tables, as illustrated in Figure 14.11 for Millwright work.

Initial Development of Simple Slotting Tables

Establishing and maintaining an effective reference library of slotting tables with a uniform structure requires contributions from all planners in order to cover the various classes of equipment and skills nec-

Instructions for Recommending Benchmarks
Identify jobs performed by your crew that occur frequently and are familiar to you.
Determine the most effective crew size.
Record each selected job in the appropriate interval on the Benchmark Solicitation Form (Figure 14.13). Put the job in the appropriate crew size column and in the increment row representing your best judgment. Work in pencil, as you may wish to change an estimate based on the relativity of one job to another. Use of this relativity is the concept behind the slotting techniques.
Offer a recommended benchmark job in as many slots as possible, but do not force a job into a wrong slot. It is better not to use a job or to leave an increment blank, than to force a job into an improper increment.
Submissions will be compared to those from other participants and as necessary will be spot-checked by work measurement. Therefore, do not hesitate to offer your best insights.

Figure 14.12

Using Foreman Estimates
To Building the Slotting Library

Craft:			Suggested Benchmark Job				Maintenance Management Program Benchmark Development Sheet
Elapsed Time							
Slot	Allowed Hours	Range From-To	Units	One Person Crew	Two Person Crew	Three Person Crew	Four Person Crew
A	0.30	0 to 20	Minute				
B	0.50	20 to 40	Minute				
C	0.00	41 to 50	Minute				
D	1.00	51 to 1-1/4*	Min-Hr				
E	1.50	1-1/4 to 1-3/4*	Hours				
F	2.00	1-3/4 to 2-1/2*	Hours				
G	3.00	2-1/2 to 3-1/2*	Hours				
H	4.00	3-1/2 to 5*	Hours				
I	6.00	5 to 7*	Hours				
J	8.00	7 to 9*	Hours				
K	10.00	9 to 11*	Hours				
L	12.00	11 to 14*	Hours				
M	16.00	1-3/4 to 2-1/4*	Shifts				
N	20.00	2-1/4 to 2-3/4*	Shifts				
O	24.00	2-3/4 to 3-1/2*	Shifts				
P	32.00	3-1/2 to 4-1/2*	Shifts				
Q	40.00	4-1/2 to 5-1/2*	Shifts				

*Not Including: - Use the Higher Slot

Figure 14.13

ecessary to maintain them. Initial tables can be developed in the following manner.

Solicit up-front input involving all planners and supervisors (as well as technicians). Explain job slotting and its usage. Then enlist participation with the technique described in Figures 14.12 and 14.13.

This approach provides a quick start-up with reasonable initial accuracy and consistency. Resultant benchmarks are subsequently refined as the planning and estimating process matures. The refinement process incorporates the following technique:

- ☐ Each week an exception report is generated via the CMMIS. For each planner, the report lists the top ten jobs for which actual labor-hours differed from the estimated labor-hours by the largest percentage, regardless of sign (plus or minus).
- ☐ The involved planners should evaluate each of their respective top ten jobs to determine if adjustment of the estimate is necessary. If a planner decides that an estimate was wrong, the estimate should be changed in the database for future application. On the other hand, if it is decided that the source of the variance was poor performance by crew or supervisor, or some avoidable delay that will be precluded in the future; the estimate should not be revised.

BUILDING AN ESTIMATE

There is no substitute for task knowledge when it comes to estimating. Regardless of the measurement technique, familiarity with operating equipment and the jobs required to maintain them is essential. Planners bring some of the necessary familiarity to the job, whereas the balance is acquired through formal and on-the-job training. Craft background is the ideal starting point because it enables Planners to visualize how the job should be performed. If planners do not know the job well enough to visualize it, they should talk through the job with a maintenance craftsman or supervisor who is familiar with it.

At times planners observe a job as it is performed. Although this approach will not contribute to effective job performance on this occurrence of the job, it will facilitate development of a good planned job package for future occurrences. Such observations expand planner knowledge of plant processes and equipment, and inherent skills of individual techni-

cians comprising teams that the planner supports. Tremendous advantages accrue when planners are *seen to be involved* in proper preparation for effective job performance. Good planned job packages build credibility and appreciation among planners, supervisors, technicians, managers, and internal customers.

The elements of planning (description, scope, job steps, parts, special tools, heavy equipment, access, work platforms, information, hazard assessment, permission, skill sets, waste disposal) influence the estimating process. If any one of these elements is not reflected in estimates, jobs will not be completed on time, assets will not be returned to service on time, overtime will be incurred, efficiency and schedule compliance will suffer, and customer satisfaction will fall short of expectations.

Effective work order descriptions are also critical to accurate estimating. "Fix Pump" is not adequate information on which to establish an appropriate estimate. The description must clarify whether the pump requires a simple Adjustment to stop a leak, Replacement of a check valve, or possibly Replacement of the entire unit. Requestors may not know requirements when originating work orders, but planners must know specific requirements before estimating jobs.

Scope Creep occurs when scope changes as jobs progress. Actual time appears to differ from Estimate when internal customers request undocumented work while technicians are in the area. On the surface such requests seem reasonable enough. However, customers have little concept of other commitments dependent upon the labor capacity the given technician represents.

Estimates for planned jobs must be pre-posted. Jobs are not in quantified backlog until estimated. Estimates for emergency jobs cannot be reliably estimated without feedback when work actually performed differs from work requested. Such estimates are therefore post-applied so that appropriate credit is given within Efficiency Reports. Complications or extra work encountered must be fed back by technicians.

Properly-developed standards are based upon experienced, properly-trained technicians working deliberately without undue exertion for the period of time between authorized breaks (approximately two hours). Experience, age, gender, and health of individual technicians are not reflected in proper standards. To begin with, planners must promptly quantify the backlog of planned jobs. This may be weeks before the job will be scheduled. Therefore, at the time of estimating, they have limited knowledge as to whom the job will eventually be assigned.

Any discussion of job time requirements is incomplete without considering job quality. Standards must include sufficient time for workers, as described above, to perform a quality job in the first instance. Supervisors must assure that timely job completion is not achieved at the sacrifice of job quality.

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Coordination with Operations

Coordination is a weekly cooperative process among Maintenance, Operations, and other involved functions. It is designed to achieve weekly consensus—between custodians of the assets and providers of asset reliability—regarding the most effective deployment of available maintenance resources in order to gain ongoing reliability of capacity.

Coordination and scheduling are requisite to economically optimal maintenance effectiveness and reliability. The goal is stable, long-term, and short-term operational reliability—considering reliability requirements of equipment as well as timely response to needs of internal as well as external customers. Loss of production capacity is minimized and maintenance work is completed when needed, in a safe, environmentally-conscious manner.

This chapter (Coordination) and the next chapter (Scheduling) address the expectation of work to be completed during the forthcoming week. Without a coordinated (agreed upon) and detailed schedule no expectation has been established. We all accomplish more when working to achieve a challenging and realistic expectation.

Maintenance is designed to optimize operational uptime; if not on a day-to-day basis, certainly on a month-to-month basis. Although short-term, band-aid approaches are often favored by Production in order to meet daily or weekly output targets, they do not result in optimized, continuing, and stable output. One of the most important duties for maintenance professionals is to continually emphasize this reality. The weekly coordination meeting is the best forum for this purpose.

Due to common pressures of daily and weekly output and reliability demands; Operations and Maintenance each has its own agenda.

Production is concerned about meeting targets for daily output during the current week. Maintenance is concerned about on-going preservation of optimized and stable asset capacity so that future output demands can be met. Of necessity, therefore, coordination efforts must pursue optimization between near-term and long-term demands. The vehicle is a Weekly Coordination Meeting—a forum within which the parties can continually balance the near/long-term dichotomy.

Coordination also encompasses all logistical efforts that must be accomplished for each individual job before it is put forth for coordination with the internal customer (Operations). Within these up front efforts, all resources required to perform each job are assembled and coordinated through harmonious efforts of Maintenance Planning, Purchasing, Receiving, and Stores. Only when Assembly is complete, does the job become Ready to Schedule.

During the Coordination Meeting, the full list of Ready work orders is reviewed and force-ranked by order of importance. Operations screens out jobs for which associated assets cannot be released. Within limits of available maintenance resources, the parties then select jobs for the approaching schedule week, thereby achieving consensus as to the most critical work to be performed. The parties then address questions surrounding “when.” If access to an asset is an issue, both Operations and Maintenance must agree upon a day and time when Operations can release the asset and Maintenance can provide the necessary resources. Their agreement constitutes a contract between Operations and Maintenance with mutual understanding that should schedule slippage (non-compliance) become absolutely necessary, the least important jobs are to be sacrificed, not the most important (as agreed to).

The Rule Is

Maintenance and Operations must communicate, collaborate, and coordinate at all times! Planners should view liaison with Operations as a permanent relationship. They should learn and take interest in the problems of their internal customers; remaining abreast of operating workloads, short- and long-term plans, and priorities. Help Operations think far enough in advance to facilitate effective planning and scheduling of maintenance resources in optimal support of Operating Plans and Reliability (long- and short-term).

Never Neglect Customer Relations!

Consider the relative importance of the various jobs from all perspectives. However, that of the internal customer must ultimately prevail. The importance ranking is carried over to the schedule with the most important jobs being placed at the top to emphasize that they are not to be missed.

SET UP OF THE COORDINATION PROCESS

Begin with a logical determination of the maintenance schedule week. It may not be the same as the payroll week. Many maintenance organizations select Friday through Thursday because the greatest amount of access to equipment occurs over the weekend—this would suggest the schedule week begin with Saturday. However, shop fabrication is often performed on Friday in preparation for the weekend work. This scenario leads to the Friday through Thursday logic. If this logic is not ideal for your situation, give the same level of thought in development of your own logic. Schedule Compliance will be off to a strong start.

Determine and firmly establish when the Coordination Meeting is to be held. Insight to the Operating Schedule for next week must be available prior to the meeting. Time must remain after the meeting to detail and post the schedule. Thursday morning is therefore frequently selected.

Establish mandatory attendees, as well as backup, and optional attendees. The number of attendees is dramatically reduced when Operations establishes a Maintenance Liaison. This position might be a Process Engineer or a Senior Supervisor. Prior to the meeting, Maintenance Liaisons bring themselves abreast of the relative importance of all open work orders originated with the Operating Department so that they can effectively represent the entire department.

One or more senior managers should attend the coordination meeting at least once a month.

PREPARATION FOR THE WEEKLY COORDINATION MEETING

The Planner's preparation for the meeting proceeds in essentially the following sequence of steps:

- ☐ Planners determine the quantity of resources available to work on scheduled backlog relief and PM/PdM routines that are

coming due during the approaching schedule week. This effort provides insight necessary to hold schedule promises within limits of labor resources. This limit reflects budgets for straight time, overtime, and contract support, detailed by skill and experience. This is the Work Program process.

- ☐ Planners then sort through Open Backlog to:
 - Separate jobs Ready to Schedule from jobs in other Status codes.
 - Review available Planned Job Packages for the Ready jobs to confirm that all requirements have been considered in determination of “ready.” This includes final (in some cases “hands on”) confirmation that materials, parts, special tools, safety instructions, and permit requirements are indeed available for job performance.
- ☐ Next, sort ready jobs by requesting organization (primary sort), in order of Equipment Criticality (secondary sort), and Aged Priority or start date required to meet Requested Completion Date (tertiary sort).
- ☐ The planner then links Ready jobs to be performed on the same equipment or in the same proximity. This effort often avoids multiple outages and saves labor resources. Such opportunities must be put forth in the coordination meeting if they are to be included within agreements reached.

In production environments where uptime is essential, coordination maximizes the use of windows of opportunity to accomplish work whenever and wherever equipment is not in use. Windows of opportunity are times where the equipment is not in service or not needed by operations, for example, during changeovers and shift change. Where opportune windows do not present themselves, downtime for planned maintenance must be scheduled to minimize disruption of production.

The coordination challenge in Building Maintenance does not differ much from Plant Maintenance. In hospitals, for instance, servicing of some systems, areas, or rooms must wait until convenient for the medical staff or patient.

- ☐ Jobs should not be put forth for scheduling if required resources (i.e., materials) are not on-hand. “Ready” means

that all requirements are at hand and set to be performed whenever Operations can release the associated assets and Maintenance can allocate the necessary labor resources.

Except on rare and essential occasions, do not build potential upsets into the schedule even if required materials will presumably be received sometime during the schedule week. There should be ample jobs in the backlog that are truly ready. Why risk upsets to the schedule?

- ❑ One day before the coordination meeting, planners are ready to issue current, well-organized, and relevant Ready Backlog Reports to Operating, Maintenance, and other functions requesting support. Ideally, distribution is electronic.
- ❑ Now the preparation burden shifts to the internal customers. Individual requestors within each major process area or department must reach accord regarding relative importance of each Ready work order comprising their combined backlog. In the interest of a controlled and reasonably brief coordination meeting, it is best if agreement within each unit can be reached prior to the coordination meeting. That way, the meeting can be focused upon reaching accord between, not within, the several requesting organizations.
- ❑ Meanwhile, Operations develops the Operating Schedule for the coming week. They bring it to the coordination meeting so near-term demands for output and longer term demands for future reliability of operating capacity reliability can be optimized. When consensus is reached, there should be very few legitimate reasons for resource diversion to unscheduled work (true emergencies only). The ultimate operating approvers of the schedule are the logical authority regarding any urgency constituting a legitimate schedule break.

When emergencies (failures) do occur, RCFA should answer the basic question, "How did we allow this to happen? Where is the hole in our PM/PdM program?" The answer may be that the program is fine—the shortfall was in PM/PdM Schedule Compliance. Access to the equipment was denied when the routine was due. Two weeks later, failure occurred. Use these instances to reinforce conduct of PM/PdM as designed—in other words, as a controlled experiment!

Conduct of the Weekly Coordination Meeting

As Maintenance enters the meeting, the only information available comes from backlog features of the CMMIS (description, priority, requested completion date, age of the request and ideally equipment criticality). At the end of the meeting, all parties know which jobs are to be performed the following week, their agreed upon order of importance, and the day and time agreed upon. Within the meeting:

- ❑ Attendees establish equitable allocation of available maintenance resources to each cost center that has requested support during the upcoming schedule week. Allocations are based on varying business needs, demands of customers (internal as well as external), and other criteria. Total labor-hours allocated cannot exceed the quantity budgeted and available for the coming week. If more work absolutely needs to be accomplished, additions will have to come from overtime, contracting, or transfer from other internal organizational units. In this case, attendees need to prepare a joint presentation to management requesting variance to current budget limitations.
- ❑ Operations, Maintenance, and other attending functions present their cases regarding the most important jobs from their perspective:
 - Maintenance campaigns for all identified Corrective WOs stemming from PM/PdM inspections. If those jobs are not performed promptly, the Return on PM/PdM Investment can be lost. This is of particular concern in regard to predictive inspections, which do not trigger corrective alarms until associated equipment condition is at the low end of acceptable control limits.
 - Schedule of PM/PdM inspections and routines at their predetermined frequencies are next on the agenda of Maintenance (distinct from the agenda of Operations). Be sure that PM/PdM routines of all crafts are considered (i.e., electrical as well as mechanical). Do not take an asset down for electrical PM/PdM in one week and expect Operations to grant access again the following week for mechanical PM/PdM. For the same reason, whenever access is granted, look forward a few weeks to see if any other routines for the same

asset are coming due. Perhaps they should be performed early to take advantage of scheduled downtime agreements, thereby avoiding redundant shutdowns. Also consult the Ready Backlog for jobs on that asset that can be done at the same time (2 jobs with 1 downtime).

- Operational priorities come next (in an ideal world). In reality, the top operational priorities will likely be interspersed with Corrective and PM/PdM work.
 - Be certain to deliver proactive rather than reactive messages. Too often, PMs are found at the bottom of the schedule, which communicates a reactive message (e.g., "Here are some trivial jobs to be used for fill-in as time allows"). Wrong Message! Proactive messages must be continually delivered if the organizational culture is to become reliability focused.
 - All parties should be concerned about jobs approaching their requested completion dates. Any jobs that cannot be scheduled to meet those dates should be discussed with the Originator in the context of importance established by designated attendees. If originators continually fail to receive timely response, they lose faith in the coordination process and revert to their reactive habit of "crying wolf". This crying wolf becomes dramatically worse when the Ready Backlog starts to increase past the control limits of 2–4 weeks. The experience of the customers is that jobs are put in and never come out since higher priority jobs always displace them.
- ❑ Based upon importance to the overall operation with consideration of short- and long-term wellness, the parties reach consensus regarding jobs to be scheduled next week:
- Operations agree to make the equipment available in the agreed-upon state, so work can be performed as scheduled.
 - Maintenance agrees to perform the work as scheduled, starting on time and finishing on time. Therefore, reliable estimates of outage duration must be made.
 - Resources should not be diverted to unscheduled work except in the case of true emergencies.

This contractual process (the give and take of the Weekly Operations/Maintenance Coordination Meeting) becomes more binding when either the Manager of Operations or Maintenance chairs initial meetings. Managers should establish meeting structure, demonstrate essential toughness of the questioning process, and emphasize the participative commitment that should emerge. Once the pattern is established, the Planner/Scheduler becomes the Chairperson. However, each of the functional managers should continue to attend at least once a month. The General Manager ought to attend at least once a quarter.

As the coordination process is streamlined, the meeting requires less than one hour per week. All participants will find that their attendance time is extremely well spent. Operational personnel come to realize that the best way to get work properly performed in timely fashion is through the formal planning, coordination, and scheduling process.

Agenda for the Weekly Coordination Meeting

The coordination meeting normally falls into six sections:

1. Outlook of Schedule Compliance for the Current Week

Because the current schedule has not quite ended, this is an outlook discussion.

- ☐ How are we doing against this week's schedule? Followed by exploration of the non-compliance issues.
- ☐ What jobs were not completed as scheduled, and why?
 - Did Operations deny equipment access?
 - Did Maintenance fail to get to them?
 - Did unscheduled jobs break into the schedule and were they necessary?
- ☐ What jobs need to be carried over to the new schedule?

Such questions highlight underlying reasons for poor schedule compliance. When the General Manager or some other senior manager raises the questions, the process becomes more effective. Problem resolution is speedier and the potential of scheduling is captured. Therefore, one or more senior managers should attend the coordination meeting at least once a month and as needed.

2. *Outline of Operating Schedule for the Coming Week*

In order to identify windows of opportunity for performance of maintenance -reliability work, all parties need perspective as to when each process line is and is not scheduled to run. Discussion would highlight critical jobs and any work for new products and customers.

3. *Comparison of Important Jobs—Perceptions*

Operations and Maintenance each comes to the meeting with functionally influenced ideas as to what jobs are most important to be performed during the coming week. Even between the individual Operating groups there are conflicts of interest. With the target for Ready Backlog being two-to-four weeks, not everybody will leave the meeting with all of their desires fulfilled. Fair allocation of resources is sought, but optimal interest of the entity must ultimately prevail. When dissatisfaction persists week after week, the apparent inadequacy of resources must be explored. Resolution stems from forced ranking one-job-at-a-time. Repeatedly ask, "What is the next most important job in the overall interest of the business entity?" Every represented organization has the opportunity to make its case. If the group cannot reach consensus, a higher level of management must be brought into the process.

4. *Search for Operating Windows to Accommodate Agreed Upon Jobs*

When a window is not found, the associated job is removed from consideration for the coming week. A provisional scheduling feature provides for these jobs (see Chapter 16, Scheduling).

5. *The Contract is Established*

Operations agree to release the asset when and in the condition agreed to. Maintenance agrees to provide necessary resources to start, perform, and complete the job at the time both parties have committed to.

6. *Discussion of Other Backlog Jobs to be Prepared*

Expediently Operations is commonly interested in work orders that were not yet in the Ready state due to lack of parts, engineering, approval, budget, etc. Therefore, these work orders were not presented for consideration within next week's schedule. This step is their opportunity to request that Maintenance move these jobs forward as quickly as possible.

At the conclusion of the Weekly Coordination Meeting,

Planner/Schedulers have all essential information and agreements. Upon return to their workstations, they are ready to lay out detailed schedules for approaching next week; these schedules reflect agreements reached between crucial parties.

Planners should view liaison with operations as a permanent relationship. They should learn and take interest in the problems of their internal customers, remaining cognizant of operating workloads, short- and long-term plans, and priorities. By these means, planners can help operations think far enough in advance to facilitate effective planning and scheduling of maintenance resources in optimal support of operating plans.

Maintenance work must always be coordinated and scheduled to optimize the balance between minimal adverse impact on the operating schedule and effective utilization of maintenance resources. This optimization requires coordination with Production Planning to link the maintenance schedule with the operating schedule. This synchronization is illustrated in Figure 15.1.

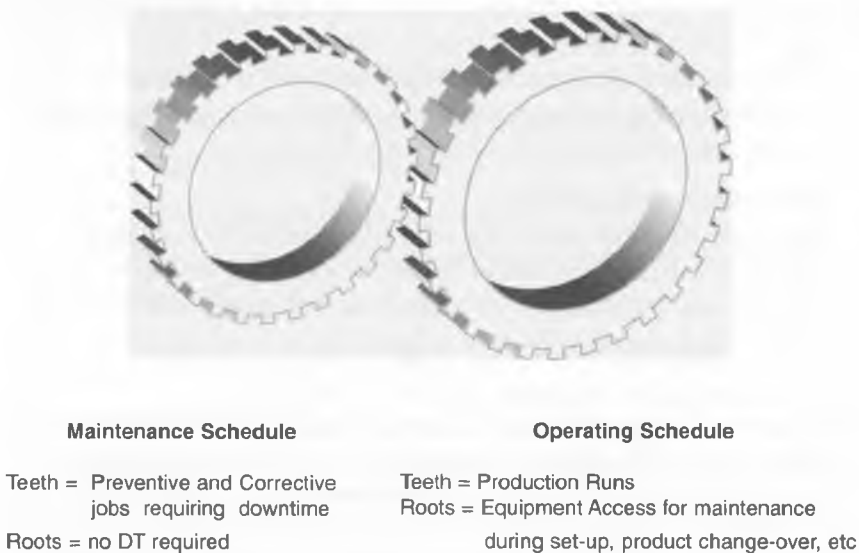


Figure 15.1 Synchronization of Production and Maintenance Schedules

Scheduling Maintenance Work

THE WEEKLY EXPECTATION

Once jobs have been planned, availability of required materials and other resources have been assured, and coordination with internal customers has been achieved; work scheduling can begin.

Scheduling (when to do the job) is the process by which all resources required for performance of specific jobs are allocated, coordinated, and synchronized at the proper time and place, with necessary access, so that work can be executed with minimal delay and completed by the agreed upon date, within estimated budget. The schedule establishes when jobs will be done and what resources can best be applied to their performance. Schedules establish and document management's expectation for what work is to be accomplished with resources to be funded during the schedule week.

Resources include manpower, materials, tools, and special equipment. Access refers to assets being prepared and accessible to be worked upon – in safe locked out state, with necessary precautions taken, permits obtained, and any specialized documentation, drawings, and other information on hand. Proper time relates to job start, duration of execution, and job completion within the time frame agreed upon with the internal customer during the weekly coordination meeting.

When properly scheduled, maintenance work economically optimizes the balance between impact on operating schedules and effective utilization of maintenance resources. Maintenance and Operating Schedules must be linked and supportive, each to the other.

Scheduling is the hub from which all maintenance activity is executed. Upon start up of any new maintenance management installation,

scheduling is the “point function” and marketing arm of the installation; it yields the earliest tangible results evident to Operations (often within weeks). Production personnel resist change to their historical reactive culture until they experience benefit accruing to them. In fact, the return on investment of planning and scheduling comes from the schedule and its ability to draw all the elements of the maintenance job at the right time and to the right place—allowing the job to be done with minimal downtime and maximizing the use of the maintenance resources.

By contrast, preventive maintenance, job planning, and root cause failure analysis require months of focused effort before they yield measurable results. In the meantime, internal customers ask, “What are we receiving from this new system?” Program success demands that this question is not allowed to linger.

All organizations and individuals perform better and accomplish more when working to accomplish clearly established, well communicated, and published expectations that are fair, yet challenging. Without a weekly schedule, there are no definitive expectations as to what is to be accomplished with the resources for which payroll checks will be drawn. Without proactive expectations, the reactive demands that typically arise are the only work performed. Equipment failure dictates workload! By contrast, scheduled, proactive maintenance work limits equipment failure. Although they can be engulfing, urgent tasks are seldom the most important tasks when viewed from a perspective of overall business objectives. Keep the horse in front of the cart! Otherwise, asset failures devour the bulk of available maintenance resources. Consequently, crucial proactive maintenance routines are not performed when they are due. Reliable output, quality, customer service, safety, energy conservation, and environmental protection suffer the consequences.

Thinking that we work best when under pressure, we subconsciously procrastinate until pressure forces us into reaction (Management by Crisis). The actuality is: although initial response may be faster, the work performed is less effective and less reliable due to lack of preparation, resulting in the need for hasty judgments. Performing the wrong things faster accomplishes nothing. The tyranny of urgency lies in its distortion of priority. Failures require instant reactive repair whereas preventive and predictive inspections, and planned backlog relief, rarely need to be performed today. However urgency alone (without consideration of importance) cannot be allowed to dictate how vital resources are to be consumed. Excellence can be described as “performing the right things properly.”

One way or another, maintenance resources will perform the work to which they are assigned. The question is, "Are they to be deployed effectively through proactive scheduling, or are they to be ineffectively consumed at time of failure"? Either management schedules how it intends for vital maintenance resources to be applied or equipment failures and impatient internal customers dictate how those same resources are consumed and wasted on poorly conceived work. Prompt performance of the wrong jobs will not yield reliability!

Either you schedule the team yourself, or you allow equipment failure and impatient customers to dictate how and when resources are consumed (and wasted).

Operational control of maintenance, though much slower reacting, is similar to the thermostat principle illustrated in Figure 16.1.

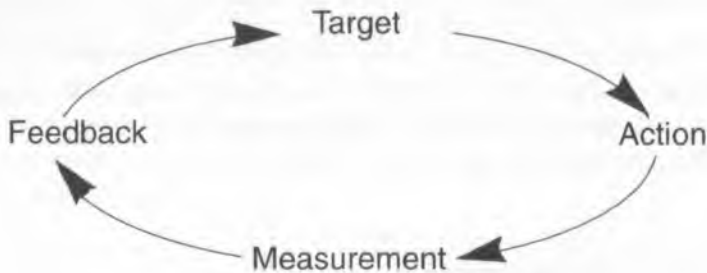


Figure 16.1 The Thermostat Principle

The fundamental requirement is a target (the schedule) against which to control, followed by action (execution of the schedule) to achieve targeted results. Schedule Compliance (results measured against intent) provides feedback by which to address deviations (reasons for non-compliance). Managers and supervisors must schedule precisely, proceed positively, and persistently pursue weekly targets. Using results, reports, and metrics (such as Schedule Compliance), planners, supervisors, managers, and internal customers can continuously improve the planning, coordination, and scheduling process.

Without effective scheduling and high schedule compliance,

maintenance is haphazard, costly, and ineffective. Promises are irregularly met, thereby causing frequent problems for Operations and for external customers expecting deliveries. When this happens, Operations become increasingly reluctant to release their assets for proactive maintenance, thereby jeopardizing sustained reliability.

Schedules and Schedule Compliance are among the most important objectives of Maintenance. Managers should know specifically what they expect maintenance resources to accomplish throughout the schedule week. Supervisors need to remain abreast of progress relative to schedule expectations. The overall challenge is twofold—create a maintenance operation which is both responsive to the customer while achieving intrinsic efficient. Schedules are a device for lining up jobs waiting to be performed so that operational reliability is served and optimal utilization of maintenance labor is achieved. Four abilities are essential:

- ☐ Mutual determination of the comparative importance of various jobs ready to be scheduled.
- ☐ Schedule focus upon those jobs deemed to be most important.
- ☐ Concentrated execution of jobs in compliance with schedule.
- ☐ Perseverance Do Not Quit! Achieve the highest possible level of Schedule Compliance despite interruption (Schedule Breakers) encountered.

The Scope of Scheduling

Via the scheduling process, maintenance resources are brought together in terms of time and importance. Labor, materials, and equipment needs are identified on paper and provided for before resources are committed to the field.

Schedules are essential for periods of full production as well as periods of shutdown (windows of downtime, down days, major outages, and annual turnarounds). To minimize disruptions to the operating schedule, maximize effective utilization of downtime windows of opportunity. This is the **“If Down—Do!”** concept. Promote it!

Critical Path Analysis is required to properly prepare for and control the many interrelated activities associated with Major Shutdowns (see Chapter 18, *Planning Major Maintenance Projects*). Within this chapter, our focus is weekly maintenance schedules. Further discussions will

therefore refer to a GANTT Bar format.

Experience shows that increased productivity is achieved when maintenance personnel know tomorrow's assignments before they leave work today. Knowing about work in advance enables a craftsman to make mental preparation. With good scheduling, the maintenance supervisor is provided with a list of all work to be performed by members of his team, both individually and as a unit. Every evening, the supervisor can tell each team member his or her assignments for the next day. Normal weekly scheduling includes:

- ☐ Preparation of a Weekly Schedule reflecting mutually agreed upon expectations. Agreements must assure that all preventive and predictive routines will be accomplished within established windows of time.
- ☐ Matching next week's demand for service with resources available after accounting for all indirect commitments (vacations, absence, training, meetings, etc.).
- ☐ Bringing together in precise timing the elements of a successful maintenance job: labor; tools, equipment, work platforms, materials, parts and supplies, information, engineering data and reference drawings, custody of the unit being serviced, and authorizations, permits, and statutory permissions.
- ☐ Allocation of specific technicians to specific scheduled jobs. The allocation must take into consideration the skills possessed by individual technicians as well as exposure to specific jobs required to enforce the training and development of specific individuals.
- ☐ Ensuring that responsible supervisors receive and understand planned job packages assembled for each scheduled job.

The resultant document is a strategic Master Schedule for the forthcoming schedule week. It represents agreed-upon expectations, detailed by day, by job, and by individual technician assigned.

Tactical adjustments may be necessary as the week unfolds. While transitioning from a reactive environment to a proactive environment, daily adjustments can be burdensome and may require Planner involvement. However, as schedule compliance matures, required updat-

ing throughout the week becomes minimal and can be performed by supervisors, freeing Planners/Schedulers to focus on preparation of work to be performed during the following weeks.

The Scheduling Format

Although scheduling principles are simple, their effective application is not quite so easy because several tasks must be performed concurrently in order to develop a viable weekly schedule. The scheduling format advocated within this book is presented in Figure 16.2. The schedule should be fully developed but, in the interest of simplicity and clarity, only a few jobs are illustrated in the figure as presented.

Additional pages can be added to Job Loading and Job Scheduling sections (lower portions of Figure 16.2), but the Labor Deployment Scheme (upper portion) must be visually worked and displayed on page one. The resultant Labor Deployment Scheme (full) can be superimposed on top of each additional page.

Begin with deliberate selection of the Schedule Week. On what day does the week begin and end? Proper selection will contribute significantly to schedule compliance. Notice the unusual sequence of days (Friday through Thursday) in Figure 16.2. When Maintenance and Operations meet, typically they first discuss work to be completed over the weekend. Further discussions progress to preparatory work that can be performed on Friday, thereby avoiding use of the limited, weekend, downtime window for fabrication and other preparatory tasks. Hence, a schedule week beginning on Friday and ending on Thursday makes sense in many situations, although not in all. Each organization needs to define its own schedule week.

Also consider when the coordination meeting would be held. The Operating Schedule must be available before the coordination meeting can be effective. Scheduling and schedule compliance will have much better support if the selected schedule week makes practical sense to all involved parties.

A master scheduling format is established for each crew responsible to each maintenance supervisor (approximately 8-to-12 persons). The narrow columns under each day reflect eight one-hour periods of straight time plus four hours of potential overtime. Time-line bars within the job scheduling section show duration hours (i.e., 8 hours). Bars in the labor-deployment section indicate labor-hours distributed to members of the crew (i.e., the same 8 hours duration but for each of two team members equals 16 labor-hours). Be careful here! There is a tendency to show

Company Schedule

WEEKLY MAINTENANCE SCHEDULE

Task	Wk #	Job Description	Req. Crew Size	Crew Clock	Min. Hour	Assigned Crew	Worker Dependency Schedule						
							Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	
							Job Loading						
Line	Wk #	Job Description	Req. Crew Size	Crew Clock	Min. Hour	Assigned Crew	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
1	1010	Check Run Line #1	4	1	1								
2	1020	Replace Limit Switches	4	2	4	ABB							
3	1030	Replace Limit Switches - In #1	4	1	2	C							
4	1040	Connect Power to Line #3	4	2	4	CAA							
5	1050	Check Run Line #3	4	2	4	GAT							
6	1060	Check Run Line #3	4	2	4	GAT							
7	1070	Check Run Line #3	4	2	4	GAT							
8	1080	Check Run Line #3	4	2	4	GAT							
9	1090	Check Run Line #3	4	2	4	GAT							
10	1100	Check Run Line #3	4	2	4	GAT							
11	1110	Check Run Line #3	4	2	4	GAT							
12	1120	Check Run Line #3	4	2	4	GAT							
13	1130	Check Run Line #3	4	2	4	GAT							
14	1140	Check Run Line #3	4	2	4	GAT							
15	1150	Check Run Line #3	4	2	4	GAT							
16	1160	Check Run Line #3	4	2	4	GAT							
17	1170	Check Run Line #3	4	2	4	GAT							
18	1180	Check Run Line #3	4	2	4	GAT							
19	1190	Check Run Line #3	4	2	4	GAT							
20	1200	Check Run Line #3	4	2	4	GAT							
21	1210	Check Run Line #3	4	2	4	GAT							
22	1220	Check Run Line #3	4	2	4	GAT							
23	1230	Check Run Line #3	4	2	4	GAT							
24	1240	Check Run Line #3	4	2	4	GAT							
25	1250	Check Run Line #3	4	2	4	GAT							
26	1260	Check Run Line #3	4	2	4	GAT							
27	1270	Check Run Line #3	4	2	4	GAT							
28	1280	Check Run Line #3	4	2	4	GAT							
29	1290	Check Run Line #3	4	2	4	GAT							
30	1300	Check Run Line #3	4	2	4	GAT							
31	1310	Check Run Line #3	4	2	4	GAT							
32	1320	Check Run Line #3	4	2	4	GAT							
33	1330	Check Run Line #3	4	2	4	GAT							
34	1340	Check Run Line #3	4	2	4	GAT							
35	1350	Check Run Line #3	4	2	4	GAT							
36	1360	Check Run Line #3	4	2	4	GAT							
37	1370	Check Run Line #3	4	2	4	GAT							
38	1380	Check Run Line #3	4	2	4	GAT							
39	1390	Check Run Line #3	4	2	4	GAT							
40	1400	Check Run Line #3	4	2	4	GAT							
41	1410	Check Run Line #3	4	2	4	GAT							
42	1420	Check Run Line #3	4	2	4	GAT							
43	1430	Check Run Line #3	4	2	4	GAT							
44	1440	Check Run Line #3	4	2	4	GAT							
45	1450	Check Run Line #3	4	2	4	GAT							
46	1460	Check Run Line #3	4	2	4	GAT							
47	1470	Check Run Line #3	4	2	4	GAT							
48	1480	Check Run Line #3	4	2	4	GAT							
49	1490	Check Run Line #3	4	2	4	GAT							
50	1500	Check Run Line #3	4	2	4	GAT							
51	1510	Check Run Line #3	4	2	4	GAT							
52	1520	Check Run Line #3	4	2	4	GAT							
53	1530	Check Run Line #3	4	2	4	GAT							
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58	1580	Check Run Line #3	4	2	4	GAT							
59	1590	Check Run Line #3	4	2	4	GAT							
60	1600	Check Run Line #3	4	2	4	GAT							
61	1610	Check Run Line #3	4	2	4	GAT							
62	1620	Check Run Line #3	4	2	4	GAT							
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80	1800	Check Run Line #3	4	2	4	GAT							
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89	1890	Check Run Line #3	4	2	4	GAT							
90	1900	Check Run Line #3	4	2	4	GAT							
91	1910	Check Run Line #3	4	2	4	GAT							
92	1920	Check Run Line #3	4	2	4	GAT							
93	1930	Check Run Line #3	4	2	4	GAT							
94	1940	Check Run Line #3	4	2	4	GAT							
95	1950	Check Run Line #3	4	2	4	GAT							
96	1960	Check Run Line #3	4	2	4	GAT							
97	1970	Check Run Line #3	4	2	4	GAT							
98	1980	Check Run Line #3	4	2	4	GAT							
99	1990	Check Run Line #3	4	2	4	GAT							
100	2000	Check Run Line #3	4	2	4	GAT							

Total Man-Hour Schedule							Man-Hour Summary						
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Figure 16.2 Weekly Maintenance Schedule

16 hours in the job schedule section, which would imply a two-shift job rather than a one-shift job.

Next, notice that the schedule format consists of six sections (four major and two supplemental) as described below.

1. **Available Labor Resources.** The upper section lists team members by name, alpha reference code, and skill. Individuals comprising each crew are pre-posted (upper left of the weekly schedule). Should any individual have a fixed commitment to indirect activities such as training, the pre-committed capacity is posted at the appropriate time period (see employees **K** and **L** on Wednesday afternoon). Scheduling of committed capacity to a job at a time when the resource is not available for direct assignment is thus avoided. Pre-commitments are easily forgotten at inopportune times.
2. **Job Load.** The lower left section lists work orders to be completed during the schedule week, as agreed upon by Maintenance and Operations to best meet near-term Operating needs as well as long-term Reliability needs to assure on-going asset capacity. Jobs are listed in order of agreed upon importance stemming from the coordination meeting. The listing is, therefore, not chronological.
3. **Job Schedule (Primary).** The lower right section displays timelines for loaded jobs throughout the schedule week (in bar chart format). They are based on meaningful estimates of required duration and agreed upon access to equipment. Technicians to whom scheduled jobs are allocated are cross referenced in the Assigned Crew column.
4. **Labor Deployment.** The upper right section is used to show (also in bar chart format) optimal allocation of labor resources to specific scheduled jobs. Line item numbers for the scheduled jobs to which technicians are allocated are superimposed upon the appropriate time-line bars.
5. **Provisional Job Load.** This section is immediately below the Primary Job Load. It provides for provisional loading of ten (10%) to fifteen (15 %) percent additional work. No matter how strong schedule agreement is between Maintenance and Production, access to one or more primary jobs is frequently denied as the week enfolds. Provisional jobs are ready as substi-

tute work orders for such occasions. They must be equally well prepared as primary jobs in order to be legitimate substitutions. If performed during the schedule week, they are credited in calculation of Schedule Compliance.

6. **Contingency Job Load.** This section is at the very bottom of the format beneath Provisional Job Load. It provides for contingency scheduling of jobs that participants in the coordination meeting may have deemed to be more important than some primary jobs. However, expectations were such that involved assets could not be released at any time during the scheduled week. It was agreed, however, that should a window of opportunity be presented for any of these jobs, they would be legitimate substitutions for a primary job of relatively low importance at the bottom of the primary schedule. They too must be equally well prepared as primary jobs and also receive Schedule Compliance credit if performed.

When completed, the weekly schedule documents the full expectation for work to be accomplished by Maintenance during the ensuing schedule week. Jobs are listed in order of importance. Necessary resources are allocated by job, day of the week, and individuals to whom allocated for assignment by the appropriate supervisor on the day each job is scheduled.

INSTRUCTIONS FOR PREPARING SCHEDULES

At the beginning of the weekly scheduling process, planners retrieve a copy of the master scheduling format for each crew to be scheduled; they proceed through the following steps:

- ☐ Fill in the week's beginning date.
- ☐ Determine known absences (e.g., vacations) for the specific schedule week. Block them with a code established for each nature of absence, from resources expected to be available for assignment. Sources of this information include vacation schedules and discussion with supervisors.
- ☐ Plan strategy on a weekly basis. Provide the big picture as to which jobs are most important to complete and how team

resources are to be assigned throughout the schedule week.

- ☐ Review all jobs in the backlog starting with incomplete jobs from current or previous schedules. Focus on the most important jobs as agreed upon in the weekly coordination meeting.
- ☐ Load all three sections of the schedule—Primary, Provisional, and Contingency.
- ☐ Individual jobs should not be scheduled until all required parts and materials, tools and special equipment, and asset access are available in required quantities.
- ☐ Block the job schedule timeline to show scheduled start, duration, and finish. Each block represents one hour. The format, as illustrated, provides for a shift up to twelve hours, including four hours overtime.
- ☐ Using a recognizable key, note the end of a preceding job (precedent) and the beginning of a succeeding job to be performed in conjunction (perhaps electricians must complete the preceding job before mechanics can begin the succeeding job). Visual connection on the schedule reminds the first supervisor that schedule deviation by his crew dictates timely notification to the second supervisor as well as to involved planners and operating requestors.
- ☐ Scheduling each available technician for a full day of productive work throughout the schedule week, load sufficient jobs to consume all available labor-hours. When finished, all resources available for the week should be fully scheduled, without voids but also without overload.
- ☐ As jobs are scheduled, allocate specific technicians to each job. Initially, the type of work to be allocated to specific technicians is a cooperative effort between planner and supervisor. In a short period of time, however, planners can perform the allocation themselves, thus freeing supervisors for on-the-job leadership.
- ☐ When schedules are not detailed by individual, but taken only to the level of a full load of identified work for a given team or skill, too much work tends to be loaded on some individuals whereas too little is loaded on others. Individual team members are not clones of each other, even if they are multi-

skilled. When this situation is allowed to happen, the resultant schedule is not feasible.

- ☐ This allocation process minimizes the risk of scheduling an individual to two jobs at the same time and clearly shows full or partial allocation of each crew member across the week. Duplicate allocation does occur periodically, usually at the most inopportune times.
- ☐ Utilize individual skills to the greatest extent possible. Provide craftsmen with challenge and opportunity to grow. As necessary, allocate specific persons based on required skills or equipment knowledge as necessary, but do not ignore individual training needs.
- ☐ Provide job exposure to individuals to reinforce their training so they gain experience and expand their skills. Sometimes two people must be allocated to a one-person job: one to teach, one to learn.
- ☐ In making allocations, also balance equipment specialization of individuals with their need to gain broad facility familiarity and knowledge.
- ☐ Cross-reference each scheduled job to allocated team members by noting their reference codes in the Assigned To column and shading the time period abreast of each allocated individual. The period shaded in the Labor Allocation section corresponds to the period shaded in the Job Schedule section.
- ☐ Cross-reference each individual assignment to the appropriate job by superimposing the line-item number of the job in the shaded allocation period. Line-Item numbers require fewer digits than work order numbers (1 or 2 versus 4 or 5).
- ☐ Keep the majority of crews scheduled to important work, which should be started and completed without interruption.
- ☐ Approximately 10 to 15% of scheduled labor-hours should be allocated to the least important jobs at the bottom of the schedule. These are the jobs that can be sacrificed when necessitated by urgent demands. Allocate them to a few good workers who are flexible, can stop and resume jobs, be reinstructed, and be reassigned to emergencies with minimal loss of efficiency and without morale drop. They are the best

troubleshooters to be pulled in response to schedule breaks (urgencies).

When laying out a schedule of jobs for the week, the essentials for each job are duration of the job, when it is to be performed (start and finish), crew size and labor-hours required, and specific crew members to be assigned. Start and finish are particularly important when shutdown of production processes and coordination of multiple resources are involved. Job Schedules are defined in **Duration Hours**. Labor-hours equals Duration-hours multiplied by Crew Size. Do not confuse these two variables. Duration is used for job scheduling. Labor-Hours are used for Labor Deployment.

- ☐ Balance work loaded with labor-hours available. Load what can be accomplished, not necessarily what needs to be accomplished.
- ☐ Integrate preventive/predictive maintenance into the overall schedule to assure that all routines are accomplished at established frequencies. PM/PdM routines falling due during the schedule week and Corrective work orders should be loaded to the top of the schedule. This delivers the essential proactive message that nothing is more important to on-going reliability (safety, regulatory compliance, quality, and output capacity) than PM/PdM inspections and resultant Corrective Work Orders. So often proactive routines appear at the bottom of schedules, which is a very reactive message implying "get to them as time allows." **Wrong Message!**

Determine the most logical time of day to schedule PPMs. Often, the beginning of each shift is the period of heaviest operating problems. It may not be the best time to schedule PMs. On the other hand, it is not advisable to leave them until the end of the day either. They may not be performed that day at all. Therefore, consider late morning or early afternoon as the most effective time to schedule PMs.

Each year preventive, predictive, and programmed routines are forward loaded in balanced weekly manner into a 52-Week PM/PdM Schedule which is then reflected in Annual Operating Plans. This step avoids PM/PdM peaks and valleys (that would otherwise overload maintenance crews) and related access time on some weeks, and under-load

other weeks. Access for performing proactive routines should therefore be nearly automatic. How to handle inevitable exceptions is discussed below.

- ☐ Carry over jobs that were scheduled for the previous week but not completed should be shown next. It is logical that any work orders deemed sufficiently important to be scheduled the previous week but not actually completed should be accomplished the following week—especially if they had been started.
- ☐ Next to be scheduled are jobs established within the weekly coordination meeting. They are listed in descending order of agreed-upon importance until all available labor-hours have been committed. The order (sequence) may not be chronological within the week. If Production indicates that the asset can not be released until late in the week, but the job is nevertheless the most important, the job appears at the top of the schedule immediately after the proactive routines. Top posture in the loading sequence indicates the job commitment is the most crucial of the entire week. **“Do not fail to perform this job as scheduled!”**
- ☐ The least important jobs are loaded to the bottom of the Primary Schedule. When absolutely necessary to divert scheduled resources to true emergencies, these least important jobs can be sacrificed. If an Emergency Response Crew has been organizationally established, schedule breaks should be minimal.

Order of Importance facilitates adjustment whenever essential Schedule Breaks occur. Sacrifice the least important jobs that appear at the bottom of the schedule. Protect the most important jobs at the top of the schedule. Schedule Compliance shortfalls are acceptable, provided critical jobs are completed as scheduled.

Note that order of importance is not equivalent to priority. Priority Code fades once order of importance is agreed upon between the parties.

- ☐ Load the Provisional Schedule. Given the contractual agreement reached in the coordination meeting, denial of equipment access should not occur. However, periodically it does.

Therefore, add ten-to-fifteen percent additional labor hours as the Provisional Schedule (10%–15% of subtotal A in the labor-hour column). This practice informs supervisors of the work orders to be substituted for jobs and crews impacted by denial of equipment access. No labor resources should be allocated to jobs on the Provisional Schedule.

- ☐ Also look forward in the PM/PdM Schedule to identify routines that can be performed early. Add these to the provisional schedule to be substituted if access is denied for performance of a PM/PdM routine. This will provide an opportunity to catch the delinquent routine in a future week without a PM/PdM spike. Doing all work on a single asset at one time usually increases productivity.
- ☐ Load the Contingency Schedule. Use the contingency section of the schedule for jobs of high importance requiring equipment access not expected to be available during the schedule week. If availability should occur, however, these jobs are more important than some jobs on the primary or provisional schedule. No labor resources should be allocated to jobs on the Contingency Schedule.
- ☐ Do not add Provisional or Contingency labor-hours to Total Scheduled Labor-Hours at the bottom of the schedule format (Sub Total "A"). However, if the related jobs are performed as the schedule week unfolds, the associated labor-hours are to be included in Total Scheduled Hours Completed when calculating Schedule Compliance (see Chapter 18).

Figure 16.3 on the next page summarizes the sequence of job loading by work nature.

- ☐ Maximize off-site job preparation and post-startup efforts in order to minimize downtime.
- ☐ Search for ways to save labor-hours and minimize downtime by linking multiple jobs on the same equipment. Avoid duplicate shutdowns by assembling all work requiring common equipment access. Do not shut equipment down for mechanical work on one week and then again the following week for electrical work. Do not shut equipment down one week for project work and the following week for PM/PdM.

Order Within Schedule	Clarification
Corrective Work Orders	Derived from previous PM/PdM Inspections
Delinquent PM/PdM Inspections	Carried forward from previous weeks
PM/PdM Inspections	Those coming due during the schedule week
Other Carry Over Jobs	Not completed as scheduled in previous weeks
Most Important Jobs Established in Weekly Coordination Meeting	In agreed upon order of importance (even if scheduled late in the week)
Less Important Jobs	That can be sacrificed when demanded by Emergency/Urgent response
Provisional Jobs (+ 10%)	Backup if access to Primary Jobs is denied
Contingency Jobs	To be substituted if unanticipated access develops to critical assets

Figure 16.3 Sequence of Job Loading By Work Nature

- ☐ Eliminate unnecessary travel. Search for opportunities where assignments can take advantage of jobs in the same location, jobs using the same tools or materials, and jobs needing the same skills.
- ☐ Think about crew balancing delays on multi-man jobs. Often small jobs can be performed by taking advantage of crew balancing delays (enforced idle time) contained within larger multi-person jobs being performed in the same proximity. Seldom can all members of a multi-person crew work continuously throughout duration of the job. See Allowances in Chapter 14.
- ☐ Schedule multi-person jobs as the first job in the morning whenever possible. This assures everyone is available to start the job at the same time.
- ☐ When scheduling multi-person jobs later in the day, consider previous assignments. Don't assign one person to a one-hour job and the helper to a two-hour job. There is no way that they will both be available to start the two-person job concurrently.

- ☐ Save minor indoor jobs for severe temperatures and inclement weather.
- ☐ Review Planned Job Packages to confirm that they are sufficiently complete for execution, including parts and materials, special tools, safety instructions, permit requirements, and equipment access.
- ☐ Prior to the end of the current week, finalize and post final schedules for the forthcoming week?by 3PM Thursday if the new schedule week begins on Friday. Issue a schedule for each maintenance crew responsible to a given supervisor. The schedules show how team resources are to be assigned throughout the schedule week. Experience shows that increased productivity is achieved when personnel know tomorrow's assignments before leaving work today. Knowing assignments in advance enables technicians to make mental preparation.
- ☐ Provide Operations with copies of schedules. Ensure that they are studied, understood, approved, and accepted by all concerned parties.
- ☐ Confirm and document that all agreed-upon commitments are understood by Operations as well as Maintenance. Approval means that a contract has been reached whereby Operations and Maintenance commit to their joint-schedule for release of Operating assets and deployment of Maintenance resources.
- ☐ Ensure that responsible maintenance supervisors receive and understand the Planned Job Packages for each scheduled job. Nothing can be allowed to fall through the cracks due to misinterpretation of intent or meaning.
- ☐ This is the point at which responsibility transitions from Planner to Supervisor.
- ☐ In turn, the same level of transition must take place between supervisor and technician at the time of assignment. Nothing can be allowed to be lost within the transition. Note the difference between allocation of technicians to jobs by the Planner/Scheduler and actual assignment of jobs to technicians by the supervisor.

- ☐ Associated Planned Job Packages are delivered to and reviewed with responsible supervisors to assure that nothing falls through the cracks due to misinterpretation of intent or meaning. These consultations form the point at which responsibility transitions from planner to supervisor.

When all sections are brought together as a cohesive Weekly Schedule, a complete picture for the coming week evolves as a document for which, through mutual participation and contribution, all parties accept ownership. Available Maintenance resources are fully committed and Operations has committed to all necessary access to equipment.

Any urgent work required or requested (there is a difference) will result in an overload upon maintenance resources. Such work can only be performed at sacrifice of one or more scheduled jobs unless additional overtime is approved. Otherwise, displaced jobs will have to be carried over to the next schedule week. Jobs displaced will be one of the jobs scheduled within the work center that initiated the schedule-break. Accordingly, requested schedule breaks require approval of the associated Departmental Manager, the Maintenance Manager, or the overall Operations Manager.

Transition from Preparation to Execution

Scheduled jobs are now ready to be transitioned from Preparation to Execution. Transitional steps include:

- ☐ At this point, the Weekly Master Schedule becomes a document of which all parties, through mutual contribution, accept ownership.
- ☐ Operations must advise planners at the earliest possible moment if they are unable to release equipment as scheduled. Similarly, Maintenance must notify and consult with Operations about any problems encountered or anticipated before associated commitments are missed.
- ☐ Planners and supervisors must remain abreast of schedule status, and detect when a job runs into trouble milestone. Planners are responsible for the associated communication and coordination.

- ❑ As the week unfolds, tactical adjustments may be necessary on a daily basis. As necessary, the schedule for the balance of the week is updated each evening. While transitioning from a reactive to a proactive culture, updating will be burdensome and will likely need planner support. As schedule compliance matures, updating becomes minimal and can be performed by the supervisor, allowing planners to focus on future weeks.
- ❑ Limit the number of people authorized to break the agreed upon schedule. Three-to-five is a reasonable number. When Schedule Breaks are necessary, the least important jobs at the bottom of the schedule should be the ones sacrificed, not the most important jobs at the top of the schedule. Rigorously enforce this logic. Managers and internal customers will be more acceptable of schedule misses if the most important jobs are not among the missed promises.
- ❑ Monitor compliance to weekly expectation. Measurement of compliance to daily schedules conveys a reactive rather than proactive message. Twenty-four hour lead-time does not provide for adequate preparation of many maintenance jobs.
- ❑ When jobs are completed, supervisors return the associated planned job package together with appropriate feedback, including status results for calculation of Schedule Compliance. Feedback also includes what actually happened, what failed, and ideas for improvement. Such feedback is vital to meaningful Equipment History, refinement of Planned Job Packages, and effective Root Cause Failure Analysis.
- ❑ Verify that the job was done according to the plan. When a job deviates from description, plan, or schedule, determine Why. Were all the parts and materials listed in the Bill of Material applied? Were all the required specialized tools and equipment identified in the planned job package? Were required reference documents and drawings provided and current? Were all required permits provided or at least identified? Were all arrangements with Operations fulfilled in timely and effective manner? Keep track of all the reasons. If the reasons reappear frequently, there is probably something missing from the operations or maintenance business system. Avoid blame by looking for what is missing.

- ☐ Update Planned Job Packages based upon Feedback.

OTHER APPROACHES TO SCHEDULING

The scheduling format, process, and procedures presented within this book are espoused by the authors. Some notable authorities espouse other approaches worthy of recognition and brief comparison to what has been proffered. In Appendix E, Pete Little of MPACT Learning Center presents a paper supportive of one such alternative.

There are essentially four points of contention between approaches. In Figure 16.4 we identify and compare the points; with defense of the approach presented throughout this chapter.

Point of Contention	Position of		Author's Belief
	Authors	Others	
Time Period	Weekly	Daily	Proactive vs. Reactive Message
Resources Scheduled	All	Progressive	Cultural Change Requires Bold Action
Resource Allocation Depth	Individual	Crew	Individuals not Clones of Each Other
Period of Sched. Compl.	Weekly	Daily	Time Required for Proper Preparation
Numerator of Sch. Compl.	Sch. Hrs.	Actual Hrs.	Apples to Oranges *
	Completed	Charged	
∞ Schedule Compliance = Scheduled Hours Completed / Scheduled Hours Scheduled			

**Pete Little of MPACT Learning Center has contributed a paper supportive of one other approach (see Appendices).*

Figure 16.4 Comparing Schedule Approaches*

ADDITIONAL OFFERINGS

SIMPLIFIED PROCESS OF SCHEDULING

1. Block out people known to be off the schedule (vacations for example).
2. Block out meetings (safety meeting).
3. Schedule jobs where production has given you a specific time.

4. Schedule PM.
5. Schedule your critical resources. Be sure to schedule critical resources to insure that they are available when the job is scheduled.
6. Schedule jobs in order of importance.
7. Review for opportunities to increase execution efficiency, either by reducing indirect time or working continuously on predecessor or successor jobs.

Specific Means by which Scheduling Yields Productivity Increases

1. Collisions of labor, material, tooling, and unit to be serviced are far cheaper to resolve on paper than on the shop floor.
2. A reasonable amount of work is expected each day. Workers are freed from a "hurry" atmosphere one day and a "kill time" atmosphere the next. Given a reasonable amount of work, most mechanics will attempt to complete it each day.
3. Emergencies are easier to handle because you know who is available, and at what time. Most emergencies can wait a short time. That short time will buy you productivity by allowing the worker to complete the previous job.
4. Management's expectation of how much work is a reasonable day's work is given to the mechanic in advance.
5. The rule of management control is: To control the whole, control the parts. You must manage at the time and at the level where you can have a useful impact.
6. Uncover many hidden operations problems. The solution to these problems (many are small) will increase productivity.
7. Execution of a planned job steps will remind the worker what to take care of ahead of time and will minimize downtime and job duration.

Scheduling Uncovers Operational Problems

Once scheduling is established, areas are highlighted where mechanics cannot do their job due to problems outside their control and other problems still carried over from the old way of doing business.

These previously hidden (or unpublicized) problems suddenly come into the foreground:

- ☐ Mechanics often get pulled off to work on non-productive activities
- ☐ Stock room contributes to off-schedule conditions regularly
- ☐ Failure to put equipment back into service when promised
- ☐ Inability to get control of units when needed for schedule
- ☐ Quality becomes a big issue
- ☐ Lack of cross-training causes clashes of resources
- ☐ Failure to meet standards and the immediate conclusion that it (the scheduling system) must be wrong
- ☐ Inability to handle emergencies, excess absenteeism, unusual peak load conditions

1. The first part of the document is a list of the names of the members of the committee.

2. The second part of the document is a list of the names of the members of the committee.

3. The third part of the document is a list of the names of the members of the committee.

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20. The twentieth part of the document is a list of the names of the members of the committee.

Job Execution and Supervision

Jobs have been well prepared and scheduled. Schedules and Planned Job Packages have been conveyed from planners to supervisors. It is Friday morning (or whenever your Schedule Week begins). Supervisors are now responsible for tactical execution of scheduled jobs.

LINKING THE SCHEDULE TO SUPERVISOR RESPONSIBILITIES

The schedule provides a strategic depiction of Maintenance activity throughout the schedule week. The next job is always ready and known to each technician. The schedule facilitates the ability to monitor crew progress and to determine when scheduled jobs are falling short of expectation; and intervention is in order.

Supervisors are responsible for leadership of job execution. Being responsible for a crew with several jobs in concurrent progress, supervisors are able to visit each job site no more frequently than once per hour to organize jobs about to begin, check on jobs in progress, and check proper wrap up of job completion. Those jobs in progress must be observed repeatedly to determine if support or action by the supervisor or other intervention is in order. The schedule is a vehicle to which progress can be compared. Without a target (expectation), how can supervisors assess current status? The schedule enables supervisors to judge when exceptions fall outside of reasonable boundaries and intervention is in order. Without the benefit of a visually documented expectation, Supervisors are left to rely on mental judgments as to whether or not jobs are falling behind commitments. A comprehensive schedule facilitates

supervisory effectiveness.

Staying abreast of job status relative to schedule enables the supervisor to take corrective action before problems become serious. In addition to the weekly schedule, monitoring is further enhanced when a detailed job timeline is included within the planned job package; supervisors can make a better informed judgment. If the schedule indicates that disassembly should be complete, the supervisor can clearly determine if job progress is on schedule or not. If the amount of work completed is within reasonable boundaries, no intervention is needed. However, if the supervisor finds that an important job is falling behind schedule, tactical decisions (larger crew, overtime, contractor support) can be made to correct the situation before a major shortfall develops. Intervention while the job is still in process can make a profound difference in downtime and equipment availability. The scenario reflects effective supervisory leadership. Excuses for missed scheduled does not. The horse is out of the barn!

When jobs go awry, parts and materials, tools and equipment, or lack of operational adherence to coordinated commitments are common culprits. If problems stem from shortfalls in planned job packages, then provide appropriate feedback to planners. This feedback enables improvement of the package, which precludes repetition of the same shortfalls in the future.

Supervisor Time-Line throughout Execution

Supervisor responsibilities fall into the following timeline as jobs flow from final preparation through to job completion and wrap up. Weekly Schedules and associated Planned Job Packages have been transitioned from planners to supervisors. This transition is more than delivery to supervisor desks or forwarding to supervisor terminals. This is a sit down, face-to-face exchange. No preparation is to be lost in transition from planner to supervisor or in subsequent transition from supervisor to assigned technician.

Prior to Job Start

- ☐ Check preparedness for each day's work. Despite all preparation performed by planner/schedulers, supervisors confirm that all supportive actions of others are performed as agreed upon in order to assure effective execution and resultant operational reliability.

- ❑ Supervisors make the actual job assignments (distinct from allocations) to specific team members in accord with the labor resource deployment scheme. At times, tactical deviations may be necessary, but in most cases the labor deployment scheme is followed because planners have thought through who might be the “most experienced and familiar” technician to perform the job as well as who needs exposure to the job to entrench recent skills training or to expand familiarity with assets throughout the facility.

During Job Execution

- ❑ Be in attendance at crew job sites (2/3 of the day) to provide effective execution, follow-up, and feedback. Ensure that crews have planned job packages open at job sites and regularly reference them, making full use of all preparatory efforts in order to perform jobs safely, effectively (quality), and efficiently. The packages have been prepared for good reason. Use them! Constant effort is needed to improve work methods, completeness, neatness, quality, safety, and efficiency. Crews are encouraged to feedback ideas for improvement of planned job packages.
- ❑ Remain abreast of progress relative to scheduled milestones. Look for exceptions to detect when jobs are running into trouble. Take corrective action before milestones are missed and problems become serious. Early intervention yields major favorable impact in terms of schedule compliance, uptime, quality, and output.
 - Reverse or limit delays, overruns, and interruptions.
 - Increase crew-size, work overtime, or add contractor support. The discrete use of overtime to protect adherence to the Schedule can be very cost effective.
 - When jobs go awry, materials, parts, tools, or access denial are common culprits.
- ❑ Control the time lost between jobs, at breaks, and during shift changes.
- ❑ As the week unfolds, tactical adjustments may be necessary on a daily basis. Coordinate schedule adjustments as dictated by

“real” operating needs and changing resource constraints. Urgent interruptions do occur despite how well schedules are constructed and coordinated. However, potential schedule breaks should require approval of at least one Operating and one Maintenance manager who participated in the weekly schedule coordination meeting. Everybody cannot be authorized to override agreements reached therein. The department experiencing the schedule break will have to sacrifice one of its important scheduled jobs. Perhaps that department will learn to better honor essential PM/PdM routines.

- ☐ Record the description and duration of schedule breaks at bottom of the schedule.
- ☐ Keep team members fully aware of assignment changes. Also keep operating supervisors abreast of schedule revisions.
- ☐ Notify Operations as early as possible about any pending disruptions such as inability to provide maintenance resources as scheduled. Normally, simple reassignments of technicians do not impact operations.
- ☐ In like fashion, Operations must alert Maintenance as early as possible if they will be unable to release equipment as scheduled.

The above supervisory scenario reflects true management. After-the-fact excuses do not. *“The horse is out of the barn.”* Scheduling allows the supervisor to see where the job should be. The only way to improve a plant is to intervene while the job is going on. Recapping the problem jobs at the end of the month will never improve a shop.

After Job Completion

- ☐ Every afternoon, inform each team member as to assignments for the next day.
- ☐ Extended or delayed jobs must be carried over to the next day, particularly if they are already started. However, if the delayed job has not been started, it should not bump any job of greater importance (higher on the schedule).

- ☐ Schedule adherence or protection is usually a cost-effective use of overtime.
- ☐ If not started, they should not bump any job of greater importance (higher on the schedule).
- ☐ Due to urgent (high priority) work orders that break the schedule, supervisors must review, adjust, and update schedules each evening for the remainder of the schedule week. While transitioning from a reactive to a proactive culture, the updating can be burdensome and need planner support. Get beyond this stage as quickly impossible. Until you do, the planner will be spending too much time on scheduling, to the detriment of planning. However, as schedule compliance matures, updating becomes minimal and can be performed by the supervisor allowing planners to focus on future weeks.
- ☐ Despite essential schedule breaks, always strive to optimize schedule compliance. When schedule compliance does falter, every effort must be made to protect as much of the original schedule as possible. Adjustments to schedules should always be made in a manner whereby the most important jobs are completed by the end of the week. Because scheduled jobs are listed in order of importance, the jobs to be protected are those highest on the schedule.
- ☐ Selected scheduled jobs should sometimes be sacrificed to take advantage of unexpected access (downtime windows of opportunity) to perform planned jobs listed on the provisional schedule.
- ☐ As each job is finished, collect the planned job package, assure reasonably accurate distribution of time and materials, record appropriate feedback, and mark the schedule for calculation of compliance.
- ☐ Verify that each job has been completed to customer satisfaction and performed according to plan (quality, thoroughness, clean up, feedback). When a job deviates from description, plan, or schedule, determine **"Why."**
- ☐ Were all parts and materials listed in the Bill of Material applied to the job?

- ☐ Were all required specialized tools and equipment identified in the planned job package?
- ☐ Were required reference documents and drawings provided and current?
- ☐ Were all required permits provided or at least identified?
- ☐ Were all arrangements with Operations fulfilled in a timely and effective manner?
 - Did any other problem interfere with schedule compliance?
- ☐ Return planned job packages together with appropriate feed back. Feedback includes: what actual work was performed, if different from plan; what failed; and ideas for improvement. Such feedback is vital to meaningful Equipment History, refinement of Planned Job Packages, and effective Root Cause Failure Analysis.
- ☐ At the end of the schedule week, review schedule compliance with all technicians under your responsibility.
- ☐ The planner calculates schedule compliance, determines reasons for non-compliance, and updates planned job packages. This constructive step focuses on opportunities for future improvement, not the fixing of blame.
- ☐ At the end of schedule week, Managers and/or Superintendents should review schedule compliance with all supervisors reporting to them.

THE MORNING MEETING

With the introduction of the morning meeting, there are now two meetings to be clarified in relation to each other. As previously discussed, the Schedule Coordination meeting takes place once a week—late in the current week, shortly before the start of the coming schedule week. The Morning meeting occurs every day that the facility is in operation. The coordination meeting is proactive whereas the morning meeting is reactive. The coordination meeting is focused on sustained reliability. The morning meeting is focused on daily and currently week output. The maintenance planning function chairs the coordination meeting. Operating management chairs the morning meeting. The two meetings must be kept in harmony. Jointly, they must find an optimum between their two focuses: near term output and sustained long term output.

The meeting is typically conducted about one hour into the day shift. The duration is usually thirty-to-forty-five minutes. This timing allows participants to come abreast of results during the previous evening and night shifts, and experience how well or poorly start up of the current day shift went. As necessary, these inputs are factored into Schedule Adjustments for the current day and balance of the schedule week.

The meeting is focused on resolving immediate problems and desired results while protecting the weekly schedule as much as possible. "Who made the mistake" is not at issue. Tactics are discussed and resources realigned as necessary. An upbeat attitude, oriented toward problem solving, is the tenor of the meeting.

Not everyone can be authorized to override agreements reached therein. Any proposed adjustments to the weekly schedule require approval of at least one Operating and one Maintenance participant at the weekly coordination meeting. These are the individual that set and agreed on the most important jobs for Maintenance to perform during the schedule week. The department experiencing the schedule break will have to sacrifice one of its important scheduled jobs.

The Morning meeting is especially essential if the organization still operates in a reactive environment within a reactive culture. Once the culture transitions from reactive to proactive, the Morning meeting has less prominence and the Weekly Schedule Coordination meeting has more.

Ideas to Make Morning Meetings More Efficient¹

Agenda: The agenda is the road map for the meeting. Without it the meeting will meander (but you might not notice it because there is no agenda).

Attendees: Who is in the room is almost as important as what is discussed. Be ruthless about who you invite and be sure they attend.

Accountability: Holding people accountable for what they promise or what you've requested is important to the smooth functioning of the whole shutdown. People tend to avoid accountability. Once they get a taste of accountability without blame, they will realize that their growth and development is linked to the accountability that they take on.

Rule of meetings: The more comfortable the room, the longer the meeting. Serve coffee or donuts and the meeting will last an hour longer for the same agenda items. Consider removing the chairs for the morning meeting!

¹ Adapted from *Managing Maintenance Shutdowns* by Joel Levitt

1. The first part of the paper discusses the importance of the study of the history of the English language. It is argued that the study of the history of the English language is essential for a full understanding of the language and its development. The paper then discusses the various factors that have influenced the development of the English language, including the influence of other languages, the influence of social and cultural changes, and the influence of technological advances.

2. The second part of the paper discusses the importance of the study of the history of the English language. It is argued that the study of the history of the English language is essential for a full understanding of the language and its development. The paper then discusses the various factors that have influenced the development of the English language, including the influence of other languages, the influence of social and cultural changes, and the influence of technological advances.

3. The third part of the paper discusses the importance of the study of the history of the English language. It is argued that the study of the history of the English language is essential for a full understanding of the language and its development. The paper then discusses the various factors that have influenced the development of the English language, including the influence of other languages, the influence of social and cultural changes, and the influence of technological advances.

4. The fourth part of the paper discusses the importance of the study of the history of the English language. It is argued that the study of the history of the English language is essential for a full understanding of the language and its development. The paper then discusses the various factors that have influenced the development of the English language, including the influence of other languages, the influence of social and cultural changes, and the influence of technological advances.

Job Feedback, Close Out, Analysis, and Schedule Compliance

Once jobs have been completed, assets have been returned to operational control (permit completed and locks removed), and Planner/Schedulers have received feedback from technicians and supervisors, jobs are ready to be Closed Out in the CMMIS.

JOB FEEDBACK

Proper close out with all essential information is dependent upon proper feedback regarding work performed. The degree of required feedback varies with the nature of the job. Labor-Hours worked by individuals on each job are required in all cases. A single entry into the Labor Distribution System should feed CMMIS as well as Accounting Systems. Materials and Parts used must be charged on all unplanned jobs. A single entry into a Stock Requisition or a Direct Purchase Order should also feed both CMMIS and Accounting. In regard to planned jobs, Parts and Materials have already been charged to both systems via the Work Order Planning Module of CMMIS. Further entries are necessary only if additional items were needed or some planned items were returned to Stock or Vendor.

Proper feedback also relates to Description of work performed, which often differs from the original description in the Work Request—especially in the case of unplanned jobs. However, planned jobs may also require improved (after the fact) description. If planning was effective, such need should be minimal.

If there were other shortfalls or necessary corrections to the Planned Job Package, they should be posted to Work Order free text within CMMIS (permit, safety, reference materials, environmental, additional job steps, special tools required, interruptions encountered, etc.).

If the job contained inspection aspects, the feedback should include conditions found. If corrective actions were taken, they should be described. If Corrective Work Orders were originated, they should be cross referenced to the work order from which they originated.

A complete picture of each work order is dependent upon comprehensive feedback. Without it, CMMIS databases and resultant informational outputs become ineffective in terms of reliability, completeness, currency, and usefulness as itemized below:

- ☐ Completeness and Accuracy of Equipment History
- ☐ Refinement of Planned Job Packages
- ☐ Analysis of Failure Root Causes
- ☐ Optimization of PM/PdM Routines
- ☐ Effectiveness of Management Reports
- ☐ Completeness and Accuracy of Accounting Values

JOB CLOSE OUT

Planner/Schedules are responsible for Job Close Out. However, they should have some administrative support in this regard. Much of it is routine and should not be allowed to consume valuable Planner/Scheduler capacity. Their primary duties are too critical to be marginalized by routine duties.

In performance of job close out, the Planner/Schedulers and/or their administrative assistants:

- ☐ Collect completed work orders including associated feedback and enter them or direct them to be entered into the CMMIS, ensuring that all fields are filled in with valid codes, comments are understood (contact the writer if not), and detail is sufficient to support future analysis.
- ☐ Close completed work orders to proper Accounts and to Equipment History promptly, to preclude inappropriate future charges. Distribution should be electronic.

- ☐ Verify that all feedback has effectively been captured.
- ☐ Reconcile urgent work order completions with Schedule Break work orders recorded at schedule bottom.
- ☐ Verify that: a) leftover parts and materials have been returned to Stores or Vendors with credit to associated work orders and b) special tools have been returned to the Tool Crib in good working order. Care is required on this point because maintenance technicians tend to “squirrel” such items in their private caches, especially in reactive environments with insufficient materials support. Responsibility for the disposition of leftover items falls upon individuals shown below:
 - Rental Equipment (Planners). Avoid unnecessary charges.
 - Equipment and Components for Rebuild (Planners)
 - Direct Purchase items (Supervisor)
 - Stock Items requisitioned (Technicians)
 - Special Tools (Technicians)
 - Free Stock Items (Technicians)

ANALYSIS

Planner/Schedulers compare Actual Job Results to Planned Job Packages in order to refine the plans for future application. One approach used to refine job estimates is Weekly Top Ten Reports. For each Planner, these reports identify the ten jobs completed each week for which Actual Labor Hours charged differed from Estimated Labor-Hours by the greatest percentage regardless of sign (+/-). Planners review their respective top ten each week to determine what refinements might be appropriate.

If actual hours exceed estimate, it does not necessarily mean the estimate is wrong. Crew performance may have been poor; which would be a problem to be addressed by supervision. Actual work may have differed from planned work. The Planned Job Package may have been deficient (improve the package). Operations may not have followed through on commitments (strengthen coordination). If there are no other explanations, then change the estimate!

Maintenance/Reliability Engineers focus their efforts on high maintenance assets, as identified in Equipment History. The information gained facilitates root cause failure analysis, optimization of PM/PdM pro-

grams, determination of economic asset life, repair/replace decisions, etc.

Management Reports lead to decisions regarding effectiveness of control systems, budget adequacy, resource sizing, and correlation between PM/PdM Schedule Compliance with uptime, capacity, reliability, and quality.

SCHEDULE COMPLIANCE

Maintenance supervisors keep planner/schedulers abreast of job completions and failures to meet schedule throughout the schedule week. At week's end, planners calculate three metrics: Schedule Compliance, Performance, and Effectiveness. Then they prepare associated trend charts and reports (including reasons for non-compliance). Intent of the non-compliance reports is not to fix blame, but to identify constructive actions that improve compliance in the future.

Schedules are not rigid. They represent an agreed upon objective for supervisors to pursue with their teams. Accordingly, supervisors are authorized to make necessary tactical decisions as essential, urgent, schedule breaks arise. The expectation for schedule compliance is 85%-to-90%; not 100%. These percentages assume that all resources are fully allocated to scheduled work.

Non-compliance is especially serious when Maintenance fails to bring capacity back on-line as scheduled and promised. Non-compliance of this nature diminishes future cooperative, schedule coordination. In the future, there will be reluctance to release assets for scheduled maintenance.

REASONS FOR NON-COMPLIANCE

When maintenance crews encounter schedule compliance problems outside their control, the reasons should be captured and studied for trends, leading to constructive actions. An appropriate coding structure is provided in Figure 18.1.

Reason for Schedule Non-Compliance	Suggested Code
Operations may fail to release equipment as previously agreed to.	FR
Excessive emergencies. Technicians are pulled off scheduled jobs to work on less important jobs without consultation or communication with maintenance leadership. Commonly found where operating personnel are allowed to redirect maintenance personnel rather than going through the work management process.	EE
Poor assignments of technicians. Completions are bound to fall short of expectations if schedules are not feasible from the outset. Schedules often are not feasible when they are not detailed to the level of individual assignments.	PA
Insufficient technician capacity. Lack of cross training contributes to shortage of essential resources.	IC
Stock-outs are frequent. Inaccurate inventory control regularly contributes to compliance shortfalls. Quantities shown to be in inventory are not correct, and excessive time is spent at the storeroom because required materials and parts have not been pre-requisitioned and delivered to the job site.	SO
Planning packages do not reflect reality. Parts are incorrect, job steps are incomplete or wrong, and lockouts, specifications and regulations are not documented.	PP
Failure to meet estimated job durations means that some scheduled jobs are not completed by end of the scheduled week. The immediate conclusion is that the schedule must be wrong. Planners must not jump to the conclusion that the estimate was wrong. It may have been, but the crew may have performed poorly, or another identified problem may have occurred.	LE
Excessive absenteeism and simultaneous peak loads.	EA

Figure 18.1 Reasons for Non-Compliance

CALCULATION OF SCHEDULE COMPLIANCE

The scheduling format presented in Chapter 16 provides two columns and two sub-totals (A and B) for collecting information to calculate Schedule Compliance. The formula is:

Schedule Compliance

= Scheduled Labor-Hours Completed / Scheduled Labor-Hours Scheduled

= B/A

The right column of the format accumulates the numerator. The fourth column from the left accumulates the denominator. Cells B and A reflect totals for inserting into the formula. Cell A represents all work scheduled and expected to be accomplished. It does not include either contingency or provisional jobs. Cell B reflects all work fully or partially completed. It does include any work substituted and completed from provisional or contingency schedules. This logic is reflected in vertical positioning of the two sub-totals (Cell A at the bottom of the primary schedule and Cell B at the bottom of the total schedule).

If 1000 labor-hours were scheduled, 1000 would show in Cell A. If 800 labor-hours were completed, 800 would show in Cell B and Schedule Compliance would equal 80%.

Two related metrics are Schedule Performance and Schedule Effectiveness:

Schedule Performance

= Scheduled Labor-Hours Completed / Total Labor-Hours Available

Schedule Performance adds available hours that were not scheduled into the denominator of the equation. They may have been left out intentionally as provision for breakdown work or simply not fully scheduled. In the above example, if 800 labor-hours of scheduled assignments were completed, but 1200 labor-hours were available, then:

Schedule Performance

= 800 Scheduled Labor-Hours Completed / 1200 Labor-Hours Available
= 75%

Schedule Effectiveness is the most inclusive of the three scheduling metrics. It excludes any completed indirect labor-hours from the numerator of the equation—such as vacations, training, and meetings. Continuing with the same example, if 600 hours of scheduled work completed were direct work, with the other 200 hours being indirect commitments, then:

Schedule Effectiveness

= Scheduled Direct Labor-Hours Completed / Total Labor-Hours Available

Schedule Effectiveness

= 600 Scheduled Direct Labor-Hours Completed / 1200 Total Labor-Hours Available
= 50%

Assume this table lists the jobs scheduled and completed.

Work Order	Job Description	Req. Skill	Crew Size	Clock Hours	Labor Hours	Hours completed
4	Clean sensor housings 1-4	M	1	4	4	4
21	Replace belts, check sheaves W7	M	2	4	8	8
19	Move debris from #2 and clean up	Any	2	8	16	16
14	Check source of oil on floor by #2	M	1	2	2	2
39	Track down ground loop	E1	1	6	6	6
5	Replace Limit Switches J8	E1	2	5	10	-
37	Fix broken vent for Unit 4	M	1	4	4	4
6	Weld Gusset on #3 where damaged	M	2	8	16	16
18	Paint #3 where repaired	Any	2	8	16	16
7	Replace Cut-offs #1	M	2	16	32	-
P24	Align new hopper to J8	M	2	8	16	-
41	Repair tank where thin	M	4	4	16	16
46	Weld attachment points per drawing D65	M	2	8	16	-
9	Bracket for inspection scope - #4	M	2	8	16	16
10	Repair crack in pipe (temp fix)	M/M	2	16	32	32
40	Replace hold downs #1 Press	M	3	2	6	-
P11	Connect Power to Line #3 run diagnostics	E1/E2	2	8	16	16
12	Check Run Line #3	M	2	4	8	8
51	Reset scanner for new input press #1	E1	1	8	8	8
	Fix intermittent service fault	E1	1	4	4	4
	Total Hours on the schedule				252	172

Schedule Compliance = Scheduled Labor-Hours Completed / Labor-Hours Scheduled

73%	172	252
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Schedule Performance In this case we add in the 80 hours of emergency !

Schedule Performance = Scheduled Labor-Hours Completed / Total Labor-Hours Available

51.8%	172	332
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Schedule Effectiveness

Schedule Effectiveness = Scheduled Labor Hours Completed / Total Labor Hours Available + Indirect hours

30.7 %	172	560
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Figure 18-2 Schedule Success

The three metrics combined provide a comprehensive picture of scheduling success (see Figure 18.2):

- ☐ Schedule Compliance measures the percent of scheduled labor-hours completed during the schedule week.
- ☐ Schedule Performance measures the percent of labor-hours available to be scheduled that were indeed scheduled and completed during the schedule week.
- ☐ Schedule Effectiveness measures the percent of total labor-hours worked that were scheduled to direct work (vs. indirect work) and completed during the schedule week.

Under no circumstance are Schedule Breaks (urgent jobs that broke into the schedule) credited to any of the three schedule compliance metrics. However, schedule breaks are credited when Crew Efficiency is calculated. Efficiency is almost always higher when a job has been properly planned and scheduled.

The three metrics are needed because Schedule Compliance can easily be high if only a small portion of departmental capacity is scheduled. Performance will highlight this shortfall. Compliance and Performance can both be high if inflated by a lot of indirect work added to the schedule, e.g., training, infrastructure assignments, loaned to engineering, and so on.

Compliance is essentially automatic (100%) for indirect assignments, thereby inflating Compliance and Performance.

If indirect assignments are not scheduled, the Schedule Effectiveness metric is not needed.

Computerized Maintenance Management Information System (CMMIS) in Support of Planning, Scheduling, and Coordination

Fast, flexible access to reliable, current, and comprehensive information is vital if planning, coordination, and scheduling are to be effective. Although these functions were performed moderately well for many years without computer support, it is no longer an economically sound decision to manage a function as critical as maintenance without on-line informational support provided by a sound Computerized Maintenance Management Information System—people and programs simultaneously inputting, updating, and retrieving information. Only such systems and network satisfy the informational immediacy required. They provide easy access to large amounts of data, thereby enhancing world-class preparation for maintenance execution and enabling levels of analysis that would be too time consuming if performed manually. Such information is vital if management is to control the maintenance function on the basis of knowledge rather than intuition.

Although the generally accepted term for maintenance computer systems is Computerized Maintenance Management Systems (CMMS), the authors have used Computerized Maintenance Management

Information Systems (CMMIS) because such systems are not comprehensive systems for the management of maintenance. They are simply an information system (tool) to facilitate management of the function—only one of twenty-one building blocks of Maintenance/Reliability (see Introduction).

Computerized informational support is essential to full realization of maintenance/reliability excellence potential. However, CMMIS systems in isolation accomplish *nothing* and are not justifiable. The bottom-line impact results from actions taken on the basis of information extracted, but not from the systems in isolation.

Information derived from CMMIS systems support and foster realization of the following bottom-line contributions:

- ☐ Efficiency of maintenance resources (both hourly and salaried)
- ☐ Improvement of responsive service to needs of internal customers
- ☐ Improvement of asset reliability yielding equipment uptime and capacity assurance
- ☐ Improved product quality and delivery performance for external customers
- ☐ Lower unit costs, increased profitability, and business continuance

WHERE DOES YOUR WORK COME FROM?

Several sources are internal (PM tasks, corrective maintenance) and the rest are external to the maintenance department. A study of your information flow might show where requests for work could fall through the cracks. It would also show where you are not capturing the minimum amount of information to effectively dispatch maintenance technicians (see Figure 19.1).

All requests for service should converge in the maintenance control center and be reviewed by a single person or group. If you use a team structure, then the team point person on planning and scheduling should review the requests. Emergencies should be handled directly (through the supervisor). The goal of the system is to serve the customer efficiently with minimum overhead while maintaining control.

Needs = Incoming Work

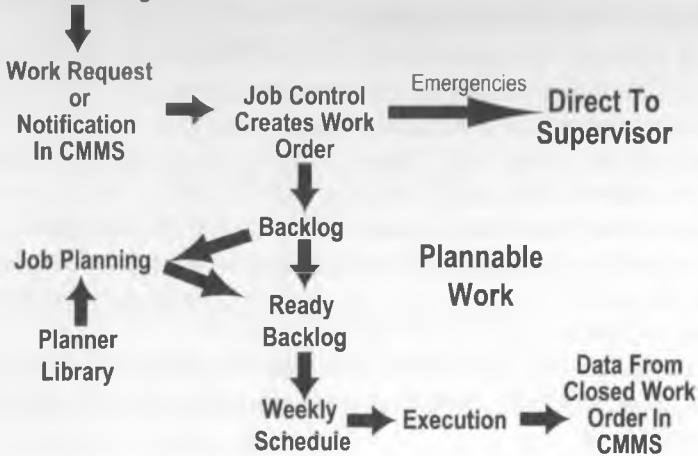


Figure 19.1 Work Flow

Function of Job Control

All services requested from all sources go through the in-box (electronic or other). These include incoming E-mails, phone messages, notes, and radio calls from all maintenance users and PM/PdMs that are due to be performed, as identified within the CMMIS. Time/date stamping should take place when a job enters the in-box. The in-box should be scanned frequently to assure that all work request and work orders are progressing appropriately.

Triage: Emergency jobs are promptly dispatched to the floor and the time of dispatch is recorded. After the true emergencies are processed, all other jobs are progressed through the planning, coordination, and scheduling process.

With proper application of the CMMIS, the Maintenance Control Center becomes the "gatekeeper" to maintenance/reliability services.

Planner/Schedulers rely on the CMMIS for most of their informational needs. It is not the intention of this book to discuss the many features and characteristics essential to fulfill informational needs of all CMMIS users. We will, however, explore the features pertinent to work identification, backlog management, job planning, material procurement, logistical coordination, and weekly scheduling—as discussed below.

CMMIS Characteristics Essential to Planning, Coordination and Scheduling

A common complaint about many CMMISs is, "It is not user-friendly." The statement is often true, and the function to which the system is least user-friendly is Planning and Scheduling—the singular function that uses the system more than any other. Too many keystrokes are required to process P&S daily routines. Any function, whether a power user or casual user, soon loses commitment if their required input is not as simple and quick as possible. They rightly believe that computerized systems should make life easier, not harder. This does not minimize the importance of collecting essential information.

To effectively support the functions discussed throughout this book, the chosen CMMIS (hardware and software) must offer the following characteristics:

- ❑ On-Line, Real-Time, and User-Friendly so all parties can carry their share of information input and retrieval. If the system does not meet this requirement, planner/schedulers are left to perform all input and output. Planners become little more than clerks. Although it is a sound investment to take skilled technicians off their tools to become planner/schedulers or supervisors, it is a very unsound decision to relegate those workers to the routine clerical entry of everybody's requests and inquiries.

Distribution of the input responsibility is nothing new. When work order systems first came into use over 60 years ago, individuals requesting maintenance support were required to prepare and submit written work orders. Today's equivalent is a computerized Work Order Request with requestor-required fields.

- ❑ Backlog Management with which to control all open work orders from origination through preparation, execution, and completion. Backlog must be current, complete, accurate, and pure:
 - Current = Real-time
 - Complete = All jobs that need to be performed are captured
 - Accurate = Management can rely on information presented

- Pure — All work shown still desired. No jobs shown have been completed but never closed out, no duplicates and no jobs you never intend to do.
- ☐ Macro-Planning features to facilitate development of Work Programs to keep resources in balance with workload.
 - Weekly available labor-hours by crew
 - Requirements for PM/PdM
 - A company-wide or maintenance-wide calendar is useful
 - Projection of urgent work order demands
 - Resultant resources available for backlog relief.
- ☐ Sound, comprehensive work order control features, including an effective coding structure:
 - Requestor field to keep internal customers apprised as to current status of their requests. Ideally, requestors should be able to access this information themselves, on-line.
 - Date of Origin field to facilitate aging of work orders.
 - Asset/Equipment code to facilitate the assembly of all ready-to-go work orders that might be performed during access to a given asset.
 - Priority code with an aging feature to prevent individual jobs in the backlog from being indefinitely deferred. Medium and even low priority jobs are to be completed within a reasonable time frame.
 - Responsible Planner field so that planners are able to separate their workloads from the complete backlog.
 - Job Status code to facilitate planner efforts to keep all work orders moving to completion rather than allowing them to bog down in a state of limbo. This is the planner's code. It brings order out of chaos. It is the most important code in the entire process.
 - Assigned Team field to facilitate preparation of a weekly schedule fully deploying resources reporting to each supervisor.
 - Condition Required code because there is regular need to separate work that can be performed at any time versus work

requiring asset downtime of various duration (e.g., a few hrs., a weekend, periodic programmed access, annual shut down).

☐ Effective Materials Management with:

- Bill Of Materials (BOM) by asset
- Cross references between manufacturer part number, vendor part number, and storeroom item number
- Reliable replenishment of authorized stock
- Prompt processing of purchase order requests for direct purchases
- Prompt and reliable notification of receipts
- Reservation (allocation) of inventory item units to specific planned jobs and release (de-allocation) of same as needed
- System capability to store, retrieve, modify, and copy previously developed job plans and estimates from history or planner libraries (ability to cut and paste is useful); associated information includes:
 - Job Steps
 - Bills of Material
 - Job Estimates

☐ Ready access, without backing out of planning module, to information in other modules:

- Full bill of materials by asset
- Current stock availability by SKU with reservation capability
- Safety and environmental procedures, instructions, permits, and authorization (e.g., hot work, isolation, lockout/tag out)
- Drawings and other reference documents with provision for automatic attachment to planned job packages.

☐ Scheduling Features

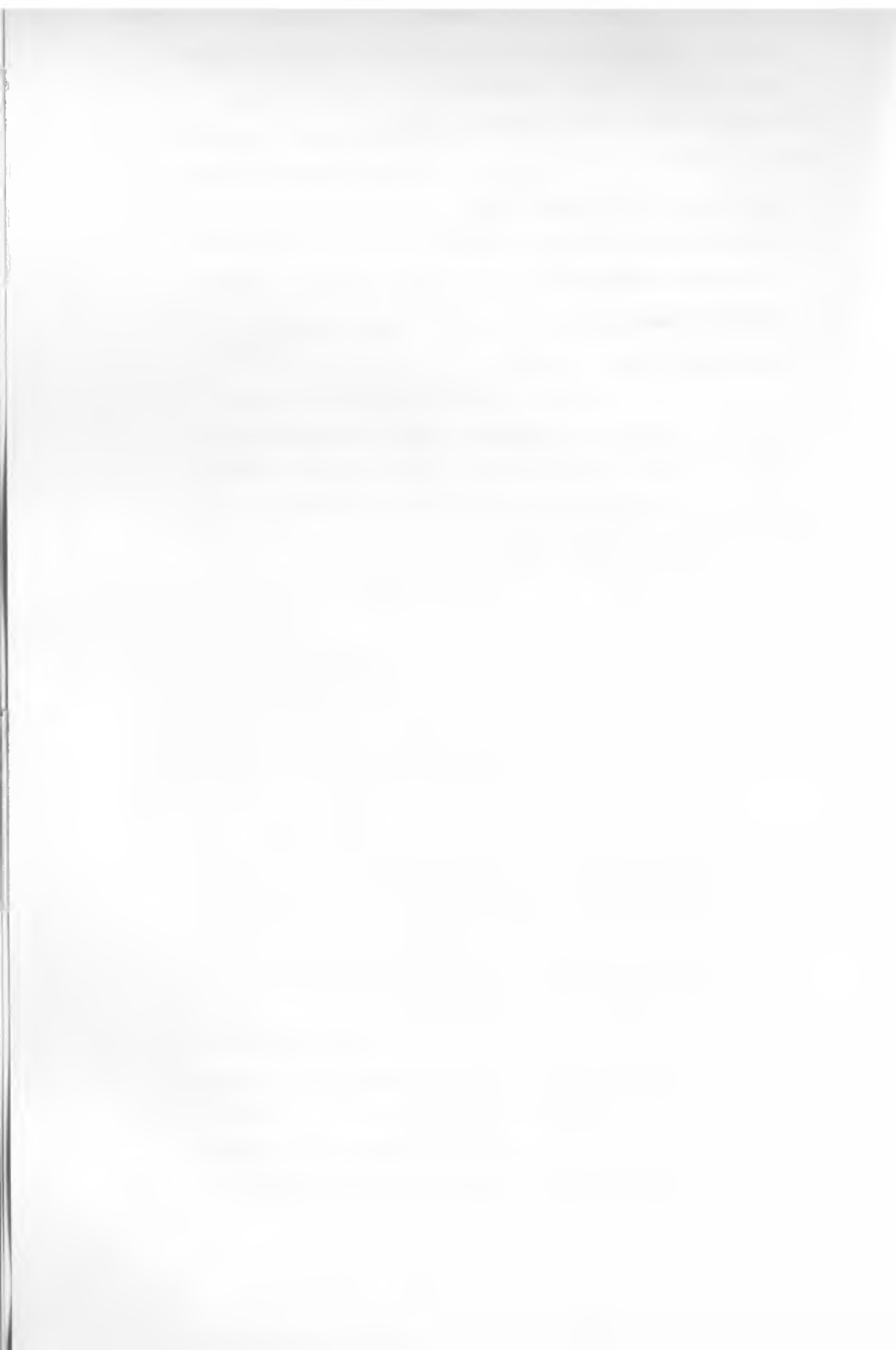
- Weekly scheduling by: crew, job, day of week, and individual to whom job is to be assigned
- Annualized leveling of PM/PdMs
- Notification when PM/PdM routines come due

- Ability to call forward nearly due PM/PdM routines to take advantage of access to equipment

☐ Linkage to Project Management Software

☐ Report Features including:

- Backlog by Planner and Status
- Work Programs
- Schedule Compliance
- Crew Efficiency
- Age of Reserved Inventory



Planning Major Maintenance Projects

In addition to the thousands of day-to-day and week-to-week work orders that Maintenance performs throughout the year, Maintenance also participates in major efforts that encompass dozens, even hundreds, of tasks within a continuum of time. These efforts may encompass capital projects as well as major maintenance shutdowns (such as two-to-three-week process turnarounds commonly performed annually in many industries). This chapter addresses planning, coordination, scheduling, and management control of efforts of such magnitude.

Throughout our discussions of Weekly Maintenance Scheduling, we have applied Gantt Bar Charts. They are effective for display of sequential jobs in series, with linear relationships. However, shutdowns for major maintenance or for installation of capital projects require more comprehensive preparation than that associated with normal weekly maintenance. The principle differences lie in work scope, required resources, a limited time frame, and associated expansion of coordination required. Although the basic process presented in previous chapters remains essentially unchanged, the magnitude of the differences dictates a more advanced project management technique.

Bar charts have limited ability to show interrelationships between jobs or job steps. Major efforts comprising many jobs, with diverse relationships, dictate application of computerized network diagrams to identify the critical path through simultaneous parallel activities. Accordingly, throughout this chapter, the reference will be to Critical Path network diagrams and project management software that encompass planning, scheduling, and control features (both progress and financial). A brief introduction to Critical Path Analysis (CPA) is provided in Chapter 21.

THE PLANNING PROCESS FOR MAJOR MAINTENANCE EFFORTS

For discussion purpose, we need to establish a typical scenario. Assume an annual turnaround intended to perform major programmed maintenance that must be performed once a year in order to bring reliability of a production process back to economic viability. During the turnaround, management also wants one or more capital projects to be performed. Perhaps some preventive/predictive maintenance routines should also be performed to take advantage of the extended downtime. More than likely various members of the organization have additional thoughts as to other work to be accomplished, some of which are already reflected in backlog, and others that have not been previously identified.

Despite popular thinking, much of the work to be performed within annual turnarounds is common year-to-year. There are always differences, but the annual tasks are far more repetitive than unique (70% to 80% is the same). This statement is especially true in relation to the many associated shutdowns, disassemblies, reassemblies, and startups (return to service). These repetitive annual tasks should be captured and preserved to facilitate the annual preparation process.

Assume that any capital projects to be installed during the period of shutdown have successfully passed through the Capital Project Appropriation, Design, and Approval Process.

1. Start of the Planning Process

Planning for programmed shutdowns begins at the Project Review Meeting conducted at the conclusion of each previous shutdown. Lessons Learned are reflected in all project documentation so that the same mistakes or shortfalls will not be repeated.

2. Shutdown Kick Off Meeting

Several months (4-to-6) prior to major shutdowns, site managers call together key members of their staff to discuss the upcoming shutdown event. These people will constitute the Shutdown Steering Committee.

Maintenance Managers contribute the Master Critical Path Network of the repetitive activities that occur each year, thus providing a starting template. They also suggest additional maintenance projects for consideration. Project Engineering Managers introduce Capital Projects that will be ready for installation. Operating Managers most likely have

process improvements to be considered, etc. The Controller attends to gain early awareness of financial magnitude. Other functional managers attend as appropriate.

Based upon these inputs, the Steering Committee develops a preliminary statement of project objectives and deliverables. They agree upon individuals for assignment to the Project Management Team, as well as key stakeholders to be solicited for input. They also establish a date for the first meeting of the Project Management Team.

To qualify as members of the Senior Steering Committee, individual candidates should be directly affected by the outcome of the shutdown, be responsible for approval of funding, or be a decision maker regarding scope and timing of the shutdown. At a minimum, the Project Management Team (distinct from the Senior Steering Committee) must include a:

- ☐ Project/Shutdown Manager (Czar of the overall endeavor)
- ☐ Project Engineer to coordinate the capital work and to provide technical counsel
- ☐ Project Superintendent to lead the execution of all jobs to be performed as part of the shutdown (including contracted work)
- ☐ Project Logistics Coordinator to lead procurement and expediting of all required capital equipment, parts, materials, rental equipment, and contractor support
- ☐ Critical Path Analyst (engineer or senior P/S) to develop the network, record all progress, identify any slippages, identify impact thereof, and make dynamic recommendations for recovery to the Shutdown Manager

These five positions need to be full-time assignments beginning at least four months before a major shutdown and for nearly another month after shutdown to complete all responsibilities associated with project feedback, documentation, reporting, and administration.

3. Formation of the Project Team

The individuals named above are immediately alerted of their assignments and of the date for the first Project Management meeting. Additionally, they are directed to assemble their individual thoughts

regarding jobs or problems to be addressed within the scope of the shut-down efforts.

4. Project Announcement with Tentative Schedule Date and Request for Input

An announcement is sent to all identified stakeholders. It lists all jobs already identified and requests that everybody use their best insight to identify additional candidate jobs and problems or opportunities for consideration. The intent is for all expectations to be considered during project development, in order to avoid scope changes after project initiation.

5. First Meeting of the Project Team

Several purposes of the meeting include assembling a list of all candidate jobs, initial screening for jobs to be excluded with reasons supportive of their inclusion, identifying long lead time items, separating jobs already engineered or planned from those that most still be planned, and assigning responsibilities for required preparation and execution of inclusive jobs. To be incorporated are jobs and related activities to be performed prior to shutdown, as part of the execution workload, as one of the restart up activities to transitioning assets back to Operations, plus those activities that can be performed post startup in order to minimize required downtime.

The project management team then identifies other members of the project team, establishes an organizational structure for the project with a chain of command. Some of the positions that may of necessity be added to the team include:

- ☐ Purchasing Agent/Buyer to support the Project Logistics Coordinator
- ☐ Stores/Receiving Representative
- ☐ Shutdown Accountant to support the Steering Committee and the Project Management Team; this accountant will work in close harmony with the Critical Path Engineering-Analyst
- ☐ Process Coordinator who will be the primary point of liaison with Operations (lockouts, etc.)
- ☐ Maintenance Planner/Scheduler who will be the primary point of coordination with all craft teams (internal as well as external

contractors)

- ☐ Engineers associated with specific capital jobs
- ☐ Other technical support personnel associated with safety, environment, quality, regulatory, etc.
- ☐ Supervisors of Execution Crews (internal and contractor) responsible to the Project Superintendent (all shifts to be covered)

Make sure that team members know their individual roles in the overall project. Many will likely require training regarding input to and use of project network information.

6. Job Planning

All planning steps discussed in previous chapters are equally important in regard to each job constituting the overall shutdown. Determine general manpower requirements segregated into in-house versus contract personnel. Of particular importance to the CPM Network are required duration, skill sets, crew size, labor-hours, precedent jobs, and job cost.

Depending on the risk associated with schedule delinquency, three estimates may be made (optimistic, probable, and pessimistic), especially for activities on the critical path. Studies show that pessimistic estimates best reflect actuality when the activity is performed the first time. As previously discussed, most job overruns result from preparatory short-falls. Therefore, provided that "lessons learned" are documented, probable estimates become reliable on future occurrences of each job.

Gather all PM/PdMs and backlog jobs that may be candidates for performance during the shutdown or while project work is being performed in a specific area of the facility. Look forward in the PM/PdM Schedule for tasks that might be accelerated while downtime and access are available.

Make provision for parts, materials, and equipment to arrive with sufficient time for quality checks prior to installation.

Establish acquisition activities for all specialized equipment required (cranes, man-lifts, scaffolds, compressors, welding machines, etc.).

Initiate required P.O.s and WOs for in-house work as well as contractor work, parts and materials, rental equipment, etc.

Assemble and review drawings, specifications, wiring diagrams, and other reference documents. Add them to appropriate Planned Job Packages.

Secure permits covering safety, fire, and regulatory requirements (local, state, and federal).

Secure a list of all contract workers and arrange for safety orientations.

Note: Not all the above preparation will be completed prior to steps listed below.

7. Project Team Workshop

This workshop is the first effort to draw the project network. The initial network is commonly developed on a wall of brown craft paper with index card and pins for each event connected by lines for each activity. The multi-function team cooperates to identify precedents (events from which activities emanate).

8. Load all Gathered Information into Project

Management Software

This effort is performed by the Critical Path Analyst with data entry support attached to the Project Team.

Available labor resource by skill plus work day/work week parameters must also be loaded to the software at this point.

- ☐ Build work quality into the project. Add Hold Points into the network. As significant installations are completed, work is held for quality inspection. Such inspections might include vibration readings, amp readings, or other Non-Destructive Tests. Holds are released with a sign-off by the Process Coordinator or other designated representative of the appropriate internal customer.
- ☐ The last aspect of quality is control. Quality control includes making random inspections, performing Non-Destructive Testing of welds, and job site supervision. The control aspects must be into the shutdown plan with scheduled activities like pressure testing, vibration checks, and amp readings.

9. Establish the Initial Computerized Network

Given required inputs, project management software is programmed to perform this task. However, the members of the Project Management Team must collectively review, troubleshoot, adjust, and refine the output as discussed in Step 10.

The initial network is shown in relative time (elapsed project days from the kick-off event — day zero). When the kick-off date is established and plugged in, the software automatically posts elapsed project days to each event (earliest and latest event times). See Chapter 21 for further understanding.

As the network is finalized, the software also presents the project in linear bar-chart format, with an associated resource demand chart by which to level peak demands, and a project cash flow. The Resource Demand Chart is extrapolated from the bar chart version of the network. It is used to level peak demands as feasible within limits of Float available off the critical path. The critical path is first identified assuming no resource restrictions. The software then seeks to shave peak demands by delaying activities that are off the critical path. Shaving can be done only to the limit of available float, so as not to change and lengthen the critical path.

Although the network diagram is the better tool for analysis, the bar chart is more readable for many people on the periphery of the project.

10. Present the Project Network to the Project Team

The Project Team continues to refine the Network. This is an opportunity to think the project through, on paper or computer. Find as many potential shortfalls now, not during live project execution.

Can all proposed jobs be accomplished without extending overall project duration, as shown in the Master Critical Path Network for this specific programmed turnaround? What if additional resources are added? Should an extended downtime window be requested or should some jobs be rejected in order to have an acceptable shutdown duration?

11. Develop Project Budget

From labor estimates, contractor requirements, material and parts estimates, and required equipment rentals, establish a comprehensive budget for the overall project. Justifications, appropriations, procurement contracts, and budgets for capital projects encompassed within the shut-

down were processed significantly prior to this point.

The overall shutdown/project budget is the accumulation of financial requirement to achieve each event throughout the network. These financial requirements are also attached to the network. Thereby, the network is also a financial control tool as the shutdown progresses. Head off adverse trends and/or management surprises.

12. Contractor Support

When selecting contractors, check that they can meet commitments. Contractors go to great length to make a gain work. Look at the depth of their organization. Would they become over committed if they were to win a string of proposals?

When good contractors are found, the Project Manager must be a fair customer and avoid negotiating deadlines, milestones, or budgets that are unreasonable. Reliable contractors are worthy of treatment as partners.

13. Present Project Network to Steering Committee and Key Stakeholders

Achieve understanding and concurrence. This is the appropriate time to explain why some requests had to be excluded from this particular shutdown.

Safety thrives when the Steering Committee requires that safety is to be discussed at every level and at every step throughout the shutdown.

14. Obtain Management Approval and Funding for the Project as Finalized

15. Prepare Unique Packets of Project Reference Documents for Each Individual Supervisor

☐ General Section:

- Turnaround Objectives
- Turnaround Organizational Chart
- Overall Turnaround Schedule

- General List of supervisor responsibilities to the overall project and helpful reminders
 - Progress report forms with instructions for their use
 - Instructions regarding contractor daily time reporting
 - List of telephone and beeper numbers for all personnel key to the Project
- ☐ Supervisor/Team Specific Section:
- Planned Job Packages including Work Orders of all jobs to be supervised
 - Daily Schedule excerpts from the overall Project Schedule

16. Begin Project Preparation in Compliance with the Project Network.

Build quality of work into shutdown plans. W.E. Deming (the foremost quality expert of the 20th century) preached that the greatest motivator for workers is “pride in a job well done.” Accordingly, a key to quality in shutdown work is to plan and manage work so that maintenance workers have the opportunity to feel such pride and job satisfaction. The shutdown plan must insure that they have the right tools, the right parts, etc., and are not pulled off jobs. Quality flows from removal of the barriers thereto!

Quality control also includes random inspections, non-destructive testing of welds, pressure testing, vibration checks, amp readings, and effective job site supervision.

Project planning must also address the four steps of Risk Management (risk identification, risk quantification, risk response, and risk vigilance) at each activity of the project network:

- ☐ Risk identification. Is there risk here; within or outside of team control?
- ☐ Risk quantification. What would be the impact on successful completion of the shutdown/project? What is the likelihood of the risky event happening?
- ☐ Risk response. What response would be necessary? How much will it cost? How likely is it that the response will eliminate the risk? Is the response likely to introduce its own risk? Can we

insurance against the risk in some manner?

☐ Risk Vigilance. How can the shutdown be organized and managed so when a risk becomes apparent it is immediately communicated with sufficient time for reaction and response?

Risk includes procurement and delivery problems, scope creep, unavailability of anticipated staffing or essential skills, poor estimates, poor communication and resultant misunderstandings, contractor deficiencies, weather, accidents, environmental upsets, design errors and omissions, etc.

Many shutdowns require 1000 or more parts and materials from dozens of vendors. The Project Logistics Coordinator supported by the Purchasing Agent/Buyer and the Stores/Receiving Representative are key players. They are responsible for shutdown logistics from ordering through tracking, shipping, expediting, delivery, receiving, stocking or storage, issuing, and delivery to the work site. All storage must be performed in a manner that will not degrade the associated items. Information systems (CMMIS or Materials Management System) must support effective execution of logistical responsibilities. Answers to numerous questions must be retrievable by item:

☐ Status

- Where is it?
- Has it arrived?
- Did it pass inspection?
- When will it arrive?
- Can you pick it up?
- Can you deliver it?
- Can you dispose of this?
- Did we get rid of it?
- Did we return it?

☐ Logistical support also encompasses provision for necessary:

- Toilets
- Safety Facilities (eye wash, decontamination, etc.)

- Services (electric, air, water)
- Storage (trailors)
- Transportation (personnel, materials, equipment)
- Waste Removal
- Welding Tanks

17. Begin Project Execution in Compliance with the Project Network

Execution of the project or shutdown begins. The first step is literally to shut down the plant! It seems simple, but executed properly this step is the key that makes the shutdown occur safely and smoothly. Considerations include:

- Must we be concerned about product in process?
 - Must materials in process be dumped?
 - What hazards must be addressed?
 - What must be decontaminated (pipes, valves, fittings, pumps)?
 - Steps to isolate the plant, area, or system(s)?
 - Run down, bleed or empty services (compressed air, feed gasses, water, capacitors).
 - Hand over to shutdown team
- ☐ Continuously monitor and post progress relative to the network plan. This enables the software to provide real-time alerts when events are delinquent or expenditures exceed budget. Identify slippage promptly. Focus on and protect the critical path, but keep an eye on the float limitations of other paths.
 - ☐ Take corrective actions and expedite as necessary. Add resources (overtime or additional contract support) to overcome shortfalls and recover to plan.
 - ☐ Track differences between what was planned and what is actually happening. This specifically includes tracking:
 - Adherence to start and finish dates for activities on or near

the critical path

- How actual costs are meeting estimated costs
- Whether actual resource utilization is meeting planned requirements

- ☐ Resist Scope Creep. It is not uncommon for work scope within shutdowns to increase by 10% or more after the shutdown plan has been finalized (approved, funded, and duration set). How many times has it been said, "While we are doing this, it would be easy to also do this other thing?" All requests for additional work must be approved by the Project/Shutdown Manager and be rigorously scrutinized. Some changes may not be avoidable. Sometimes in the middle of the shutdown the need for essential, additional work is discovered. Still, the overall project must be protected and the work force must be protected from disruptions. Such increases to scope cannot be avoided. Promptly communicate them to the Steering Committee. Protect management from surprises. Document these occurrences so their impact is evident during the Acceptance of additional work and the impact on cost and duration of the shutdown should be documented. Documentation of additions to scope and their impact will also facilitate post mortem of the shutdown. A Change Order Process should be in place to facilitate control of scope creep.
- ☐ Always Emphasize Safety. Shutdown work is inherently more dangerous than routine maintenance work.
 - People who may not be familiar with the plant (contractors)
 - Work performed infrequently (vessel entry)
 - Heavy work (construction)
 - Open and disassembled equipment
 - Unstable work site, with new hazards introduced every day
 - Induced hazards (cutting, burning, open lines, and tank entry)
 - Pressure to complete on time
 - Heavy overtime and fatigue
 - Many days without time off
 - Around the clock work

- Extreme danger right after start-up due to errors in design or execution of modifications and repairs

Think through all safety aspects of the shutdown at every step of the network.

- ☐ Provide for data collection that, due to hazards, can only be obtained during a major shutdown. Use this shutdown to begin preparation for the next shutdown. This might include wiring or identifying old spares that would give you an edge on the next repair on the unit or shutdown.

- Dimensional information
- Sketches of electrical wiring that is normally concealed
- Count of gear teeth (useful for vibration analysis)
- NDT on piping (after removal of insulation)
- NDT on vessels and tanks (after removal of insulation)
- Eddy current testing on tubes
- Identify and sketch parts for older equipment for which prints are no longer available
- Transformer oil samples that cannot be safely collected at any other time
- Sleeve and bearing wear

- ☐ Finally, the last step of every shutdown is start-up (returning the processes to operation). Like shutdown, start-up must be tailored to each plant:

- De-isolate systems.
- Return services to on-line status. Remove all blinds.
- Run plant function tests.
- Create punch list of problems indicated from function tests, inspections, and incomplete work.
- Complete punch list items that will have an impact on start up.
- Re-commission and re-introduce inventory to system in a structured way.

- Verify run up to full product output and quality.
- Stabilize full production.

18. Project Completion and Close-Out

- ☐ Cleanup Job Sites.
- ☐ Complete Punch Lists and Obtain End-User Acceptances.
- ☐ Start Up, perform Job Quality Assurance items, and Turn Over Impacted Assets back to Operational Control with required Life/Safety Testing.
- ☐ Deliver to appropriate library files, all “As Installed” drawings, and other contracted deliverables associated with capital projects (e.g., recommended PM/PdM Routines, BOMs, and Recommended Spare Parts to be carried in Inventory).
- ☐ Post Completion Review and Project Closeout Report. This is a debriefing to identify Lessons Learned in order preclude their repetition. The Report should be written as soon after completion as possible to document what happened, what didn’t happen, and how problems were addressed.

19. Project Review (Six Months after Completion)

- ☐ This step applies only to capital projects. Actual financials realized should be compared to projections upon which each project was justified. Failure to deliver expectations damages credibility, which appropriately influences approval of future capital requests.

SHUTDOWN METRICS

Shutdown success is measureable in a number of ways. Some common measures are:

- ☐ **Budget:** Budgeted spending to actual spending or budgeted spending for the total shutdown, portions thereof, and by account.
- ☐ **Duration:** How many days did the total shutdown last, com-

pared with the number of days scheduled? If specific dates are of importance, then how many days early or late was the plant brought back on-line?

- ☐ **Impact on Production:** Total days from last good production to first stable production of quality product.
- ☐ **Jobs Completed:** In terms of total scheduled labor-hours completed divided by total labor-hours completed.
- ☐ **Safety/ Environment:** The usual safety measure is lost time due to safety per million labor-hours. For an environmental metric, look at the number and type of incidents.

WHO SHOULD APPLY PROJECT MANAGEMENT TECHNIQUES TO MAJOR MAINTENANCE PROJECTS?

A single designated Project Manager is essential for all Major Shutdowns or Projects in any environment. Throughout the year, Senior Maintenance Planners usually handle minor shutdowns and small projects that are expensed. But the individual selected as Project Manager for a capital project or major turnaround for required annual maintenance is usually a middle-manager or an engineer. If internal maintenance resources are to be utilized, whoever is designated as Project Manager needs the support of a Senior Maintenance Planner. Required procedures and proper methods; required parts, materials, tools, and equipment; and access to the asset must be established and assembled into Planned Job Packages. These requirements don't come from a project manager or project engineer. They come from a Maintenance Planner-Scheduler.

However, the requirements of turnaround planning are too significant to load upon a planner already consumed by week-to-week demands. Although the annual turnaround may occur only once a year, the associated preparation workload is intense for at least four months. During that period, it is not practical for a single individual to concurrently plan and schedule normal weekly/daily efforts as well as major shutdowns—at least not effectively. One of the two responsibilities always suffers and planning falls significantly short of essential need.

Therefore, the Turnaround Planner-Scheduler is usually not the same person who prepares for the effective execution of maintenance work throughout the year. Planning and Scheduling for major shutdowns

is clearly a specialty. Planning for next year's turnaround begins with debriefing this year's turnaround, which is the identification of things that went wrong or could be improved upon. These items are immediately documented to avoid repetition in future years. The intense planning process begins at least four months prior to upcoming turnarounds. Even this amount of lead-time is not sufficient when the sources of major equipment and parts are located in Europe, Asia, or India.

Network Analysis - Development and Depiction

Major shutdowns and projects require the development and application of network diagrams to identify and convey interrelationships between the dozens, if not hundreds, of activities (in series and in parallel) comprising the overall effort.

Our intent here is only to provide an introduction to Network Analysis. This is not an attempt to train the reader in application of the tool using either of the two common approaches to network analysis (Critical Path Method Analysis—CPM; and Project Evaluation Review Technique—PERT). Several excellent publications offer a far more comprehensive coverage than we hope to provide here. We recommend that all planner/schedulers be exposed to an appropriate offering. Their preparatory thought process and analytical skills will be dramatically expanded by the exposure.

All discussion within this chapter uses terminology and techniques associated with CPM, which is the network method most commonly used by maintenance planner/schedulers. The popular software they use is *Microsoft Project*. Project Engineers and Project Managers responsible for major capital projects tend to use Primavera software, which is more comprehensive but more difficult to use.

When first exposed to network analysis, even experienced managers, engineers, planner/schedulers, supervisors, and technicians are amazed at opportunities to compress, optimize, and control major projects, identified by use of the network process. The more important it is to

complete shutdowns or projects in the shortest time period, the more essential application of network analysis becomes.

GENERAL DISCUSSION OF THE CRITICAL PATH TECHNIQUE

Elsewhere in this book, Planning has been characterized as “how to perform” and Scheduling as “when to perform.” CPM contains aspects of both planning and scheduling as well as coordinating, budgeting, and overall project management and control. It is a technique for showing relationships between all activities and events comprising a major effort or total project.

The overall network contains **events** (or milestones achieved) throughout the overall project and **activities** that must be completed before specific events can be achieved. The network also depicts events that must be achieved before certain activities can begin (e.g., a roof must be erected before it can be papered and shingled).

When all events and activities have been identified, an initial network of the overall project can be sketched. Early input from all stakeholders is critical to development of an all-inclusive and reliable network.

A planner, engineer, or project manager must then plan and estimate each identified activity by determining required skill, duration, crew size, and labor-hours. With that information, the project network can be developed further. Durations throughout the project can be expressed in any unit of time selected. Throughout this chapter, we will use days.

The most time consuming path leading to each event establishes the earliest day of the overall project that each event can be achieved (Earliest Event Time). By pursuing this process throughout the network, the earliest event time for the event that represents project completion is determined. This is the shortest period of time in which the project can be completed.

By working back through that path by which the shortest project time was derived, the Critical Path is identified. There is no Float on the Critical Path. All other paths have some amount of Float (slack-time, elbow room, leeway) at one or more points of the path.

The key to completing projects on schedule is to keep track of progress throughout the network, detect developing problems promptly, and take appropriate corrective actions. By monitoring the earliest start times of activities and earliest realization times of events, project man-

agers can remain abreast of overall project progress and take necessary action as soon as possible to protect Earliest Completion Time and stay within project budget. Accordingly, CPM facilitates project tracking by providing alarms when:

- ☐ Earliest Activity Start Times or Earliest Event Times are missed, whether they are on or off the Critical Path
- ☐ Available Float has been reduced
- ☐ Financial Commitment (Project to Date) exceeds budget projection

Example of a Completed CPM Network (Simple)

Figure 21.1 illustrates a completed CPM network. Event numbering, activity duration, earliest event times, critical path, and latest event time are all illustrated. Duration can be stated in any chosen unit of time. We will use "Project Days" throughout this discussion.

Critical Path Terms, Definitions, and Symbols

- ☐ An **Event** results from completion of an activity or group of activities culminating in a definitive and recognizable accomplishment (e.g., completion of site work or completion of foundation). Events are represented by circles or other shapes divided into quadrants. The bottom quadrant contains the event number. See Figure 21.1.
- ☐ An **Activity** is a specific action (task, job, work order) that must be performed to achieve a specific objective (an Event). One or more activities emanating from one or more predecessor events may be required to achieve a specific subsequent Event. Activities are represented by Arrows emanating from a predecessor Tail Event and concluding in a Head Event. The nomenclature for the first activity of the network shown is *Activity 0-1*. Activity Duration is posted above the Activity Arrow (10 days). Event 0 is the tail event and Event 1 is the head event.

Each activity has a defined beginning and ending. Activity 1-2 can begin as soon as Event 1 is achieved. It must be completed before Event 2 is achieved. All activities must originate from an

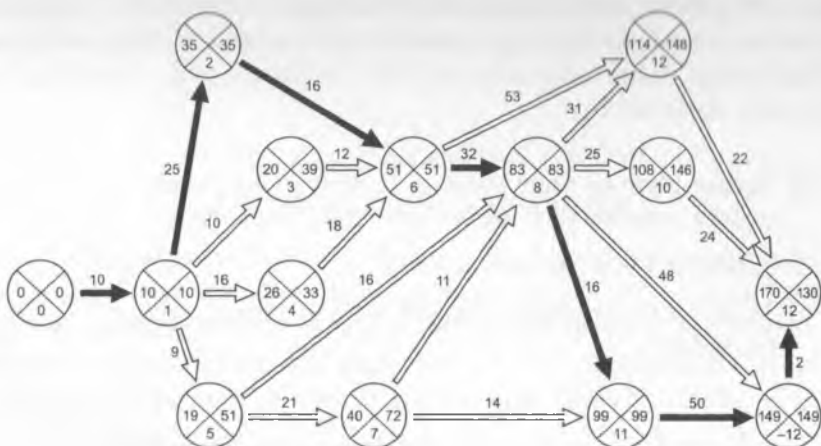


Figure 21.1 Critical Path Schedule

event. They cannot originate from the middle of another activity. When this situation seems to occur, most likely the envisioned activity is really two activities with an additional event separating them. The better activities are defined, the easier they are to manage.

Each activity should represent only one resource—some portion (one or more individuals) of a specific crew responsible to one supervisor).

- **Earliest Event Time (ET)** is the earliest day of the project that all activities leading into the event can be completed. All activities leading into the same event must be checked in order to determine the latest to complete. Add activity duration to Earliest Event Time of the preceding event (Tail Event) to determine the earliest day of the project that each activity can be completed. The greatest result determines Earliest Event Time for the event in question. Work the network left to right, in numerical sequence. Check Event 2 before Event 3, before Event 4, etc. We will use Event 6 to clarify this explanation:

Activity	Tail Event ET	Duration	Project Day of Completion	Head Event ET
2-6	35	+	16 =	51
3-6	20	+	12 =	32
4-6	26	+	18 =	44

- ❑ Critical Path is determined by calculating Earliest Event Times for all events of the network. The Earliest Event Time for the event that reflects completion of the project is also Total Duration of the overall project. See Event 17. Total duration of the project is 170 days. Activities constituting the longest path through the project are the Critical Path (shown by solid arrows). Slippage in any activity on the Critical Path will result in Project Completion being delayed; because there is no Float along the Critical Path.
- ❑ Latest Event Time (LT) is the latest point (day) of the project that an event can be achieved without extending duration and completion of the overall project. Latest Event Times are determined by working the network backward (right to left) in much the same fashion as it was worked forward (left to right) to determine Earliest Event Times. There is no need to calculate LT for events on the critical path because there is no float. Therefore ET and LT are equal. Just post the same number in the right-hand quadrant.

All LTs off the critical path must be calculated. Work in descending order of event numbers. Again check all paths back to each event. Subtract activity duration from the Latest Event Time of the succeeding event. The smallest result determines Latest Event Time. If the result for an activity yielding a larger number were to be posted, there would be at least one path through which completion could not be reached on schedule. Project duration would be extended. We will use Event 7 for clarification:

Activity	Head Event LT	Duration	Project Day of Completion	Tail Event ET
7-8	83	- 11 =	72	72
7-11	99	- 14 =	85	

- **Float** is the amount of delay that can be absorbed at each event before it adversely impacts subsequent activities, events, and possibly project completion date. CPM defines three forms of float, allowing project managers and planners to quickly assess the impact of project delays:
- **Total Float** is the amount of delay in activity start or increase in activity duration that can be absorbed without delaying project completion.
- **Free Float** is the amount of delay in activity start or increase in activity duration that can be absorbed without impinging upon the timely start of any immediate successor activity. This would occur if the delayed activity caused Earliest Event Time to be missed.
- **Independent Float** is the amount of delay in activity start or the increase in activity duration that can be absorbed without diminishing float available to any other activity (predecessor or successor).

Again, there is no Float on the Critical Path. Earliest and Latest Event Times are equal for all Events on the Critical Path.

- **Dangling Events and Dummy Activities**— A Dangling Event is one that hangs out alone and does not close to the final event (indirectly or directly). This is unacceptable. All events and activities must have connection to the Final Event. If there is no real work remaining to be performed, connect the dangling event (directly or indirectly) to the final event by inserting a dummy activity. Dummy Activities have zero duration since they represent no true work.

Network to Bar Chart Conversion and Leveling of Labor Requirements (Simple)

Figure 21.2 illustrates both techniques. Although required staffing in the example is minimal, the concepts are conveyed. It is assumed that each of the activities require only a one-person crew with multiple skills (all activities can be performed by a multi-skilled technician).

The Work Force Leveling Chart shows two periods that require a maximum work force of five technicians. However, scrutiny of the Bar Chart Schedule indicates the first peak can be shaved to four technicians by delaying the start of Activity 3-6 for twelve days. Likewise, the second peak can be shaved by delaying Activity 8-10 for twenty-one days. Neither action delays scheduled completion of the overall project.

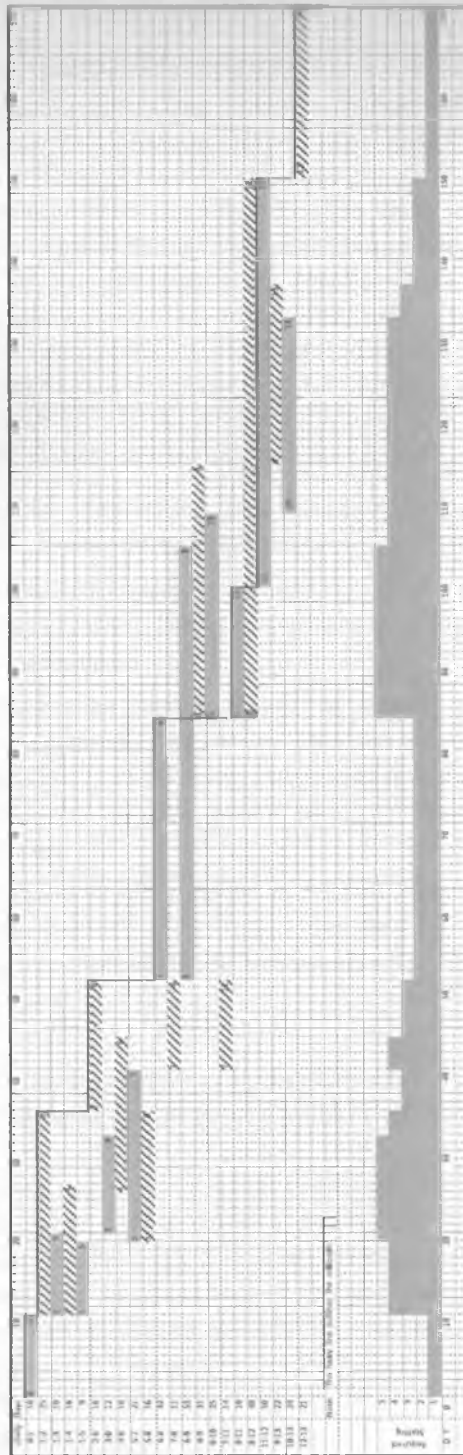


Figure 21.2 Compinted Bar Chart Schedule and work Force Leveling Chart

Table of Appendices

Title	Page
A. Job Description – Manager of Maintenance Support Services	253
B. Recognizing the Pitfalls in Planning for Others	255
C. Job Description – Maintenance Planner/Scheduler	259
D. Labor Library – Example	265
E. Alternate Approach to Scheduling	267
F. Typical Sources of Planned Maintenance Work	275
G. Planner Activity Log Sheet	277
H. Determination Worksheet Ratio of Technicians to Planner/Schedulers	279
I. Assessment – Maintenance Organization	281
J. Assessment – Maintenance Computer Information Support	287
K. Assessment – Maintenance Work/Job Planning	289
L. Assessment – Maintenance Work Measurement	295
M. Assessment – Maintenance Material Support & Control	297
N. Assessment – Maintenance Coordination & Scheduling	303
O. Assessment Summary & Comparison to Benchmarks	309

1000
1000

APPENDIX A

POSITION DESCRIPTION

Title: Manager of Maintenance Support Services

Often Maintenance Managers have a line versus staff orientation. Very likely they were once Master Mechanics. Their strength is still at the job site. Others are more oriented to work preparation and administration. Small organizations cannot justify a position to manage the support staff. Larger organizations can.

Reports To: Maintenance Manager

Position Scope: Planning and Scheduling, Maintenance Engineering, Administrative Support, and Maintenance Purchasing & Stores (if those functions are in the maintenance organization).

Responsibilities and Duties: The Manager of Maintenance Support Services is the primary:

- Champion and Chief Advocate of the Maintenance Excellence process
- Marketer and Seller of Maintenance Control throughout the organization
- As manager of all maintenance support services (staff not line), they are is the Chief Planner/Scheduler, chief of maintenance engineering, CMMIS administrator and head librarian for maintenance and planning technical libraries
- Owner and Controller of the CMMIS and as such:
 - Controls system security
 - Steers and continuously seeks system improvements (coding, user friendliness, application and reporting)
 - Trainer of all parties regarding their responsibilities to the system and their usage thereof
 - Enforcer of proper system input and usage by confrontation of the abusers

- Leader of Data Base, Job Plan, and Job Estimate refinement
- Quality Assurance & Control leader of the CMMIS, Planning, and Scheduling, and RCM (PPM and Root Cause Analysis) processes
- Analyzer of system information and trends of improvement and maintenance metrics including the balance of resources with workload
- Develops and carries forth the consensus issues and recommendations of the Maintenance Support Services Team
- May be responsible for the maintenance store room and for purchasing
- Executes special studies requested by management or self generated

APPENDIX B

RECOGNIZING THE PITFALLS IN PLANNING FOR OTHERS

A. Problems

A.1 Competitive Feelings

Two or three groups plan maintenance jobs (supervisors, technicians and planners). Each thinks their own efforts are the best. The thoughts go as follows: If we execute their plans well, they gain/we lose. If they look good, we look bad by comparison. Such feelings may produce reactions that out right sabotage planner efforts and the transition from a reactive to proactive environment.

A.2 Communication Problems

It is believed there is more room for error, distortion, and misdirection when plans formulated by one group need to be transmitted in their entirety to another group for execution .

A.3 Less Understanding

Doers report greater difficulty in understanding the plans that others assigned to them than with their own plans.

A.4 Less Flexibility

A high percentage of doers think that their own plans are more flexible than plans assigned to them.

A.5 Poor Use of Available Manpower

Doers think that their own plans make better utilization of available manpower.

A.6 Less Commitment

Some believe that persons not involved in planning for their own jobs are less committed to seeing that job planning yields positive results.

A.7 Less Effort to Make the Plan Work

Most maintenance technicians believe they can best plan their own work. This conviction is acted out by expending great effort to ensure that their belief is validated by unfavorable outcome of planner efforts.

A.8 Less Sense of Accomplishment

There is less sense of achievement when others define job plans.

B. Possible Remedial Actions

B.1 Minimize Competition

Planners brief supervisors, supervisors brief mechanics, make the assignment and convey planned job packages to the mechanics. Supervisors and planners must avoid creating a situation in which technicians see themselves in a zero-sum game. If the plans succeed, the doers must not lose status, prestige, power, or material benefits. They must experience the benefits of effective planning. Conditions must be established in which supervisors, technicians and planners share common goals. The division of labor should be seen as benefiting each of the three partners.

B.2 Improve Communications

Maintenance technicians must have and believe they have ample opportunity to question supervisors for clarification of plans. Supervisors and planners need to assess whether or not they are overestimating or underestimating the ability of others to comprehend plans and judge whether they are transmitting too much or too little, with or without adequate documentation and visuals.

B.3 Promote Understanding

Understanding of the plan can be fostered by ensuring that the plan itself has been created in a way to minimize its ambiguities. Repeating instructions may increase reliability of understanding. Ask the receiver to paraphrase the instructions back. Promote feedback. Use the Planned Job Survey (presented here in the appendix). Instructions need to be simple enough to be understood by the least capable doer. If he can understand the stages of the plan, all others can also understand it.

B.4 Explain the Importance

Tactical decisions must be made based upon conditions encountered. However, those decisions are to be made by the supervisor. Technicians should be encouraged to recommend essential changes to the supervisor, but not to make them unilaterally. The safety, environmental, and customer communication necessities of this view must be communicated for understanding.

B.5 Maximize Effective Use of Available Manpower

Utilize the scheduling format presented in Chapter 14. There is a thorough approach to scheduling.

B.6 Promote Commitment

Doers can be consulted at various stages in the planning process. Wherever possible, the ideas of the doers can be incorporated in the plans; or when

such ideas are unusable, the reasons can be discussed with doers. Also their feedback is reflected in refined job packages, thereby increasing ownership.

11.7 Promote Confirmatory Behavior

Planners and supervisors need to communicate confidence in accuracy and expectation of plans together with reasons for their confidence, while technician reservations must be openly discussed and resolved. Although the plans may be sound, if the technicians are left to believe the plans are unrealistic, they may behave in a way to confirm their own doubts. Proper supervision should assure that no negative actions are allowed to persist.

11.8 Foster a Sense of Accomplishment

Plot, post and communicate the trends against baselines and goals. Recognize the small improvements. It won't take long before they add up to major improvement. Give credit to the technicians. Performance of individual members working as a group improves the most when they receive constructive information about their individual efforts as well as the group's success as a whole, particularly if the expectations are challenging. Equally useful is meaningful feedback from a member of the team that will improve the future effectiveness of the entire team.



APPENDIX C

JOB DESCRIPTION

Title: Maintenance Planner/Scheduler

The job of Planner/Scheduler is critical to effective maintenance. It is essential that the positions be staffed by personnel with required aptitudes and temperament. Planners should be assigned 100% to their function. Planners can be used to back up maintenance supervisors when absent for one reason or another. However, someone in turn must back up the planner. Leaving the planner role uncovered must not be condoned. It conveys lack of commitment to the function and implies the function is not essential (and in the next downsizing can be done without).

Reports to: Maintenance Manager

Position Scope: The primary role of the maintenance planner is to improve work force productivity and work quality by anticipating and eliminating potential delays through planning and coordination of labor, parts and material, tools and equipment, permissions, specialized documentation and equipment access.

Planner/schedulers are responsible for the planning and scheduling of all maintenance work performed in the area to which they are assigned. They maintain liaison and coordination between the production and maintenance organizations; and maintain appropriate records and files to permit meaningful analysis and reporting of results or work done.

Responsibilities and Duties: A planner/scheduler has the following duties:

1. Is the principal contact and liaison person between the maintenance department and the plant departments served by maintenance and ensures that the assigned production areas or functions receive prompt, efficient, and quality service from the maintenance function, and ensures the maintenance function is given the opportunity to provide this service.
2. Is responsible for long-range as well as short-range planning. Long-range planning involves the regular analysis of backlog relative to available resources. These two basic variables must be kept in balance if a proactive maintenance environment is to be established and sustained.
3. The planner receives all work orders from the requesting departments of the assigned areas, except for emergency work that is requested of the

appropriate maintenance supervisor for immediate attention.

4. Makes any additional sketches diagrams, etc., necessary to clarify the intent of the work order.
5. Reviews and screens each work order to see that it is has been properly filled out:
 - Requested Work is clearly described
 - Check if the priority and requested completion date are realistic and provide practical lead-time
 - Charge numbers and other coding are complete and accurate
 - Authorization is proper
 - Discusses the details with the originating department as appropriate
6. Assures that work requested is needed. If need is questioned and not readily resolved with production or requesting personnel, refers the work order to the maintenance manager.
7. Reviews with engineering those work orders requiring engineering design.
8. Maintenance work orders which can be planned out but which require participation by shop or other crews are copied (cross-order) and provided to the appropriate planner for planning of the supplemental work.
9. Examines jobs to be done and determines scope and best way to accomplish the work. Consults with requester or maintenance supervisor when necessary.
10. Obtains blueprints, drawings, instructional manuals and special procedures, as needed, from files or other sources. Makes any additional sketches, diagrams, etc., necessary to clarify the intent of the work order.
11. Identifies and obtains (requisitions, orders, kits as appropriate and in keeping with procedures) determinable materials, entering material needs on the work order. Determines if critical items are in stock by verifying availability with stores and reserving same.
12. Ensures the safety needs are given a top priority in work planning and scheduling.
13. Estimates jobs, showing sequence of steps, the number of mechanics, and required man-hours for each step.
14. Estimates the cost of each work order in terms of direct labor, materials required, and total cost.
15. Maintains backlog files of work orders awaiting scheduling in accordance with their priority and requested completion date. Those jobs unplanned, requiring engineering, waiting for materials, waiting for down-time, etc., are filed accordingly. When ready for scheduling, work orders are filed by supervisor by required completion day. Ideally, this filing is accomplished within a computerized system.

16. Once a job is planned and estimated, prior to scheduling, verifies the availability of parts, materials and special tools required for its execution.
17. Planner should have complete knowledge of each department's PM workload in order to better schedule work orders in that area.
18. Reviews schedule status and forecast of manpower availability on a regular basis.
19. Develops a maintenance work schedule for the maintenance area supervisor. From the backlog files for each crew, selects a group of work orders with manpower requirements matching the capability of the identified work forces, taking into account any work carry over from jobs previously scheduled. Identifies any special skill requirements and makes such arrangements with the responsible planners.
20. Allocates manpower and coordinates these requirements through maintenance supervisors and the maintenance manager.
21. In the selection of jobs for scheduling, meeting the deadlines established by the requesting department and maintaining preventive maintenance schedules is essential. If any work orders cannot be scheduled within the time expectation, the requesting department management and requester are promptly notified so that appropriate action can be taken to get the work done in a satisfactory and timely manner.
22. Attends meetings with the production planning department and participates in the overall plant scheduling of the following week's work, and negotiates for downtime "windows" during which preventive and corrective maintenance requiring downtime can be performed. Finalizes own schedules for which they are responsible, ensuring that the work scheduled balances with the man-hours available so that a full day's work is provided each person.
23. Recommends equipment to be included in preventive maintenance programs.
24. Schedules preventive maintenance and other planned work in coordination with production and maintenance supervisors.
25. On the basis of firm work schedules, coordinates requisition of all predetermined parts, materials, and special tools, and ensures that equipment to be worked on will be available and ready. Arranges for any safety inspection, fire and standby watch.
26. Issues approved schedules together with relevant work orders and other planning documents to area supervisors. Discusses "planning packages" as necessary with special instructions or considerations to be observed in the execution of the jobs and reviews new jobs coming up in the future. (All work orders except emergencies come through the planner.)

27. Follows up to ensure the completed schedules and work orders are returned at the proper time. Carefully reviews completed schedules and corresponding work orders returned by the maintenance area supervisors monitors work order progress, and prepares associated reports .
28. In accordance with standard practice, provides the clerks with all documents for reporting or closeout.
29. Remains abreast of accumulated cost to standing work orders versus budget.
30. Promotes the conservation of energy.
31. Maintains close contact with other planners to ensure coordination of complex multi-skill field and shop jobs.
32. Schedules weekly meetings with production and maintenance supervisors concerned with the areas for which they are responsible, consulting them regarding facilities or equipment to be maintained. Makes recommendations to production concerning long-range maintenance needs and, in collaboration with production, prepares a weekly forecast of all jobs expected to be scheduled for the following week.
33. Maintains a list of planned work orders requiring equipment to be down, so that some or all can be performed in the event of an unscheduled. These lists, by asset, are reviewed and updated weekly.
34. Develops a file of standard work orders (plans) for regularly recurring repair jobs, based on historical experience, to simplify the planning process.
35. Reviews with the maintenance area supervisors the actual labor expended versus estimated labor and material used for completed jobs, in order to determine corrective measures needed to improve the accuracy of estimating and improving methods of doing work.
36. Job Estimates are continually refined and thereby reflect improving accuracy and consistency; representing fair as well as challenging expectancies.
37. Assists maintenance and production management in periodically analyzing costs and, where necessary, recommends corrective action needed to reduce maintenance costs.
38. Keeps the maintenance manager properly informed on all abnormal or critical situations and seeks advice on matters outside of the planner's knowledge or authority.
39. Make recommendations for system improvement.
40. Maintains necessary records and files and prepares and distributes meaningful and accurate control reports.
41. Performs other tasks and special assignments as requested by the maintenance manager.

42. Develops and maintains planner reference systems (library) including a file of Planned Job Packages for recurring jobs, plus labor and material libraries for each equipment center.
43. Recommend additions to authorized stock.

Measurement of Position Performance

Planner/schedulers are measured in the performance of their duties and execution of assigned responsibilities by:

1. Labor effectiveness of the area to which assigned as measured by work sampling.
2. Orderly placement of work orders on the schedule(s) and their completion within the specified time frame.
3. Jobs worked and completed per schedule.
4. Improvement in mean time between equipment failures.
5. The accuracy of estimates of labor and material.
6. The improved use of labor and improved plant condition as expressed by a reduction in emergencies, overtime hours worked and in unscheduled labor worked, and reduction of contractor support.
7. The timely and accurate preparation and distribution of meaningful control reports.

Position Goals

1. To ensure that production areas or functions that they serve receive prompt, efficient, and quality service from the maintenance function enabling them to operate at a high level of efficiency.
2. To ensure the maintenance function is given the opportunity to provide production with the service it requires.
3. Accurately define and estimate work requests.
4. Properly prepare and distribute meaningful control reports.
5. To meet the needs of customers, both internally and externally. Our future is in the hands of our customers. We will be courteous, helpful, and responsive to them always. We want our people to be focused on our/their customers (internal and external).

Relationships

1. Reports to the maintenance manager.

2. Works closely with production supervision.
3. Works closely with maintenance supervisors.
4. Works closely with stores and purchasing personnel.
5. Maintains good working relationships with other organizational units in the plant.

Requirements, Qualifications, and Selection Criteria

1. Mechanical/electrical background necessary and tech school background desired.
2. Adequate craft knowledge to estimate labor hours and materials and to visualize the job to be performed.
3. Good oral/written communication skills and possession of tact.
4. Good administrative and mathematical skills with willingness to handle paperwork.
5. Have or able to acquire a working knowledge of personal computers in a reasonable training period (typing skills helpful)
6. Good planning and organizational skills.
7. Ability to understand what constitutes good instructions.
8. Able to read blueprints and shop drawings.
9. Sketching ability.
10. Understanding of the proper use of work orders, priorities, scheduling, etc.
11. Ability to keep multiple jobs in controlled motion—simultaneously.
12. Ability to bring about order—from chaos.
13. Orientation and commitment to customer service.
14. Style and capability commanding respect within both maintenance and operating organizations

Exclusions from Planner responsibilities are:

- Involvement in daily emergency and urgent requests. There is no opportunity to plan such work. If any planning is to be accomplished, the Planner must focus on tomorrow and beyond.
- Daily assignment of individual mechanics to specific jobs. However, the Planner should develop the recommended manpower deployment plan for assignment of individuals to specific jobs—this is a preparatory exercise.
- Maintenance Engineering including development of the PM/PdM system, analysis of Equipment History and re-engineering points of repetitive equipment failure.

APPENDIX D

Labor Library

Job Data				Labor/Material Library									
Job No. Type 01				Desc: GROTNES RIM ROLL 600									
W/I	Div	Line	Type Work								Status		
Job													
Req No	Task Sequence		Labor Estimates										
				MR M/HR		TK M/HR		PF M/HR		EL M/HR		SF M/HR	M/HR
0.10	Remove Top Hydromotor		2	2.00									
0.20	Remove Top Roll Shaft		2	5.00	1	0.50							
0.30	Remove Top Front Trunion		2	2.00									
0.40	Remove Lower Hydromotor		2	2.00				1	0.50				
0.50	Remove Lower Roll Shaft		2	5.00	1	0.50							
0.60	Remove Front Trunion		2	2.00									
0.70	Remove Flotork		2	1.00				1	0.50				
0.80	Remove Main Hyd Pump		1	1.00				1	0.50				
0.90	Remove Small Hyd Pump		1	1.00				1	0.50				
1.00	Remove Main Drive Motor		2	3.00	1	1.00				1	0.80		
1.10	Remove Rim Loader		2	2.00				1	0.80	1	0.50		
1.20	Change Bearings in Crank		2	16.00									
1.30	Remove Loader Control Arm		1	1.50									
1.40	Remove Small Drive Motor		2	2.00						1	0.50		
1.50	Remove Pitman Arm		1	0.60									
1.60	Remove Guide Roll Assem.		2	3.00				1	0.50				
1.70	Replace Guide Roll Assem.		2	3.00				1	0.50				
1.80	Replace Pitman Arm		1	0.80									
1.90	Replace Small Drive Motor		2	3.00						1	1.00		
2.00	Replace Loader Control Arm		1	1.50									
2.10	Replace Rim Loader		2	4.00				1	0.70	1	0.80		
2.20	Replace Main Drive Motor		2	3.00	1	1.00				1	1.00		
2.30	Replace Small Hyd Pump		1	1.00				1	0.70				
2.40	Replace Main Hyd Pump		2	1.50				1	0.70				
2.50	Replace Rotac		2	1.50				1	1.00				
2.60	Replace Ft Lower Trunion		2	2.00									
2.70	Replace Lower Roll Shaft		2	6.00	1	0.60							
2.80	Replace Lower Hydromotor		2	2.00				1	0.50				
2.90	Replace Top Ft Trunion		2	2.00									
3.00	Replace Top Roll Shaft		2	6.00	1	0.50							
3.10	Replace Top Hydromotor		2	2.00				1	0.50				
SEQ			Material Estimates										
0.1001		0.001	Vickers Hydromotor							45M110A-11C-10			
0.2001	334663	0.001	GR-44 Roll Shaft							SAME			
0.2002	509055	0.002	Torrington Roller Bearing							HJ-12415448			
0.2003	334616	0.002	Gromes Race							A-3508-107			
0.2004	508604	0.002	Timken Cone							HM-926747			
0.2005	508602	0.002	Timken Cup							HM-926710			
0.2006	507506	0.004	Garlock Klezure Seal							53X3548			
0.2007	334646	0.001	Sleeve							A-3508-106			
0.2008	508045	0.001	Timken Lock Nut							TAN-124			
0.2009	508043	0.001	Timken Lock Washer							TW-24			
0.201	525027	0.001	Sier Bath Cpling 2 In Bore							1/2 In Key			
0.3001		0.001	Front Upper Trunion							C-3508-120			
0.3002		0.002	Trunion Sleeve							A-3508-170			
0.4001		0.001	Vickers Hydromotor							45M110A-11C-10			

Note: These estimates account only for Direct Work

ALTERNATIVE APPROACH TO SCHEDULING

About the Contributor

Pete Little is a highly regarded consultant and trainer who has served the maintenance population well for nearly fifty years. His approach when initiating Coordination and Scheduling differs from that espoused within this text. He has authorized us to share his thoughts with readers of this book. Some organizations may find that Pete's approach fits their local environment.

Pete begins with a before and after description of a Weekly Coordination Meeting. He then presents some Scheduling Precepts and a "walk before running" approach to Weekly Scheduling.

Maintenance Scheduling

by Pete L. Little, PE

The following two Weekly Coordination Meetings took place at the same manufacturing plant that employs about 100 maintenance technicians. The organization was endeavoring to activate Weekly Maintenance Scheduling in a historically Reactive Maintenance Environment. The scenarios are frequently encountered.

Initial Experience Chaos

At 9:50 Thursday morning people began drifting into a mandatory weekly maintenance coordination meeting to begin at 10:00 AM. The meeting actually began at 10:12. Seven department managers, five maintenance supervisors, three maintenance planners, the Plant Engineer, and the Plant Manager were in attendance. A maintenance planner passed around copies of a fourteen-page list of all work orders for all departments to everyone in the room, sorted by craft, and within craft by department.

The Inspection Department Manager asked if he could go first. He started thumbing through the work order list, identifying work orders he wanted to be included on the following week's schedule. A number of his requests generated planner responses such as:

- ☐ "That one is waiting for materials".
- ☐ That one requires downtime - will you have #3 down next week?"

Many of the work orders required significant discussion, which were of little interest to other attendees.

At 10:31 the Inspection Department Manager was through. One down—six to go. The Plant Manager excused himself to make an “important call”. The Processing Department Manager left five minutes later.

You get the picture. The meeting ended at 11:46, with an over-ambitious schedule requiring 125% of maintenance technician hours to be available during the schedule week. In actuality, only 15% the schedule work was accomplished.

Not a very assuring performance. No way to build belief throughout the organization that the new coordination and scheduling process will generate reliable production output.

Now, jump ahead five months. A Maintenance Coordinator position had been established within each production department. Maintenance Supervisors still scheduled their own crews based upon priorities established in the Weekly Coordination Meeting.

Controlled Coordination Meeting

At 10:00AM sharp, a senior supervisor functioning as the Maintenance Coordinator for Processing Department entered the Plant Engineer's office. He brought with him his copy of the work order backlog for specifically related to Processing. It was sorted first by status—Ready, Downtime Required, Waiting on Materials—and within status by priority and age. He had received the backlog list Wednesday afternoon, and had already gone over it with the Process Manager. He had highlighted the jobs that the department really needed if at all possible, and those which would be “nice to have.”

Waiting for him were the Plant Engineer and the three craft planners, each with a copy of the same backlog list. The Coordinator began identifying “Ready” work orders that the Maintenance Department had identified as its “must haves.” Most had been generated from preventive maintenance inspections and would cause downtime or other serious problems. From the two lists (Processing and Maintenance), the three planners each noted those work orders to be performed by the craft he had plans for; starting with work orders to be performed over the coming weekend, followed by those to be performed during the following week.

The Processing Coordinator left the meeting and the Inspection Department Coordinator joined the meeting at 10:15. The process was repeated until all seven production departments had met with the Maintenance staff. The entire process took about ninety minutes. The resulting schedules contained estimated hours equivalent to 71% of weekly straight time hours for all maintenance technicians on all shifts.

By the end of the schedule week, 91% of those scheduled hours had been completed. The organization had a coordinating/scheduling process that generated reliability.

How did this plant transition from chaos to control? Described below is the approach I led the organization through. The scene is the Maintenance Department conference room, Wednesday afternoon. I played the managers roll, but he was present together with all Maintenance Supervisors.

Speaker	Statement/Question/Response
Mngr.	“Good afternoon. We need to have a heart-to-heart discussion. We are going to do things a little differently from what you're used to, and my purpose for this meeting is to explain what those differences will be. I am going to start by describing our new approach to scheduling, which will be the responsibility of each of you for your respective team. Before you leave work Friday afternoon, I want each of you to give me, in writing, your schedule of work orders you will do next week—Monday through Friday, not on the weekend. On the schedule, I want you to show work order number, job description, department, machine number if any, craft, crew size, and total labor-hours for each job. You will not impress me or gain points if you put a large number of work orders on the schedule; and you will not earn my displeasure if you put very few work orders on the schedule—even if you put only one work order on it. It's your call. Remember: get it to me in writing by Friday afternoon. On Friday of next week, I will meet with each of you individually to see which work orders on your schedule you completed. Did I forget anything? Oh yeah, one more thing: if you don't complete all the jobs on your schedule, you are fired!” Any Questions? Well, let me ask you some questions.”

Mngr.	"John (Supv. #1), how many jobs do you think you will put on your schedule?"
John	"Uh, I guess about five."
Mngr.	"Really? How about you Mary (Supv. #2)?"
Mary	"One."
Mngr.	"Good for you, Mary! John, if your job is in jeopardy, why would you schedule more than one job?"
Mngr.	"Mary, what will be the characteristics of the job are you going to put on your schedule?"
Mary	"Well, it will be one I am sure I can do: I already have the parts, and if the guy I'm counting on to do it is out sick, it will be a job someone else can do."
Mngr.	"That's the right answer."
Mngr.	"Bill (Supv. #3), 'when are you going to start on your scheduled jobs?'"
Bill	"Make that 'job'—singular. I'll start on it Monday morning. And if I'm covered up Monday, I'll do it Tuesday."
Mngr.	"In other words, you're not going to take a chance by waiting until the last minute. Good!"
Mngr.	"Chris (Supv. #4), what if the vendor promises your parts will be here by Wednesday—will you schedule it?"
Chris	"No way!"
Mngr.	"How about if it requires downtime?"
Chris	"No way!"
Mngr.	"You people are getting the hang of this! Understand that your schedule is not the only work you will do next week. You'll do all the things you normally do, but you WILL do your scheduled jobs—er, job. By the way, DO put PM inspections on your schedule as long as you know you can do them."
Mngr.	"OK, now let me change signals on you: you will not be fired if you don't complete your schedule. But I wanted to make the point that your schedule is that important, and you should treat it as that important. And even though I won't fire you for not completing your schedule, I'll tell you that if you can't complete a one-job schedule, you won't rate very high on my list of good supervisors. Here is why I'm doing this: I want you to learn how to schedule, and complete your schedule. I know that up to now your schedule has mostly been a wasted effort—many jobs, few completions. Your credibility with Operations is nil. We have to change that."

Mngr.	"Next week, after you've completed this schedule, I want you to come up with a new schedule for the following week. Add another job or two. Again, you won't impress me with big numbers. You'll keep adding a job or two each week, always putting jobs you know you can complete on the schedule."
Mngr.	"Eventually, as you keep adding jobs, you won't be able to complete them all. When that happens we'll investigate the reasons why. Once we identify a reason, we can do something about it. In that way, we'll gradually increase our schedule, complete our PMs, reduce emergencies and downtime, and gain the efficiency of working on well-planned jobs."

Precepts

Work to perform jobs that have been planned, coordinated, and scheduled jobs is far more effective than work to perform unplanned and emergency jobs. The more that jobs are properly prepared, the more that craft capacity is generated for the performance of pro-active work (PM/PdM and Corrective Maintenance) that: reduces failures and associated downtime, increases effective capacity, and improves quality, as well as credibility of the Maintenance Department.

1. Organization

Maintenance organizational structure has a significant bearing on scheduling success. A Functional Maintenance Organization features a Scheduled Crew for Backlog Relief, a PM/PdM Crew which performs only inspections, and an Emergency Crew which handles all legitimate emergencies and non-emergencies that cannot wait to be scheduled and, in so doing, shields the other two crews from interruption.

2. Planning

Effective scheduling is dependent upon thorough planning, assuring that all materials are on hand before a job is marked "ready to schedule," identifying special tool requirements, and realistically estimating labor-hour demand on each craft is provided.

Planners don't come into the world knowing how to plan. Being a good technician does not automatically make a good planner. Use the teaching opportunities of "start small and grow" scheduling to improve planner capabilities.

Interruptions are the bane of maintenance effectiveness. When Maintenance pulls off scheduled jobs to handle emergencies, it is necessary to secure the area, parts, and tools; add additional round trips; and upon return— get

back into the diagnostic process to figure out where the interrupted job was at.

Other major sources of interruptions include return trips for parts or special tools, and finding and conferring with the supervisor or requestor. Good planning reduces most of these interruptions. A designated Emergency Response Team, practically eliminates interruptions.

3. Coordination

It makes no sense whatsoever to hold a weekly coordination meeting lasting almost two hours, which ties up numerous key people, and results in very limited scheduled work being completed the following week. Talk about wasted effort!

Secondly, the idea to "schedule a lot to get more done" is ludicrous. It requires a lot of work for little gain. Additionally, completing almost none of the schedule destroys whatever credibility the Maintenance Department has and it makes it virtually impossible to remain abreast of status for specific work orders.

4. Scheduling

However, effective Scheduling is extremely important. The secret is to: Start small and grow over time — crawling before walking, before running.

It is far better to have a short schedule and complete most of it than to have a long schedule and complete almost none of it.

Start with a few well-planned jobs, that you know you can complete, and give them highest priority except for true emergency work—then do them all!

When preparing the schedule, don't try to specify who to assign to each job, nor when each job will be performed. That level of scheduling can come after becoming more proficient.

Don't schedule jobs waiting on materials or requiring downtime. You can do such jobs if possible, just don't schedule them. The schedule is a commitment to your customers and yourself. Honor that commitment by always doing all the work on the schedule.

Add a few hours to the schedule each week. Initially, do not involve Operations—you pick the jobs while you're just crawling.

As the schedule grows, begin asking your customers, "Is there any single job you'd really like to have us do next week?" Don't let them load you up. Explain that you're still learning to walk. As you add their jobs and complete them, your credibility will grow.

When only one job is scheduled, the goal for percent schedule completion has to be 100%. Once the pointy is reached where the full schedule cannot be completed, the goal becomes and remains 90% schedule completion. The

term commonly used is "Schedule Compliance". If compliance drops below 90%, you are over-scheduling. Reduce the percent of available resources scheduled. Always complete 90% or more of what is scheduled.

Any maintenance department that is scheduling 70% and completing 90% of that schedule has good planning and supervision, a good work order system, the proper organizational structure, an excellent PM/PdM program, and well-trained technicians—in short, Maintenance Excellence. It is not achieved by accident. Like a long freight train that slowly picks up speed and momentum, rescheduling drives the Reliability/Maintenance Excellence Management train to World-Class.

5. Analysis

When a job is not completed as scheduled, perform an "accident investigation":

- ☐ Was it poor planning? If so, review it with the Planner. Turn it into a teaching opportunity so he/she learns from the experience.
- ☐ Did a Maintenance Supervisor put off a scheduled job until it was too late within the week to complete it? Another teaching opportunity.
- ☐ Did the Plant Manager "pull-rank" and force a job or project that "bumped" a scheduled job? If so, recognize that he is the boss but point out how it hurt productivity and the importance of that even he keep "schedule-breaks" to a minimum.

If there are only one or two scheduled jobs that are not performed as scheduled, learn, correct, and improve! If you over schedule and fail to complete thirty jobs, there's no way to figure out what went wrong.

Author's Rebuttal

Remember that Pete is offering a transitional, start-up strategy. That strategy is largely consistent with the text of this book. There are five primary differences:

- ☐ The book advocates "full scheduling of all available maintenance resources." The schedule establishes a managerial expectancy for weekly output. Lack of a published expectancy is a major contributor to a reactive culture. Transition from reactive to pro-active culture requires bold change. A fully loaded schedule fits the need!
- ☐ In many facilities the various operating departments are vying for the same maintenance resources. There is need for the various "

customers” to be in a common meeting in order to reach mutual agreement regarding the “most important application of maintenance resources” for the common good of the overall enterprise.

- ☐ In most situations, the most important work to be performed requires downtime. They should be the first jobs scheduled.
- ☐ Supervisors should be at each day’s job sites, leading their teams. Planners should perform all preparation for next week, including scheduling.
- ☐ Even multi-skilled technicians are not clones. Individual skills and aptitudes must be taken into consideration when allocating labor resources to specific jobs. Otherwise, it is highly likely that schedules contain too much work for a specific few and far too little work for the many. In such cases, schedules are not realistic.

Despite these differences, Pete’s strategy may be a better fit for given situations. If so, **Go for It!** Commit to your belief and make it work.

APPENDIX F**TYPICAL SOURCES OF PLANNED MAINTENANCE WORK**

Sources of Planned Work	Percent of Total Planned Work
Results of PM inspections	30%
Scheduled component replacements	20%
Overhauls/rebuilds	15%
Internal customer input (Operators and Supervisors)	10%
Engineering project support	8%
Safety work	5%
Analysis of repair history	5%
Management directed work	4%
Service requests	2%
Accident damage	<u>1%</u>
Total Planned Work	100%



PLANNER ACTIVITY LOG SHEET

Name: _____

Date: _____

Time	Act. Code	Time	Act. Code	Time	Act. Code	Time	Act. Code	Time	Act. Code
5:30		8:05		10:40		1:15		3:50	
5:35		8:10		10:45		1:20		3:55	
5:40		8:15		10:50		1:25		4:00	
5:45		8:20		10:55		1:30		4:05	
5:50		8:25		11:00		1:35		4:10	
5:55		8:30		11:05		1:40		4:15	
6:00		8:35		11:10		1:45		4:20	
6:05		8:40		11:15		1:50		4:25	
6:10		8:45		11:20		1:55		4:30	
6:15		8:50		11:25		2:00		4:35	
6:20		8:55		11:30		2:05		4:40	
6:25		9:00		11:35		2:10		4:45	
6:30		9:05		11:40		2:15		4:50	
6:35		9:10		11:45		2:20		4:55	
6:40		9:15		11:50		2:25		5:00	
6:45		9:20		11:55		2:30		5:05	
6:50		9:25		12:00		2:35		5:10	
6:55		9:30		12:05		2:40		5:15	
7:00		9:35		12:10		2:45		5:20	
7:05		9:40		12:15		2:50		5:25	
7:10		9:45		12:20		2:55		5:30	
7:15		9:50		12:25		3:00		5:35	
7:20		9:55		12:30		3:05		5:40	
7:25		10:00		12:35		3:10		5:45	
7:30		10:05		12:40		3:15		5:50	
7:35		10:10		12:45		3:20		5:55	
7:40		10:15		12:50		3:25		6:00	
7:45		10:20		12:55		3:30		6:05	
7:50		10:25		1:00		3:35		6:10	
7:55		10:30		1:05		3:40		6:15	
8:00		10:35		1:10		3:45		6:20	

Activity Codes: DS = Daily Scheduling ME = Material Expediting MT = Meetings JP = Job Planning
 WP = Weekly Planning EN = Engineering PR = Problem Resolution
 MO = Material Ordering CO = Counseling FO = Schedule Follow-up
 AA = Avoidable Activity UD = Unavoidable Delay



219

APPENDIX H
DETERMINATION WORKSHEET
RATIO OF TECHNICIANS TO PLANNER/SCHEDULERS

	Points Assigned
Planning and Scheduling Structure:	
➤ Separate from material coordinating (vertical structure) - 1 point	
➤ Combined (horizontal structure) - 2 points	()
Number of Crafts Coordinated	
➤ One - 1 point	
➤ Two - 2 points	
➤ Three - 3 points	
➤ Four - 4 points	()
Level of Planning	
➤ Craft and general description with schedule - 1 point	
➤ Craft, general instructions, special tools, and major materials with schedule - 3 points	
➤ Craft, specific instructions, tools, materials, prints, and schedule - 5 points	
➤ All the above plus work methods described - 7 points	()
Level of Estimating	
➤ Estimates or historical averages - 1 point	
➤ Slotting against benchmarks or labor library - 3 points	
➤ Analytical estimating - 5 points	
➤ Engineered Standards - 7 points	()
Inappropriate Responsibilities (additive)	
Sourcing - 1 point	
Procuring - 1 point	
Expediting - 1 point	
Receiving - 1 point	
Stocking - 1 point	
Picking and kiting - 1 point	
Staging and securing - 1 point	
Delivery to scheduled job site - 1 point	()

Total Points: _____

1000

1000

1000

1000

1000

1000

1000

1000

1000

1000

1000

APPENDIX I ASSESSMENT - MAINTENANCE ORGANIZATION

This series of appendices (I through O) is extracted from a broad Maintenance Assessment Process covering all twenty-one essential building blocks depicted as "The Maintenance Arch" (Introduction). The complete process is available through the authors. Appendix section N has scoring information from the authors database.

Assessment : Organization Structure and Size

Success of maintenance operations begins with a sound and efficient organizational structure, properly designed and adequately staffed with competent personnel working together. The organization may be designed to reflect management's attitudes and objectives, or it can be allowed to evolve in response to requirements. The former approach is more readily controlled and can generally be expected to yield better results.

The Maintenance organization should be structured for responsiveness to facility needs while ensuring intrinsic functional effectiveness through the systemic use of labor, material, equipment, capital, and technological resources.

Response Statements

Numerical Assessment

1 ST	2 ND
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1. A current and complete organization chart is available and posted showing every position from the Maintenance Manager down, with reporting relationships, and includes supervisors, engineers, planners, clerks, mechanics, janitors, stock and tool attendants, guards, etc.

The table of organization is clear as to who is responsible for which crafts or areas of plant and all personnel know where they fit within the organization. Periodic reviews are made to determine if structure is still appropriate considering departmental obligations and objectives.

- Current, complete, effective and posted. (2 points)
- Available but lacking or not posted information. (1 point)
- Not available. (0 points)

() ()

2. Maintenance is an excellent career opportunity. It carries status, recognition and rewards sufficient to attract high caliber, young people on a career basis:

- Salaries, wages, and other rewards are sufficient to attract, retain, and motivate competent maintenance personnel (salaried and hourly). (1 point)
- Career opportunities elsewhere in the organization are not blocked by extended maintenance assignment. (1 point)

() ()

() ()

3. The organizational concept allows Maintenance personnel to function as efficiently as possible:

- There is a clear-cut separation between line and staff. The three (3) distinct functions within maintenance responsibilities (work execution, planning and scheduling, and maintenance engineering) are recognized and segregated, yet coordinated. As long as these duties are delegated, effectively performed, and posted, full credit is given even if the functions are not performed by different people.
 - Both planning and engineering separate from line (3 points)
 - Planning separate (2 points)
 - Engineering separate (1 point)
 - Line only (0 points) () ()
 - The line structure is responsive to the triple nature of maintenance work (regular and periodic routine responsibilities, response to emergency and urgent demands, and timely backlog relief). Similar and closely-related activities are collected into logical groups that facilitate the efficient execution of required work. () ()
 - PPM routines occur on schedule. (0-2 points) () ()
 - Legitimate emergencies are responded to promptly. (0-2 points) () ()
 - Backlog jobs are completed within requested timing. (0-2 points) () ()
- There is a realistic balance between popular concepts of participative management and traditional controls structure. (0-2 points) () ()
- The line structure balances resource decentralization and centralization. To minimize response time and maximize equipment availability, area crews are decentralized as far as practical, but functional shop crews are centralized to assure effective utilization of all resources, and timely backlog relief. Small plants normally do not fragment crews, therefore, all points are awarded if crews are centralized. (1 point) () ()
- Maintenance responsiveness can be characterized as:
 - Always prompt to job; immediately responds after one call. (4 points)
 - Usually responds immediately. (3 points)
 - Usually responds within a reasonable amount of time. (2 points)
 - Sometimes prompt; sometimes two calls required. (1 point)
 - Never prompt despite repeated requests. (0 points) () ()

- The concept of process teams with mutual support between Maintenance and Production personnel is part of organizational philosophy and direction. This is the future of operating/maintenance organizational structure, particularly for heavily automated process industries. (1 point) () ()
 - The number of organizational levels is at the necessary minimum. Decision can be made and carried out at the lowest appropriate level. Communication and team work are facilitated. (1 point) () ()
 - "High-Involvement" work methods (Quality Circles, Work Teams or Self-Management Groups) are evident throughout Maintenance. (1 point) () ()
4. Supervisors are not so overloaded with administrative or other duties that performance of their direct supervisory duties is hampered. (1 point) () ()
 5. Maintenance planning and scheduling is a formal and disciplined operation with one person in charge and with sufficient capacity; approximately one Planner/Scheduler for each twenty mechanic positions authorized. (0-2 points) () ()
 6. Coordination of outside contractors is also adequately provided for. Each million dollars of such contracts, on an ongoing annual basis, warrants an additional position (planner, contract administrator or project engineer). (1 point) () ()
 7. Sufficient maintenance engineering capacity exists, with approximately one maintenance engineer position (not necessarily degreed) for every forty craft positions. The primary mission of the maintenance engineer is to determine causes of failures and recommend corrective action. Only by engineering failures out of equipment can the maintenance function truly fulfill its mission and contribute toward improving profitability of the company. If enough maintenance engineering capacity exists, either primary or collateral duty, and the function is effectively performed then full credit will be given. (0-2 points) () ()
 8. Sufficient clerical support exists, with one clerk for every forty wage positions. This allows planners to spend their time planning, and supervisors supervising. Clerks are effective from the standpoint of cost, dexterity, and aptitude. (1 point) () ()

9. Supervisory span of control is within manageable limits; one direct supervisor for eight to twelve mechanics:
– Between 5 and 12 to 1 (2 points) () ()
– Between 13 and 15 (1 point)
– Above 15 or below 5 (0 points)
10. Hourly workers are adequately supervised on dark and weekend shifts. (1 point) () ()
11. Supervisory effectiveness is not diluted by policies or practices preventing direct instruction from foremen to workers. (1 point) () ()
12. An Area Supervisor plan is in effect and supervisors are given across-the-board responsibility in their areas regardless of the organizational source of needed skills or capacity. (1 point) () ()
13. All supervisors possess a copy of his own and their crew's job descriptions (up-to-date) as well as those of their immediate manager and others with whom they have close working contact:
– All of the above (2 points) () ()
– Several of the above (1 point)
– None of the above (0 points)
14. Positions with incumbents retiring within two years have been identified and potential replacements are undergoing appropriate training or an appropriate plan of action is in progress to prevent deterioration of mechanic skills and responsiveness. (1 point) () ()
15. The number of recognized crafts and jurisdictional agreements are suitable to plant needs. Non-union plants, omit jurisdictional agreements. (1 point) () ()
16. There are sufficient cross-trained mechanics and general helpers to balance craft workloads. (1 point) () ()
17. Routine and urgent repair work is adequately controlled:
– Area assigned mechanics are not excessive. If no area assigned mechanics, full credit is given. (1 point) () ()
– Fixed routine assignments charging to "standing" work orders are not excessive. (1 point) () ()
– Personnel allocated to emergency/urgent response do not exceed 20% of the total maintenance work force.(1 point) () ()
- Maintenance forces assigned to shifts without supervisors or lead men are sized to cover only emergency/urgent or routine work which is best performed by unsupervised shifts due to better equipment. (1 point) () ()

18. The Maintenance organization structure is supportive of daily operation, internal as well as external to the department.

- Communication channels and procedures are designed for clear and quick transmission of information. (1 point)
- Relationships with staff departments (such as Purchasing, Stores, Plant Engineering, and Accounting) are clearly defined. (1 point)
- Changes in systems, procedures, and people relations are readily assimilated. (1 point)

() ()

() ()

() ()

19. Operating departments have designated a specific individual as their maintenance liaison. (1 point)

() ()

20. There is an effective training program for Maintenance Planners:

- Formal classroom training (1 point)
- On-the-job training (1 point)
- Refresher training (1 point)

() ()

() ()

() ()

21. It encompasses:

- The Planner's role within the overall maintenance management program. (1 point)
- Job breakdown and analysis. (1 point)
- Materials Requirements determination. (1 point)
- Job estimating. (1 point)
- Weekly scheduling detailed by worker, by job, by day. (1 point)
- Communication, coordination and follow-up. (1 point)
- Expediting. (1 point)
- Planning for major shutdowns, turnarounds, outages. (1 point)
- Computer and/or CMMIS training. (1 point)

() ()

() ()

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() ()

() ()

() ()

() ()

() ()

() ()

Element Summary:

Score: Points Awarded / Potential Points
÷ Potential Points

$\frac{()}{60}$ $\frac{()}{60}$

= Current State of Effectiveness-Organization
(Carry to 3 places)

0. 0.

Points Lost: 60 Potential Points – _____ Points Awarded = _____



APPENDIX J

Assessment— Maintenance Computer Information Support

Fast, flexible access to current information is vital if managers are to effectively control the Maintenance function. Good manual control systems have partially fulfilled the informational need for years, but computer support is essential if the full potential of the Maintenance Control System is to be realized. Batch processing is no longer good enough. Major decisions hinge on the availability of current information.

Reliable computer power is essential in order to provide immediate comprehensive information, enabling managers to base their actions on information instead of intuition. Users need fast, flexible access to current information, which they can look at from several perspectives.

Only on-line transaction processing systems and networks—people and programs simultaneously retrieving and updating information—satisfy the immediacy today's environment requires.

Response Statements	Numerical Assessment	
	1 ST	2 ND
1. The maintenance management system is computerized:		
– On-line (2 points)		
– Batch (1 point)		
– Not computerized (0 points)	()	()
2. The computerized system integrates all maintenance informational needs:		
– Work Order Control (1 point)	()	()
– Work Planning (1 point)	()	()
– Work Measurement (1 point)	()	()
– Material Support and Control (1 point)	()	()
– Scheduling and Coordination (1 point)	()	()
– Preventive/Predictive Maintenance (1 point)	()	()
– Equipment History (1 point)	()	()
– Training Records and Scheduling (1 point)	()	()
– Budgetary Control (1 point)	()	()
– Cost Distribution (1 point)	()	()
– Management Reporting and Control (1 point)	()	()
– Supervisory/Administrative Support (1 point)	()	()
– Project Control (1 point)	()	()
– Downtime Reporting (1 point)	()	()
– Statistical Analysis (1 point)	()	()
– Trend Reporting (1 point)	()	()

3. The system satisfies real needs:
- The right maintenance information is available to all levels of the organization, consistent with their needs. (0-2 points) () ()
 - All personnel who need maintenance/material information are able to obtain it. (0-2 points) () ()
 - Information is complete and reliable. (0-2 points) () ()
 - Information and access to it, either on-line or in report format, is timely..(0-2 points) () ()
 - System capabilities align well with user responsibilities. (0-2 points) () ()
 - System security effectively controls who has access to specific system functions and levels. (0-2 points) () ()
 - Ad hoc report generation is available. (1 point) () ()
4. Record files are complete and easy to locate. (0-2 points) () ()
5. Monthly reports (i.e., budgetary) are promptly available relative to month-end:
- One day or less (2 points)
 - Two to five workdays (1 point)
 - More than one week (0 points) () ()

Element Summary:

Points Awarded	()	()
÷ Potential Points	35	35

= Current State of Effectiveness-Computer Support (Carry to 3 places)	0.	0.
--	----	----

Points Lost: 35 Potential Points - _____ Points Awarded = _____

APPENDIX K

Assessment- Maintenance Job Planning

Sound planning is a prime requisite to effective control of the maintenance function. Maintenance planning is the advance preparation for selected jobs so they can be performed efficiently. It is the process of analyzing each job to determine such factors as what work is to be performed; the specific steps and sequence required; and for each step, the skills, skill level, labor-hours, parts, materials, drawings, special tools, equipment required with estimated costs.

Backlog measurement is a key planning tool to ascertain the degree to which maintenance keeps up with the amount of work requested and to adjust the size and make-up of the work force accordingly.

Response Statements

Numerical Assessment 1ST 2ND

1. The flow of work orders from originator to maintenance and within maintenance is clearly established and understood by all concerned parties. There are no piles of unprocessed work orders at any point in the system. This applies to emergency and standing work orders as well as backlog work orders. Work orders in the various stages of processing and execution are systemically filed (manual or computerized). (0-2 points) () ()
2. The portion of each crew's capacity which is consumed by emergency response and routine duties, and which therefore is not available for backlog relief is known. (0-2 points) () ()
3. Backlog data by priority code are effectively used for macro planning of resource requirements.
 - ☐ At least once per period (3 points)
 - ☐ At least once per quarter (2 points)
 - ☐ On an ad hoc basis (1 point)
 - ☐ Not used (0 points) () ()
4. Planning predicts the workload of each skill group in sufficient detail to permit early identification of pending shortfalls, excesses, and imbalances; and in time to take appropriate corrective action. (0-2 points) () ()
5. Backlog data is effectively utilized:
 - ☐ Maintenance uses backlog to track how workload is changing. (0-2 points) () ()
 - ☐ The purpose of tracking work backlog is understood by all concerned parties. (0-2 points) () ()

- ☐ Through the use of work backlog, maintenance determines the needs for outside contractors.
- During normal operations. (1 point) () ()
 - During shutdowns. (1 point) () ()
 - On major projects. (1 point) () ()
6. "Ready Backlog" is distinguishable from backlog pending engineering, planning, material, shutdown, etc. (0-2 points) () ()
7. Backlog information is reliable and routinely utilized to make decisions regarding:
- Scheduling priority (1 point) () ()
 - Internal resources vs. contractor (1 point) () ()
 - Overtime (1 point) () ()
 - Revise equipment operating plan (1 point) () ()
 - Personnel reassignment or temporary help (1 point) () ()
8. How many weeks of "Ready Backlog" currently exist?
- ☐ 3.0 to 5.0 weeks (4 points)
 - ☐ 2.0 to 2.9 weeks (3 points)
 - ☐ 5.1 to 7.0 weeks (2 points)
 - ☐ Below 2.0 or above 7.0 weeks (1 point)
 - ☐ Unknown (0 points) () ()
9. How many weeks of "Total Backlog" currently exist, excluding annual turnabout?
- ☐ 6.0 to 8.0 weeks (3 points)
 - ☐ 4.0 to 5.9 weeks (2 points)
 - ☐ Below 4.0 or above 8.0 weeks (1 point)
 - ☐ Unknown (0 points) () ()
10. Policy specifies how much detailed planning of selected maintenance work will be done. A target has been established internally. (1 point) () ()
11. There is enough maintenance planning capacity that all work requiring planning can be planned. (0-2 points) () ()
12. The concept of maintenance planning is fully accepted by maintenance and operating supervision. (0-2 points) () ()
13. Planning procedures are well documented.
- ☐ All normal repetitive procedures are documented. (2 points)
 - ☐ At least half of repetitive procedures are documented. (1 point)
 - ☐ No documentation. (0 points) () ()

14. Incumbent planners have been effectively trained in their responsibilities.

- ☐ Combination of in-plant and outside training (2 points) () ()
- ☐ In-plant training only (1 point)
- ☐ No training (0 points)

15. Current systems readily determine current status of all open work orders:

- ☐ Completed work is identified. (1 point) () ()
- ☐ Scheduled work is identified. (1 point) () ()
- ☐ Ready Backlog is identified. (1 point) () ()
- ☐ Ready Backlog requiring downtime is identified. (1 point) () ()
- ☐ Waiting for shutdown is identified. (1 point) () ()
- ☐ Waiting for material is identified. (1 point) () ()
- ☐ Waiting for planning is identified. (1 point) () ()
- ☐ Waiting for engineering is identified. (1 point) () ()
- ☐ Waiting for approval is identified. (1 point) () ()

16. Backlog is well organized or easily retrievable (computerized systems).

Work orders can be located by:

- ☐ Assigned crew (1 point) () ()
- ☐ Priority (1 point) () ()
- ☐ Associated purchase order numbers (1 point) () ()
- ☐ Equipment (1 point) () ()
- ☐ Originator (1 point) () ()

17. Work is requested with sufficient lead time (2 to 4 weeks) to allow effective planning:

- ☐ Over 90% of the requests (5 points)
- ☐ 80% to 90% of the requests (4 points)
- ☐ 70% to 79% of the requests (3 points)
- ☐ 60 to 69% of the requests (2 points)
- ☐ 50 to 59% of the requests (1 point)
- ☐ Less than 50% of the requests (0 points) () ()

18. Coverage with planned work is high and is evident in the form of

"Ready to Schedule" work orders listing work content by craft,

materials, special tools and equipment, multi craft sequencing,

required man-hours, crew size, and job site access:

- ☐ Over 90% (5 points)
- ☐ 75% to 90% (4 points)
- ☐ 60% to 74% (3 points)
- ☐ 40% to 59% (2 points)
- ☐ 20% to 39% (1 point)
- ☐ Less than 20% (0 points) () ()

19. All shutdown work is preplanned.
- ☐ At least 90% of shutdown work (2 points)
 - ☐ Greater than 75% of shutdown work (1 point)
 - ☐ None (0 points) () ()
20. To improve control, large jobs (over 40 hours) are broken into logical segments with separate work orders for each. (0-2 points) () ()
21. Planners utilize effective methods, including:
- ☐ Discussion with originator to establish objectives. (1 point) () ()
 - ☐ Visiting job site to thoroughly determine job requirements. (1 point) () ()
 - ☐ Determining safety/tag out/permit requirements. (1 point) () ()
 - ☐ Discussion of job with maintenance supervision regarding necessity and correctness. (1 point) () ()
 - ☐ Analysis and breakdown of job into steps and labor requirements. (1 point) () ()
 - ☐ Determining all material requirements and preparing bills of material showing locations of reserved items. (1 point) () ()
 - ☐ Obtaining or preparing necessary prints and sketches. (1 point) () ()
 - ☐ Identifying and providing all special tools and equipment required to complete the job. (1 point) () ()
 - ☐ Coordinating shop work with field requirements. (1 point) () ()
22. A well-organized work order materials cage, under lock and key, contains necessary materials for all jobs which will be scheduled within the next few weeks. All materials are tagged with the work order number.
- ☐ Adequate security and control exists. (1 point) () ()
 - ☐ All materials are tagged with work order number. (1 point) () ()
23. Work measurement is part of the planning process. Accuracy is progressive.
- ☐ Measurement is used regularly to define backlog, schedule jobs, measure efficiency and refine planning. (2 points)
 - ☐ Measurement is applied but not effectively used for all purposes. (1 point)
 - ☐ No measurement is being done. (0 points) () ()
24. Reference files are maintained to minimize planner time demands associated with repetitive jobs and effort:
- ☐ Previous job plans. (1 point) () ()
 - ☐ Material libraries. (1 point) () ()
 - ☐ Labor libraries. (1 point) () ()
 - ☐ Standard bills of material for repetitive jobs, overhauls, and rebuild. (1 point) () ()

25. Policy specifies the characteristics of jobs which must be
costed prior to authorization. (1 point) () ()
26. Planners spend their day in essentially the following proportions:
- ☐ Site visitation (10-20%) (1 point) () ()
 - ☐ Job breakdown (10-15%) (1 point) () ()
 - ☐ Estimating (5-10%) (1 point) () ()
 - ☐ Material requirements (10-20%) (1 point) () ()
 - ☐ Material expediting (0-10%) (1 point) () ()
 - ☐ Inter/intra-coordination (10-20%) (1 point) () ()
 - ☐ Scheduling (10-20%) (1 point) () ()
 - ☐ Follow-up and reporting (5-15%) (1 point) () ()
 - ☐ Administration (0-10%) (1 point) () ()
27. The use of formal planning has successfully reduced downtime
durations. (0-2 points) () ()
28. Maintenance crews prefer to work on jobs which have been planned.
(0-2 points) () ()
29. Maintenance supervisors substantiate planning effectiveness.
(0-2 points) () ()

Element Summary:

Points Awarded () ()
÷ Potential Points 100 100

= Current State of Effectiveness-Work Planning 0. 0.
(Carry to 3 places)

Points Lost: 100 Potential Points - _____ Points Awarded = _____



APPENDIX L

ASSESSMENT—MAINTENANCE WORK MEASUREMENT

Work measurement is essential to the entire planning, scheduling and control process. Sophistication of work measurement technique varies with the maturity of the maintenance management installation.

Response Statements	Numerical Assessment	
	1 ST	2 ND
1.0 Work standards/estimates for maintenance work are established by:		
☐ Engineered job standards. (5 points)		
☐ Averages or estimates adjusted by work sampling. (4 points)		
☐ Analytical estimates. (3 points)		
☐ Historical averages. (2 points)		
☐ Supervisor/planner estimates. (1 point)		
☐ No measurement applied. (0 points)	()	()
2. New jobs are estimated using slotting techniques in comparison to engineered benchmarks. (1 point)	()	()
3. Estimates are automatically posted to repetitive jobs by the computer. (1 point)	()	()
4. Planners can recover their logic and buildup of established standards. (1 point)	()	()
5. Work measurement coverage is high:		
☐ Over 85% of jobs (5 points)		
☐ 70% to 84% (4 points)		
☐ 50% to 69% (3 points)		
☐ 30 to 49% (2 points)		
☐ 1 to 29% (1 point)		
☐ No work measurement (0 points)	()	()
6. Standards/estimates are pre-applied but when feedback indicates a change of work scope, adjustments are post-applied. (1 point)	()	()
7. Work measurement is a vital aspect of the managerial control process:		
☐ The size and makeup of the maintenance work force has been determined by careful measurement of the type and amount of work to be done—using work programs or a similar analytical format. (1 point)	()	()
☐ Management requests measurement of workload and required work force before force adjustments (up or down) are made. (1 point)	()	()

- ☐ Performance reports are regularly issued for the total department and each supervisor. (1 point) () ()
- ☐ Standards/estimates are effectively used to help control major jobs. (1 point) () ()
- ☐ The maintenance work force understands and accepts the purpose of periodic manpower productivity measurement. (1 point) () ()
8. Records are maintained to facilitate modification of standards/estimates as needed. Comparisons of actual to estimated hours are regularly made and analyzed. One group of estimates, or estimates by one planner may be inconsistent with others. (0-2 points) () ()
9. Methods and standards are integral to the maintenance management program:
- ☐ There is sufficient confidence in and/or understanding of the standards/estimates that they are posted to the crew copy of the work order. (1 point) () ()
- ☐ A full-scale methods effort is in place, with field review of job activities by planners, engineers, and supervisors. (1 point) () ()
- ☐ Responsibility for researching and selecting best methods, tools, materials, safety requirements and standards/estimates is specifically assigned to one or more persons. (1 point) () ()
- ☐ Proper tools and machine feeds and speeds are established for Machine Shop and Tool Room jobs. (1 point) () ()
- ☐ Work sampling and other productivity studies focus attention on the elimination or reduction of delays, non-productive time, and productivity losses incurred for various reasons – including contractual agreement interpretation. (1 point) () ()
- ☐ Shutdowns and turnarounds are studied to find means to reduce their cost and duration, thereby improving on-stream time and profitability. (1 point) () ()
10. Maintenance is covered by an incentive program:
- ☐ Plant-Wide Plan on Overall Operating Results (3 points)
- ☐ Small Group Plan (2 points)
- ☐ Individual Plan (1 point)
- ☐ No Plan (0 points) () ()

Element Summary:

Points Awarded () ()
 ÷ Potential Points 30 30

= Current State of Effectiveness-Work Measurement 0. 0.
 (Carry to 3 places)

Points Lost: 30 Potential Points- Points Awarded =

APPENDIX M

ASSESSMENT—MAINTENANCE MATERIAL SUPPORT & CONTROL

For maintenance to fulfill its mission, materials must be available when needed. Material availability and control of inventory can be conflicting objectives without managerial policy which best satisfies the overall needs of the plant, establishes a system for optimization, and calls for continual review and control.

Response Statements	Numerical Assessment	
	1 ST	2 ND
1. Purchasing, inventory and stores procedures covering stock, spare parts, special purchases, and "in-house" manufactured items are well documented by written instructions which have been effectively distributed to the involved organization(s). (1 point)	()	()
2. The supply room is sufficiently stocked to meet day-to-day needs as determined by the following service levels:		
☐ Insurance Spares (100%) - (1 point)	()	()
☐ Other Critical Spares (98%) - (1 point)	()	()
☐ Standard Replacement Parts (95%) - (1 point)	()	()
☐ Fasteners and Fittings (90%) - (1 point)	()	()
☐ Small Tools (90%) - (1 point)	()	()
☐ M&R Supplies (85%) - (1 point)	()	()
3. There is a perpetual inventory system in place with activity systematically recorded and reported to maintenance, purchasing and accounting for use in the management of inventory.		
☐ Computerized (2 points)		
☐ Manual (1 point)		
☐ No system (0 points)	()	()
4. Inventory control practices are effective:		
☐ Control increases with inventory value using A-B-C categorization. (1 point)	()	()
☐ Cycle counting is used to preserve reliability of inventory records. (1 point)	()	()
☐ Free issue of low-value items, such as fasteners, is effectively Integrated into the system. (1 point)	()	()
5. An intelligent part numbering system is utilized. (1 point)	()	()
6. Usage records are employed to determine stocking levels, order points, and order quantities. (1 point)	()	()

7. Maintenance, Purchasing, Accounting, and the Supply Room work together to assure availability of necessary parts, elimination of obsolete parts, adjustment of stocking levels, minimal lost time by craftsmen, etc. (1 point) () ()
- ☐ Maintenance is regularly consulted regarding stocking parameters generated by the system. (1 point) () ()
 - ☐ Human judgment can override the system on an item-by-item basis. (1 point) () ()
 - ☐ Maintenance influences the type and quantity of repair materials to be set up in stock. (1 point) () ()
 - ☐ A system exists which flags frequently ordered parts which must be authorized for stocking. (1 point) () ()
 - ☐ Stocking levels (min/max), reorder points and order quantities are set up and maintained:
 - Automatically (2 points)
 - Manually (1 point)
 - Not set up (0 points) () ()
8. There is an established procedure to review obsolescence, scrap, excessive quantities, etc. (2 points) () ()
- ☐ Zero and low activity items are reviewed at least once a year. (1 point) () ()
 - ☐ Stores does not remove maintenance items from authorized stock without Maintenance agreement. (1 point) () ()
9. Inventory categories are defined and segregated when analyzing inventory value and turns. (1 point) () ()
10. Inventory turns by category are reasonable. (1 point) () ()
11. Management supports the Scientific inventory Control (SIC) formula to optimize inventory management. Artificial decisions and targets are not allowed to destroy the system. Order points and quantities are calculated on an economic basis. (1 point) () ()
12. SIC formula parameters for EOQ calculation (carrying cost, purchase order cost, delivery time, etc.) are current and periodically updated. (1 point) () ()
13. A computerized inventory control system automatically reorders when on-hand quantity drops to or below reorder point. (1 point) () ()
14. There is an established insurance/critical spares program containing review provisions. They are protected from discard due to low usage. (1 point) () ()
15. Withdrawal procedures are established and enforced. (0-2 points) () ()

16. An approved work order indicating which cost center to charge is sufficient authorization to withdraw stock materials. (1 point) () ()
17. Studies are periodically made of stock outages to keep tabs on levels and trends. (1 point) () ()
18. Maintenance has ready access to up-to-date material and stock information, including: part numbers, stock numbers, bin locations, economic order quantities, reorder points, etc.
- ☐ Computerized (2 points) () ()
 - ☐ Cataloged (1 point) () ()
 - ☐ No Ready Access (0 points) () ()
19. Bill of Material information is cross-referenced by point-of-use. (1 pt) () ()
20. Attribute scanning on key words of the description are available to help find the desired items. (1 point) () ()
21. Procedures exist for reserving and kiting stock parts for planned jobs:
- ☐ Reservation/Allocation (1 point) () ()
 - ☐ De-Allocation (1 point) () ()
 - ☐ Kiting or Pre-staging (1 point) () ()
 - ☐ Secured Staging to Avoid "Borrowing" from One Job for Another (1 point) () ()
22. Unit dollar values are readily available and routinely used in planning and work order approval efforts. (1 point) () ()
23. The cost of material consumed in the repair of specific units of equipment can be easily obtained. (1 point) () ()
24. Critical spare part lists exist for all important equipment. (1 point) () ()
25. There are control procedures in use for all company-owned tools and supplies, such as drills, special saws, ladders, files, gloves, etc.
- ☐ Such items are identified. (1 point) () ()
 - ☐ Kept in designated places. (1 point) () ()
 - ☐ Checked in and out of secured tool crib to authorized personnel using a tool check or similar system. (1 point) () ()
26. Special tools are kept in good repair and in service. (1 point) () ()
27. There are standard listings of those tools which are company provided and those which are employee provided. (1 point) () ()

28. Satellite inventories of shop-based materials, such as fasteners, O-rings, lumber, and so forth:
- ☐ Exist only by plan. There are no hidden satellite stocks throughout the plant. (1 point) () ()
 - ☐ Are under Stores control procedures. (1 point) () ()
 - ☐ Are located close to point-to-use. (1 point) () ()
 - ☐ Are limited to low value, fast-moving items. (1 point) () ()
29. Adequate material handling resources are available to stores to and from work sites. (1 point) () ()
30. There are also provisions for the mobilized delivery of urgent material needs (i.e., policies relieve mechanics from the necessity to leave their job site to obtain parts or material while engaged in urgent repairs), thereby reducing material-related travel time by the crew involved in work performance. (1 point) () ()
31. Maintenance takes necessary lead time into consideration when requisitioning:
- ☐ Stores Items (1 point) () ()
 - ☐ Purchased Items (1 point) () ()
32. Practices and procedures between Maintenance and Purchasing are clear and effective. (1 point) () ()
- ☐ Maintenance is allowed to specify individual manufacturers or suppliers when deemed necessary. (1 point) () ()
 - ☐ Maintenance is allowed to specify method of shipment when necessary. (1 point) () ()
 - ☐ Purchasing reviews any change in original specs with Maintenance and/or Engineering. They do not make material or equipment substitutions without consulting technically knowledgeable resources. (1 point) () ()
 - ☐ Purchasing routinely, and when requested, follows up and expedites purchase orders. (1 point) () ()
 - ☐ There is an effective system for tracking Purchase Requisitions from generation to receipt by originator. (1 point) () ()
 - ☐ The volume of Emergency Purchase Orders is "reasonable". (1 pt) () ()
 - ☐ Production is not permitted to purchase direct and to maintain duplicate inventory of unit spares of special parts. (1 point) () ()
 - ☐ Maintenance is permitted to purchase direct locally when necessary. (1 point) () ()
 - ☐ When appropriate, Maintenance is permitted to contract work to local shops with Purchasing's cooperation. (1 point) () ()

33. Blanket and system contracts, and blanket and system orders are effectively employed to minimize redundant paper work and administrative effort:

- | | |
|---|---------|
| ☐ Blanket Contracts (i.e. repetitive service) (1 point) | () () |
| ☐ System Contracts (i.e. service contractors negotiated by GO) (1 point) | () () |
| ☐ Blanket Orders (i.e. parts price agreement) (1 point) | () () |
| ☐ System Orders (i.e. parts costs negotiated by GO) (1 point) | () () |

34. Support systems for purchasing are effective:

- | | |
|---|---------|
| ☐ Quality performance measures have been established for materials. (1 point) | () () |
| ☐ Purchase Order tracking is functional. (1 point) | () () |
| ☐ A vendor rating system is in place and vendors are aware of their responsibility to provide materials of acceptable quality on time. (1 point) | () () |
| ☐ Procedures are in place which alert requisitioners regarding receipt of their materials. (1 point) | () () |
| ☐ Buyers are evaluated on the performance of vendors which they select. (1 point) | () () |
| ☐ There is an established interface/partnership with vendors to improve material quality, cost, availability, and standardization at minimal carrying expense. (1 point) | () () |

35. There is an effective salvage, reclamation and unit repair program in place and operated as a profit center. (1 point) () ()

- | | |
|--|---------|
| ☐ Lowest cost to the company governs the repair/replace decision. (1 point) | () () |
| ☐ Inventory controls are adequate & are under Stores control. (1 pt) | () () |
| ☐ Storage is adequate. (1 point) | () () |
| ☐ Reclaimed items are returned to Stores' custody. (1 point) | () () |
| ☐ Stores accepts returned items in less-than-unit quantities for credit. (1 point) | () () |
| ☐ Work Orders for the salvage/repair of a replaced part is generated by the technician who replaced it. (1 point) | () () |

36. There is appropriate storage provision for left over materials (short lengths, etc.), scrap, and waste. (1 point) () ()

37. A two-bin system is utilized by Stores to facilitate issue while maintaining control on high-volume, low-cost items. (1 point) () ()

38. Storeroom layout and procedures maximize order fulfillment service and functional efficiency:

- ☐ Parts are identified with stock (computer ID) numbers. (2 points) () ()
- ☐ Stock storage is environmental controlled as needed. (1 point) () ()
- ☐ If needed, rotating equipment in stock is regularly rotated to avoid development of bearing flat spots. This is controlled by a PM-type revolving file. (1 point) () ()
- ☐ ABC analysis has been applied to the arrangement of stock storage to optimize efficiency of stock retrieval. (1 point) () ()
- ☐ Stock pick lists are arranged to facilitate picking. (1 point) () ()
- ☐ Computer control will accept no more than two storage locations for any one item in any given storeroom. The impact of this is to keep inventory from becoming too spread out and impossible to control. (Note: MP2 allows more than two storage locations; therefore award no points) (1 point) () ()
- ☐ QA inspection is required, in varying degrees based on criticality, for all materials received for stock or special orders. (1 point) () ()
- ☐ Bar coding is utilized to facilitate tracking of Purchase Orders, Stock Requisitions, parts, and progress through the receipt/inspection/stocking/issue process. (1 point) () ()

39. In Central Stores, parts are stored randomly to maximize storage density. (1 point)

- ☐ In satellite stores, parts are stored by equipment to facilitate retrieval. (1 point) () ()
- ☐ Cannibalization from a serviceable unit (generally not installed) to another serviceable unit (generally installed) is against established policy. When necessary in an emergency situation, the necessary work orders, Stores requisitions, and/or purchase orders are initiated to restore the cannibalized unit to a serviceable state. (1 point) () ()

Element Summary:

Points Awarded () ()
 ÷ Potential Points 100 100

Current State of Effectiveness—Material Support 0. 0.
 (Carry to 3 places)

Points Lost: 100 Potential Points - ____ Points Awarded = ____

APPENDIX N

ASSESSMENT—MAINTENANCE SCHEDULING & COORDINATION

Scheduling and associated coordination is the process by which all required resources are made available at the proper time and place, with necessary access, so that work can be started and processed to completion with minimal delay and within established time frames. The established time frames reflect available manpower and the relative importance of work to be performed. Resources encompass manpower, materials, tools, equipment, and reference materials. Access refers to equipment in accessible and safe (locked out) state, including any necessary permits.

Response Statements	Numerical Assessment	
	1 ST	2 ND
1. The criticality of effective scheduling is recognized and supported throughout the organization. (0-2 points)	()	()
2. Scheduling procedures and timing are well documented and distributed. (0-2 points)	()	()
3. Work activity is analyzed by priority code to determine trends of man-hours spent on scheduled work. (0-2 points)	()	()
4. The relationship of backlogged work and backlog relief experience is regularly analyzed and reviewed with management.		
☐ At least monthly (2 points)		
☐ At least quarterly (1 point)		
☐ Never or rarely reviewed (0 points)	()	()
5. No more than 20% of the backlogged work orders on the MP2 Work Order Aging Report are over 60 days old. (1 point)	()	()
6. The system of work prioritization effectively distinguishes between legitimate rush jobs, and those which can be planned.		
☐ The system for prioritizing work orders has been thoroughly developed in consultation with operating personnel. (0-2 points)	()	()
☐ The system reflects equipment importance as well as individual job importance. (0-2 points)	()	()
☐ All work orders are assigned a priority code. (0-2 points)	()	()
☐ Work order age is also considered in the process of selecting backlog to be scheduled. (0-2 points)	()	()
☐ Operating/Maintenance liaisons are individualized and effectively utilized to establish finite priorities prior to each scheduling period. (0-2 pts)	()	()
☐ There is seldom any confusion about the relative priorities of what work Maintenance should be doing. (0-2 points)	()	()

- ☐ Originators are not allowed to "cry wolf" to get their work performed more quickly. The Plant Manager supports this policy and the Maintenance manager polices it. (0-2 points) () ()
- ☐ The Plant Manager and other senior members of management do not abuse the privileges of rank to override the system. (1 point) () ()
7. Scheduled job start, completion and crew assignments are published and distributed. (0-2 points) () ()
8. Schedule performance is measured and reported. Compliance and percent of man-hours scheduled are calculated, analyzed, plotted, published, and regularly reviewed by management. (1 point) () ()
- ☐ Both graphs and data are shared or posted within the department and with production plus other customers. (3 point)
- ☐ Either graphs or data are shared or posted within the department and with production plus other customers. (2 point)
- ☐ Shared or posted only within the department. (1 point)
- ☐ Not posted or shared; only required KPI report is sent to GO. (0 pts) () ()
9. Percent of labor resource scheduled is high, include PLANNED and PPM work only:
- ☐ 90% to 100% (10 points)
- ☐ 85% to 89% (9 points)
- ☐ 80% to 84% (8 points)
- ☐ 75% to 79% (7 points)
- ☐ 70% to 74% (6 points)
- ☐ 60% to 69% (5 points)
- ☐ 50% to 59% (4 points)
- ☐ 40% to 49% (3 points)
- ☐ 30% to 39% (2 points)
- ☐ 1% to 29% (1 point)
- ☐ No measurable scheduling (0 points) () ()
10. Schedule compliance, substantiated by the KPI report, is high:
- ☐ 90% to 100% (10 points)
- ☐ 85% to 89% (9 points)
- ☐ 80% to 84% (8 points)
- ☐ 75% to 79% (7 points)
- ☐ 70% to 74% (6 points)
- ☐ 60% to 69% (5 points)
- ☐ 50% to 59% (4 points)
- ☐ 40% to 49% (3 points)
- ☐ 30% to 39% (2 points)
- ☐ 1% to 29% (1 point)
- ☐ No measurable scheduling (0 points) () ()

11. Procedures for requesting, coordinating, and controlling supporting services such as mobile equipment, rigging, transportation, etc., are effective. (0-2 pts) () ()
12. Maintenance schedules are closely coordinated with Production schedules, and availability dates for repair of Production equipment are agreed upon in advance.
- ☐ Initial and final coordination each week. (3 points)
 - ☐ Weekly (2 points)
 - ☐ Occasionally (1 point)
 - ☐ No coordination or agreement on schedules (0 points) () ()
13. There are effective procedures for assigning work to specific mechanics. (0-2 points) () ()
- ☐ Where a work assignment board is appropriate, the supervisor maintains enough work in each mechanic's slot to keep them gainfully assigned. (0-2 points) () ()
14. Those procedures include consistent effort to assign mechanics to work which matches their skill classification and which is supportive of the OJT needs. (0-2 points) () ()
15. All types of backlog work (repair, changes, and additions) are scheduled but only those jobs with all material on hand appear on the current schedule. (0-2 points) () ()
16. The weekly schedule is reviewed, finalized, and approved during a joint maintenance/production meeting held late in the week so that both weekend work and next week's work can be addressed. Production liaisons, Maintenance Supervisors, and Maintenance Planners attend regularly. Others attend as warranted. Schedules are written, approved by the Plant Manager or designee, and then distributed. (0-2 points) () ()
17. Production and Maintenance diligently work together toward high schedule compliance:
- ☐ Production and Maintenance reach agreement on the weekly schedule before work is performed. (0-2 points) () ()
 - ☐ Production consistently makes equipment available according to the approved schedule. (0-2 points) () ()
 - ☐ Maintenance continues to coordinate throughout the schedule week to assure that all agreements and arrangements are still valid. Nothing is assumed even though the schedule has been approved. (0-2 points) () ()

- ☐ Maintenance Supervisors make every effort to follow schedule every day. (0-2 points) () ()
- ☐ Jobs-in-progress are reviewed against the schedule every day. (0-2 points) () ()
- ☐ Causes of deviation from schedules and deviation from estimates are determined and addressed. (0-2 points) () ()
- ☐ Production personnel do not assign work to Maintenance craftsmen. Even craftsmen assigned to certain production are as receive regular instructions from their Maintenance Supervisor, and respond to production requests only in the event a machine goes down. (0-2 points) () ()
- ☐ Analysis is made of work orders to determine the degree to which priority is abused. (0-2 points) () ()
- ☐ Analysis is made of completed work orders to determine the degree to which the priority implied completion date is met. (1 point) () ()

18. Production, production scheduling, and maintenance all contribute to adherence to the PM schedule.

- ☐ Production notifies Maintenance, with as much advance notice as possible, of any maintenance or other downtime (over a few minutes) which occurs due to changes in the production schedule. (0-2 pts) () ()
- ☐ Unexpected equipment shutdowns. (0-2 points) () ()
- ☐ Production planning is totally involved by scheduling production equipment downtime for PM. They take advantage of product changeovers and other necessary outages to help keep the PM program on schedule with the least demand on near-term production capacity. (0-2 points) () ()
- ☐ Maintenance, with notification by Production, and through use of backlog by machine, takes advantage of downtime to do any possible jobs, including backlog and PM inspections. (0-2 points) () ()
- ☐ Schedule provision is made to take advantage of those "windows". Schedule compliance does not suffer. Plan "B" is accomplished, rather than Plan "A". (0-2 points) () ()

19. The Maintenance organization recognizes and accepts accountability (partial) for meeting Production schedules. (0-2 points) () ()

20. Maintenance management regularly attends and contributes to plant operating and planning meetings. (0-2 points) () ()

21. Peak loads and resource "valleys" are balanced:

- ☐ Shutdowns and/or other major projects are planned sufficiently in advance to permit effective allocation or procurement of manpower, materials, and tools. (0-2 points) () ()
- ☐ Concurrent peak loads (overhauls and construction) are avoided if at all possible. (1 point) () ()
- ☐ Vacation schedules are determined far ahead, and in keeping with maintenance planning of known peak loads. (1 point) () ()

Element Summary:

Points Awarded () ()
 ÷ Potential Points 100 100

Current State of Effectiveness-Coordination and Scheduling 0.____ 0.____
 (Carry to 3 places)

Points Lost: 100 Potential Points - _____ Points Awarded = _____



APPENDIX O

ASSESSMENT – SUMMARY & COMPARISON TO BENCHMARKS

	Organ.	Comp. Supp.	Planning	Measurement	Material Supp.	Scheduling	Composite
Potential Pts.	60	35	100	30	100	100	425
Pts. Awarded							
Score *							
Data Base Low	0.180	0.000	0.006	0.000	0.130	0.090	0.078
Data Base Avg	0.625	0.500	0.382	0.288	0.584	0.418	0.475
Data Base High	0.950	0.929	0.850	0.717	0.890	0.880	0.877
Proactive Target	0.700	0.750	0.700	0.600	0.700	0.800	0.720
World Class Goal	0.850	0.850	0.800	0.750	0.800	0.850	0.820

*Score = Points Awarded/Potential Points



Glossary

Maintenance preparatory terms (planning, estimating, procurement, coordination, and scheduling).

Autonomous Maintenance: Routine maintenance and PMs are carried out by operators in independent groups. These groups, which may include maintenance workers, solve problems without management intervention. The maintenance department is only officially called for bigger problems that require more resources, technology, or downtime.

Asset: A machine, building, or system.

Asset number: A unique number necessary to identify an asset.

Backlog: All plannable work not yet complete.

Call Back Job: Job to which maintenance must return because the asset failed again for the same reason because the job was not performed properly the first time.

Charge rate: This is the rate in dollars charged for a mechanic's time. Rates used may be base direct wages or fully loaded to include benefits and overhead (such as supervision, clerical support, shop tools, truck expenses, supplies). Rates used may be average for a maintenance group or specific for a given mechanic.

Corrective maintenance: Restorative maintenance activity initiated as a result of finding during a scheduled inspection.

Deferred maintenance: This is known work that should be performed but is put off indefinitely, usually due to budgetary constraints.

DIN crew: Do It Now. A portion of the maintenance crew designated to perform urgent work.

Emergency work: Maintenance work requiring immediate response from the maintenance staff. Usually associated with some kind of danger, safety, damage, or failure of critical production equipment.

Failure Code: Why did the part fail (broken, worn through, bent, etc.)?

Feedback: Information provided by assigned mechanic or supervisor that will enhance future planned job packages.

Inspectors: Those technicians with primary responsibility for PM/PdM inspections.

Intrusive task (also interruptive task): Any inspection that interrupts normal operation of an asset.

MTBF: Mean time between failures. Metric used to evaluate reliability of an asset.

MTTR: Mean time to repair. Metric used to evaluate the effectiveness of job preparation. A trend line is necessary to convey improvement.

Maintenance: The dictionary definition is "the act of holding or keeping in a preserved state."

Maintenance Prevention: Maintenance-free designs resulting from increased effectiveness in initial design of equipment.

PCR: Planned Component Replacement based upon known life of component.

PM: Preventive Maintenance. A series of tasks to extend the life of an asset or to detect that an asset has reached a point of critical wear and is about to fail.

PM Clock: The parameter that initiates a PM routine to be scheduled.

PM frequency: Time interval driving PM inspections.

PdM: Predictive Maintenance techniques that inspect asset condition relative to a known point of failure, thereby predicting when failure will occur.

Priority: The relative urgency of a job.

Proactive: Action before a stimulus. Preparation for effective execution prior to failure.

Root cause (and root cause analysis): The underlying cause of a problem.

Routine work: Work of a known content at a known frequency.

Task: Single item in a step-by-step procedure.

Technical Library: Repository of all maintenance information filed by asset; including maintenance manuals, drawings, repair history, vendor catalogs, shop manuals, etc.

Work Order: Written authorization to proceed with a repair or other activity to be performed by the maintenance organization.

Work request: Formal request to have work performed. Subsequently transformed into a Work Order upon authorization.

Index

- ABC analysis 127–130
- activities 244–245, 248, 277
- adjusted averages 145–146
- administrators 58, 101
- allowances 156–160
- analysis 213–214, 273
- analytical estimating 151–169
- assembly 111–112
- assessment 104–105, 281–309
- averages 144–146

- backlog, 38, 39, 69–76, 78–81, 89–90, 147, 174
- benchmarking 142, 309
- benefits 19–23
- bill of materials 61, 126
- budget 24, 233–234
- building maintenance ratio 85

- calendar 82
- capacity 35, 61
- capacity assurance 2
- capital programs 91
- close out 212–213, 240
- CMMIS 9, 37, 58, 65, 91, 98, 135, 211, 219–225
- communication 47–49, 255–256
- compliance 30, 178, 214–218, 237–240
- computer information support 287–288
- computerized maintenance management information system see CMMIS
- consignment 131
- construction trade estimates 143
- contingencies 189
- contractors 234
- control spans 55–56, 59–60
- coordinated maintenance 14
- coordination 30–31, 47–49, 99, 171–180, 272, 303–307
- costs vii
- craftsmen 44
- critical path analysis 184, 232, 243–250
- criticality 174
- cycle counting 130–131

- database 27–28
- deferred maintenance 3–5, 71, 90–91
- deliverables 23
- direct liaisons 48
- direct work 77
- documents 234–235
- downtime 13
- duration 192

- economic order quantity (EOQ) 124
- effectiveness 76, 218
- efficiency 147–148
- emergency group 38–39
- engineers 57–58
- environment, proactive 25–26
- equipment 113
- equipment history 108
- equipment maintenance ratio 85
- estimating 151–169
- events 244–245, 248
- excellence 69, 77,
- execution 112, 197–198

- fixed provision 153
- float 248
- functional control spans 55–56
- functional responsibility 50

- Gantt Bar 185
- goals 263
- gross estimates 143

- historical averages 144–145

- indirect liaisons 48
- informational database 27–28
- inspections 87–89, 92, 101
- inventory 121–122, 124, 126

- JIT 125–127
- job assessment 104–105
- job control 221
- job creep 106
- job execution 31, 203–209

job feedback 211–212
job load 188, 194–195
job package 111–112
job planning 18–19, 101–117, 159, 231–232, 289–293
job preparation 51
job requirements 109–110
job schedule 188
job sequence 152
job slotting 162–163
job status 73
job steps 109–110
job supervision 203–209
just-in-time 125–127

kick off meeting 228–229

labor 188
labor library 154–156
lead time 123–124
liaisons 48–49
Little, Pete 267–274
log sheet 277

maintenance 67
 activities 35–40
 calendar 82
 costs vii
 excellence 7–10
 manager 41
 schedule 180
 staffing 83–94
 status vii–xvi
 support services 41, 53, 253–254
major projects 227–242
master schedule 81–82
materials coordinator 51
materials library 98
materials management 61–62, 121–125
materials support 28–29, 119–136, 28–29, 297–302
measurement 137–149
meetings 71, 173–180, 208–209, 228–229, 268–273

national contracts 131–132
necessity 119–121
network analysis 243–250

operating schedule 175
organization 26–27, 35–40, 42, 271, 281–285

overtime 72

Pareto's Law 127–130
performance 218, 263
PERT 243
planned work 38
planners 42–43, 58–62, 65–66, 101, 116–117, 173–174, 259–264, 279
planning 30–31, 41–35, 66–67, 101–117, 227–242, 271–272
PM/PdM 36–37, 39, 69, 70, 76, 81–82, 87–90
preparation 13, 25–34, 43–45, 51, 82, 95–100, 112, 197
prioritization 103–104
proactive maintenance 4, 8, 11–12
process 114–115
procurement 98
production 180
productivity 200
project evaluation review technique 43
project team 229–233
provisional schedule 193
purchase order requests 98, 135
purchasing 23

rate of use 123
reactive maintenance 2–7, 11, 14, 61
rebuildable parts 136
relationships 263–264
reliability 3, 7–10, 29–30, 62–63, 67
reorder point 124
replacement cost 85
required maintenance 92
research 107–108
reserve stock 124–125
resource demand 233
responsibilities 55–63, 135–136, 203–208, 264
review 240
roles 55–63, 135–136
root cause failure analysis 30
routine crew 38

safety 113–114
safety stock 124–125
schedule compliance 37, 178, 214–218
scheduled maintenance 14
schedulers 58–62, 259–264, 279
schedules 81–82
scheduling 30–31, 49–52, 147, 181–202, 267–274, 303–307

scoping 104–105
screening 101–104
shutdown 240
SIC 123–125
slotting 161–167
staffing 83–94
statistical inventory control 123–125
status codes 74
statutory permits 115–116
stock keeping units (SKUs) 123, 126
stores 23
supervision 203–209
supervisors 44–45, 56–57, 102, 203–208
support services 41–42, 53, 67, 253–254

team leaders 44–45
temporary staffing 92–94
time estimating 160–167
timelines 10, 12, 204
travel time 154
trends 80

urgent response 90

validation 75
vendors 131–135
visualizing 117

weekly coordination meeting 173–180
weekly schedule 99–100, 185–187
work flow 221
work measurement 137–149, 151–169.
295–296
work orders 37, 74, 101–102
work preparation 43–45, 95–100
work programs 77–82
work request 27–28, 101–104
workload 83–84, 88
world class maintenance 95

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